Volume I.
Introduction.

We presuppose, in our considerations, the natural numbers $1, 2, 3, \ldots$ and the rules according to which calculations are made with these numbers as known and given. The fundamental arithmetical operations, the so-called four species, are addition, multiplication, to which, as repetition, exponentiation belongs, subtraction, and division. The first two are called the direct operations of calculation; they are distinguished by the fact that in the realm of the natural numbers they can be carried out without restriction. The first of the indirect or inverse operations, subtraction, can be carried out only when the minuend is greater than the subtrahend.

The problem of division can be understood in two ways. In the first elementary interpretation one asks how often the divisor is contained in the dividend. A number is not contained in a smaller one. But if the dividend is equal to or greater than the divisor, then the answer to the question gives a quotient and, in most cases, a remainder which is smaller than the divisor. If no remainder is left, one says that the division comes out exactly, or that the dividend is divisible by the divisor, or that the divisor is a factor or divisor of the number which forms the dividend.

This problem, which already presents itself at the first stages of the art of calculation, leads to a more deeply lying distinction among numbers, which is the foundation of all number theory.

Since a factor can never be greater than the number of which it is a factor, every number has only a finite number of factors. Every number is divisible by 1 and by itself, and a number which has no other divisor is called a prime number. The number 1 itself is customarily not designated a prime number, for reasons of expediency. If two numbers are divisible by a third, then the sum and the difference of the first two are also divisible by the third; and if one number is divisible by a second and this second by a third, then the first is also divisible by the third.

Two numbers always have the common divisor 1. If they have no other common divisor, as for example the pairs of numbers 5 and 7, or 21 and 38, then the two numbers are called relative prime numbers, or also numbers prime to one another.

Among the common divisors of any two given numbers one will be the greatest, and it is a very important problem to find this greatest common divisor. This is accomplished by a procedure known under the name algorithm of the greatest common divisor, which already occurs in Euclid.[1]

Let a, a_1 be the two numbers; take the larger of them, say a, as dividend and the smaller a_1 as divisor, and determine the quotient q_1 and, if the division does not come out exactly, the remainder a_2, so that

$$a = q_1 a_1 + a_2,$$

and $a_2 < a_1$. Every common divisor of a and a_1 is then also a common divisor of a_1 and a_2, and conversely. If one proceeds with a_1, a_2 just as with a, a_1, and puts, when the division does not come out exactly,

$$a_1 = q_2 a_2 + a_3,$$

then again $a_3 < a_2$, and every common divisor of a_1 and a_2 is also a common divisor of a_2 and a_3, and conversely. If one continues dividing in this way, then, since the numbers $a, a_1, a_2, a_3, \ldots$ continually decrease, the division must necessarily come out exactly after a finite number of steps; that is, at last a pair of equations

$$a_{\nu-2} = q_{\nu-1} a_{\nu-1} + a_\nu, \qquad a_{\nu-1} = q_\nu a_\nu$$

must occur. Then a_ν is a common divisor of all preceding a's, hence also of a and a_1, and every common divisor of a and a_1 is a divisor of a_ν. Thus a_ν is the greatest common divisor of a and a_1, and at the same time we have arrived at the theorem that every common divisor of two numbers must divide their greatest common divisor.

[1] Elements, Book VII, 11, vol. II of Heiberg's edition.

If a and a_1 are relatively prime numbers, then the last divisor is $a_\nu = 1$. Multiplying, under this assumption, the preceding equations by an arbitrary number b, it follows that b is the greatest common divisor of ab and $a_1 b$, and hence that every common divisor of ab and a_1, or of a and $a_1 b$, must be a divisor of b. Thus if a_1 is relatively prime to a and to b, it is also relatively prime to ab; and from the special assumption that a_1 is a prime number it follows that a product can be divisible by a prime number only when at least one of the factors is divisible by it. Thus, if the product ab is divisible by a_1 and a_1 is relatively prime to a, then b must be divisible by a_1.

Let a, b be any two numbers with greatest common divisor d, and let $a = da'$, $b = db'$. Then a' and b' are relatively prime to one another. Every number m which is at once a multiple of a and of b has the form $m = am' = da'm'$, and in this $a'm'$ must be a multiple of b'; therefore m' too must be a multiple of b'. That is, every number which is at once a multiple of a and of b is divisible by $a'b'd$. This number $a'b'd$, which itself is a common multiple of a and b, is therefore called the least common multiple of a and b.

If a number is divisible by two relatively prime numbers, then it is also divisible by their product; and if, therefore, a number m is divisible by several numbers of which each two are relatively prime to one another, then it is also divisible by the product of all these numbers.

On this rests the proof of the important theorem that a number m can always be represented, and in only one way, as a product of prime numbers. For since a divisor cannot be greater than the dividend, m can certainly be divisible by only a finite number of prime numbers. If a is one of these prime numbers and a^α the highest power of a which divides m, then α too is a definite number. Let likewise $b^\beta, c^\gamma, \ldots$ be the highest powers of the remaining prime numbers $b, c, \ldots$ which divide m. Then, since any two of the numbers $a^\alpha, b^\beta, c^\gamma, \ldots$ are relatively prime, m must also be divisible by the product $a^\alpha b^\beta c^\gamma \cdots$; and it must be equal to this product, since otherwise another prime number, or a higher power of one of the prime numbers $a, b, c, \ldots$, would have to divide m.

We now give an overview of the extensions of the concept of number which are necessary in mathematics and which were gradually introduced.

By a manifold or set, or by the abbreviated sign M, we understand a system of objects or elements of any kind, which is so delimited and complete in itself that, for every arbitrary object, it is completely determined whether it belongs to the system or not, regardless of whether we are able, in each particular case, actually to make the decision or not.

A set is called ordered when, of any two different elements, one that is fully determined in itself always counts as the greater, and in such a way that from $a > b$, $b > c$ one always has $a > c$. If $a > b$ and $b > c$, or $a < b$ and $b < c$, we say that b lies between a and c.

The natural numbers form an ordered set; between two successive elements of it there lies no further element. Such a manifold is called discrete. An ordered set with the property that between any two elements further elements can always be found is called dense. One can form a dense set by combining the natural numbers in pairs and regarding these pairs as elements of a set. These pairs shall be called fractions and denoted by $m : n$ or m/n, and two such fractions $m : n$ and $m' : n'$ are set equal when $mn' = nm'$. If all fractions equal to one another are combined into one element, one obtains a manifold which is ordered once one further stipulates that $m : n$ is greater than $m' : n'$ when $mn' > nm'$. That this manifold is dense is seen as follows: if $\mu = m : n$, $\mu' = m' : n'$ are two fractions and $\mu > \mu'$, then, with h arbitrary, one can put

$$\frac{hmn'}{hnn'}, \qquad \frac{hm'n}{hnn'},$$

and therein $hmn' > hm'n$. One can always take h so large that further numbers lie between hmn' and $hm'n$, and if p is such a number, then $p : hnn'$ lies between μ and μ'.

The points of a straight line can also be regarded as an ordered set if by greater and smaller one understands some indication of position, for instance farther right and farther left, or higher and lower.

A division of an ordered set M into two parts A, B of such a kind that every element a of A is smaller than every element b of B is called a cut in M and is suitably denoted by (A, B). Such a cut arises when one selects an element μ of M, assigns all smaller elements to A, all larger elements to B, and assigns μ itself at will to A or to B. Strictly speaking, according as one does the one or the other, two cuts arise, but we shall always regard them as equal. If, in a cut (A, B), either A contains a greatest element or B a least element μ, then we say that μ produces the cut (A, B). But it may also occur that neither A has a greatest nor B a least element.

If every cut in a dense set is produced by a definite element μ, then the set is called continuous.

Continuity as well as density are properties which, by the nature of the matter, are inaccessible to our sense perception; consequently they also cannot be demonstrated with rigor in things of the external world, in spatial magnitudes, intervals of time, or masses, however much they may seem to lie in the essence of our intuition. But pure conceptual systems can very well be constructed to which density without continuity, or also density and continuity, belong.[2]

This very abstract way of looking at the matter gives us the assurance that the assumption of a continuous set contains no contradiction, and that such sets at least exist in the realm of thought. Geometry, and also analysis, which has always willingly attached itself to geometrical intuition, long tacitly assumed the existence of continuous sets, for example in the points of a straight line or of any other connected line-segment, as a kind of axiom. The difference between a dense and a continuous set, which lies at the basis of the distinction between commensurable and incommensurable segments, was also not missed by the ancients.[3]

We too shall in what follows not renounce the aid of geometrical intuition, and shall, for example, without hesitation regard the points of a straight line as a continuous set.

An ordered set M is called measurable under the following assumptions. Addition and multiplication are generally executable in M, and likewise the subtraction of a smaller element from a greater one; that is, from any two elements a, b (which may also be identical) a new element $a + b$ of M can be derived according to a definite rule, in such a way that $a + b$ is greater than a and greater than b, and the familiar laws of addition expressed in the formulae

$$a + b = b + a, \qquad (a + b) + c = a + (b + c)$$

hold. Moreover, for two elements a, c, of which the second is the greater, a third element b can be found such that $a + b = c$, which is also expressed by the sign of subtraction $b = c - a$. Thus two unequal elements of M always have a definite difference. From these assumptions it follows that a sum becomes greater when one of its summands becomes greater. The repeated, say m-fold, addition of the same element a is called multiplication, and its result is denoted by ma.

Finally there is added one more assumption, namely that, for every given a, a sufficiently high multiple ma is greater than any other element b given in advance. Among the elements of a measurable set there is therefore no greatest element.

In a dense measurable set there is also no least element. For if a were the least and b an arbitrary element, then between b and $b + a$ no elements could lie; for if $b < c < b + a$, then it would follow from the definition of measurability that $a' = c - b$ would be smaller than a. Conversely it also follows that a measurable set in which no least element occurs is dense. For if a and $a + b$ are any two elements, one need only add to a an element smaller than b in order to obtain an element between a and $a + b$.

If a set is continuous, then the assumptions for measurability can be simplified still further, because then subtraction is a consequence of addition. For if a and c are two elements of a continuous ordered set M in which addition exists, and if $c > a$, then one obtains a cut (A, B) in M by assigning to A all elements x for which $a + x < c$, and to B those for which $a + x > c$. This cut is produced by an element b, for which $a + b = c$ must hold.

[2] This was proved by Dedekind, to whom we owe in general the definition of continuity given above. Compare Dedekind's writings, *Stetigkeit und irrationale Zahlen*, Braunschweig 1872, 1892, and *Was sind und was sollen die Zahlen?*, Braunschweig 1888, 1893. Other manifolds to which continuity belongs were formed by Weierstrass and G. Cantor.

[3] Euclid, Elements, Book X.

The natural numbers form, according to our definition, a measurable set in which a least element, namely 1, occurs. The manifold of rational fractions is likewise measurable when addition and subtraction are explained according to the known rules of fractional calculation. Especially important, and as it were typical for the measurable sets, is the manifold of rectilinear segments, or lengths of a line, which are added simply by laying them end to end. Quantities of material, compared by means of the balance, and intervals of time, measured by the clock, also furnish examples of measurable sets. The manner of measuring does not lie in the nature of the manifold itself, but is put into it by the thinking observer; thus, for example, it would be just as admissible to understand by the sum $a + b$ of two segments a and b the hypotenuse of a right triangle with legs a, b, instead of, as is usually assumed, the segment composed from a and b by laying them end to end.

In order to make the continuous set B of cuts in the manifold R of rational fractions into a measurable one, observe first the following. If $\alpha = (A, B)$ is an element in B and μ is an arbitrarily given rational fraction, then one can always choose an element a' in A so that $a' + \mu = b'$ is contained in B. For if one chooses two arbitrary elements a, b, then one can determine the natural number m so that $m\mu > b - a$, hence $a + m\mu$ is contained in B. If then h is the smallest integer for which $a + h\mu$ is contained in B, then $a + (h - 1)\mu = a'$ is in A and hence $a' + \mu = b'$ is in B.

We now understand by the sum $(A, B) + (A', B') = (A'', B'')$, or $\alpha + \alpha' = \alpha''$, the cut in R which is obtained by assigning a rational fraction a'' to A'' only when an a in A and an a' in A' exist such that $a'' \leq a + a'$. In fact (A'', B'') is a cut; for if a'' is contained in A'', the same holds for every smaller fraction, and there are fractions which are in A'' and fractions which are not in A'', namely fractions of the forms $a + a'$ and $b + b'$. Further, α'' is greater than α and than α'. For A'' first contains all of A, and if a' is an arbitrary fraction in A', one can choose a in A so that the element $a + a'$ of A' is contained in B; hence A'' is more comprehensive than A. The cuts produced by rational fractions μ, μ' give, by addition, the cut produced by $\mu + \mu'$.

We now pass to the definition of ratios, which since antiquity have been regarded as the foundation of the theory of numbers, and here follow Euclid first of all.[4]

If one combines the elements of a measurable set M into pairs, and regards these pairs themselves as elements, then a new manifold arises. We denote such a pair by $a : b$, or also by a/b; but, when a and b are different elements, we distinguish $a : b$ from $b : a$, and call a the numerator and b the denominator of $a : b$. We shall call these pairs ratios, and shall now order and make measurable this new set.

Assume first that two whole numbers m, n exist such that $na = mb$, as is always the case, for example, when M is the system of natural numbers, or when a, b are two commensurable segments. Then, if p, q are two other whole numbers, $qa = pb$ holds if and only if $mq = np$. The pair of numbers p, q is completely determined by this requirement if the condition is added that p, q are to be relatively prime. Then, if h is an arbitrary whole number, $m = hp$, $n = hq$. In this case we call the ratio $a : b$ rational and set it equal to the rational fraction $m : n$ or $p : q$. Accordingly these rational fractions may be regarded as ratios of whole numbers.

All rational ratios equal to one another form a rational number, and the rational numbers form, like the rational fractions, an ordered, dense, and measurable manifold. The natural numbers themselves are inserted into the manifold of rational numbers if by the natural number m one understands the ratio $m : 1$.

We now return to any measurable manifold M and take from it any two elements a and b. If one chooses, which is always possible, two natural numbers m, n so that $na > mb$, then the ratio $a : b$ is called greater than the rational ratio $m : n$, or

$$\frac{a}{b} > \frac{m}{n};$$

[4] Elements, Book V.

and if $m : n = p : q$, then also $a : b > p : q$. Similarly, if $n'a < m'b$, then

$$\frac{a}{b} < \frac{m'}{n'}.$$

If $a : b > m : n$, then one can insert one, and consequently also arbitrarily many, rational numbers $m_1 : n_1$ such that

$$\frac{a}{b} > \frac{m_1}{n_1} > \frac{m}{n}.$$

For this choose an arbitrary whole number k and determine the whole number h so that $h(na-mb) > kb$, which is always possible. Then

$$\frac{a}{b} > \frac{hm+k}{hn} > \frac{m}{n}.$$

And in exactly the same way, if $n'a < m'b$ and $h'(m'b - n'a) > k'a$, one obtains

$$\frac{a}{b} < \frac{h'm'}{h'n' + k'} < \frac{m'}{n'}.$$

If now $a : b$ and $\alpha : \beta$ are any two ratios, which for brevity we shall also denote by ϵ, ϵ', and whose elements belong to the same or to different manifolds, then two cases are possible: either no rational ratio μ lies between ϵ and ϵ', or a rational ratio lies between them.

In the first case the two ratios ϵ and ϵ' are called equal. One sees that two ratios which are equal to a third are also equal to one another; for if $\epsilon < \mu < \epsilon'$, then every other ratio ϵ'' is either smaller than, equal to, or greater than μ. If ϵ'' is equal to μ, then rational ratios lie both between ϵ and ϵ'' and between ϵ'' and ϵ'. If, however, ϵ'' is smaller than μ, then μ lies between ϵ'' and ϵ', and if ϵ'' is greater than μ, then μ lies between ϵ and ϵ''; hence ϵ'' cannot at the same time be equal to ϵ and equal to ϵ'.

In the second case the ratios ϵ, ϵ' are called unequal. Then either $\epsilon < \mu < \epsilon'$ or $\epsilon > \mu' > \epsilon'$ must hold, and these two cases exclude one another; for from $\epsilon < \mu < \epsilon'$ and $\epsilon > \mu' > \epsilon'$ it would follow that $\epsilon < \mu'$ and consequently $\mu < \mu'$. The relation of magnitude remains unchanged when ϵ or ϵ' is replaced by an equal element. In the case $\epsilon < \mu < \epsilon'$ the ratio ϵ is called smaller than ϵ', and in the opposite case greater. Between two unequal ratios one can insert an arbitrary number of rational ratios.

If now we collect all ratios equal to one another, we obtain a generic concept which we designate as a number in the general sense of the word. A number is therefore a name or sign for a certain manifold whose elements are precisely the ratios equal to one of them.[5] Under this concept of number are included the rational ratios, and hence also the natural numbers as the ratios $m : 1$; they form the rational numbers. Numbers which do not arise from rational ratios are called irrational numbers. According to what has been developed so far, the numbers form an ordered set, and their order can be fixed if, for each number, any one of the ratios contained under it is chosen as representative.

Of two ratios with the same denominator and unequal numerators, that one is greater whose numerator is greater; and of two ratios with the same numerator and unequal denominators, that one is smaller whose denominator is greater. For if a, a', b are arbitrary elements of a measurable set and $a' > a$, choose first a whole number n such that $na > b$ and $n(a' - a) > b$. Then take m to be the least whole number satisfying $mb > na$; then $na < mb$, but $mb < na'$. Thus

$$\frac{a}{b} < \frac{m}{n} < \frac{a'}{b},$$

and hence $a : b < a' : b$. In exactly the same way one proves that if $b' > b$, then $a : b > a : b'$. This theorem is also expressed by saying that a ratio increases with the numerator and decreases with increasing denominator.

[5]The natural numbers too can be traced back in a simple and consistent way to the generic concept.

From this one easily obtains the following theorem. If a, b, c, d are elements of the same measurable set and $a : b = c : d$, then also $a : c = b : d$. For suppose that $a : c < b : d$; then there would have to be two whole numbers m, n such that

$$na < mc, \qquad nb > md.$$

Then, however, by the theorem just proved, $na : nb < mc : md$, and hence $a : b < c : d$, contrary to the supposition.

To this is now attached the following principal theorem. If of four quantities a, b, c, d any three are given arbitrarily from a continuous measurable set, then the fourth can be determined in the same set so that $a : b = c : d$.

The theorem is an immediate consequence of the assumed continuity. For if, for example, an element x of M is put into A or into B according as $x : b$ is smaller or greater than $c : d$, then a cut is obtained, which is produced by an element a satisfying $a : b = c : d$. But this theorem also holds in certain non-continuous sets, for example for the rational fractions.

It follows that as representatives of two numbers one can always choose two ratios whose elements belong to the same manifold and have the same arbitrarily chosen denominator. The addition is then defined by

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}.$$

This rule contains as a special case the addition of rational fractions; and to justify it generally it remains only to show that if $a : c = a' : c'$ and $b : c = b' : c'$, then also $(a + b) : c = (a' + b') : c'$. We prove this by showing that if $a : c = a' : c'$ and $(a + b) : c > (a' + b') : c'$, then also $b : c > b' : c'$ must hold. Let therefore, with m, n two whole numbers,

$$\frac{a+b}{c} > \frac{m}{n} > \frac{a'+b'}{c'}.$$

Then $n(a + b) > mc$ and hence

$$\frac{b}{c} > \frac{mc - na}{nc}.$$

On the other hand, from the supposition $a : c = a' : c'$ it follows easily that

$$\frac{mc - na}{nc} = \frac{mc' - na'}{nc'},$$

and therefore $b : c > b' : c'$, as was required to be proved.

It is thus shown that the numbers, as we have defined them, also form a measurable set. If a and c are taken from a continuous manifold, then, with fixed c, the ratios $a : c$ also form a continuous set; hence, since continuous sets exist at all, it follows that the numbers too form a continuous set.

If $\alpha, \beta, \gamma, \delta$ are now numbers, then from the proportion $\alpha : \beta = \gamma : \delta$ any one of the four numbers can be determined by the three others given. If one puts $\delta = 1$ and seeks α, one obtains the multiplication $\alpha = \beta\gamma$; the commutativity of the factors is a consequence of the theorem that from $\alpha : \beta = \gamma : \delta$ there follows $\alpha : \gamma = \beta : \delta$. If one seeks γ, one obtains division; and from the definition of addition given above there follows the fundamental formula $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$. The four fundamental operations are therefore executable in the domain of numbers, with the sole restriction that in subtraction the subtrahend must be smaller than the minuend.

The construction of a cut in the sequence of numbers always supplies the proof for the existence of a number satisfying prescribed requirements. Thus one obtains a cut (A, B) by putting into A all and only those numbers whose square is smaller than a prescribed number α; to this cut there corresponds a definite number whose square is equal to α and which is denoted by $\sqrt{\alpha}$. In this way the existence of square roots is proved.

The numerical sequences introduced by G. Cantor for the definition of irrational numbers can also be referred back to cuts.[6]

According to Cantor, by a numerical sequence one is to understand any unbounded system of numbers ordered in a definite way

$$S = x_1, x_2, x_3, x_4, \dots$$

having the property that there is a definite number g below which none of the numbers of S sinks, and that, if δ is any chosen number and x_n, x_m are different elements of S, then $x_n - x_m$ or $x_m - x_n$ always remains smaller than δ when m and n have exceeded a sufficiently large number.

There are always numbers which are not exceeded by the numbers of such a sequence; for, once δ has been chosen and n suitably determined, x_m, whatever the value of m, never exceeds the greatest of the numbers $x_1, x_2, \dots, x_n, x_n + \delta$. One now obtains a cut (A, B) by throwing into B those numbers which, when n has a sufficiently high value, are no longer exceeded by any x_n, and all other numbers, which are therefore exceeded by infinitely many x_n, into A. If this cut is produced by the number α, then, however small ε may be, there are always infinitely many numbers x_n between $\alpha - \varepsilon$ and α; and one may say that this number α, completely determined by S, is produced by the numerical sequence S. According to Cantor the numerical sequence S is directly the definition of the number α. Of course one and the same number α can be produced by very different numerical sequences. These numerical sequences, however, are all to be regarded as equal to one another. Among other things, one may take all numbers of S to be rational fractions. Conversely, it is also easily shown that for every given number α one can always give numerical sequences S by which α is produced, so that the totality of the numerical sequences S likewise forms a continuous set.

In the explanation of the fundamental operations an inconvenient restriction has appeared in subtraction, from which we can free ourselves by introducing negative numbers.

Let x denote any element of the number system defined so far, which we shall now call the system of positive numbers. We take this number system a second time and, to distinguish it, denote in this second system, which is to be called the system of negative numbers, every element by $-x$. This second system we now order in exactly the reverse way from the first, so that wherever in the system x "greater" stands, in the system $-x$ "smaller" is put, and conversely. Addition and subtraction in $-x$ are explained just as in x, so that

$$(-x) + (-y) = -(x + y), \qquad (-x) - (-y) = -(x - y)$$

shall hold.

But we wish to order these two number systems together in such a way that every $-x$ is to be smaller than every x. We thus obtain an ordered set in which there is no greatest and no least element. This system is also in general continuous; only the single cut $(-x, x)$ is produced by no element, and here, where the two systems meet, a violation of continuity is still present. To restore continuity we must therefore add, corresponding to the cut $(-x, x)$, a number zero, or 0, which is precisely defined by this cut. Then we have an ordered continuous set unbounded on both sides, the complete series of real numbers.

In this enlarged number domain we now explain addition generally by putting, by definition,

$$x + (-x) = 0, \qquad x + 0 = x,$$

$$x + (-y) = x - y \quad \text{if } x > y, \qquad x + (-y) = -(y - x) \quad \text{if } y > x.$$

With this explanation of addition, if z_1, z_2, z_3 are any three numbers of the whole number domain, the laws

$$z_1 + z_2 = z_2 + z_1, \qquad (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

[6]Cantor, *Ueber die Ausdehnung eines Satzes aus der Theorie der trigonometrischen Reihen.* Mathematische Annalen, vol. 5 (1872); compare also Heine, *Elemente der Functionenlehre.* Journal fuer Mathematik, vol. 74 (1872).

hold, which are called the commutative and the associative laws. One forms the sum of an arbitrary number of summands by uniting, at will and successively, two summands at a time into a sum. Subtraction no longer needs special consideration if one explains $z_1 - z_2$ by $z_1 + (-z_2)$ and sets $-(-z) = z$.

The number line is represented intuitively by points, by laying off from a fixed point of a straight line, denoted by 0, the positive numbers as segments on one side, say the right, and the negative numbers on the other, the left, side. The image of the sum of two segments $z_1 + z_2$ is obtained by laying off from the point z_1 the segment of length $+z_2$ to the right or to the left, according as z_2 is positive or negative.

Multiplication and division are introduced into the enlarged number domain by the equations

$$x(-y) = (-x)y = -xy, \qquad (-x)(-y) = xy, \qquad 0x = 0.$$

Division, as the inverse of multiplication, is always possible except when the divisor is zero.

A further extension of the concept of number consists in the introduction of complex quantities. We combine every two numbers of the complete number line into pairs (x, y), and regard two such pairs (x, y) and (a, b) as equal only when $x = a$, $y = b$. These pairs of numbers form a manifold whose elements cannot indeed be ordered, but with which the arithmetical operations of addition, subtraction, multiplication, and division are to be performed according to the following rules:

$$(x, y) + (a, b) = (x + a, y + b),$$

$$(x, y)(a, b) = (xa - yb, xb + ya).$$

We also stipulate that $(x, 0) = x$, which does not contradict these equations. The pair (x, y) is equal to 0 only when x and y are both zero. For abbreviation we denote $(0, 1)$ by i. The equations above then give the consequences

$$(x, 0)(0, 1) = (0, x) = ix, \qquad (x, 0) + (0, y) = (x, y) = x + yi,$$

$$x + yi + a + bi = (x + a) + (y + b)i,$$

and the inverse of addition

$$x + yi - (a + bi) = (x - a) + i(y - b),$$

and further multiplication

$$(x + yi)(a + bi) = xa - yb + i(xb + ya), \qquad i^2 = -1,$$

$$(x + yi)(x - yi) = x^2 + y^2,$$

$$\frac{x + yi}{a + bi} = \frac{ax + by + i(ay - bx)}{a^2 + b^2},$$

by which division is defined, except when $a + bi = 0$.

Numbers of the form ix are called purely imaginary numbers, and i the imaginary unit. The numbers $a + bi$ are called imaginary or complex. The systems of real and of purely imaginary numbers are contained among them as special cases. Thus, in the domain of complex numbers $x + yi$, the fundamental operations can be carried out without restriction, except for division by zero; and calculation with real numbers is a special case.

Following Gauss, one represents the complex numbers $z = x + yi$ geometrically by the points of a plane, laying down a rectangular system of coordinates and regarding the point with coordinates x, y as the image of the numerical value z. The points of the x-axis represent the real numbers x in the way discussed above. The points of the y-axis are the images of the purely imaginary numbers yi. The origin of coordinates is the image of the number 0. The radius vector from the zero point to the point z has the numerical value $\rho = \sqrt{x^2 + y^2}$, which is called the absolute value, or the magnitude, or, according to the older terminology, the modulus of the complex number z.

The only number having absolute value 0 is 0. Every positive number occurs as the absolute value of infinitely many complex numbers, and the image-points of all numbers with the same absolute value lie on a circle whose center is at the origin of coordinates.

Two imaginary numbers which differ only by the sign of i, thus $x + yi$ and $x - yi$, are called conjugate imaginary. Their product is the square of the absolute value of either of them. If one of two conjugate imaginary numbers is zero, then the other is also zero; hence in every correct numerical equation one may replace i by $-i$ without disturbing its correctness.

We shall also record the often used theorem that the absolute value of the sum of two non-zero complex numbers is never greater than the sum of the absolute values of the summands, and is equal to it only when the ratio, or quotient, of the two summands is real and positive. Namely, let

$$z = x + yi, \qquad c = a + bi, \qquad Z = z + c = (x + a) + (y + b)i,$$

$$\rho^2 = x^2 + y^2, \qquad r^2 = a^2 + b^2, \qquad R^2 = (x + a)^2 + (y + b)^2.$$

Then

$$(\rho + r - R)(\rho + r + R) = (\rho + r)^2 - R^2 = 2(\rho r - ax - by),$$

which is surely positive when $ax + by < \rho r$. If however $ax + by > \rho r$, then it would follow that

$$\rho^2 r^2 - (ax + by)^2 = (ay - bx)^2 < 0,$$

which is impossible; and equality occurs only when $ay - bx = 0$. Thus $\rho + r \geq R$, and only in the special case $\rho + r = R$ when $ay - bx = 0$ and $ax + by > 0$. This is the assertion.

In the geometric representation this theorem says that in a triangle one side is smaller than the sum of the two others. The theorem has the further consequence that the absolute value of a sum cannot be smaller than the difference of the absolute values of the summands. For applying the preceding theorem to the sum $c = Z - z$, one obtains $r \leq R + \rho$, or

$$R \geq r - \rho.$$

Equality here occurs only when the quotient $z : c$ is real and negative.

It remains to say a word about the most important aid of algebra, literal calculation. The use of this aid is so general that one sometimes uses the word literal calculation outright as synonymous with algebra. We presuppose as known the rules according to which calculations are made with such literal expressions. Equations between literal expressions can be of two kinds. Either they are so-called identities, that is, the two expressions set equal to one another can, by applying the rules of calculation, be transformed so that both expressions agree exactly. From such literal equations one then obtains correct numerical equations when the letters are replaced by arbitrary real or complex numbers, provided that the demand to divide by zero does not arise. The letters in such equations are often also designated as variables, because one may imagine, without ever coming to a contradiction, that other and again other numerical values are successively put for the letters.

Another kind of equations between literal expressions does not have this character of identity. They contain rather a requirement imposed on the numbers which may be substituted for the letters without committing an error. Algebra has the task of determining numerical values which satisfy such a requirement, that is, of solving the equation. In these equations the letters are also designated as "unknowns." Very often in one and the same equation letters of two kinds occur: some for which arbitrary numerical values are to be put, and others whose numerical value is first to be determined.

First Section. Rational Functions.

§ 1. Whole functions.

The next object of consideration is whole rational functions, or, briefly, whole functions of one variable. By these we understand expressions of the following form:

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n, \tag{1}$$

where n is an integral exponent, the degree of the function $f(x)$. The degree is a natural number. Sometimes, however, it is also useful to speak of whole rational functions of degree 0, by which a constant is understood. x is called the variable, $a_0, a_1, a_2 \ldots a_{n-1}, a_n$ the coefficients. Both x and $a_0, a_1, \ldots, a_n$ are symbols for indeterminate quantities, with which one proceeds according to the rules of literal calculation, and for which definite numerical values may also be substituted under some circumstances (cf. the Introduction).

If the function $f(x)$ is written in the way shown in (1), we call it ordered according to descending powers of x. The summands may be arranged in any other order; for example, the function may also be ordered according to ascending powers of x:

$$f(x) = a_n + a_{n-1}x + \cdots + a_1x^{n-1} + a_0x^n.$$

By addition, subtraction, and multiplication of whole rational functions there again arise whole rational functions. The rules of literal calculation directly give the laws of formation of these new functions.

In addition, the coefficient of any power x^ν in the sum is equal to the sum of the coefficients of x^ν in the individual summands. The degree of the function that arises is equal to the highest of the degrees of the summands, and can decrease only in the special case in which the highest degree occurs in several summands and the sum of the coefficients of the highest powers is zero.

In the multiplication of two or more whole rational functions there arises a function whose degree is equal to the sum of the degrees of the factors.

To survey the law of formation of the coefficients for the product, put

$$\begin{aligned} A(x) &= a_0x^m + a_1x^{m-1} + a_2x^{m-2} + \cdots, \\ B(x) &= b_0x^n + b_1x^{n-1} + b_2x^{n-2} + \cdots, \end{aligned} \tag{2}$$

$$A(x)B(x) = C(x) = c_0x^{m+n} + c_1x^{m+n-1} + c_2x^{m+n-2} + \cdots \tag{3}$$

and obtain

$$c_0 = a_0b_0,$$

$$c_1 = a_0b_1 + a_1b_0,$$

$$c_2 = a_0b_2 + a_1b_1 + a_2b_0,$$

$$\cdots,$$

or, in general notation, if ν denotes one of the numbers $0, 1, 2, \ldots, m+n$,

$$c_\nu = a_0b_\nu + a_1b_{\nu-1} + a_2b_{\nu-2} + \cdots + a_{\nu-1}b_1 + a_\nu b_0, \tag{4}$$

where all a whose index is greater than m, and all b whose index is greater than n, are to be set equal to zero. For c_ν is the coefficient of $x^{m+n-\nu}$, and hence $c_\nu x^{m+n-\nu}$ arises by multiplying all terms of the form $a_\mu x^{m-\mu}$ by all terms of the form $b_{\nu-\mu}x^{n-\nu+\mu}$ and then adding. If a_0 and $b_0 = 1$, then also $c_0 = 1$.

If in all factors of such a product the highest power of x has coefficient 1, then in the product too the coefficient of the highest power is 1.

From these rules for calculating with whole functions it follows that the rules of calculation expressed in formulae such as $ab = ba$, $(ab)c = a(bc)$, $(a+b)c = ac + bc$ and the like also hold when a, b, c are whole functions of x, and in the sense that whole functions are to be regarded as equal only when equal powers have equal coefficients.

By whole functions of several variables we understand sums of products of whole, positive powers of the variables $x, y, z \dots$ with arbitrary coefficients that count as constant. One can think of these functions as arising from functions of one variable x when the coefficients $a_0, a_1 \dots a_n$ occurring in them are themselves regarded as whole rational functions of other variables $y, z \dots$. Thus functions of m variables arise from functions of $m-1$ variables. All terms of such a function are of the form $x^r y^s z^t \cdots$, multiplied by a coefficient; and if the function is properly collected, then each combination of the exponents $r, s, t \dots$ occurs only once. Two whole functions so ordered are regarded as equal only when they contain the same products $x^r y^s z^t \cdots$ with the same coefficients.

§ 2. A theorem of Gauss.

We shall at once make an application of the multiplication rule for two whole rational functions, in order to prove a theorem of Gauss which will be useful to us later, but which here is to serve as an introduction to the mode of calculation and as an example.[7]

We consider here the case in which the coefficients in the functions $A(x), B(x)$ are whole numbers, so that, by (4), the coefficients of $C(x) = A(x)B(x)$ are also whole numbers.

If all integral coefficients $a_0, a_1, \dots, a_m$ of a function $A(x)$ have no common divisor, then the function $A(x)$ is called an original or primitive function, and the theorem we wish to prove is:

If $A(x)$ and $B(x)$ are primitive functions, then their product $C(x)$ is also a primitive function.

The proof follows almost immediately from looking at formula (4).

For if all coefficients $c_0, c_1, c_2 \dots c_{m+n}$, as given in (4), had a common divisor greater than 1, then there would also have to be at least one prime number which divides all these coefficients. Let p be such a prime number. By the assumption that $A(x), B(x)$ are primitive, this prime can divide neither all $a_0, a_1 \dots a_m$ nor all $b_0, b_1 \dots b_n$.

Suppose now that p divides

$$a_0, a_1 \dots a_{r-1}, \quad \text{but not } a_r,$$

and divides

$$b_0, b_1 \dots b_{s-1}, \quad \text{but not } b_s.$$

Then, by assumption, $r \leq m$, and r may also be zero if p already does not divide a_0. Likewise $s \leq n$.

If now, by (4), we form c_{r+s} and arrange it in the following way,

$$\begin{aligned} c_{r+s} = a_r b_s + a_{r-1} b_{s+1} + a_{r-2} b_{s+2} + \cdots \\ + a_{r+1} b_{s-1} + a_{r+2} b_{s-2} + \cdots , \end{aligned} \tag{1}$$

one sees at once that c_{r+s} cannot be divisible by p, as had been assumed; for the first term $a_r b_s$ is not divisible by p, while all other terms, since they are multiplied by one of the coefficients $a_0, a_1 \dots a_{r-1}, b_0, b_1 \dots b_{s-1}$, are divisible by p. Thus the assumption that $C(x)$ is not primitive while $A(x)$ and $B(x)$ are primitive leads to a contradiction.

This theorem can be transferred to functions of several variables. We call a whole rational function of m variables with integral coefficients original or primitive if the coefficients have no common divisor, and we state the theorem that the product of two primitive functions is again a primitive function.

[7] Gauss, *Disquisitiones arithmeticae*, Art. 42.

To see its truth, we need only take, in the argument carried out above, the coefficients $a_0, a_1, a_2 \ldots$, $b_0, b_1, b_2 \ldots$ not as whole numbers, but as whole rational functions of $m-1$ variables $y, z \ldots$, and call such a function divisible by a prime number p when all its coefficients are divisible by p. If we then suppose that the theorem to be proved has already been proved for functions of $m-1$ variables, formula (1) gives its correctness for functions of m variables, hence its general validity by complete induction, or by the conclusion from $m-1$ to m.

An imprimitive function is one whose integral coefficients all have a common divisor. We call this divisor the divisor of the function. Then we can also express the theorem just proved as follows:

The divisor of a product of two whole functions is equal to the product of the divisors of the two functions.

For if PA and QB are two whole functions with divisors P and Q, then A and B are primitive functions. Hence $AB = C$ is also a primitive function, and PQ is the divisor of the function $PA \cdot QB = PQC$.

Insofar as it refers to functions of one variable, we can give the theorem the following form, without changing its content essentially; in this form it is especially useful. Let

$$\varphi(x) = x^m + \alpha_1 x^{m-1} + \alpha_2 x^{m-2} + \cdots + \alpha_m,$$

$$\psi(x) = x^n + \beta_1 x^{n-1} + \beta_2 x^{n-2} + \cdots + \beta_n$$

be two whole rational functions in which the highest powers of x have coefficient 1, while the remaining coefficients are rational numbers. Then, in the product

$$\varphi(x)\psi(x) = x^{m+n} + \gamma_1 x^{m+n-1} + \gamma_2 x^{m+n-2} + \cdots + \gamma_{m+n},$$

the coefficients γ cannot all be whole numbers if the coefficients α, β in $\varphi(x)$ and $\psi(x)$ are not all whole numbers.

For let the smallest common denominator of the coefficients α of φ be denoted by a_0, and that of the coefficients β of ψ by b_0. Then $a_0\varphi(x) = A(x)$ and $b_0\psi(x) = B(x)$ are primitive functions of x. If the coefficients γ were whole numbers, the product $a_0 b_0 \varphi(x)\psi(x)$ would have divisor $a_0 b_0$ and, if a_0 and b_0 were not both equal to 1, would therefore not be primitive; but this would contradict the theorem proved above.

§ 3. Division.

Let, as before,

$$\begin{aligned} A &= A(x) = a_0 x^m + a_1 x^{m-1} + \cdots, \\ B &= B(x) = b_0 x^n + b_1 x^{n-1} + \cdots \end{aligned} \tag{1}$$

be two whole rational functions of x; but now suppose that $m \geq n$ and that a_0 and b_0 are different from zero. Then the difference

$$A - \frac{a_0}{b_0} x^{m-n} B \tag{2}$$

is also a whole rational function of x, but its degree is smaller than m, since the highest power in the two terms of the difference has the same coefficient and therefore cancels. We put this difference equal to

$$A' = A'(x) = a_0' x^{m'} + a_1' x^{m'-1} + \cdots, \qquad m' < m. \tag{3}$$

If now m' is still $\geq n$, then in (2) we can put A' in place of A and repeat the same reasoning.

This gives a chain of equations:

$$\begin{aligned} A - \frac{a_0}{b_0} x^{m-n} B &= A', \\ A' - \frac{a_0'}{b_0} x^{m'-n} B &= A'', \\ A'' - \frac{a_0''}{b_0} x^{m''-n} B &= A''', \\ &\cdots, \end{aligned} \tag{4}$$

and this chain can be continued until the degree of the function that has arisen is smaller than n. Since in the sequence of functions $A, A', A'' \dots$ the degree of each following function is lowered by at least one unit, the chain of equations (4) consists of at most $m - n + 1$ members; but it may also contain fewer members when several powers cancel simultaneously. If we add all equations (4), denote the last of the functions $A, A' \dots$ by C, and, for abbreviation, set

$$Q = \frac{a_0}{b_0} x^{m-n} + \frac{a_0'}{b_0} x^{m'-n} + \cdots, \tag{5}$$

so that Q also is a whole rational function of x, then there follows

$$A = QB + C. \tag{6}$$

The operation just described, by which the functions Q, C are found from A, B, is called division. A is the dividend, B the divisor, C the remainder, and Q the quotient. The degree of the remainder is always lower than the degree of the divisor.

The coefficients of the functions Q and C are composed from the coefficients a and b by addition, subtraction, multiplication, and division. In the denominator, however, only powers of b_0 occur; and hence if $b_0 = 1$, the coefficients of Q and C are whole functions of the a and b. The highest power of b_0 which can occur in the denominator is the $(m - n + 1)$st, since in the chain (4) the factor b_0 is added once in the denominator in each following equation. In special cases, however, the highest power of b_0 in all denominators may be a lower one.

Take, for example, for A a function of the third degree,

$$f(x) = a_0 x^3 + a_1 x^2 + a_2 x + a_3, \tag{7}$$

and for B the so-called first derivative of $f(x)$, which is of the second degree,

$$f'(x) = 3a_0 x^2 + 2a_1 x + a_2. \tag{8}$$

Then one obtains

$$Q = \frac{1}{3} x + \frac{a_1}{9a_0}, \tag{9}$$

$$C = \frac{6a_0 a_2 - 2a_1^2}{9a_0} x + \frac{9a_0 a_3 - a_1 a_2}{9a_0}. \tag{10}$$

How the calculation prescribed in the equations (4) is expediently arranged may here be assumed as known from the elements. We draw attention to the analogy between this calculation and division in the decimal number system. A function $f(x)$ represents a number written decimally if the coefficients $a_0, a_1 \dots$ are whole numbers between zero, inclusive, and 10, exclusive, and $x = 10$ is set. If one also permits numbers greater than 10 among the coefficients, then one can represent a number by $f(x)$ in various ways. This is used in the division procedure in order to avoid fractional and negative coefficients in the results in the simplest way.

§ 4. Division by a linear function.

We shall apply the general rule for division to one special case. Let the dividend be

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n, \tag{1}$$

arbitrary, while the divisor is of the first degree, or, as one also says, linear. We shall also suppose the coefficient of the first power of x in the divisor to be 1, and therefore take the divisor in the simple form $(x - \alpha)$, where α remains arbitrary. In this case the quotient Q is of degree $n - 1$, and the remainder C of degree 0, that is, independent of x. Thus set

$$f(x) = (x - \alpha)Q + C. \tag{2}$$

Then C no longer contains x, and if we set

$$Q = q_0 x^{n-1} + q_1 x^{n-2} + \cdots + q_{n-2} x + q_{n-1}, \tag{3}$$

there follows from (2)

$$\begin{aligned} f(x) = q_0 x^n &+ q_1 x^{n-1} + \cdots + q_{n-2} x^2 + q_{n-1} x + C \\ &- \alpha q_0 x^{n-1} - \cdots - \alpha q_{n-3} x^2 - \alpha q_{n-2} x - \alpha q_{n-1}, \end{aligned} \tag{4}$$

and by comparison with (1):

$$\begin{aligned} q_0 &= a_0, \\ q_1 - \alpha q_0 &= a_1, \\ q_2 - \alpha q_1 &= a_2, \\ &\cdots, \\ q_{n-1} - \alpha q_{n-2} &= a_{n-1}, \\ C - \alpha q_{n-1} &= a_n. \end{aligned} \tag{5}$$

From this one obtains

$$\begin{aligned} q_0 &= a_0, \\ q_1 &= a_0 \alpha + a_1, \\ q_2 &= a_0 \alpha^2 + a_1 \alpha + a_2, \\ &\cdots, \\ q_{n-1} &= a_0 \alpha^{n-1} + a_1 \alpha^{n-2} + \cdots + a_{n-1}, \\ C &= a_0 \alpha^n + a_1 \alpha^{n-1} + \cdots + a_{n-1} \alpha + a_n = f(\alpha). \end{aligned} \tag{6}$$

C arises from $f(x)$ by setting $x = \alpha$, and therefore can also be denoted by $f(\alpha)$.

Accordingly we also have the formula

$$\frac{f(x) - f(\alpha)}{x - \alpha} = Q(x), \tag{7}$$

where $Q(x)$ is a whole function of degree $n - 1$.

If in the expressions (6) we put the sign x in place of the indeterminate quantity α, there arises from this a sequence of whole rational functions of x, which, if for simplicity we put $a_0 = 1$, we write as follows:

$$\begin{aligned} f_0 &= 1, \\ f_1 &= x + a_1, \\ f_2 &= x^2 + a_1 x + a_2, \\ &\cdots, \\ f_{n-1} &= x^{n-1} + a_1 x^{n-2} + a_2 x^{n-3} + \cdots + a_{n-1}. \end{aligned} \tag{8}$$

These functions $f_0, f_1 \dots f_{n-1}$ will later render us good service. For the present we add the following remarks.

According to (8), one can express the powers $1, x, x^2 \dots x^{n-1}$ of x linearly in terms of the functions $f_0, f_1 \dots f_{n-1}$, and indeed in such a way that only integral rational combinations of the a occur in the coefficients, for instance

$$1 = f_0,$$

$$x = f_1 - a_1 f_0,$$

$$x^2 = f_2 - a_1 f_1 + (a_1^2 - a_2) f_0,$$

$$\cdots .$$

From this it follows that every whole rational function of x whose degree is not greater than $n-1$ can likewise be expressed linearly by $f_0, f_1, \dots, f_{n-1}$ in the form

$$y_0 f_0 + y_1 f_1 + \cdots + y_{n-1} f_{n-1}, \tag{9}$$

where the coefficients $y_0, y_1 \dots y_{n-1}$ are independent of x.

Thus, if $F(x)$ is any whole rational function of x, then by § 3, regarding $f(x)$ as divisor, one can set

$$F(x) = Qf(x) + y_0 f_0 + y_1 f_1 + \cdots + y_{n-1} f_{n-1}, \tag{10}$$

where Q is also a whole rational function of x.

For the recurrent calculation of the functions $f_r(x)$, relation (8) gives

$$f_r(x) - x f_{r-1}(x) = a_r. \tag{11}$$

§ 5. Rational fractional functions; divisibility.

If $F(x)$ and $f(x)$ are two integral rational functions of x, then the fraction

$$\frac{F(x)}{f(x)} \tag{1}$$

is called a *fractional rational*, or also briefly a *fractional* or *rational function* of x.

If the degree of the numerator is lower than the degree of the denominator, then the function is called a *proper fraction*; in the opposite case it is called an *improper fraction*.

According to § 3 the integral rational functions Q and $\varphi(x)$ can be so determined that

$$F(x) = Qf(x) + \varphi(x) \tag{2}$$

and the degree of $\varphi(x)$ is smaller than the degree of $f(x)$; consequently

$$\frac{F(x)}{f(x)} = Q + \frac{\varphi(x)}{f(x)}, \tag{3}$$

and hence the theorem:

Every fractional function can be decomposed into the sum of an integral function and a proper fractional function.

If n is the degree of $f(x)$, then the degree of $\varphi(x)$ is at most $n-1$; it may also be lower. In particular, the case may occur in which $\varphi(x)$ vanishes identically.

In this case the function $F(x)$ is said to be divisible by $f(x)$. The function

$$\frac{F(x)}{f(x)}$$

is in this case only apparently fractional; in reality it is equal to the integral function Q.

The formula

$$\frac{x^m - 1}{x - 1} = x^{m-1} + x^{m-2} + x^{m-3} + \cdots + 1$$

gives a simple example of this.

For divisibility of functions the same laws hold as for divisibility of numbers, in particular the following:

1. If the function $F(x)$ is divisible by the function $f(x)$, and $f(x)$ is divisible by a third function $\varphi(x)$, then $F(x)$ is also divisible by $\varphi(x)$.

For if

$$F = Qf, \qquad f = q\varphi,$$

where Q and q are integral rational functions, then

$$F = Qq\varphi,$$

and since Qq is an integral rational function, F is divisible by φ.

2. If $F(x)$ is divisible by $f(x)$ and Q is any integral rational function, then $QF(x)$ is also divisible by $f(x)$.

3. If $F(x)$ and $f(x)$ are divisible by $\varphi(x)$, then $F(x) \pm f(x)$ is also divisible by $\varphi(x)$, or more generally:

4. If $F_1, F_2 \ldots$ are divisible by $f(x)$ and $Q_1, Q_2 \ldots$ are arbitrary integral rational functions, then

$$Q_1F_1 + Q_2F_2 + \cdots$$

is also divisible by $f(x)$.

The last theorem includes the two preceding ones and is proved simply as follows.

If $F_1, F_2 \dots$ are divisible by f, then the integral rational functions $\Phi_1, \Phi_2 \dots$ can be determined so that

$$F_1 = \Phi_1 f, \qquad F_2 = \Phi_2 f, \dots$$

and consequently

$$Q_1 F_1 + Q_2 F_2 + \cdots = (Q_1 \Phi_1 + Q_2 \Phi_2 + \cdots) f.$$

Since $Q_1 \Phi_1 + Q_2 \Phi_2 + \cdots$ is an integral rational function, the theorem is proved.

5. Every function is divisible by itself.

With respect to divisibility or indivisibility of functions nothing is changed if the functions are multiplied by arbitrary factors independent of x.

A non-zero quantity independent of x may be regarded as a function of degree zero. If we call such a quantity a constant, we may say:

6. Every function is divisible by every constant.

If a function is divisible by another, then the degree of the quotient is equal to the difference between the degree of the dividend and the degree of the divisor. The former therefore cannot be smaller than the latter. If the degrees are equal, the quotient is a constant, and hence:

7. If, of two integral rational functions of the same degree, one is divisible by the other, then they differ from one another only by a constant factor, and the second is also divisible by the first.

8. By § 4, the necessary and sufficient condition that the function $f(x)$ be divisible by the linear function $x - \alpha$ is that $f(\alpha) = 0$.

§ 6. Greatest common divisor.

It is a problem of fundamental importance to decide whether two integral rational functions have, besides the constants, any further common divisor. The solution is found by the algorithm of the greatest common divisor in exactly the same way as the corresponding question concerning divisibility of integers is answered. See the Introduction.

Let

$$f(x) = A, \qquad \varphi(x) = A' \tag{1}$$

be two given functions; the degree of $\varphi(x)$ is supposed to be lower, or at least not higher, than that of $f(x)$.

We divide A by A' and denote the remainder, whose degree is lower than the degree of A', by A'', thus:

$$A = Q'A' + A'';$$

then we divide A' by A'' and denote the remainder by A''', and continue in this way in forming the sequence of functions

$$A, A', A'', A''', \dots, \tag{2}$$

whose degrees $n, n', n'', n''' \dots$ always decrease and therefore, after a finite number of divisions, necessarily go down to zero. Let the last remainder of degree zero, and therefore a constant, be $A^{(\nu)}$. Then we have the chain of equations

$$\begin{aligned} A &= Q'A' + A'', \\ A' &= Q''A'' + A''', \\ &\cdots \\ A^{(\nu-3)} &= Q^{(\nu-2)} A^{(\nu-2)} + A^{(\nu-1)}, \\ A^{(\nu-2)} &= Q^{(\nu-1)} A^{(\nu-1)} + A^{(\nu)}, \end{aligned} \tag{3}$$

where $A^{(\nu)}$ is a constant.

If now A, A' have any common divisor, then by the theorems of the preceding paragraph, as is seen from the equations (3), it is also a divisor of A'', A''', A'''' and so on down to $A^{(\nu)}$. If $A^{(\nu)}$ is a non-zero constant, then no divisor of $f(x)$ and $\varphi(x)$ depending on x can exist. Such functions are called *prime to one another*, or *relatively prime*. One also says, taking account only of divisors depending on x, that the functions have no common divisor.

The condition that the two functions have a common divisor is therefore

$$A^{(\nu)} = 0. \tag{4}$$

In this case $A^{(\nu-1)}$ is a divisor of $A^{(\nu-2)}$, as the last equation (3) shows, and by the preceding equation it is a divisor of $A^{(\nu-3)}$, and so on, hence also a divisor of A and A', that is, of f and φ. Conversely, every common divisor of A, A' is also a divisor of $A^{(\nu-1)}$; accordingly $A^{(\nu-1)}$ is called the *greatest common divisor* of f and φ. The algorithm (3) shows that $A^{(\nu)}$ or $A^{(\nu-1)}$ can be derived from the coefficients of f and φ by the rational operations of arithmetic, and in such a manner that division is always only by the coefficients of the highest powers of x in the functions $A, A', A'', \ldots$ which are not zero.

We shall apply these considerations to two examples.

First let

$$\begin{aligned} A &= a_0x^2 + a_1x + a_2, \\ B &= b_0x^2 + b_1x + b_2 \end{aligned} \tag{5}$$

be two functions of the second degree, so that a_0, b_0 are non-zero.

The first step is to form the equation

$$A = \frac{a_0}{b_0}B + C, \tag{6}$$

where

$$C = c_0x + c_1 \tag{7}$$

and

$$c_0 = \frac{a_1b_0 - b_1a_0}{b_0}, \qquad c_1 = \frac{a_2b_0 - b_2a_0}{b_0}. \tag{8}$$

If $c_0 = 0$, so that C is constant, then A, B have a common factor only if also $c_1 = 0$; and then A is divisible by B, that is, A and B differ only by a constant factor. If however c_0 is different from zero, we continue the calculation by putting

$$B = QC + D,$$

where, according to § 4, if there $\alpha = -c_1 : c_0$ is put,

$$D = \frac{b_0c_1^2 - b_1c_0c_1 + b_2c_0^2}{c_0^2}. \tag{9}$$

If D is non-zero, then A, B have no common divisor. If however $D = 0$, then C is the greatest common factor of A and B.

Substituting the values c_0, c_1 from (8) into (9), and suppressing the denominator $b_0c_0^2$, one obtains the condition for the existence of a common divisor C of A and B in the form

$$\begin{aligned} a_0^2b_2^2 + a_2^2b_0^2 - 2a_0a_2b_0b_2 - a_1a_2b_0b_1 - a_0a_1b_1b_2 \\ + a_0a_2b_1^2 + a_1^2b_0b_2 = 0 \end{aligned} \tag{10}$$

or

$$(a_0b_2 - b_0a_2)^2 + (a_0b_1 - a_1b_0)(a_2b_1 - a_1b_2) = 0. \tag{11}$$

This condition is also fulfilled when c_0 and c_1 are both zero, and is therefore the necessary and sufficient condition that A, B have a common divisor.

The left hand side of (10) or (11) is called the *resultant* of the functions A and B.

As a second example we take the one already chosen in § 3. Put

$$\begin{aligned} f(x) &= a_0x^3 + a_1x^2 + a_2x + a_3 = A, \\ f'(x) &= 3a_0x^2 + 2a_1x + a_2 = B, \end{aligned} \tag{12}$$

and suppose a_0 different from zero. Then by § 3 we have

$$A = QB + C, \tag{13}$$

$$C = c_0x + c_1, \tag{14}$$

where

$$c_0 = \frac{6a_0a_2 - 2a_1^2}{9a_0}, \qquad c_1 = \frac{9a_0a_3 - a_1a_2}{9a_0}. \tag{15}$$

If c_0 is zero, the algorithm is already closed. If c_1 is non-zero, then A and B have no common divisor; but if c_1 is also zero, then B itself is the greatest common divisor of A and B, that is, A is divisible by B. The conditions for this are therefore

$$3a_0a_2 - a_1^2 = 0, \qquad 9a_0a_3 - a_1a_2 = 0. \tag{16}$$

If c_0 is not zero, we go one step further and put

$$B = PC + D, \tag{17}$$

where D becomes constant and has the expression

$$D = \frac{a_2c_0^2 - 2a_1c_0c_1 + 3a_0c_1^2}{c_0^2}. \tag{18}$$

If this expression is non-zero, then A and B are prime to one another; if it is zero, then A and B have the greatest common divisor C. If for c_0, c_1 one substitutes the values (15), omits the denominator, still removes the non-vanishing factor $9a_0$, and changes the sign, this condition, after a simple calculation, assumes the form

$$a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2 = 0. \tag{19}$$

It is, as one readily sees either by calculation or from (18), also fulfilled when the conditions (16) hold, and is therefore the necessary and sufficient condition that $f(x)$ and $f'(x)$ have a common divisor. The left hand side of (19), which is an integral rational and homogeneous function of the coefficients of $f(x)$, is called the *discriminant* of the function $f(x)$.

We derive one more theorem from the algorithm (3), which is often used.

From the first of these equations it follows that

$$A'' = A - Q'A', \tag{20}$$

and if this value of A'' is substituted in the following equation, one obtains

$$A''' = (1 + Q'Q'')A' - Q''A,$$

thus, denoting by p, p' integral rational functions,

$$A''' = pA + p'A'. \tag{21}$$

Substituting the expressions (20), (21) in the third equation (3), one obtains for A'''' again an expression of the form (21), and so one may continue, obtaining finally

$$A^{(\nu)} = PA + P'A', \tag{22}$$

where P, P' are integral rational functions whose coefficients are composed from the coefficients of A and A' by rational operations.

In the formula (22), $A^{(\nu)}$ is a constant. This theorem is especially important in the case where $A^{(\nu)}$ is non-zero, so that A, A' are relatively prime. If in this case we put

$$P = A^{(\nu)}F(x), \qquad P' = A^{(\nu)}\Phi(x),$$

then, after suppressing the factor $A^{(\nu)}$, the theorem can be expressed as follows:

I. If $f(x)$ and $\psi(x)$ are two integral functions without a common divisor, then two other integral functions $F(x)$ and $\Phi(x)$ can be determined which satisfy identically the equation

$$F(x)f(x) + \Phi(x)\psi(x) = 1. \tag{23}$$

The theorem can still be generalized. Multiplying (23) by an arbitrary integral function $\chi(x)$ gives

$$F(x)\chi(x)f(x) + \Phi(x)\chi(x)\psi(x) = \chi(x), \tag{24}$$

and by § 3 we can put

$$\Phi(x)\chi(x) = Q(x)f(x) + \varphi(x) \tag{25}$$

in such a way that $Q(x), \varphi(x)$ are integral rational functions of x and the degree of $\varphi(x)$ is lower than that of $f(x)$. Substituting this in (24), and putting in place of $F(x)\chi(x) + Q(x)\psi(x)$ again $F(x)$, we obtain

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x).$$

We can therefore generalize the preceding theorem as follows:

II. If $f(x), \psi(x), \chi(x)$ are given integral rational functions, and $f(x)$ and $\psi(x)$ are without a common divisor, then the integral rational functions $F(x), \varphi(x)$ can be so determined - with $\varphi(x)$ of lower degree than $f(x)$ - that the equation

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x) \tag{26}$$

is identically satisfied.

§ 7. Products of linear factors.

According to § 1, by multiplication of integral functions we again obtain such functions of higher degree, and the degree of the product is determined by the sum of the degrees of the individual factors.

If therefore we multiply n linear factors by one another, there arises a function of the nth degree, whose manner of formation we must examine somewhat more closely.

We shall take the linear factors in the simple form $x - \alpha_1$, $x - \alpha_2$, …, $x - \alpha_n$, and put, since the coefficient of the highest power is, by § 1, equal to 1,

$$\begin{aligned} f(x) &= (x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n) \\ &= x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n. \end{aligned} \tag{1}$$

A slight consideration shows the following law of formation of the coefficients $a_1, a_2 \dots a_n$:

$-a_1$ is equal to the sum of the α's; a_2 is equal to the sum of the products of every two of the α's; $-a_3$ is the sum of the products of every three of the α's; in general $(-1)^\nu a_\nu$ is the sum of the products of every ν of the quantities α, or, in formulas,

$$\begin{aligned}
-a_1 &= \sum \alpha_1, \\
+a_2 &= \sum \alpha_1\alpha_2, \\
&\cdots \\
(-1)^\nu a_\nu &= \sum \alpha_1\alpha_2\cdots\alpha_\nu, \\
&\cdots \\
(-1)^n a_n &= \alpha_1\alpha_2\cdots\alpha_n.
\end{aligned} \tag{2}$$

But to see still more clearly the correctness of this law of formation, one uses complete induction.

One first verifies the correctness in the first cases by actually carrying out the multiplication:

$$(x-\alpha_1)(x-\alpha_2) = x^2 - (\alpha_1+\alpha_2)x + \alpha_1\alpha_2,$$

$$\begin{aligned}
(x-\alpha_1)(x-\alpha_2)(x-\alpha_3) = x^3 &- (\alpha_1+\alpha_2+\alpha_3)x^2 \\
&+ (\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3)x - \alpha_1\alpha_2\alpha_3.
\end{aligned}$$

If we assume the law of formation proved for $n-1$ factors, and put

$$(x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_{n-1}) = x^{n-1} + a_1'x^{n-2} + \cdots + a_{n-1}',$$

then multiplication by $x-\alpha_n$ gives

$$\begin{aligned}
a_1 &= a_1' - \alpha_n, \\
a_2 &= a_2' - \alpha_n a_1', \\
a_3 &= a_3' - \alpha_n a_2', \\
&\cdots \\
a_\nu &= a_\nu' - \alpha_n a_{\nu-1}', \\
&\cdots \\
a_n &= -\alpha_n a_{n-1}',
\end{aligned} \tag{3}$$

and from this the correctness of the formulas (2) is immediately evident.

It is important to determine the number of terms which occur in each of the sums (2). We denote the number of terms occurring in the sum $(-1)^\nu a_\nu$ by $B_\nu^{(n)}$; it is the number of combinations without repetition of n elements to the νth class, that is, of every ν distinct ones among the n elements. To determine it, suppose first that the $B_{\nu-1}^{(n)}$ combinations to the $(\nu-1)$st class have been formed. From each of these combinations, by adding one at a time of the missing $n-\nu+1$ elements, one can derive $n-\nu+1$ combinations to the νth class. In this way however every combination is formed ν times, namely by addition of each of its elements; hence one obtains the relation

$$B_\nu^{(n)} = B_{\nu-1}^{(n)}\frac{n-\nu+1}{\nu}, \tag{4}$$

whereas $B_1^{(n)}$ plainly has the value n. Therefore, if one multiplies the equations following from (4),

$$\begin{aligned}
B_1^{(n)} &= n, \\
B_2^{(n)} &= B_1^{(n)}\frac{n-1}{2}, \\
&\cdots \\
B_\nu^{(n)} &= B_{\nu-1}^{(n)}\frac{n-\nu+1}{\nu},
\end{aligned} \tag{5}$$

it follows that

$$B_\nu^{(n)} = \frac{n(n-1)(n-2)\cdots(n-\nu+1)}{1\cdot 2\cdot 3\cdots \nu}. \tag{6}$$

In what follows we shall often, when m is an arbitrary positive integer, use the symbol

$$\Pi(m) = 1\cdot 2\cdot 3\cdots m, \qquad \Pi(0) = 1, \tag{7}$$

so that

$$\Pi(m) = m\Pi(m-1). \tag{8}$$

With the help of this symbol the expression for $B_\nu^{(n)}$ can be displayed more conveniently as

$$B_\nu^{(n)} = B_{n-\nu}^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)}, \tag{9}$$

which, if $B_0^{(n)} = 1$ is put, is also valid for $\nu = 0$ and $\nu = n$, and makes evident the invariance of $B_\nu^{(n)}$ under the interchange of ν with $n-\nu$.

The derivation of the formula (4), which we have just given according to the rules of the theory of combinations, is indeed perfectly correct and clear, but for its exact foundation it requires some reflection which cannot easily be compressed into a few words. We shall therefore subsequently prove its correctness once more by the method of complete induction.

The formulas (3), namely, give the following recurrence formulas:

$$\begin{aligned} B_1^{(n)} &= B_1^{(n-1)} + 1, \\ B_2^{(n)} &= B_2^{(n-1)} + B_1^{(n-1)}, \\ &\cdots \\ B_\nu^{(n)} &= B_\nu^{(n-1)} + B_{\nu-1}^{(n-1)}, \\ &\cdots \\ B_n^{(n)} &= B_{n-1}^{(n-1)}. \end{aligned} \tag{10}$$

Now the formula (6) can be confirmed very easily by counting for the first values of n. If one assumes it correct for $n-1$, then (10) gives

$$B_\nu^{(n)} = \frac{\Pi(n-1)}{\Pi(\nu)\Pi(n-\nu-1)} + \frac{\Pi(n-1)}{\Pi(\nu-1)\Pi(n-\nu)},$$

whence, by (8),

$$B_\nu^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)},$$

and by this the general validity of formula (9) is proved.

From the meaning of $B_\nu^{(n)}$ it follows, and it is also proved from the formulas (10) by complete induction, that the $B_\nu^{(n)}$ are positive integers.

§ 8. The binomial theorem.

If in formula (1) of the preceding paragraph we set the previously arbitrary quantities $\alpha_1, \alpha_2 \ldots \alpha_n$ equal to one another, we obtain the binomial theorem, which consists in nothing other than arranging the nth power of a binomial $x + y$ according to powers of x. Namely, if we put

$$\alpha_1 = \alpha_2 = \cdots = \alpha_n = -y,$$

then formula (2) gives

$$a_\nu = (-1)^\nu \sum \alpha_1\alpha_2\cdots\alpha_\nu = y^\nu B_\nu^{(n)},$$

where, according to the definition of the preceding paragraph, $B_\nu^{(n)}$ means the number of terms of the sum, which all become equal to one another and equal to $(-1)^\nu y^\nu$.

Accordingly one obtains

$$(x+y)^n = x^n + B_1^{(n)} x^{n-1} y + B_2^{(n)} x^{n-2} y^2 + \cdots + B_n^{(n)} y^n, \tag{1}$$

or, written out,

$$\begin{aligned}(x+y)^n &= x^n + n x^{n-1} y + \frac{n(n-1)}{1\cdot 2} x^{n-2} y^2 + \cdots + y^n \\ &= \Pi(n) \sum \frac{x^{n-\nu} y^\nu}{\Pi(n-\nu)\Pi(\nu)} = \Pi(n) \sum \frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)},\end{aligned}$$

the last sum being extended over all non-negative integral values α, β which satisfy the condition $\alpha + \beta = n$.

Accordingly the coefficients $B_\nu^{(n)}$ are called the binomial coefficients.

For reference we put here a small table of the first values of the binomial coefficients:

$n=1$	1	1						
$n=2$	1	2	1					
$n=3$	1	3	3	1				
$n=4$	1	4	6	4	1			
$n=5$	1	5	10	10	5	1		
$n=6$	1	6	15	20	15	6	1	
$n=7$	1	7	21	35	35	21	7	1

Among the various properties of the binomial coefficients we shall derive two, of which we shall afterwards make an interesting application.

From (1), replacing x, y by $1, x$ and taking $n = 0, 1, 2, 3 \ldots$, the formulas

$$\begin{aligned} 1 &= B_0^{(0)}, \\ 1 + x &= B_0^{(1)} + B_1^{(1)} x, \\ (1+x)^2 &= B_0^{(2)} + B_1^{(2)} x + B_2^{(2)} x^2, \\ &\cdots \\ (1+x)^n &= B_0^{(n)} + B_1^{(n)} x + B_2^{(n)} x^2 + \cdots + B_n^{(n)} x^n \end{aligned} \tag{2}$$

are obtained.

We now use the well-known summation formula for the geometric series:

$$1 + (1+x) + (1+x)^2 + \cdots + (1+x)^n = \frac{(1+x)^{n+1} - 1}{x},$$

where, if $(1+x)^{n+1}$ is again expanded by the binomial formula, replacing n by $n+1$ in the last formula (2) and setting $B_0^{(n+1)} = 1$,

$$\frac{(1+x)^{n+1} - 1}{x} = B_1^{(n+1)} + B_2^{(n+1)} x + \cdots + B_{n+1}^{(n+1)} x^n. \tag{3}$$

Comparing this with the sum of the right hand sides of (2), and, according to the rules of § 1, setting equal the coefficients of corresponding powers of x, one obtains

$$\begin{aligned} B_0^{(0)} + B_0^{(1)} + B_0^{(2)} + \cdots + B_0^{(n)} &= B_1^{(n+1)}, \\ B_1^{(1)} + B_1^{(2)} + \cdots + B_1^{(n)} &= B_2^{(n+1)}, \\ &\cdots \\ B_n^{(n)} &= B_{n+1}^{(n+1)}, \end{aligned} \tag{4}$$

or in general
$$B_\nu^{(\nu)} + B_\nu^{(\nu+1)} + \cdots + B_\nu^{(n)} = B_{\nu+1}^{(n+1)}. \tag{5}$$
If however the equations (2) are multiplied in order by
$$B_0^{(n)}, \quad -B_1^{(n)}, \quad +B_2^{(n)}, \ldots, \quad \pm B_n^{(n)},$$
where the upper sign is to be taken for even n and the lower for odd n, the sum of the left hand sides gives
$$B_0^{(n)} - B_1^{(n)}(1+x) + B_2^{(n)}(1+x)^2 - \cdots \pm B_n^{(n)}(1+x)^n = [1-(1+x)]^n = (-x)^n,$$
and equating the coefficients of equal powers on the right and left hand sides yields the system of formulas
$$\begin{aligned}
0 &= B_0^{(0)}B_0^{(n)} - B_0^{(1)}B_1^{(n)} + B_0^{(2)}B_2^{(n)} - \cdots \pm B_0^{(n)}B_n^{(n)},\\
0 &= -B_1^{(1)}B_1^{(n)} + B_1^{(2)}B_2^{(n)} - \cdots \pm B_1^{(n)}B_n^{(n)},\\
&\cdots\\
\pm 1 &= \pm B_n^{(n)}B_n^{(n)},
\end{aligned} \tag{6}$$
and this system of formulas can also be written in the reversed order, using the relation $B_\nu^{(n)} = B_{n-\nu}^{(n)}$, as
$$\begin{aligned}
1 &= B_0^{(n)}B_0^{(n)},\\
0 &= B_1^{(n)}B_0^{(n)} - B_0^{(n-1)}B_1^{(n)},\\
0 &= B_2^{(n)}B_0^{(n)} - B_1^{(n-1)}B_1^{(n)} + B_0^{(n-2)}B_2^{(n)},\\
&\cdots\\
0 &= B_n^{(n)}B_0^{(n)} - B_{n-1}^{(n-1)}B_1^{(n)} + B_{n-2}^{(n-2)}B_2^{(n)} - \cdots \pm B_0^{(0)}B_n^{(n)},
\end{aligned} \tag{7}$$
or, in a summarizing notation,
$$\sum_{\nu=0}^{\mu} (-1)^\nu B_{\mu-\nu}^{(n-\nu)} B_\nu^{(n)} = \begin{cases} 1, & \mu = 0,\\ 0, & \mu = 1, 2, \ldots, n. \end{cases} \tag{8}$$

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§ 9. Interpolation.

We shall make an application of the last obtained formulae to a problem from the theory of interpolation.

A whole rational function of degree n is to be determined which, for $n+1$ given values of the variable x, has prescribed values.

Thus the problem is the determination of the $n+1$ coefficients $a_0, a_1, \ldots, a_n$ in

$$f(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n \tag{1}$$

from the $n+1$ equations

$$f(\alpha_0) = A_0, \quad f(\alpha_1) = A_1, \quad \ldots, \quad f(\alpha_n) = A_n, \tag{2}$$

when $\alpha_0, \alpha_1, \ldots, \alpha_n$ are the given values of x, and $A_0, A_1, \ldots, A_n$ are the corresponding values of $f(x)$.

Equations (2) form a system of equations of the first degree for the unknowns $a_0, a_1, \ldots, a_n$, which in general can be solved by determinants, as we shall see in the next section. We shall later return to this problem, but treat it now under the special supposition that the given values of x are the first $n+1$ whole numbers $0, 1, 2, \ldots, n$.

The binomial coefficients $B_\nu^{(x)}$, as they were defined by § 7, (6), keep their good sense even when x is not a whole number, as was previously presupposed, but an arbitrary variable quantity. Then

$$B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1\cdot 2\cdot 3\cdots \nu} \tag{3}$$

is a whole rational function of degree ν in x. [8] The coefficient of x^ν is different from zero, and therefore x^ν can also be expressed by $B_\nu^{(x)}$ and by lower powers of x. Thus every whole function of degree n in x can also be linearly expressed by $B_0^{(x)}, B_1^{(x)}, \ldots, B_n^{(x)}$ in the form

$$f(x) = M_0 B_0^{(x)} + M_1 B_1^{(x)} + \cdots + M_n B_n^{(x)}, \tag{4}$$

where $M_0, M_1, \ldots, M_n$ are constants; and the function $f(x)$ is determined when these constants are determined.

Now, by our supposition, $f(0), f(1), f(2), \ldots, f(n)$ are given. Since, by (3), $B_\nu^{(x)}$ always vanishes when x has one of the values $0, 1, 2, \ldots, \nu-1$, the following linear equations for the unknowns M are obtained from (4):

$$\begin{aligned} f(0) &= M_0 B_0^{(0)}, \\ f(1) &= M_0 B_0^{(1)} + M_1 B_1^{(1)}, \\ f(2) &= M_0 B_0^{(2)} + M_1 B_1^{(2)} + M_2 B_2^{(2)}, \\ &\cdots \\ f(n) &= M_0 B_0^{(n)} + M_1 B_1^{(n)} + \cdots + M_n B_n^{(n)}. \end{aligned} \tag{5}$$

These equations are now to be solved with respect to $M_0, M_1, \ldots, M_n$, which is done very easily with the help of the final equations (8) of the preceding paragraph. The first equation (5) gives directly

$$M_0 = f(0). \tag{6}$$

[8] The meaning of these generalized binomial coefficients for the expansion of powers of a binomial does not belong here, but to analysis.

If the first equation (5) is multiplied by $B_0^{(1)}$, the second by $-B_1^{(1)}$, and they are added, one obtains, by the formula system mentioned (applied to $n = 1$),

$$-M_1 = B_0^{(1)} f(0) - B_1^{(1)} f(1),$$

and so in general, if the first ν equations (5) are multiplied in order by

$$B_0^{(\nu)}, \quad -B_1^{(\nu)}, \quad +B_2^{(\nu)}, \quad \dots, \quad \pm B_\nu^{(\nu)}$$

and then added,

$$\pm M_\nu = B_0^{(\nu)} f(0) - B_1^{(\nu)} f(1) + B_2^{(\nu)} f(2) - \cdots \pm B_\nu^{(\nu)} f(\nu), \tag{7}$$

whereby, according to (4), the function $f(x)$ is determined and the problem is solved. It is clear that, so long as we make no special supposition about the values $f(0), f(1), \dots, f(n)$, every arbitrary whole rational function of x can be represented in this form.

§ 10. Solution of the interpolation problem by differences.

The definition of the whole rational function

$$B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1 \cdot 2 \cdot 3 \cdots \nu} \tag{1}$$

gives the relation, already proved earlier for the special case of a positive integral x,

$$B_\nu^{(x+1)} = B_\nu^{(x)} + B_{\nu-1}^{(x)}, \qquad B_0^{(x+1)} = B_0^{(x)} = 1. \tag{2}$$

From this we obtain for the coefficients $M_0, M_1, \dots, M_n$ a mode of determination which is much more convenient for practical calculation than the application of the formulae of the preceding paragraph.

Namely, by the formulae (4) and (6) of § 9,

$$f(x) = f(0) + M_1 B_1^{(x)} + M_2 B_2^{(x)} + \cdots + M_n B_n^{(x)}, \tag{3}$$

and if in this expression we replace x by $x+1$ and form the difference

$$\Delta_x = f(x+1) - f(x), \tag{4}$$

then, with regard to (2),

$$\Delta_x = M_1 + M_2 B_1^{(x)} + M_3 B_2^{(x)} + \cdots + M_n B_{n-1}^{(x)}, \tag{5}$$

from which follows (since (5) is an equation of the same kind as (3), except that $n-1$ has taken the place of n)

$$M_1 = \Delta_0 = f(1) - f(0).$$

Put

$$\Delta'_x = \Delta_{x+1} - \Delta_x, \qquad \Delta''_x = \Delta'_{x+1} - \Delta'_x, \quad \dots, \tag{6}$$

then, accordingly,

$$M_0 = f(0), \quad M_1 = \Delta_0, \quad M_2 = \Delta'_0, \quad \dots, \quad M_n = \Delta_0^{(n-1)},$$

and formula (3) gives

$$f(x) = f(0) + \Delta_0 B_1^{(x)} + \Delta'_0 B_2^{(x)} + \Delta''_0 B_3^{(x)} + \cdots + \Delta_0^{(n-1)} B_n^{(x)}. \tag{7}$$

Likewise the functions $\Delta_x, \Delta'_x, \Delta''_x, \ldots$ can be expressed:

$$\begin{aligned} \Delta_x &= \Delta_0 + \Delta'_0 B_1^{(x)} + \Delta''_0 B_2^{(x)} + \cdots + \Delta_0^{(n-1)} B_{n-1}^{(x)}, \\ \Delta'_x &= \Delta'_0 + \Delta''_0 B_1^{(x)} + \cdots + \Delta_0^{(n-1)} B_{n-2}^{(x)}, \end{aligned} \tag{8}$$

and so on.

The $\Delta_x, \Delta'_x, \Delta''_x, \ldots, \Delta_x^{(n-1)}$ are whole rational functions of degrees $n-1, n-2, \ldots, 0$; the last of them is therefore constant. In order to represent them all, one needs only the values

$$f(0), \quad \Delta_0, \quad \Delta'_0, \quad \Delta''_0, \quad \ldots, \quad \Delta_0^{(n-1)},$$

which one computes most easily by arranging a table. For the case $n = 3$, for example, the table has the form

$$\begin{array}{llll} f(0) & & & \\ f(1) & \Delta_0 & & \\ f(2) & \Delta_1 & \Delta'_0 & \\ f(3) & \Delta_2 & \Delta'_1 & \Delta''_0 \end{array}$$

where the differences are to be formed diagonally.

§ 11. Arithmetic series of higher order.

The preceding considerations can also be stated in another form, which we shall need later. If

$$u_0, u_1, u_2, \ldots \tag{1}$$

is a sequence of quantities, we call

$$\Delta_0 = u_1 - u_0, \Delta_1 = u_2 - u_1, \Delta_2 = u_3 - u_2, \ldots \tag{2}$$

the sequence of its differences; and

$$\Delta'_0 = \Delta_1 - \Delta_0, \Delta'_1 = \Delta_2 - \Delta_1, \ldots \tag{3}$$

the sequence of its second differences, and so on.

It is clear that the sequence (1) is completely determined when its first term and the sequence of its first differences are given; for

$$u_1 = u_0 + \Delta_0, \quad u_2 = u_0 + \Delta_0 + \Delta_1, \quad \ldots,$$

$$u_m = u_0 + \Delta_0 + \Delta_1 + \cdots + \Delta_{m-1}.$$

Likewise the sequence (1) is completely determined when the first two terms and the sequence of its second differences are given, and so forth. The sequence (1) is called an arithmetic series of nth order when the sequence of its nth differences is constant, and hence the sequence of the $(n+1)$st differences consists entirely of zeros.

One obtains an arithmetic series of nth order by substituting the numbers $0, 1, 2, 3, \ldots$ for x in a whole function $f(x)$ of degree n. For if one puts

$$\Delta_x = f(x+1) - f(x), \quad \Delta_x^{(n-1)} = \Delta_{x+1}^{(n-2)} - \Delta_x^{(n-2)},$$

then Δ_x is of degree $n-1$, Δ'_x of degree $n-2$, and so on with respect to x, and therefore $\Delta_x^{(n-1)}$ is constant.

If now

$$u_0, u_1, u_2, \ldots$$

is an arithmetic series of nth order, then the whole series is completely determined when the n values $u_0, u_1, u_2, \ldots, u_{n-1}$ and, besides these, the constant nth difference are given. But this latter is determined if also the $(n+1)$st term u_n is given. We can therefore state the theorem:

An arithmetic series of nth order is completely determined when its first $n+1$ terms are given.

Since a function $f(x)$ of degree n is likewise completely determined by the arbitrarily given values

$$f(0), f(1), f(2), \ldots, f(n),$$

it follows that from the whole rational functions of degree n all arithmetic series of nth order are produced when in them the series of natural numbers is substituted for x.

The expression for the general term is then given by formula (7) of the preceding paragraph.

The sums of the first $m+1$ terms of an arithmetic series of nth order,

$$s_m = u_0 + u_1 + \cdots + u_m,$$

form an arithmetic series of $(n+1)$st order, since their first differences

$$s_{m+1} - s_m = u_{m+1}$$

form an arithmetic series of nth order.

Thus, with the help of formula (7) of § 10, the sum s_m can be determined in general when $s_0, s_1, \ldots, s_{n+1}$ are assumed known.

To find the generating function $F(x)$ of s_m, when $f(x)$ is the generating function of u_m, put

$$F(0) = f(0), \quad F(1) = f(0) + f(1), \quad F(2) = f(0) + f(1) + f(2), \ldots$$

and in formula (7), § 10,

$$F(x) = F(0) + D_0 B_1^{(x)} + D_0' B_2^{(x)} + \cdots$$

one has to put

$$\begin{aligned} D_0 &= F(1) - F(0) = f(1), & D_0' &= f(2) - f(1) = \Delta_1, \\ D_1 &= F(2) - F(1) = f(2), & D_1' &= f(3) - f(2) = \Delta_2, \\ D_2 &= F(3) - F(2) = f(3), & D_0'' &= \Delta_2 - \Delta_1 = \Delta_1', \end{aligned}$$

and so on. Thus one obtains

$$F(x) = f(0) + f(1)B_1^{(x)} + \Delta_1 B_2^{(x)} + \Delta_1' B_3^{(x)} + \cdots . \tag{4}$$

For example, take $f(x) = x^2$; then $F(m)$ gives the sum of the first m square numbers. Here

$$\Delta_x = 2x + 1, \qquad \Delta_x' = 2,$$

and therefore

$$F(x) = x + 3\frac{x(x-1)}{1 \cdot 2} + 2\frac{x(x-1)(x-2)}{1 \cdot 2 \cdot 3} = \frac{x(x+1)(2x+1)}{6}.$$

For $f(x) = x^3$ the same calculation gives

$$F(x) = \left(\frac{x(x+1)}{2}\right)^2.$$

Thus the sum of the first m cubes is equal to the square of the mth triangular number.

§ 12. The polynomial theorem.

In § 8 the following form of the binomial theorem was derived:

$$(x+y)^n = \Pi(n) \sum^{\alpha,\beta} \frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)}, \tag{1}$$

where the sum extends over all combinations of two numbers α, β, neither of which is negative, and which satisfy the condition

$$\alpha + \beta = n. \tag{2}$$

This form permits, first by induction, a generalization to the nth power of a polynomial:

$$(x+y+z+\cdots)^n = \Pi(n) \sum^{\alpha,\beta,\gamma,\ldots} \frac{x^\alpha y^\beta z^\gamma \cdots}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma)\cdots}, \tag{3}$$

with the determination that $\alpha, \beta, \gamma, \ldots$ run through all positive or vanishing integral values which satisfy

$$\alpha + \beta + \gamma + \cdots = n. \tag{4}$$

But to prove the correctness of this formula in general, we assume that it is proved when the polynomial contains one term fewer, as is indeed the case when the polynomial contains only two terms.

We then put

$$u = y + z + \cdots \tag{5}$$

and apply formula (1) to $(x+u)$, from which there follows

$$(x+y+z+\cdots)^n = \Pi(n) \sum^{\alpha,\nu} \frac{x^\alpha u^\nu}{\Pi(\alpha)\Pi(\nu)}, \tag{6}$$

with the restriction

$$\alpha + \nu = n. \tag{7}$$

By the assumption, however, it has already been proved that

$$u^\nu = \Pi(\nu) \sum^{\beta,\gamma,\ldots} \frac{y^\beta z^\gamma \cdots}{\Pi(\beta)\Pi(\gamma)\cdots}, \tag{8}$$

where

$$\beta + \gamma + \cdots = \nu. \tag{9}$$

When this is inserted in (6), formula (3) is obtained immediately, and (7) passes over into (4).

The coefficients

$$P^{(n)}_{\alpha,\beta,\gamma,\ldots} = \frac{\Pi(n)}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma)\cdots}, \tag{10}$$

which, according to their meaning, are whole numbers, are called the polynomial coefficients.

For example, for the third power of the trinomial one obtains

$$\begin{aligned}(x+y+z)^3 = x^3 + y^3 + z^3 + 3x^2y + 3xy^2 + 3x^2z + 3xz^2 \\ + 3y^2z + 3yz^2 + 6xyz.\end{aligned} \tag{11}$$

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§ 13. Derived Functions.

Let

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n \tag{1}$$

be an entire rational function of order n.

If in it we replace x by a binomial $x + y$, then we can apply the binomial theorem to each separate term, and can arrange the result according to descending or according to ascending powers of x or of y. We shall carry out the arrangement according to ascending powers of y. The highest power of y which occurs is the nth, and the coefficient of the zeroth power of y is the function $f(x)$ itself, as one sees by putting $y = 0$. We therefore put, denoting the other coefficients by

$$f'(x), \qquad \frac{f''(x)}{\Pi(2)}, \qquad \frac{f'''(x)}{\Pi(3)}, \ldots :$$

$$f(x+y) = f(x) + yf'(x) + \frac{y^2}{1 \cdot 2} f''(x) + \cdots = \sum_{0,n}^{\nu} \frac{y^\nu}{\Pi(\nu)} f^{(\nu)}(x). \tag{2}$$

The functions $f'(x), f''(x), f'''(x), \ldots$ are called the first, second, third, ... derivative or derived function of $f(x)$. They are entire functions of x, and $f^{(\nu)}(x)$ cannot exceed the degree $n - \nu$, since the sum of the exponents of x and y in no term exceeds the degree n.

The first derivative, which is thus the coefficient of the first power of y in the expansion of $f(x+y)$ according to ascending powers of y, is obtained by applying the binomial theorem to (1):

$$f'(x) = na_0 x^{n-1} + (n-1)a_1 x^{n-2} + (n-2)a_2 x^{n-3} + \cdots . \tag{3}$$

The principal theorem on derived functions follows from (2) if we transform x into $x + z$ or y into $y + z$:

$$f(x+y+z) = \sum_{0,n}^{\nu} \frac{y^\nu}{\Pi(\nu)} f^{(\nu)}(x+z) = \sum_{0,n}^{\nu} \frac{(y+z)^\nu}{\Pi(\nu)} f^{(\nu)}(x). \tag{4}$$

If by $f^{(\nu,\mu)}(x)$ we denote the μth derivative of $f^{(\nu)}(x)$, then by (2)

$$f^{(\nu)}(x+z) = \sum_{0,n-\nu}^{\mu} \frac{z^\mu}{\Pi(\mu)} f^{(\nu,\mu)}(x), \tag{5}$$

and by the binomial theorem

$$\frac{(y+z)^\nu}{\Pi(\nu)} = \sum^{\beta,\gamma} \frac{y^\beta z^\gamma}{\Pi(\beta)\Pi(\gamma)}, \qquad \beta + \gamma = \nu. \tag{6}$$

Substituting this in (4), one obtains

$$\sum_{0,n}^{\nu} \sum_{0,n-\nu}^{\mu} \frac{y^\nu z^\mu}{\Pi(\nu)\Pi(\mu)} f^{(\nu,\mu)}(x) = \sum^{\beta,\gamma} \frac{y^\beta z^\gamma}{\Pi(\beta)\Pi(\gamma)} f^{(\beta+\gamma)}(x). \tag{7}$$

The last sum is to be extended over all non-negative numbers β, γ whose sum does not exceed the degree n of $f(x)$. But the same number-combinations are also run through by the exponents ν, μ on the left-hand side, and comparison of the coefficients of equal powers and products gives (by § 1)

$$f^{(\nu,\mu)}(x) = f^{(\nu+\mu)}(x), \tag{8}$$

and therefore the theorem:

The μth derivative of the νth derivative is the $(\nu + \mu)$th derivative of the original function.

Thus all the higher derivatives are obtained by forming from each preceding one, according to the rule (3), the first derivative:

$$\begin{aligned} f(x) &= a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + \cdots , \\ f'(x) &= na_0x^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \cdots , \\ f''(x) &= n(n-1)a_0x^{n-2} + (n-1)(n-2)a_1x^{n-3} + \cdots . \end{aligned} \tag{9}$$

These derivatives assume a somewhat simpler form if one uses another notation, which is often in use and for certain purposes almost indispensable, and which we shall discuss in connection with this.

Because of the indeterminacy of the coefficients $a_0, a_1, \ldots, a_n$, it is clearly no restriction if we write an entire rational function of degree n in the form

$$f(x) = a_0x^n + B_1^{(n)}a_1x^{n-1} + B_2^{(n)}a_2x^{n-2} + \cdots + a_n, \tag{10}$$

or explicitly

$$f(x) = a_0x^n + na_1x^{n-1} + \frac{n(n-1)}{1 \cdot 2}a_2x^{n-2} + \cdots . \tag{11}$$

When a function $f(x)$ is represented in this way, we shall say that it is "written with the binomial coefficients."

A greater agreement is already shown by the formulae (9), which then read:

$$\begin{aligned} f(x) &= a_0x^n + na_1x^{n-1} + \frac{n(n-1)}{1 \cdot 2}a_2x^{n-2} + \cdots , \\ \frac{1}{n}f'(x) &= a_0x^{n-1} + (n-1)a_1x^{n-2} + \frac{(n-1)(n-2)}{1 \cdot 2}a_2x^{n-3} + \cdots , \\ \frac{1}{n(n-1)}f''(x) &= a_0x^{n-2} + (n-2)a_1x^{n-3} + \frac{(n-2)(n-3)}{1 \cdot 2}a_2x^{n-4} + \cdots , \end{aligned} \tag{12}$$

wherein the right-hand sides all likewise appear written with the binomial coefficients. We shall later become more acquainted with the usefulness of this notation, but already here we must emphasize that the choice of the one or the other manner of representation is nevertheless not wholly indifferent, and that the first often deserves preference. Especially in those cases in which the coefficients are numbers and the number-theoretic behavior of these coefficients is at issue, one must not overlook that through the binomial coefficients a numerical element foreign to the matter is introduced. That Gauss in the theory of quadratic forms (in the Disq. ar.) uses the notation with binomial coefficients when he represents the quadratic forms by $ax^2 + 2bxy + cy^2$, and that this notation has gained general entry, has led in number theory to an unnecessary and very regrettable complication.

§ 14. Derivative of a Product.

The derived functions which we have considered here are no others than the differential quotients known from differential calculus; however, here, where only entire rational functions are involved, we have obtained the concept without using infinitesimal calculus, from the coefficients of development of the powers of y in the function $f(x + y)$. If we denote the νth derivative of $f(x)$ by $D_\nu f$, then by (2), § 13:

$$f(x + y) = f(x) + yD_1f + \frac{y^2}{1 \cdot 2}D_2f + \frac{y^3}{1 \cdot 2 \cdot 3}D_3f + \cdots , \tag{1}$$

and from this follow at once the two fundamental rules expressed in the formulae

$$D_\nu(Cf) = CD_\nu f; \tag{2}$$

$$D_\nu(f+\varphi) = D_\nu f + D_\nu \varphi, \tag{3}$$

where C is a constant, and φ a second entire rational function of x.

A generalization of formula (2) gives the representation of the derivatives of the product $f\varphi$. Namely, if after (1) one writes, for abbreviation,

$$\begin{aligned} f(x+y) &= u_0 + yu_1 + y^2u_2 + \cdots + y^n u_n, \\ \varphi(x+y) &= v_0 + yv_1 + y^2v_2 + \cdots + y^m v_m, \end{aligned} \tag{4}$$

so that

$$u_\nu = \frac{D_\nu f}{\Pi(\nu)}, \qquad v_\nu = \frac{D_\nu \varphi}{\Pi(\nu)}, \tag{5}$$

then carrying out the multiplication of the two formulae (4), and seeking in the product the coefficient of y^ν, gives

$$\frac{D_\nu(f\varphi)}{\Pi(\nu)} = u_\nu v_0 + u_{\nu-1}v_1 + u_{\nu-2}v_2 + \cdots + u_0 v_\nu, \tag{6}$$

or

$$D_\nu(f\varphi) = \varphi D_\nu f + B_1^{(\nu)} D_{\nu-1} f D_1 \varphi + B_2^{(\nu)} D_{\nu-2} f D_2 \varphi + \cdots, \tag{7}$$

where $B_1^{(\nu)}, B_2^{(\nu)}, \ldots$ are the binomial coefficients. In a similar way one can, by using the polynomial coefficients, form the derivatives of a product of more than two factors.

We shall form the first derivative, which we now denote by D instead of by D_1, for a product of n factors. For two factors we obtain from (7)

$$D(f\varphi) = \varphi Df + fD\varphi = \varphi f'(x) + f\varphi'(x),$$

and generally, if we denote the factors by $u_1, u_2, \ldots, u_n$ and their derivatives by $u_1', u_2', \ldots, u_n'$,

$$D(u_1 u_2 \cdots u_n) = u_1' u_2 \cdots u_n + u_1 u_2' \cdots u_n + \cdots + u_1 u_2 \cdots u_n', \tag{8}$$

or more briefly,

$$\frac{D(u_1 u_2 \cdots u_n)}{u_1 u_2 \cdots u_n} = \sum \frac{Du_\nu}{u_\nu}.$$

Thus if we consider a product of linear factors

$$f(x) = (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n), \tag{9}$$

we obtain, since the first derivatives of $x-\alpha_1$, $x-\alpha_2, \ldots, x-\alpha_n$ are all equal to 1,

$$\begin{aligned} f'(x) = (x-\alpha_2)&(x-\alpha_3)\cdots(x-\alpha_n) \\ &+ (x-\alpha_1)(x-\alpha_3)\cdots(x-\alpha_n) + \cdots \\ &+ (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_{n-1}), \end{aligned} \tag{10}$$

for which one can also write

$$f'(x) = \frac{f(x)}{x-\alpha_1} + \frac{f(x)}{x-\alpha_2} + \cdots + \frac{f(x)}{x-\alpha_n}. \tag{11}$$

A very important result follows from this if x is put equal to one of the values $\alpha_1, \alpha_2, \ldots, \alpha_n$, namely

$$\begin{aligned} f'(\alpha_1) &= (\alpha_1-\alpha_2)(\alpha_1-\alpha_3)\cdots(\alpha_1-\alpha_n), \\ f'(\alpha_2) &= (\alpha_2-\alpha_1)(\alpha_2-\alpha_3)\cdots(\alpha_2-\alpha_n), \\ &\cdots \\ f'(\alpha_n) &= (\alpha_n-\alpha_1)(\alpha_n-\alpha_2)\cdots(\alpha_n-\alpha_{n-1}). \end{aligned} \tag{12}$$

§ 15. Entire Functions of Several Variables: Forms.

Up to now we have chiefly considered entire rational functions of one variable; but we cannot always restrict ourselves to this, and indeed we have already used functions of several variables above. By an entire rational function of degree n in several variables $F(x, y, z, \dots)$ we understand a sum of terms

$$\sum A_{\alpha,\beta,\gamma,\dots} x^\alpha y^\beta z^\gamma \cdots ,$$

where $\alpha, \beta, \gamma, \dots$ are non-negative integral exponents whose sum $\alpha + \beta + \gamma + \cdots$ does not exceed the value n, and in at least one term actually attains it. Thus the degree is determined by the greatest value assumed by the sum $\alpha + \beta + \gamma + \cdots$.

If the sum of the exponents $\alpha + \beta + \gamma + \cdots$ has the same value in all terms, then the function is called homogeneous.

A fundamental property of homogeneous functions of degree n is this: if all variables are multiplied by the same factor, the result is the same as if the function were multiplied by the nth power; in signs, if t denotes an arbitrary variable,

$$F(tx, ty, tz, \dots) = t^n F(x, y, z, \dots); \tag{1}$$

for if in the product $x^\alpha y^\beta z^\gamma \cdots$ the variables are replaced by $tx, ty, tz, \dots$, it receives the factor

$$t^{\alpha+\beta+\gamma+\cdots}.$$

If now $\alpha + \beta + \gamma + \cdots$ has in all terms the same value n, the factor t^n can be taken out before the sum. But if the sum $\alpha + \beta + \gamma + \cdots$ has different values in the separate terms, no such common factor can be taken out, at least not without leaving different powers of t in the separate terms.

By increasing the number of variables one can transform every non-homogeneous function into a homogeneous one of the same degree. If $m - 1$ is the number of variables in a non-homogeneous function of degree n, then we put

$$x = \frac{x_1}{x_m}, \qquad y = \frac{x_2}{x_m}, \qquad z = \frac{x_3}{x_m}, \dots,$$

and obtain in

$$x_m^n F\left(\frac{x_1}{x_m}, \frac{x_2}{x_m}, \frac{x_3}{x_m}, \dots\right)$$

an entire homogeneous function of degree n in the variables $x_1, x_2, \dots, x_m$, which we denote by

$$\Phi(x_1, x_2, \dots, x_m).$$

It is sometimes advisable to write homogeneous functions of several variables with the polynomial coefficients. We therefore put

$$\Phi(x_1, x_2, \dots, x_m) = \sum \frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} A_{\alpha_1,\alpha_2,\dots,\alpha_m} x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_m^{\alpha_m}, \tag{2}$$

where the sum is extended over all non-negative numbers satisfying the condition

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = n. \tag{3}$$

This notation, without the restriction (3), is also applicable to non-homogeneous functions.

But the homogeneous function can also be represented as

$$\Phi(x_1, x_2, \dots, x_m) = \sum A_{\nu_1,\nu_2,\dots,\nu_n} x_{\nu_1} x_{\nu_2} \cdots x_{\nu_n}, \tag{4}$$

where each of the indices $\nu_1, \nu_2, \dots, \nu_n$, independently of the others, runs through the value-row $1, 2, \dots, m$. Thus the sum (4) consists of m^n terms, but not all of them are different from one another. The product $x_{\nu_1} x_{\nu_2} \cdots x_{\nu_n}$ remains unchanged if the indices $\nu_1, \nu_2, \ldots, \nu_n$ are permuted among themselves in any way. The number of permutations of n elements is $\Pi(n)$. If among these elements $\alpha_1, \alpha_2, \ldots$ respectively are equal to one another, then the number of permutations is reduced to

$$\frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots},$$

whence it follows that in (4) any product $x_1^{\alpha_1} x_2^{\alpha_2} \cdots$ occurs exactly

$$\frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots}$$

times. If one further fixes that $A_{\nu_1,\nu_2,\ldots,\nu_n}$ shall not change when the indices are permuted in any way, then the notations (2) and (4) prove to be identical if, by collecting equal factors,

$$x_{\nu_1} x_{\nu_2} \cdots x_{\nu_n} = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_m^{\alpha_m}$$

and

$$A_{\nu_1,\nu_2,\ldots,\nu_n} = A_{\alpha_1,\alpha_2,\ldots,\alpha_m}$$

are put.

If the number of terms that occur in the function Φ [according to (2)] is denoted by (m, n), then, by first counting the terms which have the factor x_1 and then the remaining ones, which form a homogeneous function of order n of the remaining $m - 1$ variables, one finds the recursion formula

$$(m, n) = (m, n - 1) + (m - 1, n), \tag{5}$$

with whose aid one proves by complete induction the expression

$$(m, n) = \frac{m(m + 1) \cdots (m + n - 1)}{1 \cdot 2 \cdots n} = \frac{\Pi(m + n - 1)}{\Pi(n)\Pi(m - 1)} \tag{6}$$

to be correct.

Entire homogeneous functions are also called forms. According to the number of variables one distinguishes unary forms (simple powers), binary, ternary, quaternary forms. It is the binary forms that especially interest us here; their theory is essentially identical with the theory of entire rational functions of one variable. One passes from the binary forms back to these functions by regarding one of the homogeneous variables as constant, for example by giving it the value 1.

§ 16. The Derivatives of Functions of Several Variables.

In § 13 we defined the derived functions of an entire rational function of one variable. The concept is immediately transferable to functions of several variables, by forming the derivatives with respect to each variable for itself, as though it were the only one. Thus, if one writes for instance as in § 15

$$F(x, y, z, \ldots) = \sum A_{\alpha,\beta,\gamma,\ldots} x^\alpha y^\beta z^\gamma \cdots\,, \tag{1}$$

one obtains the first derivative with respect to x:

$$F'(x) = \sum \alpha A_{\alpha,\beta,\gamma,\ldots} x^{\alpha-1} y^\beta z^\gamma \cdots\,, \tag{2}$$

or with respect to y:

$$F'(y) = \sum \beta A_{\alpha,\beta,\gamma,\ldots} x^\alpha y^{\beta-1} z^\gamma \cdots \tag{3}$$

and so on. From these functions one can again form, according to the same rules, the derivatives with respect to the different variables, and thus obtains the higher derivatives.

In order to display the results more clearly, let $\Phi(x_1, x_2, \dots, x_m)$ be an entire rational function of order n in the m variables $x_1, x_2, \dots, x_m$. We replace these variables by binomials

$$x_1 + \xi_1, \quad x_2 + \xi_2, \dots, \quad x_m + \xi_m$$

and expand, in each term of the function

$$\Phi(x_1 + \xi_1, x_2 + \xi_2, \dots, x_m + \xi_m) = \Phi(x + \xi),$$

by carrying out the multiplication

$$(x_1 + \xi_1)^{\mu_1}(x_2 + \xi_2)^{\mu_2} \cdots (x_m + \xi_m)^{\mu_m}$$

according to powers of $\xi_1, \xi_2, \dots, \xi_m$. If equal powers and products of the variables ξ are collected each into one term, then for $\Phi(x + \xi)$, in the notation (2) of § 15, one obtains a representation which in differential calculus is called Taylor's expansion:

$$\Phi(x + \xi) = \sum \frac{\xi_1^{\alpha_1}\xi_2^{\alpha_2} \cdots \xi_m^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2) \cdots \Pi(\alpha_m)} D_{\alpha_1, \alpha_2, \dots, \alpha_m}\Phi. \tag{4}$$

The coefficients, which we denote by

$$D_{\alpha_1, \alpha_2, \dots, \alpha_m}\Phi,$$

are functions of the variables x and are called, when

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = \nu,$$

the derivatives of order ν of the function Φ.

They are also written, according to the notation usual in differential calculus, as

$$D_{\alpha_1, \alpha_2, \dots, \alpha_m}\Phi = \frac{\partial^\nu \Phi}{\partial x_1^{\alpha_1}\partial x_2^{\alpha_2} \cdots \partial x_m^{\alpha_m}}. \tag{5}$$

The law of formation of the derivatives can be summarized in the following propositions, where for brevity we omit the indices at the sign D.

I. If C is a constant, then

$$D(C\Phi) = CD\Phi.$$

II. If Φ and Ψ are any two functions, then

$$D(\Phi + \Psi) = D\Phi + D\Psi.$$

Both follow immediately from (4).

Thus we can easily form the derived functions in general if we know them for the special case in which Φ is a product of powers, that is, if we know

$$D_{\alpha_1, \alpha_2, \dots, \alpha_m}(x_1^{\mu_1}x_2^{\mu_2} \cdots x_m^{\mu_m}),$$

where the μ are arbitrary non-negative exponents.

But

$$(x_1 + \xi_1)^{\mu_1} = \sum^{\alpha_1} \frac{\Pi(\mu_1)}{\Pi(\alpha_1)\Pi(\mu_1 - \alpha_1)} \xi_1^{\alpha_1} x_1^{\mu_1 - \alpha_1},$$

and therefore

$$\begin{aligned}(x_1 + \xi_1)^{\mu_1} &\cdots (x_m + \xi_m)^{\mu_m} \\ &= \sum^{\alpha_1 \dots \alpha_m} \frac{\Pi(\mu_1) \cdots \Pi(\mu_m) x_1^{\mu_1 - \alpha_1} \cdots x_m^{\mu_m - \alpha_m}}{\Pi(\mu_1 - \alpha_1) \cdots \Pi(\mu_m - \alpha_m)\Pi(\alpha_1) \cdots \Pi(\alpha_m)} \xi_1^{\alpha_1} \cdots \xi_m^{\alpha_m}.\end{aligned} \tag{6}$$

Comparison with (4) and (6) therefore gives

$$D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}) \\ = \frac{\Pi(\mu_1)\cdots\Pi(\mu_m)}{\Pi(\mu_1-\alpha_1)\cdots\Pi(\mu_m-\alpha_m)} x_1^{\mu_1-\alpha_1}\cdots x_m^{\mu_m-\alpha_m}, \tag{7}$$

as long as $\alpha_1 \leq \mu_1, \ldots, \alpha_m \leq \mu_m$. On the other hand,

$$D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}) = 0 \tag{8}$$

as soon as one of the indices α is greater than the corresponding exponent μ.

Now let $\beta_1, \beta_2, \ldots, \beta_m$ denote a second system of indices, and form from the function (7) the derivative $D_{\beta_1,\beta_2,\ldots,\beta_m}$. By a repeated application of the same formulae (7) and (8) one obtains

$$D_{\beta_1,\ldots,\beta_m} D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}) = D_{\beta_1+\alpha_1,\ldots,\beta_m+\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}), \tag{9}$$

and from this, by II, follows the general validity of the theorem

$$\text{III.} \qquad D_{\beta_1,\ldots,\beta_m} D_{\alpha_1,\ldots,\alpha_m}\Phi = D_{\beta_1+\alpha_1,\ldots,\beta_m+\alpha_m}\Phi,$$

which is a generalization of the theorem (8), § 13; hence the higher derivatives can be formed by continued differentiation of the lower ones.

For the first derivatives of a function Φ,

$$D_{1,0,\ldots,0}\Phi, \quad D_{0,1,\ldots,0}\Phi, \ldots, D_{0,0,\ldots,1}\Phi,$$

we also use the shorter signs

$$\Phi'(x_1), \quad \Phi'(x_2), \ldots, \Phi'(x_m),$$

and likewise for the second derivatives

$$D_{2,0,\ldots,0}, \quad D_{1,1,\ldots,0}, \ldots$$

the signs

$$\Phi''(x_1, x_1), \quad \Phi''(x_1, x_2) = \Phi''(x_2, x_1), \ldots.$$

This notation too can be generalized and would lead to an expression corresponding to formula (4), § 15.

Because of their frequent use, we mention the formulae for the quadratic forms especially. Let

$$\Phi(x) = \sum a_{i,k} x_i x_k, \tag{10}$$

where i, k independently run through the row of numbers $1, 2, \ldots, m$ and $a_{i,k} = a_{k,i}$. Then

$$\begin{aligned} \frac{1}{2}\Phi'(x_1) &= a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,m}x_m, \\ \frac{1}{2}\Phi'(x_2) &= a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,m}x_m, \\ &\cdots \\ \frac{1}{2}\Phi'(x_m) &= a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,m}x_m, \end{aligned} \tag{11}$$

and we further put

$$\Phi(x, \xi) = \Phi(\xi, x) = \sum \xi_i \Phi'(x_i) = \sum x_i \Phi'(\xi_i). \tag{12}$$

Then

$$\Phi(x + \xi) = \Phi(x) + \Phi(x, \xi) + \Phi(\xi). \tag{13}$$

The function $\Phi(x, \xi)$ is called the polar of Φ. It is linear and homogeneous both with respect to the x and with respect to the ξ.

It can be expressed by

$$\Phi(x, \xi) = 2\sum a_{ik}\xi_i x_k \tag{14}$$

and satisfies the condition

$$\Phi(x, x) = 2\Phi(x). \tag{15}$$

§ 17. Euler's Theorem on Homogeneous Functions.

From the preceding developments one can easily derive a fundamental theorem on homogeneous functions, discovered by Euler and named after him.

We obtain it most simply from formula (4) of the preceding paragraph if, denoting by t an arbitrary variable, we put

$$\xi_1 = tx_1, \quad \xi_2 = tx_2, \ldots, \quad \xi_m = tx_m \tag{1}$$

and then apply the fundamental formula § 15 (1) for homogeneous functions. We thus obtain first:

$$(1+t)^n \Phi(x) = \sum \frac{(tx_1)^{\alpha_1}(tx_2)^{\alpha_2}\cdots(tx_m)^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi. \tag{2}$$

If one applies the binomial theorem to the left-hand side of (2) and then equates on both sides the coefficients of equal powers of t, one obtains for every $\nu = 1, 2, \ldots, n$:

$$\frac{\Pi(n)}{\Pi(n-\nu)}\Phi(x_1, x_2, \ldots, x_m) = \sum^{\alpha} \frac{\Pi(\nu)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} x_1^{\alpha_1} x_2^{\alpha_2}\cdots x_m^{\alpha_m} D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi, \tag{3}$$

where the sum is extended over all systems of values of the α satisfying the condition

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = \nu. \tag{4}$$

In this form the theorem to be proved is contained in its full generality. For the special case $\nu = 1$ we obtain the formula

$$n\Phi(x_1, x_2, \ldots, x_m) = x_1\Phi'(x_1) + x_2\Phi'(x_2) + \cdots + x_m\Phi'(x_m), \tag{5}$$

of which formula (15) of the preceding paragraph is a special case, and for $\nu = 2$:

$$n(n-1)\Phi(x_1, x_2, \ldots, x_m) = \sum_{i,k}^{m} x_i x_k \Phi''(x_i, x_k), \tag{6}$$

where the sum is to be extended from $i = 1$ to $i = m$ and from $k = 1$ to $k = m$, so that every term with unequal i, k occurs twice in the sum.

If, when the α are subject to the condition (4), we put

$$\Phi_\nu(\xi, x) = \sum \frac{\xi_1^{\alpha_1}\xi_2^{\alpha_2}\cdots\xi_m^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi, \tag{7}$$

then by (4) of § 16,

$$\Phi(x+\xi) = \Phi(x) + \Phi_1(x,\xi) + \Phi_2(x,\xi) + \cdots + \Phi_n(x,\xi), \tag{8}$$

and since the left-hand side remains unchanged when x and ξ are interchanged, the relation follows:

$$\Phi_{n-\nu}(x,\xi) = \Phi_\nu(\xi, x), \tag{9}$$

and therefore in particular

$$\Phi_n(x,\xi) = \Phi(\xi). \tag{10}$$

The function $\Phi_\nu(x,\xi)$, regarded as a function of x, is called the νth polar of the function Φ for the value-system ξ.

We shall still specialize formula (3) to the case of a binary form ($m = 2$). We denote the variables by x, y and put, for abbreviation,

$$\Phi(x,y) = u, \qquad D_{h,\nu-h}\Phi = u_h.$$

From (4) we obtain

$$\frac{\Pi(n)}{\Pi(n-\nu)} u = \sum_{0,\nu}^{h} \frac{\Pi(\nu)}{\Pi(h)\Pi(\nu-h)} u_h x^h y^{\nu-h}, \tag{9}$$

where ν can assume any value not greater than n.

Second Section.

Determinants.

§ 18. Permutations of n elements.

We consider a system of n distinct elements of any kind, for example the n digits

$$1, 2, 3, \ldots, n,$$

whose complex, in this definite arrangement, we shall denote by $\mathfrak{A}$. The elements of $\mathfrak{A}$ can be arranged in different ways, for example

$$2, 1, 3, \ldots, n.$$

The passage from one arrangement to another is called a permutation.

If we denote by $\Pi(n)$ the number of the different arrangements, which can depend only on the number n of the elements, then first

$$\Pi(1) = 1, \qquad \Pi(2) = 2.$$

To determine the number in general, we think of an n-th element as added to $n-1$ elements. In every arrangement of the $n-1$ elements the n-th element can now be put in n different places, namely before the first, between the first and the second, between the second and the third, and so on, finally after the $(n-1)$-st; and all arrangements thus obtained are distinct. Hence

$$\Pi(n) = n\Pi(n-1), \tag{1}$$

from which, by complete induction,

$$\Pi(n) = 1 \cdot 2 \cdot 3 \cdots n \tag{2}$$

is obtained, so that here the sign $\Pi(n)$ has the same meaning as in the first section (§ 7).

Any arrangement of the system $\mathfrak{A}$ will be denoted by $\mathfrak{A}'$, or more fully, if $\alpha_1, \alpha_2, \ldots, \alpha_n$ are the digits $1, 2, \ldots, n$ in some order, by

$$\mathfrak{A}' = \alpha_1, \alpha_2, \ldots, \alpha_n. \tag{3}$$

One can derive any arrangement from any other in very different ways, that is, one can carry out a permutation in different ways. Among the several possibilities, those carried out by so-called transpositions, that is, by successive interchange of only two elements, are of special interest for us. By several transpositions performed successively one can derive every arrangement, for example $\mathfrak{A}'$, from $\mathfrak{A}$. For this purpose one may proceed, for instance, by first interchanging in $\mathfrak{A}$ the element 1 with that which stands in the first place in $\mathfrak{A}'$, that is, with α_1 (unless $\alpha_1 = 1$); then, if $\alpha_2 \neq 2$, interchanging 2 with α_2, and so on.

Thus, for example, to pass from $(1, 2, 3, 4)$ to $(4, 3, 2, 1)$, one forms the arrangements

$$(1, 2, 3, 4), \qquad (4, 2, 3, 1), \qquad (4, 3, 2, 1).$$

If we briefly denote a transposition by the two interchanged digits, and therefore the interchange of 1 with 2 by $(1, 2)$, then here we have performed successively the transpositions

$$(1, 4), \quad (2, 3).$$

It is to be observed that the passage from one arrangement to a definite other one can be reached in infinitely many different ways by successive transpositions. Thus the arrangement $(1, 2, 3, 4)$ also passes into $(4, 3, 2, 1)$ by the transpositions

$$(1, 2), \quad (1, 3), \quad (2, 4), \quad (1, 2).$$

§ 19. Permutations of the first and second kind.

The $\Pi(n)$ arrangements of n elements can be separated into two kinds according to the following point of view.

From the n elements of our system one can form

$$\frac{n(n-1)}{2}$$

and no more pairs. We now assign to the n elements $1, 2, \ldots, n$, in a definite way, n real numerical values $a_1, a_2, \ldots, a_n$, and from these numerical values form the $\frac{n(n-1)}{2}$ differences

$$a_1 - a_2, \quad a_1 - a_3, \ldots,$$

where the lower index is to belong to the minuend. The product of differences

$$P = (a_1 - a_2)(a_1 - a_3)\cdots(a_1 - a_n)(a_2 - a_3)\cdots(a_2 - a_n)\cdots(a_{n-1} - a_n) \tag{1}$$

will, when $a_1, a_2, \ldots, a_n$ are mutually distinct numerical values, have a value different from zero, for example a positive one if $a_1 > a_2 > a_3 > \cdots > a_n$ was assumed.

If now we interchange the indices $1, 2, 3, \ldots, n$ in some way, and so pass, say, from $\mathfrak{A}$ to $\mathfrak{A}'$, then P passes into

$$\begin{aligned} P' = {} & (a_{\alpha_1} - a_{\alpha_2})(a_{\alpha_1} - a_{\alpha_3})\cdots(a_{\alpha_1} - a_{\alpha_n}) \\ & \cdot (a_{\alpha_2} - a_{\alpha_3})\cdots(a_{\alpha_2} - a_{\alpha_n})\cdots(a_{\alpha_{n-1}} - a_{\alpha_n}), \end{aligned} \tag{2}$$

and, apart from the sign, this product consists of the same factors as P. Thus P' is either equal to P or opposite to P.

I. We count the arrangement $\mathfrak{A}'$, and therefore also the permutation which transforms $\mathfrak{A}$ into $\mathfrak{A}'$, as of the first or of the second kind according as P' is equal to P or opposite to P; hence $\mathfrak{A}$ itself belongs to the first kind.

II. By a simple transposition (h, k), in which h, k denote any two of the digits $1, 2, \ldots, n$, both P and P' change their sign.

For the factors which contain neither h nor k are not affected by this transposition; furthermore, in P and P' there is the factor $\pm(a_h - a_k)$, and the pairs of factors

$$\pm(a_h - a_\nu)(a_k - a_\nu),$$

where ν runs through the series $1, 2, \ldots, n$ with the exception of h, k. The first factor changes its sign, while the pair of factors remains unchanged under the transposition (h, k). Hence follows:

III. The permutations of the first kind are composed of an even number of transpositions, and those of the second kind of an odd number.

The so-called identical permutation, which leaves $\mathfrak{A}$ unchanged, is therefore also to be counted as of the first kind.

From this one obtains further the conclusion: in however many different ways one may derive $\mathfrak{A}'$ from $\mathfrak{A}$ by transpositions, the number of these transpositions is in all these ways consistently even or odd, according as $\mathfrak{A}'$ belongs to the first or to the second kind.

If in all arrangements

$$\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$$

of the n elements we perform one transposition, say $(1, 2)$, then each of these arrangements passes into a definite other one, say $\mathfrak{A}$ into $\mathfrak{B}$, $\mathfrak{A}'$ into $\mathfrak{B}'$, $\mathfrak{A}''$ into $\mathfrak{B}''$, and so on; and if the same transposition is repeated once more, then $\mathfrak{B}$ again passes into $\mathfrak{A}$, $\mathfrak{B}'$ again into $\mathfrak{A}'$, and so on. Hence the arrangements $\mathfrak{B}, \mathfrak{B}', \mathfrak{B}'', \ldots$ are all distinct from one another, and consequently in their totality agree with the totality of the $\mathfrak{A}$'s. Since, as we have seen above, all difference-products

$$P, P', P'', \ldots$$

which are formed from P with the different arrangements $\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$ change their sign under a transposition, it follows that to each $\mathfrak{A}$ of the first kind there corresponds a $\mathfrak{B}$ of the second kind, and to each $\mathfrak{A}$ of the second kind a $\mathfrak{B}$ of the first kind.

IV. Accordingly the number of arrangements of the first kind is just as large as the number of arrangements of the second kind, namely

$$\frac{1}{2}\Pi(n).$$

For $n = 3$ we have the following six arrangements, of which the first horizontal row forms the first kind:

$$\begin{array}{ccc} (1,2,3), & (2,3,1), & (3,1,2), \\ (3,2,1), & (2,1,3), & (1,3,2). \end{array}$$

These theorems have here been obtained from the consideration of the product P, hence from a numerical quantity. How one can arrive at the same results without using such a function will be seen in the fourteenth section.

§ 20. Determinants.

We now consider a system of n^2 arbitrary quantities, with which the rational operations of arithmetic can be performed. For a simple notation of these quantities we choose one letter with a double index $a_i^{(k)}$, where both i and k are to run through the series of digits $1, 2, 3, \ldots, n$. For better survey we arrange these quantities in a square, so that all a's with the same upper index stand in a horizontal row, and all a's with the same lower index in a vertical row, and denote this square by Δ:

$$\Delta = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & a_3^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & a_3^{(2)} & \cdots & a_n^{(2)} \\ a_1^{(3)} & a_2^{(3)} & a_3^{(3)} & \cdots & a_n^{(3)} \\ \vdots & \vdots & \vdots & & \vdots \\ a_1^{(n)} & a_2^{(n)} & a_3^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}. \tag{1}$$

For brevity one calls the horizontal rows rows, and the vertical rows columns.

But by the square enclosed between vertical lines we shall understand not only the complex of the quantities a, but a definite arithmetical combination of these quantities, which can be calculated as soon as the a's are given numerically, and which we shall now describe.

Form the product of the terms standing in the diagonal running from upper left to lower right:

$$M = a_1^{(1)} a_2^{(2)} a_3^{(3)} \cdots a_n^{(n)}. \tag{2}$$

From this derive $\Pi(n)$ products

$$M, M', M'', \ldots$$

by permuting the lower indices, and give to each product so obtained the positive or negative sign according as the applied permutation belongs to the first or to the second kind; thus, in the notation of the preceding paragraph,

$$M' = \pm a_{\alpha_1}^{(1)} a_{\alpha_2}^{(2)} \cdots a_{\alpha_n}^{(n)}. \tag{3}$$

The sum of these products is to be Δ. The quantity Δ is called the determinant of the n^2 elements $a_i^{(k)}$, more precisely, when the distinction is necessary, an n-rowed determinant (also a determinant of the n-th degree or n-th order). The term M of this sum, that is, the product of all elements standing in the diagonal of the square, is called the principal term.

For example, if $n = 2$, we obtain

$$\Delta = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} \\ a_1^{(2)} & a_2^{(2)} \end{vmatrix} = a_1^{(1)} a_2^{(2)} - a_2^{(1)} a_1^{(2)}, \tag{4}$$

and for $n = 3$:

$$\begin{aligned} \Delta = {} & a_1^{(1)} a_2^{(2)} a_3^{(3)} + a_2^{(1)} a_3^{(2)} a_1^{(3)} + a_3^{(1)} a_1^{(2)} a_2^{(3)} \\ & - a_1^{(1)} a_3^{(2)} a_2^{(3)} - a_2^{(1)} a_1^{(2)} a_3^{(3)} - a_3^{(1)} a_2^{(2)} a_1^{(3)}. \end{aligned} \tag{5}$$

In another notation this is

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix} = ab'c'' + bc'a'' + ca'b'' - ac'b'' - ba'c'' - cb'a''. \tag{6}$$

§ 21. Principal theorems on determinants.

From the concept of the determinant the first theorems useful for application are easily obtained.

If in the product

$$M' = \pm a_{\alpha_1}^{(1)} a_{\alpha_2}^{(2)} \cdots a_{\alpha_n}^{(n)} \tag{1}$$

we rearrange the factors, its value is not changed. We can therefore also order the factors in such a way that the lower indices occur in their natural order $1, 2, \ldots, n$. The upper indices will then occur in a certain permuted order; thus M' will receive the form

$$\pm a_1^{(\beta_1)} a_2^{(\beta_2)} \cdots a_n^{(\beta_n)}, \tag{2}$$

where

$$(\beta_1, \beta_2, \ldots, \beta_n) = \mathfrak{B}$$

is, just as $(\alpha_1, \ldots, \alpha_n)$, an arrangement of the digits $1, 2, \ldots, n$. One obtains the arrangement $\mathfrak{B}$ by undoing, in the factors of M', the transpositions which led to $\mathfrak{A}$, beginning with the last, in order to recover the original arrangement in the row of the lower indices. The resulting order of the upper indices is then the arrangement $\mathfrak{B}$. It follows that $\mathfrak{B}$ belongs to the first or the second kind according as $\mathfrak{A}$ belongs to the first or the second kind, since both arise through the same number of transpositions. The totality of the $\mathfrak{B}$'s represents, just like the totality of the $\mathfrak{A}$'s, all permutations of the n elements, since two different $\mathfrak{A}$'s can never lead to the same $\mathfrak{B}$. Thus is proved:

I. The determinant Δ can also be formed by permuting the upper indices in the principal term $a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)}$ in all possible ways, assigning to every product so formed the positive or negative sign according as the applied permutation belongs to the first or second kind, and then taking the sum of all these products.

In the representation (1) of § 20 the rows are characterized by the upper indices, the columns by the lower indices. Hence this theorem may also be expressed as follows:

II. A determinant is not changed when the rows are made columns and the columns are made rows.

If, in all arrangements $\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$ of the n elements, we interchange any two elements with one another, the totality of these arrangements remains unchanged, but every arrangement of the first kind passes into an arrangement of the second kind, and conversely. If, therefore, in the terms $M, M', M'', \ldots$ of which Δ is composed, any two lower indices are interchanged, every term with positive sign passes into another which in Δ bore the negative sign, and conversely; therefore Δ changes its sign. With the help of II follows the theorem:

III. If in Δ two lower or two upper indices are interchanged with one another, the determinant changes only its sign.

Expressed somewhat differently: if two rows or two columns are interchanged with one another, the determinant changes only its sign; and from this more generally:

IV. If in a determinant the rows or the columns are permuted, the absolute value is not changed, and the sign is not changed or passes into the opposite sign according as the applied permutation belongs to the first or second kind.

From III one obtains the following fundamental theorem:

V. If in two rows or in two columns the terms standing in the same place are equal to one another (more briefly, if two rows are equal to one another), then the determinant has the value zero.

For the interchange of the two rows changes the sign by III, but on the other hand, since both rows are identical, cannot change anything; thus for Δ only the value zero remains.

One only expresses theorem V differently when one says:

VI. One obtains a vanishing determinant if the elements of one row are replaced by the corresponding elements of another row, or, shortly, if one lower or upper index is replaced by another.

§ 22. Subdeterminants.

In every term of the expanded determinant

$$\sum \pm a_{\alpha_1}^{(1)} a_{\alpha_2}^{(2)} \cdots a_{\alpha_n}^{(n)},$$

whose value we now wish to denote by A, every one of the numbers $1, 2, \dots, n$ occurs once and only once as a lower index. Thus a certain complex of terms will contain the factor $a_1^{(1)}$, another complex the factor $a_1^{(2)}$, and so on, finally a complex the factor $a_1^{(n)}$; every term of the determinant occurs in one and only one of these complexes.

If, therefore, we denote the first of these complexes by $a_1^{(1)} A_1^{(1)}$, the second by $a_1^{(2)} A_1^{(2)}$, and the last by $a_1^{(n)} A_1^{(n)}$, then the determinant can be represented in the following way:

$$A = a_1^{(1)} A_1^{(1)} + a_1^{(2)} A_1^{(2)} + \cdots + a_1^{(n)} A_1^{(n)}. \tag{1}$$

Instead of the lower index 1, we could equally well single out any other index r, and hence put

$$A = a_r^{(1)} A_r^{(1)} + a_r^{(2)} A_r^{(2)} + \cdots + a_r^{(n)} A_r^{(n)}. \tag{2}$$

Here the product $a_r^{(\nu)} A_r^{(\nu)}$ denotes the complex of all terms of the determinant which contain the factor $a_r^{(\nu)}$.

Since the same rules hold for upper as for lower indices, the determinant can also be written in the following way:

$$A = a_1^{(\mu)} A_1^{(\mu)} + a_2^{(\mu)} A_2^{(\mu)} + \cdots + a_n^{(\mu)} A_n^{(\mu)}, \tag{3}$$

where μ, too, may denote any one of the indices $1, 2, \dots, n$.

The quantities $A_r^{(\nu)}$ thereby completely defined are called the subdeterminants of the determinant A. To become precisely acquainted with their mode of formation, let us first consider the complex $a_1^{(1)} A_1^{(1)}$. It is obtained by leaving the lower index 1 unchanged in the product

$$a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)}$$

and permuting only the remaining indices $2, 3, \dots, n$ in all ways, and forming the sum of the terms so obtained with regard to the rule of signs. Hence $A_1^{(1)}$ is the $(n-1)$-rowed determinant

$$A_1^{(1)} = \begin{vmatrix} a_2^{(2)} & a_3^{(2)} & \cdots & a_n^{(2)} \\ a_2^{(3)} & a_3^{(3)} & \cdots & a_n^{(3)} \\ \vdots & \vdots & & \vdots \\ a_2^{(n)} & a_3^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{4}$$

or the determinant obtained from A by deleting, in the square representing A, the first row and the first column.

From this one easily obtains the meaning of $A_\mu^{(\nu)}$. By making $\nu-1$ interchanges of rows one can bring the ν-th row to the first, and by adding $\mu-1$ interchanges of columns one can bring the μ-th column to the first; apart from this the rows remain unchanged in their order. The determinant itself has taken the factor $(-1)^{\mu+\nu}$, and has remained unchanged in absolute value. In the row order thus changed the element $a_\mu^{(\nu)}$ has taken the place of the element $a_1^{(1)}$, and from this one infers the following rule of formation:

One obtains the subdeterminant $A_\mu^{(\nu)}$ by deleting from the square representing the determinant the two rows which cross in $a_\mu^{(\nu)}$, and adjoining the factor $(-1)^{\mu+\nu}$.

For example, for the determinant of three rows, one obtains the representation

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix} = a(b'c''-c'b'')+b(c'a''-a'c'')+c(a'b''-b'a''). \tag{5}$$

Since the lower index ν does not occur at all in $A_\mu^{(\nu)}$, $A_\mu^{(\nu)}$ is not changed if the lower index ν is replaced by another. But then, by § 21, VI, the determinant vanishes. We therefore obtain from (2) the important relation, in which μ,ν can be any two distinct digits $1,2,\dots,n$:

$$0 = a_1^{(\mu)}A_1^{(\nu)} + a_2^{(\mu)}A_2^{(\nu)} + \cdots + a_n^{(\mu)}A_n^{(\nu)}, \tag{6}$$

and likewise from (3)

$$0 = a_\mu^{(1)}A_\nu^{(1)} + a_\mu^{(2)}A_\nu^{(2)} + \cdots + a_\mu^{(n)}A_\nu^{(n)}. \tag{7}$$

For instance from (5), if a,b,c are replaced by a',b',c', there follows

$$a'(b'c''-c'b'')+b'(c'a''-a'c'')+c'(a'b''-b'a'')=0, \tag{8}$$

a formula of whose correctness one is convinced by the simplest calculation.

If we multiply the relation (6) by an arbitrary factor λ and add it to (2), we obtain the formula

$$\begin{aligned} A = (a_r^{(1)}+\lambda a_s^{(1)})A_r^{(1)} + (a_r^{(2)}+\lambda a_s^{(2)})A_r^{(2)} \\ + \cdots + (a_r^{(n)}+\lambda a_s^{(n)})A_r^{(n)}, \end{aligned} \tag{9}$$

which expresses for us the following theorem:

VII. The determinant does not change its value if to the elements of one row one adds the corresponding elements of another row multiplied by an arbitrary common factor.

The same theorem holds also for the columns. It is often used with advantage for the simplification and numerical calculation of determinants. We add the following theorems, which can be read off at once from the representations (2), (3).

VIII. If all elements of a row or of a column have a common factor, then this factor can be omitted from the row or column and put as a factor before the determinant.

For by (2) it is

$$pa_r^{(1)}A_r^{(1)} + pa_r^{(2)}A_r^{(2)} + \cdots + pa_r^{(n)}A_r^{(n)} = pA.$$

IX. If in a row or in a column all elements but one vanish, then the determinant reduces to the product of this one element and the corresponding subdeterminant.

For if $a_r^{(1)}, a_r^{(2)}, \dots, a_r^{(n)}$, with the exception of $a_r^{(\nu)}$, vanish, then by (2)

$$A = a_r^{(\nu)}A_r^{(\nu)}.$$

The value of A is then wholly independent of the $a_1^{(\nu)}, a_2^{(\nu)}, \dots, a_n^{(\nu)}$ with the exception of $a_r^{(\nu)}$.

To make an application of these theorems, we determine the value of the determinant

$$\Delta = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix},$$

where a, b, c are arbitrary quantities. If we multiply the second column by a and subtract it from the third, then multiply the first by a and subtract it from the second, then by VII

$$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ 1 & b-a & b(b-a) \\ 1 & c-a & c(c-a) \end{vmatrix},$$

and by IX and VIII

$$\Delta = (b-a)(c-a) \begin{vmatrix} 1 & b \\ 1 & c \end{vmatrix} = (b-a)(c-a)(c-b). \tag{10}$$

In the same way one can also treat the n-rowed determinant

$$\begin{vmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^{n-1} \end{vmatrix}$$

and finds its value equal to

$$(a_2 - a_1)(a_3 - a_1) \cdots (a_n - a_1)(a_3 - a_2) \cdots (a_n - a_2) \cdots (a_n - a_{n-1}). \tag{11}$$

If one orders the columns in the reverse sequence, then, according as n is even or odd,

$$\frac{n}{2}(n-1) \quad \text{or} \quad \frac{n-1}{2}n$$

interchanges are necessary; in either case this amounts to distinguishing the determinant so ordered from Δ by the factor

$$(-1)^{n(n-1)/2}.$$

It comes to the same thing if, in the $\frac{n(n-1)}{2}$ factors of the product (11), the opposite sign is given. Thus, along with (11), the equality also holds:

$$\begin{vmatrix} a_1^{n-1} & a_1^{n-2} & \cdots & a_1 & 1 \\ a_2^{n-1} & a_2^{n-2} & \cdots & a_2 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_n^{n-1} & a_n^{n-2} & \cdots & a_n & 1 \end{vmatrix} = \prod_{1 \le i < k \le n} (a_i - a_k). \tag{12}$$

We still mention here a notation for the subdeterminants which is borrowed from differential calculus and is often used with advantage, especially when derivatives are to be formed. If the quantities $a_\mu^{(\nu)}$ are regarded as independent variables, then the determinant A is only of the first degree with respect to each of them. The derivative or differential quotient taken with respect to $a_\mu^{(\nu)}$ is therefore equal to the coefficient of $a_\mu^{(\nu)}$, hence

$$\frac{\partial A}{\partial a_\mu^{(\nu)}} = A_\mu^{(\nu)}.$$

If therefore, for example, the $a_\mu^{(\nu)}$ are functions of a variable t, then A is also a function of t, and by the first rules of differential calculus one obtains the derivative of A with respect to t in the form

$$A'(t) = \sum A_\mu^{(\nu)} \frac{\partial a_\mu^{(\nu)}}{\partial t},$$

where $\partial a_\mu^{(\nu)}/\partial t$ denotes the derivative of $a_\mu^{(\nu)}$ with respect to t.

§ 23. The subdeterminants in the wider sense

We can now generalize the considerations of the preceding paragraph in the following way. Just as there we started from the problem of seeking out all terms in the developed determinant A which contain the factor $a_1^{(1)}$, so now we seek all terms which contain the factor

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)},$$

where ν may be any number not exceeding n. These terms are obtained from the principal term

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)} a_{\nu+1}^{(\nu+1)} \cdots a_n^{(n)}$$

if, in permuting the lower indices, one leaves $1, 2, \ldots, \nu$ unchanged and permutes only $\nu+1, \ldots, n$ in all possible ways, with observance of the rule of signs. Thus the aggregate of the required terms is

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)} \begin{vmatrix} a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \vdots & & \vdots \\ a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}. \tag{1}$$

I. The determinant of $n-\nu$ rows which occurs here as a factor, and which we denote by

$$A_{1,2,\ldots,\nu}^{1,2,\ldots,\nu},$$

arises from A by suppressing the first ν rows and columns.

We shall generalize this result as follows. Choose any ν elements

$$a_{\alpha_1}^{(\beta_1)}, a_{\alpha_2}^{(\beta_2)}, \ldots, a_{\alpha_\nu}^{(\beta_\nu)} \tag{3}$$

so that no two of them occur in the same row or in the same column, that is, so that neither a lower nor an upper index is repeated. We denote by

$$A_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}^{\beta_1,\beta_2,\ldots,\beta_\nu} \tag{2}$$

the aggregate of those terms of the determinant which contain the product of these elements as a factor.

By interchanging rows and columns, whereby at most the sign of the determinant is changed, one can always bring the elements (3) into the places of

$$a_1^{(1)}, a_2^{(2)}, \ldots, a_\nu^{(\nu)}.$$

Then rule I applies, and one obtains:

II. Apart from sign, $A_{\alpha_1,\ldots,\alpha_\nu}^{\beta_1,\ldots,\beta_\nu}$ is obtained as an $(n-\nu)$-rowed determinant by deleting from A all rows and all columns which meet in one of the elements (3), and by leaving the remaining rows and columns in their order.

For the determination of the sign one uses the following rule. Arrange the lower and upper indices $1, 2, \ldots, n$ in the orders

$$\alpha_1, \alpha_2, \ldots, \alpha_\nu, \alpha_{\nu+1}, \ldots, \alpha_n, \tag{4}$$

$$\beta_1, \beta_2, \ldots, \beta_\nu, \beta_{\nu+1}, \ldots, \beta_n, \tag{5}$$

where the missing $\alpha_{\nu+1}, \ldots, \alpha_n$, and likewise $\beta_{\nu+1}, \ldots, \beta_n$, are taken in increasing order.

III. The determinant of $n-\nu$ rows described in II receives the positive or negative sign according as the two arrangements (4), (5) of the digits $1, 2, \ldots, n$ both belong to the same kind or to different kinds.

Indeed, the determinant changes its sign through each interchange of two lower or two upper indices. In order to reduce the general case (2) to the special case (1), one has to make just so many transpositions of upper and lower indices that the permutations (4) and (5) both pass into the original arrangement $1, 2, 3, \dots, n$; the same number of changes of sign has then taken place.

The quantities so defined are called the ν-th subdeterminants, or subdeterminants of the ν-th order. They are represented by determinants of $n-\nu$ rows. From III follows the theorem:

IV. The subdeterminant

$$A^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu}$$

changes only its sign if two of its lower or two of its upper indices are interchanged; more generally, it remains unaltered in absolute value if the order of the indices is replaced by any other order, and changes sign or not according as this permutation is of the second or of the first kind.

If $\bar\alpha_1, \bar\alpha_2, \dots, \bar\alpha_\nu$ denotes any ordering of $\alpha_1, \alpha_2, \dots, \alpha_\nu$, then the determinant A also contains the complex of terms

$$\pm a^{(\beta_1)}_{\bar\alpha_1} a^{(\beta_2)}_{\bar\alpha_2} \cdots a^{(\beta_\nu)}_{\bar\alpha_\nu} A^{\beta_1,\dots,\beta_\nu}_{\bar\alpha_1,\dots,\bar\alpha_\nu}.$$

Collecting all these terms gives the complex

$$A^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu} \sum \pm a^{(\beta_1)}_{\alpha_1} a^{(\beta_2)}_{\alpha_2} \cdots a^{(\beta_\nu)}_{\alpha_\nu}. \tag{6}$$

The ν-rowed determinant which occurs in (6) is called the subdeterminant complementary to $A^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu}$, and is denoted by

$$B^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu}.$$

It contains exactly the rows and columns missing in $A^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu}$, and, apart from sign, agrees with a subdeterminant of order $n-\nu$. Thus the complex of terms (6) is denoted by

$$A^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu} B^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu}. \tag{7}$$

If now, for $\alpha_1, \dots, \alpha_\nu$, one chooses each combination of ν of the digits $1, 2, \dots, n$, while $\beta_1, \dots, \beta_\nu$ is held fixed, one obtains the same number of complexes of the form (7), and every term of the determinant A occurs in one and only one of them.

V. Consequently, by summing all expressions (7), one obtains the determinant A:

$$A = \sum A^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu} B^{\beta_1,\dots,\beta_\nu}_{\alpha_1,\dots,\alpha_\nu}. \tag{8}$$

Of course one may also keep the combination of the α's fixed and sum with respect to the β's. This is Laplace's theorem.

Another representation of the determinant A by the first and second subdeterminants is obtained as follows. Choose in A any two rows which intersect in an element, say $a^{(\mu)}_\nu$. In every term of A there occurs one element with lower index ν and one with upper index μ. Hence one may write

$$A = a^{(\mu)}_\nu A^{(\mu)}_\nu + \sum a^{(i)}_\nu a^{(\mu)}_k A^{\mu i}_{\nu k}, \tag{9}$$

or, according to IV,

$$A = a^{(\mu)}_\nu A^{(\mu)}_\nu - \sum a^{(i)}_\nu a^{(\mu)}_k A^{\mu i}_{\nu k}. \tag{10}$$

Here i denotes every index different from μ, and k every index different from ν. The quantity $A^{\mu i}_{\nu k}$ is at the same time the first subdeterminant, corresponding to the element $a^{(i)}_k$, of the $(n-1)$-rowed determinant $A^{(\mu)}_\nu$.

For the bordered determinant

$$U = \begin{vmatrix} a^{(1)}_1 & a^{(1)}_2 & \cdots & a^{(1)}_n & u_1 \\ a^{(2)}_1 & a^{(2)}_2 & \cdots & a^{(2)}_n & u_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a^{(n)}_1 & a^{(n)}_2 & \cdots & a^{(n)}_n & u_n \\ v_1 & v_2 & \cdots & v_n & q \end{vmatrix} \tag{11}$$

one obtains, by developing with respect to the elements of the last row and column,

$$U = qA - \sum u_i v_k A_k^{(i)}. \tag{12}$$

In the higher subdeterminants it is sometimes useful to denote them by differential quotients; for example

$$A_{\nu k}^{\mu i} = \frac{\partial^2 A}{\partial a_\nu^{(\mu)}\, \partial a_k^{(i)}}. \tag{13}$$

§ 24. Linear homogeneous equations

The principal application of determinants, from which the whole theory took its origin, is the solution of linear equations. We treat the problem at once in its most general form, since the special form gives scarcely any simplification and is easily derived afterwards from the general result.

Consider a system of m equations of the first degree in which n unknowns $x_1, x_2, \ldots, x_n$ occur homogeneously:

$$\begin{aligned} a_1^{(1)} x_1 + a_2^{(1)} x_2 + \cdots + a_n^{(1)} x_n &= 0, \\ a_1^{(2)} x_1 + a_2^{(2)} x_2 + \cdots + a_n^{(2)} x_n &= 0, \\ &\vdots \\ a_1^{(m)} x_1 + a_2^{(m)} x_2 + \cdots + a_n^{(m)} x_n &= 0. \end{aligned} \tag{1}$$

The coefficients $a_i^{(k)}$ are regarded as given quantities. For the present we make no assumption about the numbers m, n, but ask generally for all systems of values of $x_1, x_2, \ldots, x_n$ satisfying (1).

One solution is immediate, whatever the coefficients may be:

$$x_1 = 0,\ x_2 = 0, \ldots, x_n = 0. \tag{2}$$

There is also the other extreme case in which all coefficients vanish; then the equations are satisfied for arbitrary values of the unknowns.

We write the system of coefficients as the rectangular scheme

$$\begin{matrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(m)} & a_2^{(m)} & \cdots & a_n^{(m)}. \end{matrix} \tag{3}$$

Such a scheme, which by itself has no numerical meaning, is called a matrix insofar as it is regarded as the source of a greater number of determinants. The determinants arising from the matrix are obtained by deleting arbitrary rows and columns, in such a number that the elements left form a square, and then regarding this square as a determinant.

Assume that among the ν-rowed determinants of the matrix at least one is different from zero, while all $(\nu+1)$-rowed and therefore all higher determinants, if such exist, vanish. Excluding the uninteresting case in which all coefficients vanish, such a number ν always exists. Without loss of generality, we may suppose that the non-vanishing determinant is

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_\nu^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix}. \tag{4}$$

The subdeterminants of A are denoted, as before, by $A_i^{(k)}$, where i, k range from 1 to ν.

I. If first $\nu = n$, which presupposes that m is not smaller than n, then the equations (1) have no other solution than (2).

For take the first n equations. If these are multiplied successively by $A_\mu^{(1)}, A_\mu^{(2)}, \ldots, A_\mu^{(n)}$, where μ is any of the indices $1, 2, \ldots, n$, and then added, the relations of the preceding paragraph give

$$Ax_\mu = 0,$$

and since, by hypothesis, $A \neq 0$, it follows that $x_\mu = 0$.

II. In particular, if a system of n linear homogeneous equations with n unknowns has a determinant different from zero, all unknowns have the value zero. Equivalently: if such a system has a solution in which not all unknowns vanish, then the determinant of the system vanishes.

We now suppose that $\nu < n$. Since $m \geq \nu$, choose the first ν equations and write them in the form

$$\begin{gathered} a_1^{(1)} x_1 + \cdots + a_\nu^{(1)} x_\nu = -a_{\nu+1}^{(1)} x_{\nu+1} - \cdots - a_n^{(1)} x_n, \\ \vdots \\ a_1^{(\nu)} x_1 + \cdots + a_\nu^{(\nu)} x_\nu = -a_{\nu+1}^{(\nu)} x_{\nu+1} - \cdots - a_n^{(\nu)} x_n. \end{gathered} \tag{6}$$

If these equations are multiplied by the corresponding subdeterminants $A_\mu^{(1)}, \ldots, A_\mu^{(\nu)}$, and added, one obtains

$$Ax_\mu = -\sum_{h=\nu+1}^{n} C_{\mu h} x_h, \tag{7}$$

where, for abbreviation,

$$C_{\mu h} = \sum_{i=1}^{\nu} a_h^{(i)} A_\mu^{(i)}, \qquad h = \nu + 1, \ldots, n. \tag{8}$$

By the theorem on subdeterminants, $C_{\mu h}$ is the determinant obtained from (4) by replacing the μ-th column by $a_h^{(1)}, a_h^{(2)}, \ldots, a_h^{(\nu)}$. Thus (7), since $A \neq 0$, expresses $x_1, \ldots, x_\nu$ linearly in terms of $x_{\nu+1}, \ldots, x_n$ and the known coefficients.

III. The expressions (7) satisfy all equations (1), whatever values are assigned to $x_{\nu+1}, \ldots, x_n$; hence $n - \nu$ of the unknowns remain arbitrary and the other ν are dependent on them according to (7).

Indeed, writing (7) in the form

$$Ax_\mu = -\sum_{h=\nu+1}^{n} \sum_{i=1}^{\nu} x_h a_h^{(i)} A_\mu^{(i)}, \tag{9}$$

and multiplying by $a_\mu^{(k)}$ and summing over $\mu = 1, \ldots, \nu$, then adding the remaining terms, gives

$$A \sum_{\mu=1}^{n} a_\mu^{(k)} x_\mu = \sum_{h=\nu+1}^{n} x_h \left(A a_h^{(k)} - \sum_{i=1}^{\nu} \sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)} \right). \tag{11}$$

The factor of x_h on the right is, by (12) of the preceding paragraph, a $(\nu + 1)$-rowed determinant of the matrix (3), and therefore vanishes by hypothesis; if $k \leq \nu$ it vanishes because two rows coincide. Thus

$$\sum_{\mu=1}^{n} a_\mu^{(k)} x_\mu = 0, \qquad k = 1, 2, \ldots, m, \tag{13}$$

as was required.

A useful special case is $m = n - 1$, $\nu = n - 1$. Then only one of the unknowns remains arbitrary, and the ratios of the unknowns are fully determined. If the $(n - 1)$-rowed determinants of the matrix

$$\begin{matrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(n-1)} & a_2^{(n-1)} & \cdots & a_n^{(n-1)} \end{matrix} \tag{14}$$

taken with alternating sign are denoted by $A_1, A_2, \dots, A_n$, and if at least one of them is different from zero, the solution of

$$\begin{gathered} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_n^{(1)}x_n = 0, \\ \vdots \\ a_1^{(n-1)}x_1 + a_2^{(n-1)}x_2 + \cdots + a_n^{(n-1)}x_n = 0 \end{gathered} \tag{15}$$

is given by the ratios

$$x_1 : x_2 : \cdots : x_n = A_1 : A_2 : \cdots : A_n. \tag{16}$$

For $n = 3$, the system

$$ax + by + cz = 0, \qquad a'x + b'y + c'z = 0 \tag{17}$$

has the solution in ratios

$$x : y : z = bc' - cb' : ca' - ac' : ab' - ba'. \tag{18}$$

§ 25. Elimination from linear equations

It sometimes happens that, for a given system of linear equations, the question is not so much the actual determination of the unknowns as the decision whether a solution is possible; that is, the setting up of the condition equations which must hold between the coefficients if solutions, or solutions of a prescribed kind, are to exist. The setting up of these condition equations is called elimination. The solution of this problem has already been contained implicitly in what precedes; nevertheless we return explicitly to a few questions belonging here.

Let again a system of m linear homogeneous equations with n unknowns be given, and ask when the system has a solution in which not all unknowns vanish. This is always the case when $n > m$. If, however, $n \le m$, the necessary and sufficient condition for such a solution is that all n-rowed determinants of the matrix vanish. For if one of them does not vanish, the unknowns must all be zero by §24, II; but if they all vanish, one can find a number $\nu < n$ such that all $(\nu + 1)$-rowed determinants vanish while at least one ν-rowed determinant does not vanish, and then §24, III gives a solution of the required kind.

From the matrix one can form, when $n \le m$,

$$\frac{m(m-1)\cdots(m-n+1)}{1 \cdot 2 \cdots n}$$

n-rowed determinants, and this would be the number of conditions. If $n = m$, this number is 1. In general, however, although all these conditions must be fulfilled, their number is greater than necessary, because some are necessary consequences of the others.

To obtain a system of necessary, sufficient, and mutually independent conditions, make the question more precise: what are the conditions that from a system of m linear homogeneous equations with n unknowns, ν of the unknowns can be completely determined by the remaining $n - \nu$, which remain arbitrary?

Assume that $x_{\nu+1}, x_{\nu+2}, \dots, x_n$ are to remain arbitrary and that $x_1, x_2, \dots, x_\nu$ are to be determined by them, and assume

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix} \neq 0. \tag{1}$$

The unknowns $x_1, \dots, x_\nu$ are then calculated by §24, (9), and one forms the sums §24, (11), whose vanishing expresses that the given system is actually satisfied. The conditions are therefore the vanishing of the determinants

$$\begin{vmatrix} a_1^{(1)} & \cdots & a_\nu^{(1)} & a_h^{(1)} \\ \vdots & & \vdots & \vdots \\ a_1^{(\nu)} & \cdots & a_\nu^{(\nu)} & a_h^{(\nu)} \\ a_1^{(k)} & \cdots & a_\nu^{(k)} & a_h^{(k)} \end{vmatrix} = 0, \tag{2}$$

or, written otherwise,

$$Aa_h^{(k)} - \sum_{i=1}^{\nu}\sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)} = 0. \tag{3}$$

For $h = \nu+1, \dots, n$ and $k = \nu+1, \dots, m$ these give

$$(n-\nu)(m-\nu) \tag{5}$$

conditions. These conditions are mutually independent. For the left sides of (3) can, by suitable assumptions about the coefficients a, be made to take arbitrary values for each index combination, as is seen by setting all $a_h^{(i)}$, $a_\mu^{(k)}$, except the one in question, equal to zero.

§ 26. Non-homogeneous linear equations

We have simplified the problem of solving linear equations, and at the same time led it back to more general laws, by supposing the equations to be homogeneous in the unknowns. In applications, however, the unknowns often do not occur homogeneously. Although in principle the one case is not essentially different from the other, we consider the non-homogeneous case separately.

We can derive it from the homogeneous case by adding the requirement that one of the unknowns shall have the value 1. If this unknown belongs among those which the problem leaves arbitrary, there is no difficulty; if it belongs among those determined by the rest, further conditions have to be fulfilled. We treat the question independently, with a slightly changed notation, and restrict ourselves to the most important case in which the number of unknowns equals the number of equations.

Let the following system be solved for $x_1, x_2, \dots, x_n$:

$$\begin{aligned} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_n^{(1)}x_n &= y_1, \\ a_1^{(2)}x_1 + a_2^{(2)}x_2 + \cdots + a_n^{(2)}x_n &= y_2, \\ &\vdots \\ a_1^{(n)}x_1 + a_2^{(n)}x_2 + \cdots + a_n^{(n)}x_n &= y_n. \end{aligned} \tag{1}$$

The coefficients and the independent terms are regarded as given. Let

$$A = \sum \pm a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)} \tag{2}$$

be the determinant of the system, and let $A_\mu^{(k)}$ denote its first subdeterminants. Multiplying the equations successively by $A_\mu^{(1)}, A_\mu^{(2)}, \dots, A_\mu^{(n)}$ and adding, one obtains

$$Ax_\mu = \sum_{k=1}^{n} y_k A_\mu^{(k)}. \tag{4}$$

If $A \neq 0$, the values of the unknowns are uniquely determined and satisfy (1). If we put $A_\mu^{(k)} = A_{k\mu}$, the solution can be written in the form

$$\begin{aligned} Ax_1 &= A_{11}y_1 + A_{12}y_2 + \cdots + A_{1n}y_n, \\ Ax_2 &= A_{21}y_1 + A_{22}y_2 + \cdots + A_{2n}y_n, \\ &\vdots \\ Ax_n &= A_{n1}y_1 + A_{n2}y_2 + \cdots + A_{nn}y_n, \end{aligned} \tag{5}$$

which displays the analogy with the original system.

If $A = 0$, the equations (4) give no values for the x; they only give conditions which the y's must necessarily satisfy if the equations are to be soluble at all. Suppose that, besides A, all subdeterminants up to a certain order vanish, and denote by

$$B = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix} \tag{6}$$

a non-vanishing ν-rowed subdeterminant. If all $(\nu+1)$-rowed subdeterminants of A vanish, multiplication of the first ν equations by the subdeterminants $B_\mu^{(1)}, \ldots, B_\mu^{(\nu)}$ gives

$$Bx_\mu = \sum_{i=1}^{\nu} B_\mu^{(i)} y_i - \sum_{h=\nu+1}^{n} x_h \sum_{i=1}^{\nu} B_\mu^{(i)} a_h^{(i)}. \tag{7}$$

Thus ν of the unknowns are determined by the remaining ones. Substitution into (1) shows that the equations are satisfied if and only if, for $\lambda = \nu+1, \ldots, n$,

$$By_\lambda = \sum_{i=1}^{\nu} \sum_{\mu=1}^{\nu} B_\mu^{(i)} a_\mu^{(\lambda)} y_i. \tag{10}$$

These are $n - \nu$ independent conditions for the y's. If one of them is not fulfilled, the given system of equations has no solution.

If in (1) the x's are regarded not as unknowns but as variables, the y's also become variable quantities. In this interpretation the system (1) is called a linear substitution, since it mediates the passage from one system of variables to another. Only when the determinant A, now called the determinant of the substitution, is different from zero can the y's be considered independent variables. In that case (5) gives the representation of the variables x by the variables y, or the substitution inverse to (1).

§ 27. Multiplication of determinants.

The theorem which we still have to prove teaches how one can represent the product of two determinants with the same number of rows by a single determinant with the same number of rows. One is led to it most simply by considering the solution of two systems of linear equations.

Let the coefficients $a_i^{(k)}, b_h^{(k)}$, when i, h run through the sequence of numbers $1, 2, \ldots, n$, be arbitrary variable quantities; likewise let the quantities x_i, y_i, z_i be bound only by the relations

$$\sum_i a_i^{(k)} x_i = y_k, \tag{1}$$

$$\sum_k b_h^{(k)} y_k = z_h. \tag{2}$$

If the problem is now posed of determining the variables x by the variables z, this problem can be solved in two ways. By § 26, (4) one may solve equations (1) with respect to x, equations (2) with respect to y, and insert the latter expressions into the former. If A, B denote the determinants of the two systems of equations, that is,

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & & \cdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \qquad B = \begin{vmatrix} b_1^{(1)} & b_2^{(1)} & \cdots & b_n^{(1)} \\ b_1^{(2)} & b_2^{(2)} & \cdots & b_n^{(2)} \\ \cdots & \cdots & & \cdots \\ b_1^{(n)} & b_2^{(n)} & \cdots & b_n^{(n)} \end{vmatrix},$$

and if $A_i^{(k)}, B_i^{(k)}$ denote the first subdeterminants, one obtains

$$Ax_k = \sum_i A_k^{(i)} y_i, \tag{3}$$

$$By_i = \sum_h B_i^{(h)} z_h, \tag{4}$$

and, by substitution of (4) in (3),

$$ABx_k = \sum_h z_h \sum_i A_k^{(i)} B_i^{(h)}. \tag{5}$$

But one may also proceed by inserting from (1) the expressions for y into (2), whereby, if

$$c_i^{(h)} = \sum_s a_i^{(s)} b_h^{(s)} \tag{6}$$

is put, one obtains

$$\sum_i c_i^{(h)} x_i = z_h. \tag{7}$$

Now put

$$C = \begin{vmatrix} c_1^{(1)} & c_2^{(1)} & \cdots & c_n^{(1)} \\ c_1^{(2)} & c_2^{(2)} & \cdots & c_n^{(2)} \\ \cdots & \cdots & & \cdots \\ c_1^{(n)} & c_2^{(n)} & \cdots & c_n^{(n)} \end{vmatrix}. \tag{8}$$

If $C_i^{(k)}$ denotes the subdeterminants of C, the solution of (7) gives

$$Cx_k = \sum_h z_h C_k^{(h)}, \tag{9}$$

and comparison of (5) with (9) gives

$$AB = C, \tag{10}$$

and for the subdeterminants

$$C_k^{(h)} = \sum_i A_k^{(i)} B_i^{(h)}. \tag{11}$$

In this conclusion, however, there is still a gap, caused by the doubt whether (5) and (9) might not differ by a factor common to both sides. To remove this doubt we shall prove equation (10) directly; the correctness of (11) then follows.

We shall form the principal term of the determinant C from (6). Let $s_1, s_2, \ldots, s_n$ be mutually independent summation letters which run from 1 to n. Then

$$\begin{aligned} c_1^{(1)} c_2^{(2)} \cdots c_n^{(n)} &= \sum_{s_1} a_1^{(s_1)} b_1^{(s_1)} \sum_{s_2} a_2^{(s_2)} b_2^{(s_2)} \cdots \sum_{s_n} a_n^{(s_n)} b_n^{(s_n)} \\ &= \sum_{s_1, s_2, \ldots, s_n} a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)} b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)}. \end{aligned} \tag{12}$$

The permutation of the lower indices of the c corresponds to the permutation of the lower indices of the a, and if, with regard to the rule of signs, one therefore forms the determinant C, one obtains

$$\sum \pm c_1^{(1)} c_2^{(2)} \cdots c_n^{(n)} = \sum_{s_1, s_2, \ldots, s_n} b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)} \sum \pm a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)}. \tag{13}$$

Now

$$\sum \pm a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)}$$

is, by § 21, VI, always zero if the same digit occurs twice among $s_1, s_2, \ldots, s_n$; hence only those terms in (13) keep a non-zero value for which $s_1, s_2, \ldots, s_n$ is an arrangement of the indices $1, 2, \ldots, n$. This value is $+A$ or $-A$ according as this arrangement belongs to the first or the second kind (§ 21, IV).

Therefore the right-hand side of (13) becomes

$$A \sum_{s_1, s_2, \ldots, s_n} \pm b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)},$$

where the sum is extended over all permutations $s_1, s_2, \ldots, s_n$. But this sum is precisely the determinant B, and hence the formula

$$C = AB$$

is proved.

The law of formation of the elements $c_i^{(h)}$ contained in formula (6) can be expressed in words as follows. To obtain the elements of the determinant C which is the product of the two determinants A, B, one multiplies the elements of each column of A by the corresponding elements of a column of B and adds the products.

By the theorems on determinants the form of C can be varied in many ways. We record the following. If two columns in A or in B are interchanged, the elements $c_i^{(h)}$ do not change, but only interchange among themselves. If, however, two rows in A or in B are interchanged, then the $c_i^{(h)}$ change because the factors in the individual products of the sum are grouped differently; but if corresponding interchanges are made simultaneously in the rows of A and of B, the $c_i^{(h)}$ remain unchanged, since only the individual terms of the sum are thereby interchanged.

By turning the rows into columns in A or in B, or in both at once, one obtains three further different ways of forming the product of two determinants in determinant form. This can also be expressed as follows. To form the product of two determinants, one may multiply the elements of the separate rows or columns of one factor by the corresponding elements of the rows or columns of the other factor, add the products, and regard these sums of products as the elements of a new determinant.

The multiplication rule can be applied to a product of two determinants with different numbers of elements by transforming the determinant with the smaller number of rows, by the theorem § 22, IX, into another with more rows.

§ 28. Determinants of subdeterminants.

We immediately make an application here of the multiplication law of determinants. Let, as before,

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & & \cdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{1}$$

and let

$$\begin{matrix} A_1^{(1)} & A_2^{(1)} & \cdots & A_n^{(1)} \\ A_1^{(2)} & A_2^{(2)} & \cdots & A_n^{(2)} \\ \cdots & \cdots & & \cdots \\ A_1^{(n)} & A_2^{(n)} & \cdots & A_n^{(n)} \end{matrix} \tag{2}$$

be the system of subdeterminants. If we form from (2) the determinant which we shall denote by Δ, we may apply the multiplication rule to the product $A\Delta$. By § 22, (3) and (7), this gives

$$A\Delta = \begin{vmatrix} A & 0 & \cdots & 0 \\ 0 & A & \cdots & 0 \\ \cdots & \cdots & & \cdots \\ 0 & 0 & \cdots & A \end{vmatrix} = A^n,$$

and, by division by A,

$$\Delta = A^{n-1}. \tag{3}$$

Thus Δ is the $(n-1)$st power of A. In this derivation it has indeed first been assumed that A is different from zero. But since (3) is an identity with respect to the elements $a_i^{(h)}$, that is, since it also holds when these quantities are independent variables, it follows that the formula (3) still holds in the exceptional case; that is, if A vanishes, then Δ also vanishes.

This result is a special case of a more general theorem by which any determinant of the matrix (2) may be formed. Consider the ν-rowed subdeterminant

$$\Delta_\nu = \begin{vmatrix} A_1^{(1)} & A_2^{(1)} & \cdots & A_\nu^{(1)} \\ A_1^{(2)} & A_2^{(2)} & \cdots & A_\nu^{(2)} \\ \cdots & \cdots & & \cdots \\ A_1^{(\nu)} & A_2^{(\nu)} & \cdots & A_\nu^{(\nu)} \end{vmatrix}, \tag{4}$$

from which all other ν-rowed subdeterminants can be derived by permutation of the upper and lower indices. One may apply the multiplication rule by transforming Δ_ν, according to the concluding remark of the last paragraph, into an n-rowed determinant.

Put

$$\Delta_\nu = \begin{vmatrix} A_1^{(1)} & \cdots & A_\nu^{(1)} & A_{\nu+1}^{(1)} & \cdots & A_n^{(1)} \\ \cdots & & \cdots & \cdots & & \cdots \\ A_1^{(\nu)} & \cdots & A_\nu^{(\nu)} & A_{\nu+1}^{(\nu)} & \cdots & A_n^{(\nu)} \\ 0 & \cdots & 0 & 1 & \cdots & 0 \\ \cdots & & \cdots & \cdots & & \cdots \\ 0 & \cdots & 0 & 0 & \cdots & 1 \end{vmatrix}, \tag{5}$$

where $n-\nu$ rows and columns have been adjoined, the former containing only zeros except in the diagonal terms. If one now forms the product $A\Delta_\nu$, it follows that

$$A\Delta_\nu = A^\nu \begin{vmatrix} a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \cdots & & \cdots \\ a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{6}$$

and this is by Theorem IX, § 22. Dividing here by A and applying the notation of § 23, one obtains

$$\Delta_\nu = A^{\nu-1} A_{1,2,\ldots,\nu}^{1,2,\ldots,\nu}. \tag{7}$$

For $\nu = 2$ this gives the special result

$$A_1^{(1)} A_2^{(2)} - A_1^{(2)} A_2^{(1)} = A A_{1,2}^{1,2}. \tag{8}$$

We shall use formula (8) often later. In the notation by differential quotients it can be represented as

$$A\frac{\partial^2 A}{\partial a_1^{(1)}\partial a_2^{(2)}} = \frac{\partial A}{\partial a_1^{(1)}}\frac{\partial A}{\partial a_2^{(2)}} - \frac{\partial A}{\partial a_1^{(2)}}\frac{\partial A}{\partial a_2^{(1)}}. \tag{9}$$

It is especially important in the case where A is a symmetric determinant, so that $a_i^{(k)} = a_k^{(i)}$; then also

$$\frac{\partial A}{\partial a_1^{(2)}} = \frac{\partial A}{\partial a_2^{(1)}},$$

where in the differentiation no account is taken of the dependence $a_i^{(k)} = a_k^{(i)}$. Then formula (9) becomes

$$A\frac{\partial^2 A}{\partial a_1^{(1)}\partial a_2^{(2)}} = \frac{\partial A}{\partial a_1^{(1)}}\frac{\partial A}{\partial a_2^{(2)}} - \left(\frac{\partial A}{\partial a_1^{(2)}}\right)^2. \tag{10}$$

§ 29. Interpolation.

In § 9, under a special assumption, we already solved the problem of determining an integral rational function which assumes prescribed values for a sufficient number of prescribed values of the argument. We shall now solve this problem generally by applying determinants. First, however, we shall put forward the following observations.

In § 22, (12), the value of the determinant

$$\begin{vmatrix} \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \alpha_2^{n-1} & \alpha_2^{n-2} & \cdots & \alpha_2 & 1 \\ \cdots & \cdots & & \cdots & \cdots \\ \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix}$$

was found to be equal to the product of differences

$$(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)\cdots(\alpha_1 - \alpha_n)(\alpha_2 - \alpha_3)\cdots(\alpha_2 - \alpha_n)\cdots(\alpha_{n-1} - \alpha_n). \tag{1}$$

For brevity we now denote this product of differences by

$$[\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n], \tag{2}$$

so that the quantity $[\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n]$ changes its sign when two of the elements $\alpha_1, \alpha_2, \ldots, \alpha_n$ are interchanged.

We further define an integral rational function $f(x)$ of degree n in the variable x by the equation

$$f(x) = (x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n), \tag{3}$$

so that, by § 13,

$$\begin{aligned} f'(\alpha_1) &= (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)\cdots(\alpha_1 - \alpha_n), \\ f'(\alpha_2) &= (\alpha_2 - \alpha_1)(\alpha_2 - \alpha_3)\cdots(\alpha_2 - \alpha_n), \\ &\vdots \\ f'(\alpha_n) &= (\alpha_n - \alpha_1)(\alpha_n - \alpha_2)\cdots(\alpha_n - \alpha_{n-1}). \end{aligned} \tag{4}$$

In the product of all these expressions each factor of the product (1) occurs twice with opposite sign; since the number of factors of (1) is $\frac{n(n-1)}{2}$, we obtain the equation

$$f'(\alpha_1)f'(\alpha_2)\cdots f'(\alpha_n) = (-1)^{\frac{n(n-1)}{2}}[\alpha_1, \alpha_2, \ldots, \alpha_n]^2. \tag{5}$$

But with the aid of the relations (4) we may also represent our product (1) as follows:

$$\begin{aligned}[\alpha_1, \alpha_2, \ldots, \alpha_n] &= f'(\alpha_1)[\alpha_2, \alpha_3, \ldots, \alpha_n] \\ &= -f'(\alpha_2)[\alpha_1, \alpha_3, \ldots, \alpha_n] \\ &= \cdots = (-1)^{n-1} f'(\alpha_n)[\alpha_1, \alpha_2, \ldots, \alpha_{n-1}],\end{aligned} \tag{6}$$

where on the right-hand side each bracketed expression contains one element fewer than the bracketed expression on the left.

Now pose the problem of determining an integral rational function $\varphi(x)$ of degree $(n-1)$ which, for the argument values $x = \alpha_1, \alpha_2, \ldots, \alpha_n$, assumes in order the prescribed values $\varphi(\alpha_1), \varphi(\alpha_2), \ldots, \varphi(\alpha_n)$.

If one puts

$$\varphi(x) = a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n, \tag{7}$$

then the n unknown coefficients $a_1, a_2, \ldots, a_n$ are to be determined from the linear equations

$$\begin{aligned}\varphi(\alpha_1) &= a_1 \alpha_1^{n-1} + a_2 \alpha_1^{n-2} + \cdots + a_{n-1}\alpha_1 + a_n, \\ \varphi(\alpha_2) &= a_1 \alpha_2^{n-1} + a_2 \alpha_2^{n-2} + \cdots + a_{n-1}\alpha_2 + a_n, \\ &\vdots \\ \varphi(\alpha_n) &= a_1 \alpha_n^{n-1} + a_2 \alpha_n^{n-2} + \cdots + a_{n-1}\alpha_n + a_n.\end{aligned} \tag{8}$$

The determinant of this system is precisely the quantity (2), and the problem is therefore always soluble by § 26 when this quantity is different from zero, that is, when no two of the n quantities $\alpha_1, \alpha_2, \ldots, \alpha_n$ are equal.

Instead of solving the equations (8) by § 26 and substituting the expressions so found into (7), it is preferable, by § 24, II, to eliminate the homogeneous quantities $1, a_1, a_2, \ldots, a_n$ from the $n+1$ equations (7) and (8). This gives the determinant equation

$$\begin{vmatrix} \varphi(x) & x^{n-1} & x^{n-2} & \cdots & x & 1 \\ \varphi(\alpha_1) & \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \cdots & \cdots & \cdots & & \cdots & \cdots \\ \varphi(\alpha_n) & \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix} = 0. \tag{9}$$

We now expand this determinant according to the elements of the first column. For the subdeterminants which occur there we can use the notation (2), and we obtain

$$\begin{aligned}\varphi(x)[\alpha_1, \alpha_2, \ldots, \alpha_n] - \varphi(\alpha_1)[x, \alpha_2, \alpha_3, \ldots, \alpha_n] + \varphi(\alpha_2)[x, \alpha_1, \alpha_3, \ldots, \alpha_n] \\ + \cdots + (-1)^n \varphi(\alpha_n)[x, \alpha_1, \alpha_2, \ldots, \alpha_{n-1}] = 0.\end{aligned} \tag{10}$$

Now by (1)

$$[x, \alpha_2, \alpha_3, \ldots, \alpha_n] = (x-\alpha_2)(x-\alpha_3)\cdots(x-\alpha_n)[\alpha_2, \alpha_3, \ldots, \alpha_n].$$

Dividing (10) by $[\alpha_1, \alpha_2, \ldots, \alpha_n]$ and taking account of the several expressions (6), one finally obtains

$$\varphi(x) = f(x)\left\{\frac{\varphi(\alpha_1)}{(x-\alpha_1)f'(\alpha_1)} + \frac{\varphi(\alpha_2)}{(x-\alpha_2)f'(\alpha_2)} + \cdots + \frac{\varphi(\alpha_n)}{(x-\alpha_n)f'(\alpha_n)}\right\}. \tag{11}$$

This completely determines the function $\varphi(x)$. This expression for $\varphi(x)$ is called the interpolation formula of Lagrange. That it satisfies the stipulated requirements can be verified afterwards very easily by putting $x = \alpha_1, \alpha_2, \ldots, \alpha_n$. The denominators $x - \alpha_1, \ldots, x - \alpha_n$ are present in it only apparently.

Third Section. The roots of algebraic equations.

§ 30. Concept of roots. Multiple roots.

After, in the first two sections, the algebraic magnitudes had been considered more from the formal side, where the matter was identical transformations of letter-expressions in which the letters throughout could be conceived as symbols for variable magnitudes, the numerical magnitudes now come more into the foreground.

Here, in accordance with what was fixed in the Introduction, by numbers we understand real or imaginary magnitudes of the form $a + bi$, and for illustration we represent the real magnitudes by points of a straight line and the imaginary magnitudes by points of a plane. By the absolute value of an imaginary magnitude $a + bi$ we understand $\sqrt{a^2 + b^2}$ and denote it, following Weierstrass, by

$$|a + bi|.$$

Let now

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n \tag{1}$$

be an integral function of x, in which the coefficients are arbitrary real or imaginary numbers, and the first, a_0, is assumed different from zero. If α is a number which, when put for x, makes the function $f(x)$ equal to zero, that is, which satisfies the condition $f(\alpha) = 0$, then α is called a root of the equation

$$f(x) = 0.$$

We also say briefly that α is a root of $f(x)$. By § 4, $f(x)$ may then be divided by $x - \alpha$ without remainder, so that one may put

$$f(x) = (x - \alpha) f_1(x), \tag{2}$$

where $f_1(x)$ is only of the $(n - 1)$st degree and has the same first coefficient a_0 as $f(x)$, thus

$$f_1(x) = a_0 x^{n-1} + \cdots .$$

The coefficients of $f_1(x)$ are given in § 4, (6).

Every root of $f_1(x)$ is therefore at the same time a root of $f(x)$; and conversely every root of $f(x)$ is either $= \alpha$ or is a root of $f_1(x)$. If one root α of $f(x)$ is known, then the task of finding the remaining ones is reduced to the solution of an equation of the $(n - 1)$st degree.

The aim of the considerations of this section consists in proving that every function $f(x)$ of the nth degree has at least one root. This theorem is called the fundamental theorem of algebra. First we draw from (2) the important consequence:

I. An equation of the nth degree cannot have more than n roots.

For if $f(x)$ had more than n roots, then $f_1(x)$, which is only of the $(n-1)$st degree, would have more than $n - 1$ roots. But an equation of the first degree has not more than one root, whence the correctness of our theorem follows by complete induction.

Sometimes one also gives it the expression:

II. If a function $f(x)$ of the nth degree has more than n roots, then all its coefficients must be zero.

If β is a root of $f_1(x)$, then one may likewise put

$$f_1(x) = (x - \beta) f_2(x),$$

where $f_2(x) = a_0 x^{n-2} + \cdots$ is only of the $(n - 2)$nd degree. Thus, if one assumes that each of the functions obtained by this division, $f_1(x), f_2(x), \ldots$, has at least one root, one finally obtains

$$f(x) = a_0 (x - \alpha)(x - \beta) \cdots (x - \nu), \tag{3}$$

and we can therefore also express the theorem which we earlier described as the aim of our considerations as follows. It is to be proved:

A function of the nth degree can be decomposed into n linear factors.

The magnitudes $\alpha, \beta, \ldots, \nu$ are then all roots of $f(x)$, and their number is therefore n. It is, however, not excluded that the same number occurs more than once among $\alpha, \beta, \ldots, \nu$. Then $f(x)$ would have fewer than n roots, while the theorem would still remain that $f(x)$ is decomposable into n linear factors. To establish agreement, it has been agreed that, if $x - \alpha$ goes into $f(x)$ several times, α is to be counted several times among the roots, and hence that one speaks of simple, double, triple, etc. roots.

By the concept of the derived functions [§ 13, (2)], if α is an arbitrary magnitude,

$$f(x) = f(\alpha) + (x-\alpha)f'(\alpha) + \frac{(x-\alpha)^2}{1 \cdot 2} f''(\alpha) + \frac{(x-\alpha)^3}{1 \cdot 2 \cdot 3} f'''(\alpha) + \cdots . \tag{4}$$

If it is now again assumed that $f(\alpha)$ vanishes, then there results

$$f_1(x) = f'(\alpha) + \frac{x-\alpha}{1 \cdot 2} f''(\alpha) + \frac{(x-\alpha)^2}{1 \cdot 2 \cdot 3} f'''(\alpha) + \cdots ,$$

and hence

$$f_1(\alpha) = f'(\alpha).$$

Thus α is a double root of $f(x)$ when $f'(\alpha)$ vanishes simultaneously with $f(\alpha)$. Formula (4) also shows that only under this assumption is $f(x)$ divisible by $(x-\alpha)^2$. This conclusion may be carried further and leads to the theorem:

III. The necessary and sufficient condition that α be an m-fold root of $f(x)$ is that $f(\alpha), f'(\alpha), f''(\alpha), \ldots, f^{(m-1)}(\alpha)$ vanish, while $f^{(m)}(\alpha)$ does not vanish.

§ 31. Continuity of integral functions.

If, as before,

$$f(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_n$$

is an integral function of x, we first prove the following theorem:

IV. $f(x)$ becomes infinitely large at the same time as x.

This means that, if C is an arbitrarily given positive number, the positive number R can be so chosen that

$$|f(x)| > C$$

as soon as the absolute value of x becomes greater than R; or, expressed geometrically: one can describe in the x-plane a circle around the origin such that outside this circle the absolute value of $f(x)$ no longer falls below C, however large C is assumed. The theorem is evident for the case of a simple power x^n; for if r is the absolute value of x, then r^n is the absolute value of x^n, and r^n, as soon as r has become greater than 1, is greater than r and hence grows with r into the infinite.

To prove the theorem in general, let us denote the absolute values of $x, a_0, a_1, a_2, \ldots, a_n$ by $r, c_0, c_1, c_2, \ldots, c_n$. Then

$$|f(x)| = r^n \left| a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| , \tag{2}$$

and, since by the theorems proved in the Introduction the absolute value of a sum cannot be smaller than the difference and cannot be greater than the sum of the absolute values of the summands,

$$\left| a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| \geq c_0 - \left| \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| \geq c_0 - \frac{c_1}{r} - \frac{c_2}{r^2} - \cdots - \frac{c_n}{r^n}.$$

Since $c_0, c_1, \dots, c_n$ are fixed constants, one can choose R so large that, as soon as $r > R$,

$$\frac{c_1}{r} + \frac{c_2}{r^2} + \cdots + \frac{c_n}{r^n}$$

becomes arbitrarily small, for example smaller than $\frac{1}{2}c_0$. Then

$$\left| a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| > \tfrac{1}{2}c_0,$$

and hence by (2)

$$|f(x)| > \tfrac{1}{2}c_0 R^n,$$

so that, if one assumes

$$R^n > \frac{2C}{c_0}, \tag{3}$$

one has

$$|f(x)| > C.$$

To this is attached the theorem:

V. $f(x)$ is a continuous function of x.

By this the following is meant. If the absolute value of x is smaller than an arbitrary finite magnitude R, then, however small the positive magnitude ω is assumed, one can find a positive magnitude ε such that

$$|f(x+h) - f(x)| < \omega \tag{4}$$

as long as the absolute value ρ of h remains smaller than ε, and indeed in such a way that ε depends only on the choice of ω, not on x.

By § 13, (2),

$$f(x+h) - f(x) = hf'(x) + \frac{h^2}{1 \cdot 2}f''(x) + \cdots + h^n a_0. \tag{5}$$

If now we assume the absolute value of h smaller than 1, then the absolute value of the magnitude in parentheses

$$f'(x) + \frac{h}{1 \cdot 2}f''(x) + \cdots + h^{n-1}a_0 \tag{6}$$

is smaller than a certain non-zero number K which one obtains if in (6) one replaces h by 1, the coefficients $a_0, a_1, \dots, a_{n-1}$ by their absolute values and x by R, because thereby every individual term of the sum (6) is replaced by its absolute value or by a larger number. If one chooses ε so small that

$$\varepsilon K < \omega,$$

then by (5) the requirement (4) is fulfilled as long as $\rho < \varepsilon$.

We can give the theorem of the continuity of the function $f(x)$ a somewhat different expression, which is important for what follows. By the theorem that the absolute value of a sum of two terms lies, in magnitude, between the sum and the difference of the absolute values of the terms (cf. Introduction, p. 19), we obtain from (5), if we put

$$\varphi(h) = hf'(x) + \frac{h^2}{1 \cdot 2}f''(x) + \cdots + h^n a_0,$$

$$|f(x)| - |\varphi(h)| < |f(x+h)| < |f(x)| + |\varphi(h)|,$$

or

$$-|\varphi(h)| < |f(x+h)| - |f(x)| < |\varphi(h)|.$$

Now, if $\rho < \varepsilon$, then $|\varphi(h)| < \omega$, and we can therefore also say:

VI. The absolute value of the difference

$$|f(x+h)| - |f(x)|$$

is smaller than an arbitrarily given magnitude ω as soon as the absolute value of h is smaller than a sufficiently small ε.

From this we can still conclude, if we put $x = y + iz$ and assume h either real or purely imaginary, that $|f(y+zi)|$ is, for fixed z, a continuous function of y, and for fixed y, a continuous function of z.

If the coefficients $a_0, a_1, \dots, a_n$ and the variable x are real, then x may be chosen, in absolute value, so large that the sign of $f(x)$ agrees with the sign of the first term $a_0 x^n$; thus, for positive a_0 and even n, it becomes positive, while for odd n it becomes negative for negative x and positive for positive x.

For in

$$f(x) = x^n \left(a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right)$$

one can choose x so large in absolute value that the absolute value of

$$\frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}$$

falls below that of a_0, and therefore the expression in parentheses

$$a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}$$

has the sign of a_0, which it then retains for every absolutely larger x.

§ 32. Change of sign of $f(x)$. Roots of equations of odd degree and of pure equations.

On the basis of the theorems of the preceding paragraph we can prove the following important theorem.

VIII. If the coefficients of $f(x)$ are real and there exist two real values a, b of x such that $f(a)$ and $f(b)$ have opposite signs, then between a and b there is at least one root ξ of the equation $f(x) = 0$.

We shall assume that $a < b$ and that $f(a)$ is negative, $f(b)$ positive. We divide the numbers between a and b into two parts $\mathfrak{A}$ and $\mathfrak{B}$ as follows: a number α belongs to $\mathfrak{A}$ if, between α and a, including the endpoints, there is no x for which $f(x)$ becomes positive; and a number β belongs to $\mathfrak{B}$ if between β and a there is at least one value of x for which $f(x)$ becomes positive. A third possibility is plainly not possible, and every number between a and b is therefore placed in $\mathfrak{A}$ or in $\mathfrak{B}$. We also count the numbers smaller than a as belonging to $\mathfrak{A}$, and those greater than b as belonging to $\mathfrak{B}$.

If α belongs to $\mathfrak{A}$, then every x smaller than α also belongs to $\mathfrak{A}$; and if β belongs to $\mathfrak{B}$, every larger x belongs to $\mathfrak{B}$. Thus every β is greater than every α, and the two numerical domains $\mathfrak{A}, \mathfrak{B}$ form a cut (Introduction, p. 5), generated by a number ξ, such that every number α smaller than ξ belongs to $\mathfrak{A}$, and every number β greater than ξ belongs to $\mathfrak{B}$.

If now, for some number x_0, the value of $f(x_0)$ is negative or positive, then by V of the preceding paragraph a positive magnitude ε can be so determined that for every x between $x_0 - \varepsilon$ and $x_0 + \varepsilon$ the values of $f(x)$ are always negative or always positive. It follows from this that $f(\xi)$ can be neither negative nor positive and must therefore be zero. For if $f(\xi)$ were negative, then $f(x)$ would also be negative between ξ and $\xi + \varepsilon$; these values would therefore belong to $\mathfrak{A}$, although they are greater than ξ. If $f(\xi)$ were positive, then $f(x)$ would be positive as long as x lay between $\xi - \varepsilon$

and ξ; these values, although smaller than ξ, would therefore belong to $\mathfrak{B}$. Both alternatives are impossible by the definition of ξ. The same conclusion may be drawn in quite the same way when $f(a)$ is positive and $f(b)$ negative, and so our theorem is proved.

From this we draw two important consequences. If $f(\xi)$ vanishes, then an interval

$$\xi - \varepsilon < x < \xi + \varepsilon$$

can be so determined that $f(x)$ does not vanish in this interval except for $x = \xi$. Then four cases are possible with respect to the signs of $f(x)$ in the two intervals $\xi - \varepsilon < x < \xi$ and $\xi < x < \xi + \varepsilon$, which are put together in the following scheme:

	$\xi - \varepsilon < x < \xi$	$\xi < x < \xi + \varepsilon$
1.	$+$	$-$
2.	$-$	$+$
3.	$+$	$+$
4.	$-$	$-$

If $f^{(\nu)}(x)$ is the first among the derivatives of $f(x)$ which is different from zero for $x = \xi$, then the formula [§ 13, (2)]

$$f(\xi + h) = \frac{h^\nu}{\Pi(\nu)} f^{(\nu)}(\xi) + \cdots$$

shows that these four cases are distinguished by the following signs:

$$\begin{array}{lll} 1. & \nu \text{ odd}, & f^{(\nu)}(\xi) < 0, \\ 2. & \nu \text{ odd}, & f^{(\nu)}(\xi) > 0, \\ 3. & \nu \text{ even}, & f^{(\nu)}(\xi) > 0, \\ 4. & \nu \text{ even}, & f^{(\nu)}(\xi) < 0. \end{array}$$

We say that in the first case $f(x)$ passes decreasing through zero, and in the second increasing through zero. In case 3, $f(\xi) = 0$ is a minimum, in case 4 a maximum.

In particular, if $f'(\xi)$ is different from zero, so that ξ is a simple root, then the positive or negative sign of $f'(\xi)$ is the characteristic mark for the increase or decrease of the function $f(x)$ as x passes through ξ.

With these aids we still prove two further important theorems.

IX. The equation

$$x^n - a = 0$$

has, when a is a positive number, one and only one positive root.

For if we put

$$f(x) = x^n - a, \tag{1}$$

then by § 31, VII, $f(x)$ is positive for sufficiently large positive x, and negative for $x = 0$; hence there is a positive value of x, denoted by $\sqrt[n]{a}$, for which $f(x)$ vanishes. That there can be only one such value follows from the fact that x^n, as long as x remains positive, increases continually with x.

If n is odd, then also for negative a there exists one and only one negative root of (1), which is proved in the same way.

X. An equation of odd degree with real coefficients always has at least one real root.

For by theorem VII, $f(x)$, if the degree is odd, can become both positive and negative; hence, by VIII, $f(x) = 0$ has a root.

Finally we prove:

XI. A pure equation always has a root.

By a pure equation we understand an equation of the form

$$x^n - a = 0, \tag{2}$$

where a is an arbitrary complex magnitude. That this equation always has a root for real positive a has already been proved above; and for $a = 0$ it is plainly satisfied by $x = 0$.

The general proof can be divided into two parts; it is sufficient to assume first $n = 2$, and then n odd. For if the magnitude $\sqrt{a}$ and $\sqrt[n]{a}$, or $\sqrt[n]{\alpha}$ for every α, exists, then also the existence of $\sqrt[n]{a}$ for every arbitrary n is secured. It would even suffice to prove the existence of $\sqrt[n]{a}$ for the case in which n is a prime number. But from this no particular advantage is at first to be drawn.

We denote the complex magnitude a by $b + ci$ and put, since x will in general also be complex,

$$x = y + iz.$$

Then we first have to show that

$$(y + iz)^2 = b + ic, \tag{3}$$

whatever real values b, c may have, can be satisfied by real y, z. Equation (3), however, is fulfilled if

$$y^2 - z^2 = b, \qquad 2yz = c, \tag{4}$$

and hence

$$(y^2 + z^2)^2 = b^2 + c^2,$$

therefore

$$y^2 + z^2 = \sqrt{b^2 + c^2}. \tag{5}$$

By IX there always exists a positive square root $\sqrt{b^2 + c^2}$. From (4) and (5) it now follows that

$$y^2 = \frac{b + \sqrt{b^2 + c^2}}{2}, \qquad z^2 = \frac{-b + \sqrt{b^2 + c^2}}{2},$$

and the two magnitudes $\pm b + \sqrt{b^2 + c^2}$ are plainly positive, since $b^2 + c^2 > b^2$, hence also $\sqrt{b^2 + c^2} > \sqrt{b^2}$. Thus we obtain

$$y = \pm\sqrt{\frac{b + \sqrt{b^2 + c^2}}{2}}, \qquad z = \pm\sqrt{\frac{-b + \sqrt{b^2 + c^2}}{2}},$$

but one sign is determined by the other through the condition

$$2yz = c;$$

one of the two signs remains arbitrary, so that we obtain not merely one, but two opposite solutions of (3).

Further let n be odd; we take the equation to be solved in the form

$$(y + iz)^n = b + ci. \tag{6}$$

If $c = 0$, so that a is real, then we can put $z = 0$ and by IX obtain a real value of x. But if we assume c different from zero, then from (6) (Introduction, p. 19) there follows

$$(y - iz)^n = b - ci. \tag{7}$$

Multiplying the two equations (6), (7) by one another gives

$$(y^2 + z^2)^n = b^2 + c^2,$$

and hence, by IX,

$$y^2 + z^2 = \sqrt[n]{b^2 + c^2}, \tag{8}$$

which determines the absolute value of x. Furthermore, from (6) and (7) one obtains

$$\varphi(y,z) = (y-iz)^n(b+ic) - (y+iz)^n(b-ci) = 0. \tag{9}$$

Now $\varphi(y,z)$ is an integral homogeneous function of the nth degree in the two variables y, z, and indeed with real coefficients, since replacing i by $-i$ changes nothing in $\varphi(y,z)$. If it is ordered by descending powers of y, the first term is $2ciy^n$, hence, by assumption, different from zero. Putting

$$y = \lambda z,$$

we obtain from (9)

$$\varphi(\lambda, 1) = 0, \tag{10}$$

therefore an equation of the nth degree for λ, which by X certainly has a root. Once λ is determined, it follows from (8) that

$$y = \lambda\sqrt{\frac{\sqrt[n]{b^2+c^2}}{1+\lambda^2}}, \qquad z = \sqrt{\frac{\sqrt[n]{b^2+c^2}}{1+\lambda^2}}. \tag{11}$$

Through this the equations (8), (9) are in fact satisfied; from (9), however, follows

$$\frac{(y+iz)^n}{b+ic} = \frac{(y-iz)^n}{b-ci},$$

and from (8) that the common value of these two fractions is $+1$. Therefore we have

$$(y+iz)^n = \pm(b+ci), \qquad (y-iz)^n = \pm(b-ci),$$

and thus the sign of the square root in (11) can be so determined that equations (6) and (7) are fulfilled.

§ 33. Solution of pure equations by trigonometric functions.

The solvability of a pure equation can be shown far more completely and simply if one presupposes the trigonometric functions and their simplest properties as known. These functions are, it is true, foreign to algebra proper, and for that reason it is more satisfactory to answer the questions of principle without their aid, as has been done in the preceding and will also be done later. For application and for a more convenient visualization, however, we shall not dispense with this aid.

If, when b, c are real numbers, we put

$$b = r\cos\varphi, \qquad c = r\sin\varphi, \tag{1}$$

then, if r is assumed positive, the angle φ is thereby determined up to a multiple of 2π. It will be completely determined if we fix an interval of length 2π in which it is to lie, for example

$$-\pi < \varphi \leq \pi.$$

In geometrical interpretation r, φ are the polar coordinates of the point whose rectangular coordinates are b, c. The absolute value of the complex magnitude $a = b + ci$ is r, and we shall call φ the phase of this complex magnitude.

The product of two complex magnitudes

$$a = r(\cos\varphi + i\sin\varphi), \qquad a' = r'(\cos\varphi' + i\sin\varphi')$$

is obtained in the form

$$aa' = rr'[\cos(\varphi+\varphi') + i\sin(\varphi+\varphi')], \tag{3}$$

and from this, if n is an arbitrary positive whole number, or even a negative one, one obtains the theorem named after Moivre,

$$(\cos\varphi + i\sin\varphi)^n = \cos n\varphi + i\sin n\varphi, \tag{4}$$

which has led to regarding the magnitude $\cos\varphi + i\sin\varphi$ as an exponential function with imaginary exponent, and hence to setting

$$\cos\varphi + i\sin\varphi = e^{i\varphi}.$$

Thereby formulas (3) and (4) take the form

$$e^{i\varphi}e^{i\varphi'} = e^{i(\varphi+\varphi')}, \qquad (e^{i\varphi})^n = e^{in\varphi}. \tag{5}$$

If accordingly we put

$$a = r(\cos\varphi + i\sin\varphi) \tag{6}$$

and by $\sqrt[n]{r}$ understand the positive value existing and completely determined by § 32, IX, whose nth power is r, then

$$x = \sqrt[n]{r}\left(\cos\frac{\varphi}{n} + i\sin\frac{\varphi}{n}\right) \tag{7}$$

is a magnitude whose nth power is a, hence a root of the equation

$$x^n = a, \tag{8}$$

where n is understood to be an arbitrary positive whole number. But every one of the values

$$x_k = \sqrt[n]{r}\left(\cos\frac{\varphi+2k\pi}{n} + i\sin\frac{\varphi+2k\pi}{n}\right) \tag{9}$$

satisfies the same requirement, when k is an arbitrary integer. In (9), however, not more than n different values are contained, namely those obtained by putting

$$k = 0, 1, 2, \ldots, n-1; \tag{10}$$

for if k is increased by a multiple of n, the value of (9) is not changed, while the n values (10) give only different values of x, because they have different phases.

Equation (8) consequently has not merely one, but n different roots. The geometrical images of the values x_k all lie on a circle with radius $\sqrt[n]{r}$, and are separated from one another by the angle $2\pi/n$. The first of these points is obtained by dividing the given angle φ into n equal parts. The radius $\sqrt[n]{r}$ is smaller or larger than r according as r is smaller or larger than 1. The adjacent figure shows these relations for $n = 3$ under the assumption $r > 1$.

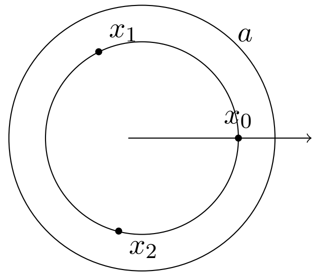


Fig. 1.

One can obtain the different values x_k by multiplying the first of them,

$$x_0 = \sqrt[n]{r}\left(\cos\frac{\varphi}{n} + i\sin\frac{\varphi}{n}\right),$$

by the different values of

$$\xi_k = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}. \tag{11}$$

The magnitudes ξ_k,

$$\xi_k^n = 1, \tag{12}$$

are roots of the equation $x^n = 1$ and are called the nth roots of unity. There are n, and only n, different values of them; for odd n only one is real, namely for $k = 0$, and for even n two are real, namely for $k = 0$ and $k = n/2$. The geometrical images of the nth roots of unity are the vertices of a regular n-gon inscribed in the circle of radius 1. Their algebraic determination is the subject of the theory of cyclotomy. If we denote by ε the value of ξ_k for $k = 1$, thus put

$$\varepsilon = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n},$$

then, by Moivre's theorem,

$$\xi_k = \varepsilon^k,$$

so that all nth roots of unity are represented as powers of one among them.

§ 34. Removal of the second term of an equation.

The certainty of the existence of the roots of pure equations puts us in a position to prove the existence of roots for equations of the second, third, and fourth degrees, or, as one also says, for quadratic, cubic, and biquadratic equations, by reducing the determination of their roots to the solution of pure equations. We shall return to this question later from various more general points of view, but here we shall briefly discuss the older methods of solution, which, even if they give little insight into the general connection of this question, are nevertheless very simple in application. They arouse the appearance as if one were dealing with a method capable of generalization to higher equations, which is not the case. First the following general remark.

Let us, for the sake of simplicity, take in the function

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \tag{1}$$

the coefficient of x^n equal to 1, which in general is obtained by division by the coefficient of x^n. Then by the simple substitution

$$x = y - \frac{a_1}{n} \tag{2}$$

we can transform $f(x)$ into another function $\varphi(y)$ of the nth degree in which the second term y^{n-1} does not occur; for, by the binomial theorem,

$$x^n = y^n - a_1y^{n-1} + \frac{n-1}{2n}a_1^2y^{n-2} + \cdots,$$

$$x^{n-1} = y^{n-1} - \frac{n-1}{n}a_1y^{n-2} + \cdots, \qquad x^{n-2} = y^{n-2} - \cdots,$$

and consequently

$$f(x) = y^n + \left(a_2 - \frac{n-1}{2n}a_1^2\right)y^{n-2} + \cdots.$$

If we actually carry out the transformation for the cases $n = 2, 3, 4$, we obtain

$$\begin{aligned} n = 2, \quad \varphi(y) &= y^2 + a, \\ n = 3, \quad \varphi(y) &= y^3 + ay + b, \\ n = 4, \quad \varphi(y) &= y^4 + ay^2 + by + c, \end{aligned} \tag{3}$$

provided that we put

$$\left.\begin{aligned}
\text{for } n=2: \quad & a &&= -\frac{a_1^2}{4} + a_2, \\
\text{for } n=3: \quad & a &&= -\frac{a_1^2}{3} + a_2, \\
& b &&= \frac{2a_1^3}{27} - \frac{a_1 a_2}{3} + a_3, \\
\text{for } n=4: \quad & a &&= -\frac{3a_1^2}{8} + a_2, \\
& b &&= \frac{a_1^3}{8} - \frac{a_1 a_2}{2} + a_3, \\
& c &&= -\frac{3a_1^4}{256} + \frac{a_1^2 a_2}{16} - \frac{a_1 a_3}{4} + a_4.
\end{aligned}\right\} \tag{4}$$

Thus the solution of the equation $f(x) = 0$ is reduced by (2) to the solution of $\varphi(y) = 0$. This first gives, for $n = 2$,

$$y = \pm\sqrt{-a},$$

and consequently

$$x = -\frac{a_1}{2} \pm \sqrt{\frac{a_1^2}{4} - a_2} = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}.$$

We therefore have two roots

$$\alpha = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}, \qquad \beta = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}. \tag{5}$$

§ 35. Cubic equations. Cardano's formula.

We take the equation of the third degree in the reduced form

$$y^3 + ay + b = 0 \tag{1}$$

and introduce two new unknowns u, v by putting

$$y = u + v. \tag{2}$$

Substitution in (1) gives

$$u^3 + v^3 + (3uv + a)(u + v) + b = 0. \tag{3}$$

We determine one of the two quantities u, v by the other according to the equation

$$3uv = -a, \tag{4}$$

and obtain from (3)

$$u^3 + v^3 = -b. \tag{5}$$

But from (4) and (5), u^3 and v^3 can be determined by a square root. Namely one obtains

$$(u^3 - v^3)^2 = (u^3 + v^3)^2 - 4u^3v^3 = b^2 + \frac{4a^3}{27}. \tag{6}$$

Put, for abbreviation,

$$R = \frac{b^2}{4} + \frac{a^3}{27}; \tag{7}$$

then

$$u^3 - v^3 = 2\sqrt{R}.$$

We give $\sqrt{R}$ one of the two signs and, according to (5), obtain

$$\begin{aligned} u^3 &= -\frac{b}{2} + \sqrt{R}, & v^3 &= -\frac{b}{2} - \sqrt{R}, \\ u &= \sqrt[3]{-\frac{b}{2} + \sqrt{R}}, & v &= \sqrt[3]{-\frac{b}{2} - \sqrt{R}}, \end{aligned} \tag{8}$$

and therefore

$$y = \sqrt[3]{-\frac{b}{2} + \sqrt{R}} + \sqrt[3]{-\frac{b}{2} - \sqrt{R}}. \tag{9}$$

Multiplication of the two expressions (8) gives

$$u^3 v^3 = -\frac{a^3}{27}$$

and shows that, if for u a determinate one among the three values of the cube root is taken,

$$v = -\frac{a}{3u} = -\frac{a}{3\sqrt[3]{-\frac{b}{2} + \sqrt{R}}} \tag{10}$$

is in any case one of the three values of $\sqrt[3]{-\frac{b}{2} - \sqrt{R}}$. But it follows from (4) that only when we take this value for v is equation (1) satisfied by $y = u + v$. Thus, corresponding to the three values of u, we obtain three roots of equation (1). From the existence of one root of the cubic equation, however, the existence of three already follows also from the fact that it has already been proved that the quadratic equation has two roots.

To derive the other two roots from the one root (9), put

$$\alpha = u + v$$

and divide $y^3 + ay + b$ by $y - \alpha$. The quotient whose roots are the two other roots β, γ of the cubic equation (1) is

$$y^2 + \alpha y + \alpha^2 + a = 0,$$

whence, by (4) of the preceding paragraph, one finds

$$y = \frac{-\alpha \pm \sqrt{-3\alpha^2 - 4a}}{2}.$$

If here one puts $\alpha = u + v$ and, by (4), $a = -3uv$, then it follows that

$$y = \frac{-(u+v) \pm \sqrt{-3(u+v)^2 + 12uv}}{2} = \frac{-(u+v) \pm \sqrt{-3}(u-v)}{2}.$$

We therefore put

$$\varepsilon = \frac{-1 + \sqrt{-3}}{2}, \tag{11}$$

from which

$$\varepsilon^3 = 1, \qquad \varepsilon^2 + \varepsilon + 1 = 0, \qquad \varepsilon^2 = \frac{-1 - \sqrt{-3}}{2} \tag{12}$$

follows. Thus one obtains the three roots of the cubic equation (1)

$$\begin{aligned} \alpha &= u + v, \\ \beta &= \varepsilon u + \varepsilon^2 v, \\ \gamma &= \varepsilon^2 u + \varepsilon v. \end{aligned} \tag{13}$$

Here ε is a third root of unity different from 1, expressed algebraically without the trigonometric functions. If one substitutes εu or $\varepsilon^2 u$ for u, and hence $\varepsilon^2 v$ or εv for v, the three quantities α, β, γ are interchanged with one another.

The expression (9) for the root of a cubic equation is called Cardano's formula.

§ 36. Cayley's expression for Cardano's formula.

Every cube root has, as we have seen, three different values, which one obtains from one of them by multiplication by $1, \varepsilon, \varepsilon^2$. Thus each of the two quantities u, v, as they are defined by (8), § 35, also has three values, and the sum $u + v$, if one takes account only of the conditions (8), has nine different values. Of these, however, only three give solutions of the cubic equation, and only by adjoining the relation (4), § 35,

$$3uv = -a \tag{1}$$

are the usable values singled out from among the nine. This is a defect of Cardano's formula, which accordingly by itself does not yet suffice to give the roots of the cubic equation unambiguously. Cayley remedied this defect by the following representation. Define two new quantities ξ, η by the equations

$$u = \xi^2\eta, \qquad v = \xi\eta^2, \tag{2}$$

from which, by multiplication and use of (1), one obtains

$$\xi\eta = \sqrt[3]{-\frac{a}{3}}. \tag{3}$$

Consequently, if this value is inserted in (2) and the expressions § 35, (8) for u, v are substituted, then

$$\xi = \sqrt[3]{\frac{3b}{2a} + \sqrt{\frac{9b^2}{4a^2} + \frac{a}{3}}}, \qquad \eta = \sqrt[3]{\frac{3b}{2a} - \sqrt{\frac{9b^2}{4a^2} + \frac{a}{3}}}. \tag{4}$$

Thus, if one now puts the root y of the cubic equation in the form

$$y = \xi\eta(\xi + \eta), \tag{5}$$

one obtains an expression which, when ξ is replaced, independently, by $\xi, \varepsilon\xi, \varepsilon^2\xi$ and η by $\eta, \varepsilon\eta, \varepsilon^2\eta$, assumes not nine but only the three values

$$\begin{gathered} \xi^2\eta + \xi\eta^2, \\ \varepsilon\xi^2\eta + \varepsilon^2\xi\eta^2, \\ \varepsilon^2\xi^2\eta + \varepsilon\xi\eta^2. \end{gathered}$$

[9]

§ 37. The biquadratic equation.

A similar path to that used in solving the cubic equation by Cardano's formula may be taken for the solution of the equation of the fourth degree

$$y^4 + ay^2 + by + c = 0. \tag{1}$$

We put

$$2y = u + v + w$$

and obtain, if we write, for abbreviation,

$$s = u^2 + v^2 + w^2, \qquad t = v^2w^2 + w^2u^2 + u^2v^2, \tag{2}$$

then

$$\begin{aligned} 4y^2 &= s + 2(vw + wu + uv), \\ 16y^4 &= s^2 + 4s(vw + wu + uv) + 4t + 8uvw(u + v + w). \end{aligned}$$

[9]Cayley, Phil. Mag. vol. XXI, 1861. Collected mathematical papers, vol. V, No. 310.

Substitution in (1) gives

$$s^2 + 4t + 4as + 16c + 8(uvw + b)(u + v + w) + 4(s + 2a)(vw + wu + uv) = 0.$$

This equation is satisfied by the assumptions

$$s + 2a = 0, \qquad uvw + b = 0, \qquad s^2 + 4t + 4as + 16c = 0.$$

With the help of the first of these three equations the third becomes

$$t = a^2 - 4c,$$

and, putting back the values (2) for s, t,

$$\begin{array}{rcl} u^2 + v^2 + w^2 & = & -2a, \\ v^2w^2 + w^2u^2 + u^2v^2 & = & a^2 - 4c, \\ uvw & = & -b. \end{array} \tag{3}$$

By § 7 these equations are satisfied if and only if u^2, v^2, w^2 are the roots of the cubic equation

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0, \tag{4}$$

and if the signs of u, v, w are so chosen that the last of equations (3) is satisfied. This last condition still allows four different determinations of signs, so that the four roots of the biquadratic equation are obtained in the following way:

$$\begin{array}{rcl} 2\alpha & = & u + v + w, \\ 2\beta & = & u - v - w, \\ 2\gamma & = & -u + v - w, \\ 2\delta & = & -u - v + w. \end{array} \tag{5}$$

Thus the solution of the biquadratic equation has been reduced to that of the cubic equation (4).

This equation is called a cubic resolvent of the biquadratic equation.

§ 38. Proof of the fundamental theorem.

We now proceed to the proof of the fundamental theorem of algebra, namely that every equation $f(x) = 0$ has at least one root. We first put forward some general propositions.

1. If S denotes an arbitrary system of real numbers all greater than a certain positive or negative number C (that is, a system which contains no infinitely negative numbers), then there exists a lower bound for the numbers S.

By a lower bound g one is to understand a number which is not undercut by any number of the system S, but which is of such a nature that, if d is an arbitrarily given positive quantity, then between g and $g + d$ (including the endpoints) there always lies at least one number of the system S, however small d may be.

The proof follows directly from the possibility of a cut $(\mathfrak{A}, \mathfrak{B})$, constructed as follows: one throws a number α into $\mathfrak{A}$ if it is not undercut by any number of the system S, and a number β into $\mathfrak{B}$ if it is undercut by at least one number in S. The number g determined by this cut is not undercut by any number in S, for otherwise there would also be numbers smaller than g, and therefore belonging to $\mathfrak{A}$, which nevertheless are undercut by numbers in S; while, on the other hand, every number β lying even ever so little above g belongs to $\mathfrak{B}$ and is therefore undercut by a number in S. These are precisely the characteristic marks of the lower bound.

2. If S' is a part of S, then the numbers S' also have a lower bound g', and this is either equal to g or greater than g.

For it cannot be smaller than g, since otherwise numbers would occur in S', and therefore also in S, which lie below g.

Now let $f(x)$ be a real function of x which has only finite values in the interval

$$a \leq x \leq b \tag{1}$$

and which is also continuous in this interval. In agreement with § 31, V and VI, this means that

$$f(x \pm h) - f(x) \tag{2}$$

falls, in absolute value, below an arbitrarily prescribed number η throughout the interval whenever the positive h is smaller than a certain number ε. For $x = a$ we take in (2) only the upper sign, and for $x = b$ only the lower sign, so as not to pass outside the interval with $x \pm h$.

For the values of such a function in the interval there is, by 1., a lower bound g, and we now prove the proposition:

3. The function $f(x)$ assumes the value g for some value ξ of the interval, so that the lower bound becomes a minimum of the function $f(x)$ in the interval.

The proof is as follows. By 2., the lower bound of the function values in a part of the interval (1) is either equal to g or greater than g.

If now $f(a) = g$, what is to be proved is true. If, however, $f(a) > g$, then by the continuity of $f(x)$ one can give an interval $a \leq x \leq a + h$ in which $f(x) > g$ remains true, and hence the lower bound of $f(x)$ is greater than g.

We now construct in the interval (1) a cut $(\mathfrak{A}, \mathfrak{B})$ of the following kind: we count a value α of the interval (1) as belonging to $\mathfrak{A}$ if the lower bound of $f(x)$ in the interval $a \leq x \leq \alpha$ is greater than g, and a value β as belonging to $\mathfrak{B}$ if the lower bound in the interval $a \leq x \leq \beta$ is equal to g.

It is possible that $\mathfrak{B}$ consists only of the single value b; but then $f(b) = g$ must hold, and our assertion is fulfilled for $x = b$. For if $f(b) > g$, one could choose a quantity g' between $f(b)$ and g, and, by continuity, an interval $b - h \leq x \leq b$ such that in this interval all function values $f(x)$ are greater than g', so that their lower bound is equal to or greater than g', and hence certainly greater than g. But since $b - h$ belongs to $\mathfrak{A}$, the lower bound is also greater than g in the interval $a \leq x \leq b - h$, and consequently in the whole interval (1), contrary to the assumption.

If therefore $f(b) > g$, the cut $(\mathfrak{A}, \mathfrak{B})$ defines a number ξ in the interior of the interval (1), and we can now show that

$$f(\xi) = g \tag{3}$$

must hold.

If $f(\xi) > g$ and $f(\xi) > g' > g$, then, by the continuity of $f(x)$, we can determine an interval

$$\xi - h \leq x \leq \xi + h$$

in which all function values $f(x)$ are greater than g', and hence their lower bound is greater than g. But since $\xi - h$ belongs to $\mathfrak{A}$, the lower bound of $f(x)$ in the whole interval $a \leq x \leq \xi + h$ is also greater than g, while $\xi + h$ belongs to $\mathfrak{B}$, and this is the contradiction.

Let $f(x)$ now be an integral rational function with arbitrary complex or real coefficients, and let the variable x also be allowed to take complex values.

By § 31, IV, if C is an arbitrary positive value, the positive quantity R can be so chosen that

$$|f(x)| > C \tag{4}$$

whenever

$$|x| \geq R. \tag{5}$$

For visualization we represent the region (R) bounded by the inequality

$$|x| < R$$

for the variable $x = y + iz$ by a circular disc of radius R in the plane of the variable $x = y + iz$.

If we take C larger than any value of $|f(x)|$ in the interior of the region (R), for example larger than $|f(0)|$, then $|f(x)|$ will certainly become smaller in the interior of (R) than on its boundary. Since now in the whole interior of (R) the function $|f(x)|$, which for abbreviation we shall now denote by X, is not negative, there is a lower bound g for the values of X, and we have to prove the proposition:

4. There exists a value ξ of x in the interior of the region (R) such that

$$|f(\xi)| = g, \tag{6}$$

so that here also the lower bound is a minimum.

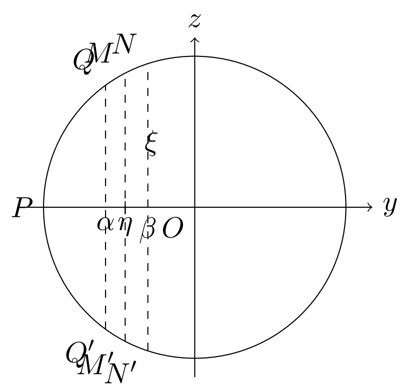
The lower bound of X in any part of the region is either equal to g or greater than g.

A domain of magnitudes determined by the inequalities

$$|x| \leq R, \qquad -R \leq y \leq \alpha \tag{7}$$

is represented in our Figure 2 by the segment $(PQ\alpha Q')$, which we shall call the segment (P, α).

Fig. 2.



We now first determine a value η of y by a cut $(\mathfrak{A}, \mathfrak{B})$ as follows: a number α between $-R$ and $+R$ is admitted into $\mathfrak{A}$ if the lower bound of X in the segment (P, α) is greater than g, and a value β is admitted into $\mathfrak{B}$ if the lower bound of X in the segment (P, β) is equal to g. This cut $(\mathfrak{A}, \mathfrak{B})$ defines a number η with the property that the lower bound of X in the region (P, γ) is greater than g if $\gamma < \eta$, and equal to g if $\gamma > \eta$.

Now, by § 31, VI,

$$|f(\eta + iz)|$$

is a continuous function of z in the interval

$$-\sqrt{R^2 - \eta^2} \leq z \leq \sqrt{R^2 - \eta^2}, \tag{8}$$

which is denoted in Figure 2 by M, M'.

Thus by 3. this function receives, for some value ζ of z in this interval, a minimum value γ, so that, if

$$\xi = \eta + i\zeta$$

is put,

$$|f(\xi)| = \gamma. \tag{9}$$

It is now further easy to see that $\gamma = g$ must be the case; for since g is the lower bound of all values of X within (R), γ cannot first of all be smaller than g.

But γ also cannot be greater than g; for all values which X assumes on the segment MM' are $\geq \gamma$. If, however, $\gamma > g$, then one can choose a quantity g' such that $\gamma > g' > g$, and, by the continuity of X proved in § 31, VI, numbers α, β can be so determined that

$$\alpha < \eta < \beta$$

and such that in the whole region $(P, \beta) - (P, \alpha) = (\alpha, \beta)$ ($QNN'Q'$ in Figure 2) X remains greater than g'. Consequently the lower bound of X in (α, β) is greater than g. Since now the lower bound of X in (P, α) is also greater than g, it is also greater than g in (P, β), which is composed of (P, α) and (α, β); but β belongs to $\mathfrak{B}$, and this would give a contradiction with the definition of $\mathfrak{B}$.

Thus there remains only the possibility that $\gamma = g$, and proposition 4 is thereby proved; namely, there is a value ξ in the interior of (R) for which

$$|f(\xi)| = g$$

holds, so that $|f(x)|$ attains a minimum value. We now prove:

5. If α is any value of x for which $f(\alpha)$ is different from zero, then h can be chosen so that

$$|f(\alpha + h)| < |f(\alpha)| \tag{10}$$

results.

It then follows that, if $f(\xi)$ were different from zero, g could not be the minimum of the function $|f(x)|$; but since this has been proved, one must have

$$f(\xi) = 0, \tag{11}$$

and ξ is a root of the equation $f(x) = 0$. The fundamental theorem will therefore be proved.

In the proof of 5. we make use of the theorem already proved (§ 32), that a pure equation always has a root.

Some of the derived functions $f'(\alpha), f''(\alpha), \dots$ may vanish; the last, $f^{(n)}(\alpha)$, which is equal to $\Pi(n)a_0$, is however different from zero. Thus suppose that $f'(\alpha), f''(\alpha), \dots, f^{(m-1)}(\alpha)$ vanish, but $f^{(m)}(\alpha)$ does not vanish. We then have, by § 13,

$$f(\alpha + h) = f(\alpha) + \frac{h^m}{\Pi(m)} f^{(m)}(\alpha) + \frac{h^{m+1}}{\Pi(m+1)} f^{(m+1)}(\alpha) + \cdots . \tag{12}$$

We choose, as is always possible by § 31, V, a positive number ε so that

$$\left| \frac{h}{m+1} \frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)} + \frac{h^2}{(m+1)(m+2)} \frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)} + \cdots \right| < 1 \tag{13}$$

whenever

$$|h| < \varepsilon, \tag{14}$$

and a positive proper fraction δ such that

$$\delta |f(\alpha)| < \left| \frac{\varepsilon^m f^{(m)}(\alpha)}{\Pi(m)} \right|, \tag{15}$$

which, if $f(\alpha)$ does not vanish, is likewise possible. We then determine h (on the basis of § 32) from the equation

$$\frac{h^m f^{(m)}(\alpha)}{\Pi(m)} = -\delta f(\alpha), \tag{16}$$

whence, by (15),

$$|h| < \varepsilon$$

follows. From (12), when the value from (16) is substituted for h^m, we now get

$$f(\alpha+h) = (1-\delta)f(\alpha) - \delta f(\alpha)\left(\frac{h}{m+1}\frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)} + \frac{h^2}{(m+1)(m+2)}\frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)} + \cdots\right),$$

and hence

$$\begin{aligned}|f(\alpha+h)| \le{}& |f(\alpha)|(1-\delta)\\ &+\delta|f(\alpha)|\left|\frac{h}{m+1}\frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)} + \frac{h^2}{(m+1)(m+2)}\frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)} + \cdots\right|.\end{aligned} \tag{17}$$

By (13), therefore,

$$|f(\alpha+h)| < |f(\alpha)|. \tag{18}$$

Thus the proof of the fundamental theorem is completed.

§ 39. Algorithm for the calculation of the roots.

The proof given in the preceding paragraph for the existence of a root of an algebraic equation, although it leaves nothing to be desired in conclusiveness, nevertheless still has the defect that it does not make clear the steps by which a root can be calculated by a convergent computational procedure. We therefore add the following considerations, intended to remedy this defect, even if only theoretically. They also give a second proof of the fundamental theorem, due in its essentials to Lipschitz (*Lehrbuch der Analysis*, vol. I, § 61 ff.; I owe a simplification to a communication of Dedekind).

If the two integral rational functions $f(x)$ and $f'(x)$ have a common divisor, then this can be found and removed by rational operations according to § 6. We may therefore assume that $f(x)$ and $f'(x)$ have no common divisor, that is, that they do not vanish simultaneously for any value of x.

We now assume that the calculation of the roots has already succeeded for a function of degree $n-1$, and accordingly suppose $f'(x)$ resolved into linear factors. Thus, if

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots, \tag{1}$$

then

$$f'(x) = nx^{n-1} + (n-1)a_1x^{n-2} + \cdots = n(x-\beta_1)(x-\beta_2)\cdots(x-\beta_{n-1}), \tag{2}$$

where $\beta_1, \beta_2, \ldots, \beta_{n-1}$ are to be regarded as known numbers, some of which may also be identical. By our assumption, the absolute values

$$|f(\beta_1)|, |f(\beta_2)|, \ldots, |f(\beta_{n-1})|, \tag{3}$$

which we denote by

$$b_1, b_2, \ldots, b_{n-1}, \tag{4}$$

are all different from zero; let the notation be so chosen that b_1 is the smallest among them, or at least exceeds none of the others. By theorem 5 of § 38 a value α of x can then be determined so that the absolute value a of $f(\alpha)$ is smaller than b_1, hence

$$a < b_1, b_2, \ldots, b_{n-1}. \tag{5}$$

We now bound a domain G for the variable x in such a way that outside this domain the absolute value of $f(x)$ is always greater than a, so that G contains all points x for which $|f(x)| < a$ (but also other points). By § 31, IV this is possible in the following way. About the origin as centre draw a circle (R) of sufficiently large radius R containing the points $\alpha, \beta_1, \beta_2, \ldots, \beta_{n-1}$, and cut out of this circle the points $\beta_1, \beta_2, \ldots, \beta_{n-1}$, at which $|f(x)|$ is certainly greater than a, by circular envelopes of sufficiently small but non-zero radii $\rho_1, \rho_2, \ldots, \rho_{n-1}$, so that within all these circles $(\rho_1), (\rho_2), \ldots, (\rho_{n-1})$ the absolute value $|f(x)|$ remains greater than a (§ 31, V). None of these circles then encloses the point α, for at that point $|f(x)| = a$. The connected piece of surface bounded by the circles $(R), (\rho_1), \ldots, (\rho_{n-1})$ is the domain G.

We now determine a sequence of numbers

$$\alpha, \alpha', \alpha'', \alpha''', \ldots$$

in such a way that the absolute values of $f(\alpha), f(\alpha'), f(\alpha''), \ldots$, which we denote by

$$a, a', a'', a''', \ldots,$$

always decrease. The points so determined all remain in the interior of the domain G, since for them $|f(x)| < a$.

For this we use a procedure quite similar to that used for the proof of theorem 5 in § 38, but simplified by the fact that $f'(\alpha)$ is already different from zero. If δ denotes a positive proper fraction, still to be determined more precisely, we put in the expansion

$$f(\alpha + h) = f(\alpha) + hf'(\alpha) + \frac{h^2}{1 \cdot 2} f''(\alpha) + \cdots \tag{6}$$

$$h = -\delta \frac{f(\alpha)}{f'(\alpha)}, \tag{7}$$

and obtain

$$f(\alpha + h) = (1 - \delta) f(\alpha) + \frac{\delta^2 f(\alpha)^2 f''(\alpha)}{1 \cdot 2\, f'(\alpha)^2} - \frac{\delta^3 f(\alpha)^3 f'''(\alpha)}{1 \cdot 2 \cdot 3\, f'(\alpha)^3} + \cdots . \tag{8}$$

Now by (2)

$$f'(\alpha) = n(\alpha - \beta_1)(\alpha - \beta_2) \cdots (\alpha - \beta_{n-1});$$

therefore, if the absolute values of $\alpha - \beta_1, \alpha - \beta_2, \ldots, \alpha - \beta_{n-1}$ are denoted by $r_1, r_2, \ldots, r_{n-1}$,

$$|f'(\alpha)| = n r_1 r_2 \cdots r_{n-1}.$$

Since α lies outside the circle of radius ρ_i drawn about β_i, we have $r_1 > \rho_1, r_2 > \rho_2, \ldots$, and consequently

$$|f'(\alpha)| > n \rho_1 \rho_2 \cdots \rho_{n-1}.$$

Thus $|f'(\alpha)|$ is greater than a positive number k independent of the position of α in G. It follows that one can choose a sufficiently large positive number Q, likewise independent of the position of α and of the proper fraction δ, so that

$$\left| \frac{f(\alpha)^2 f''(\alpha)}{1 \cdot 2\, f'(\alpha)^2} - \delta \frac{f(\alpha)^3 f'''(\alpha)}{1 \cdot 2 \cdot 3\, f'(\alpha)^3} + \cdots \right| < Q. \tag{9}$$

One obtains Q, for example, by replacing in the denominators on the left of (9) $f'(\alpha)$ by k, replacing δ by 1, replacing all the other factors by their absolute values, replacing finally the absolute value of α by the greater value R, and then making the number so obtained arbitrarily larger. From (8) one obtains

$$|f(\alpha + h)| < (1 - \delta)|f(\alpha)| + \delta^2 Q. \tag{10}$$

Assume now Q so chosen that, for every α in G,

$$2Q > |f(\alpha)| = a,$$

which is plainly compatible with the above determination. We may then put

$$\delta = \frac{a}{2Q}, \qquad h = -\frac{af(\alpha)}{2Qf'(\alpha)}, \tag{11}$$

and, denoting $\alpha + h$ by α', and denoting $|f(\alpha')|$ by a', obtain from (10)

$$a' < a\left(1 - \frac{a}{4Q}\right). \tag{12}$$

By the same procedure we derive from α', a' a second system α'', a'', then from α'', a'' a third system α''', a''', and so on; for the sequence of decreasing positive numbers $a, a', a'', \ldots$ we obtain the inequalities

$$a' < a\left(1 - \frac{a}{4Q}\right), \quad a'' < a'\left(1 - \frac{a'}{4Q}\right), \quad a''' < a''\left(1 - \frac{a''}{4Q}\right), \ldots. \tag{13}$$

We prove that this sequence approaches the limit zero. If this were not the case, a positive number ω could be given such that all $a^{(\nu)}$ remain above ω; the differences

$$\frac{a^{(\nu)}}{4Q}$$

would then all be greater than the positive proper fraction $\omega/(4Q)$, and from (13) it would follow that

$$a' < a\theta, \quad a'' < a'\theta, \quad a''' < a''\theta, \quad \ldots,$$

where $0 < \theta < 1$. By multiplication one gets

$$a^{(\nu)} < a\theta^{\nu}. \tag{14}$$

This is a contradiction, since θ^{ν} becomes infinitely small as ν increases without bound. The result so obtained is expressed by

$$\lim a^{(\nu)} = 0. \tag{15}$$

If the inequalities (13) are written in the form

$$a - a' > \frac{a^2}{4Q}, \qquad a' - a'' > \frac{a'^2}{4Q}, \qquad \ldots,$$

and added from the $(\mu+1)$-st to the $(\nu+1)$-st, one obtains

$$(a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2 < 4Q\big(a^{(\mu)} - a^{(\nu+1)}\big),$$

and therefore, when μ and ν both tend to infinity,

$$\lim\big((a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2\big) = 0. \tag{16}$$

For the sequence of numbers $\alpha, \alpha', \alpha'', \ldots$ the law of formation resulting from (11) is

$$\begin{aligned} \alpha' &= \alpha - \frac{af(\alpha)}{2Qf'(\alpha)}, \\ \alpha'' &= \alpha' - \frac{a'f(\alpha')}{2Qf'(\alpha')}, \\ &\vdots \\ \alpha^{(\nu)} &= \alpha^{(\nu-1)} - \frac{a^{(\nu-1)}f(\alpha^{(\nu-1)})}{2Qf'(\alpha^{(\nu-1)})}. \end{aligned} \tag{17}$$

If an arbitrary number of these equations are added, from the $(\mu+1)$-st to the $(\nu+1)$-st, then

$$\alpha^{(\mu)} - \alpha^{(\nu+1)} = \sum_{r=\mu}^{\nu} \frac{a^{(r)} f(\alpha^{(r)})}{2Q f'(\alpha^{(r)})}. \tag{18}$$

Since all the $|f'(\alpha^{(r)})|$ are at least equal to k, it follows for the absolute value of this difference that

$$|\alpha^{(\mu)} - \alpha^{(\nu+1)}| < \frac{(a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2}{2Qk}, \tag{19}$$

and hence, by (16),

$$\lim |\alpha^{(\mu)} - \alpha^{(\nu+1)}| = 0. \tag{20}$$

This says (according to the concept of a sequence of numbers explained in the Introduction, p. 15) that the numbers $\alpha, \alpha', \alpha'', \ldots$ approach a definite limit, which we shall denote by ξ. From the continuity of the function $f(x)$ it then follows easily that $f(\xi) = 0$; for if $|f(\xi)| > 0$, then, since the $\alpha^{(\nu)}$ come arbitrarily close to ξ, the absolute values $a^{(\nu)}$ of $f(\alpha^{(\nu)})$ could not sink below every positive value, which is precisely what we have proved for them.

§ 40. Continuity of the roots.

We close this section with the proof of the theorem:

The roots of an algebraic equation are continuous functions of the coefficients.

We have first to explain the meaning of this theorem.

Let

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots \tag{1}$$

be an integral rational function of x of degree n. By what has been proved in the preceding paragraphs, $f(x)$ can be decomposed into n linear factors, some of which may be equal to one another. Combining equal factors and denoting the roots by $\alpha, \beta, \gamma, \ldots$, we put

$$f(x) = (x-\alpha)^a (x-\beta)^b (x-\gamma)^c \cdots, \tag{2}$$

where $a, b, c, \ldots$ are positive whole numbers whose sum is equal to n.

The roots $\alpha, \beta, \gamma, \ldots$ will change with the coefficients $a_1, a_2, a_3, \ldots$; the degree of their multiplicity can also become another one.

We denote the changes of $a_1, a_2, \ldots$ by $\varepsilon_1, \varepsilon_2, \ldots$ and put

$$\varphi(x) = \varepsilon_1 x^{n-1} + \varepsilon_2 x^{n-2} + \cdots, \tag{3}$$

$$f(x) + \varphi(x) = f_1(x). \tag{4}$$

We surround the points $\alpha, \beta, \gamma, \ldots$ by regions of arbitrary smallness, but in such a way that these regions exclude one another; for instance, we enclose the points $\alpha, \beta, \gamma, \ldots$ by circular peripheries with radii $\rho, \rho', \rho'', \ldots$, and denote these regions by $(\rho), (\rho'), (\rho''), \ldots$.

If the absolute values of $\varepsilon_1, \varepsilon_2, \ldots$ lie below sufficiently small values, then we can prove first of the function $f_1(x)$

that it has no roots outside the regions $(\rho), (\rho'), (\rho''), \ldots$,

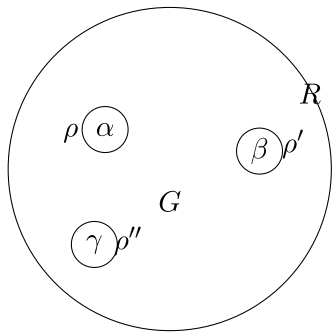
and secondly,

that the number of roots of $f_1(x)$ within (ρ) is exactly a, within (ρ') exactly b, within (ρ'') exactly c, and so on.

In the last part of the theorem, however, it must be observed that, if $f_1(x)$ has multiple roots, these must be counted according to their multiplicity.

We construct, according to §31, IV, in the x-plane a circle with radius R and with the origin as centre, which encloses the regions $(\rho), (\rho'), (\rho''), \ldots$, so that outside this circle no more roots of $f_1(x)$ lie, and we denote by G the region lying within this circle, but outside $(\rho), (\rho'), (\rho''), \ldots$. It must also be remarked that R can be assumed independently of $\varepsilon_1, \varepsilon_2, \ldots$, so long as a definite upper bound is fixed for the absolute values of these quantities.

Fig. 4.



If now x is a point of the region G, then the absolute value of $x - \alpha$ is greater than ρ, and consequently, by (2),

$$|f(x)| > \rho^a \rho'^b \rho''^c \cdots . \tag{5}$$

If now α_1 is a root of $f_1(x)$, then from (4) it follows that

$$f(\alpha_1) = -\varphi(\alpha_1), \tag{6}$$

and α_1 therefore cannot lie in the region G if one takes care that everywhere within G

$$|\varphi(x)| < \rho^a \rho'^b \rho''^c \cdots , \tag{7}$$

which is always possible by sufficient diminution of the upper bound of $\varepsilon_1, \varepsilon_2, \ldots$. Thus the first part of our assertion is proved, namely that, when the inequality (7) is satisfied, no roots of $f_1(x)$ lie outside the regions $(\rho), (\rho'), (\rho''), \ldots$.

To prove the second part, we put

$$f(x) = \psi(x)(x - \alpha)^a, \tag{8}$$

$$\psi(x) = (x - \beta)^b (x - \gamma)^c \cdots . \tag{9}$$

We now choose a positive number A, which shall however be independent of ρ, in such a way that, as long as x lies in the interior or on the periphery of the region (ρ),

$$|\psi(x)| > A \qquad (\text{for } |x - \alpha| \leq \rho) \tag{10}$$

remains true. Such a number is obtained, for example, if l is taken equal to half the least of the distances $(\alpha, \beta), (\alpha, \gamma), \ldots$ and one puts

$$A = l^{b+c+\cdots};$$

then the requirement expressed in (10) is fulfilled at least as long as ρ is smaller than l. If only one point α is present, then $\psi(x) = 1$ is to be put, and for A any proper fraction can be chosen.

Let now $\alpha_1, \alpha_2, \ldots$ be the roots of $f_1(x)$ within (ρ), and let $a_1, a_2, \ldots$ be the degrees of their multiplicity; and let $\beta_1, \beta_2, \ldots$, $b_1, b_2, \ldots$, $\gamma_1, \gamma_2, \ldots$, $c_1, c_2, \ldots$ have the same meaning for the regions $(\rho'), (\rho''), \ldots$. Then

$$n = a_1 + a_2 + \cdots + b_1 + b_2 + \cdots + c_1 + c_2 + \cdots , \tag{11}$$

since the total number of all roots must be equal to n. The function $f_1(x)$ can then be represented as

$$f_1(x) = \psi_1(x)(x-\alpha_1)^{a_1}(x-\alpha_2)^{a_2}\cdots, \tag{12}$$

where

$$\psi_1(x) = (x-\beta_1)^{b_1}(x-\beta_2)^{b_2}\cdots(x-\gamma_1)^{c_1}(x-\gamma_2)^{c_2}\cdots. \tag{13}$$

Now we can determine a positive number B independent of $\rho, \rho', \rho'', \ldots$ such that, as long as x remains within or on the boundary of (ρ),

$$|\psi_1(x)| < B \qquad (\text{for } |x-\alpha| \leq \rho). \tag{14}$$

For example, one can choose a magnitude L which is greater than twice the distance of the point α from one of the points $\beta, \gamma, \ldots$ and in every case greater than 1, and then assume

$$B > L^n.$$

Now from (4) it follows that

$$|f(x)| < |f_1(x)| + |\varphi(x)|, \tag{15}$$

hence by (8) and (12)

$$|\psi(x)|\,|x-\alpha|^a < |\psi_1(x)|\,|x-\alpha_1|^{a_1}\,|x-\alpha_2|^{a_2}\cdots + |\varphi(x)|. \tag{16}$$

If we now assume x on the periphery of ρ, and therefore put

$$|x-\alpha| = \rho,$$

then

$$|x-\alpha_1| < 2\rho, qquad|x-\alpha_2| < 2\rho, \ldots,$$

and consequently, by (10) and (16),

$$\rho^a A < (2\rho)^{a_1+a_2+\cdots} B + |\varphi(x)|. \tag{17}$$

We now choose the upper bounds for the coefficients $\varepsilon_1, \varepsilon_2, \ldots$ of φ so small that

$$|\varphi(x)| < \rho^a A'$$

where A' denotes a number which is smaller than A; then from (17) it follows that

$$A - A' < 2^a \rho^{-a+a_1+a_2+\cdots} B. \tag{18}$$

But, for sufficiently small ρ, this would no longer be possible if a were greater than the sum $a_1 + a_2 + \cdots$. Hence

$$a \leq a_1 + a_2 + \cdots. \tag{19}$$

In the same way one proves that

$$b \leq b_1 + b_2 + \cdots, \qquad c \leq c_1 + c_2 + \cdots, \qquad \ldots. \tag{20}$$

But since the sums of the left sides as well as of the right sides in the inequalities (19), (20) must be equal to n, only equality signs can hold, and therefore

$$a = a_1 + a_2 + \cdots, \quad b = b_1 + b_2 + \cdots, \quad c = c_1 + c_2 + \cdots, \quad \ldots,$$

whereby the second part of our theorem is also proved.

We shall add to the theorem just proved the following remark concerning the case of multiple roots.

The necessary and sufficient condition that $f(x) = 0$ should have multiple roots at all is that $f(x)$ and $f'(x)$ should have a common factor. Therefore, if the algorithm of the greatest common divisor is applied to these two functions according to the principles developed in the first section, the condition for common factors is obtained by setting the last constant remainder equal to zero, in the form of an equation between the coefficients,

$$F(a_1, a_2, \ldots, a_n) = 0,$$

where F is an integral rational function of the arguments $a_1, a_2, \ldots, a_n$, which is called the discriminant of f, and whose properties and mode of formation will be treated more fully in the next section. In any case, since equations of degree n without multiple roots do exist at all, F cannot vanish identically.

Consider now

$$F(a_1 + \varepsilon_1, a_2 + \varepsilon_2, \ldots, a_n + \varepsilon_n).$$

This function too, if $a_1, a_2, \ldots, a_n$ are definite and $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ are indeterminate numbers, will not vanish identically; and one can dispose of the ε's so that its value comes out different from zero, even when any arbitrary upper bound has been fixed for the absolute values of the ε's.[10]

One therefore sees from these considerations how an a-fold root α of $f(x)$, under continuous variation of the coefficients, radiates apart into a simple roots, so that from this point of view too one is justified in considering the a-fold root as having arisen from the coincidence of a simple roots.

If we do not assume the coefficient of the highest power of x equal to 1, and therefore examine the equation in the form

$$a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n = 0, \tag{21}$$

then, if $a_n = 0$, one or more of the roots are equal to zero, and equation (21) can be divided by x or by a higher power of x, whereby an equation of correspondingly lower degree arises in which the last term is different from zero. We shall therefore assume from the outset that a_n is different from zero, and now put in (21)

$$x = \frac{1}{y}. \tag{22}$$

By multiplication by y^n and division by a_n, (21) then passes into

$$y^n + \frac{a_{n-1}}{a_n} y^{n-1} + \cdots + \frac{a_1}{a_n} y + \frac{a_0}{a_n} = 0, \tag{23}$$

and we can apply to the roots of this equation the theorem just stated, by assuming $a_0 = 0$, so that one of the roots of (23) vanishes. We thus obtain the theorem:

One can take a_0 in the equation (21) so small that one of its roots becomes large beyond all bounds, while the others differ arbitrarily little from the roots of the equation of degree $n - 1$

$$a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n = 0.$$

This is also expressed by saying that, as a_0 vanishes, one of the roots of (21) becomes infinite.

If a_1 also vanishes, or a_1 and a_2, and so on, two or three roots become infinite.

This justifies the expression sometimes used, that an equation of degree n passes, by the becoming-infinite of one root, into an equation of degree $n - 1$.

[10]This will only later be rigorously justified. For the time being we presuppose it in these preliminary remarks, on which no further conclusions will be founded.

Fourth Section. Symmetric functions.

§ 41. Concept of symmetric functions. Symmetric fundamental functions.

In this section we consider whole functions, of arbitrary degree, in an arbitrary number n of variables

$$\alpha_1, \alpha_2, \ldots, \alpha_n.$$

Such a function is called symmetric if it remains unchanged when the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ are subjected to an arbitrary permutation; and it is such symmetric functions whose properties and laws of formation we now have to learn more precisely.

In order that a function $\Phi(\alpha_1, \alpha_2, \ldots, \alpha_n)$ be symmetric, it is sufficient that it not be changed when any two of the arguments $\alpha_1, \alpha_2, \ldots, \alpha_n$ are interchanged, since by § 18 all permutations can be formed by a succession of transpositions.

The function Φ will in general not be homogeneous, but will contain terms of different dimensions. If, however, one collects all terms of the same dimension, every Φ can be represented as a sum of homogeneous functions of different degrees; and, if Φ is to be symmetric, then each homogeneous constituent of a definite degree must by itself be symmetric, since the dimensions of the terms are not changed by the permutations of the variables.

Accordingly we may confine ourselves to the consideration of homogeneous symmetric functions, from which all the others can be put together.

After this it is easy to give the general form of a symmetric function. We obtain it if, in one term of such a function,

$$\alpha_1^{\nu_1} \alpha_2^{\nu_2} \cdots \alpha_n^{\nu_n},$$

as in the formation of determinants, we permute the lower indices in all possible ways and take the sum of all terms so formed. Such a function may be called an elementary constituent of a symmetric function. If we take several such elementary constituents, multiply them by arbitrary factors independent of the α's, and add them, then we obtain the most general symmetric function. The number of terms of one of these elementary constituents is $n!$ when the exponents $\nu_1, \nu_2, \ldots, \nu_n$ are all different; but if one and the same exponent ν occurs several times, the permutations which give no different terms are to be omitted. For example, for three variables, if ν_1, ν_2, ν_3 are different, an elementary constituent is

$$\alpha_1^{\nu_1}\alpha_2^{\nu_2}\alpha_3^{\nu_3} + \alpha_1^{\nu_1}\alpha_2^{\nu_3}\alpha_3^{\nu_2} + \alpha_1^{\nu_2}\alpha_2^{\nu_1}\alpha_3^{\nu_3} + \alpha_1^{\nu_2}\alpha_2^{\nu_3}\alpha_3^{\nu_1} + \alpha_1^{\nu_3}\alpha_2^{\nu_1}\alpha_3^{\nu_2} + \alpha_1^{\nu_3}\alpha_2^{\nu_2}\alpha_3^{\nu_1},$$

whereas, if $\nu_2 = \nu_3$,

$$\alpha_1^{\nu_1}\alpha_2^{\nu_2}\alpha_3^{\nu_2} + \alpha_1^{\nu_2}\alpha_2^{\nu_1}\alpha_3^{\nu_2} + \alpha_1^{\nu_2}\alpha_2^{\nu_2}\alpha_3^{\nu_1}.$$

The simplest example of a symmetric function is the sum of the variables

$$\alpha_1 + \alpha_2 + \alpha_3 + \cdots + \alpha_n.$$

Likewise the product

$$\alpha_1 \alpha_2 \cdots \alpha_n$$

belongs here.

These two are the extreme cases of a series of symmetric functions which we call the symmetric fundamental functions and obtain as follows.

The product

$$f(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n) \tag{1}$$

is, whatever x may be, provided only that x is independent of the α's, a symmetric function of $\alpha_1, \alpha_2, \ldots, \alpha_n$. If therefore we carry out the multiplication of the several factors and arrange according to powers of x, then the coefficients of the separate powers of x are likewise symmetric functions. For they change no more under interchange of the α's than does the function $f(x)$ (§ 1). We put

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n. \tag{2}$$

The functions $a_1, a_2, \ldots, a_n$ are the coefficients of the algebraic equation whose roots are $\alpha_1, \alpha_2, \ldots, \alpha_n$. They are called the symmetric fundamental functions. We already formed these symmetric fundamental functions in § 7; as we recall, a_1 is the sum of the α's taken with negative sign, a_2 the sum of the products two at a time, a_3 the sum of the products three at a time taken with negative sign, and so on; finally a_n is the product of all the α's with positive or negative sign according as n is even or odd. We write, for short,

$$\begin{aligned} a_1 &= -\sum \alpha_1, \\ a_2 &= +\sum \alpha_1 \alpha_2, \\ a_3 &= -\sum \alpha_1 \alpha_2 \alpha_3, \\ &\ \vdots \\ a_n &= \pm \alpha_1 \alpha_2 \cdots \alpha_n. \end{aligned} \tag{3}$$

The goal of our considerations is the proof of the principal theorem that all symmetric functions of the α's can be expressed rationally by the fundamental functions.

§ 42. The power sums.

We occupy ourselves first with another special kind of symmetric functions, the power sums. Namely, if ν denotes any positive integral exponent, then

$$s_\nu = \alpha_1^\nu + \alpha_2^\nu + \cdots + \alpha_n^\nu \tag{1}$$

obviously belongs to the symmetric functions, and s_ν is called the νth power sum. We shall derive formulae according to which the power sums are expressible by the fundamental functions.

In what follows, whenever $\varphi(x)$ is any function of x, we always denote by

$$S[\varphi(\alpha)]$$

the sum obtained by replacing x in $\varphi(x)$ successively by each of the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ and adding the functions so formed, thus

$$S[\varphi(\alpha)] = \varphi(\alpha_1) + \varphi(\alpha_2) + \cdots + \varphi(\alpha_n).$$

Accordingly, for example,

$$S[\alpha^\nu] = s_\nu$$

is the νth power sum.

We now divide $f(x)$, by § 4, by an arbitrary linear function $x - \alpha$ and obtain

$$\frac{f(x) - f(\alpha)}{x - \alpha} = x^{n-1} + f_1(\alpha) x^{n-2} + f_2(\alpha) x^{n-3} + \cdots + f_{n-1}(\alpha), \tag{2}$$

where, by § 4, (8),

$$\begin{aligned} f_1(\alpha) &= \alpha + a_1, \\ f_2(\alpha) &= \alpha^2 + a_1 \alpha + a_2, \\ &\ \vdots \\ f_{n-1}(\alpha) &= \alpha^{n-1} + a_1 \alpha^{n-2} + \cdots + a_{n-1}. \end{aligned} \tag{3}$$

We put in (2), for α, each of the quantities $\alpha_1, \alpha_2, \ldots, \alpha_n$, whereby $f(\alpha) = 0$, and form the sum S. The left hand side then gives, by § 14, (11),

$$nx^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \cdots + a_{n-1}, \tag{4}$$

whereas the right hand side becomes

$$nx^{n-1} + x^{n-2}S[f_1(\alpha)] + x^{n-3}S[f_2(\alpha)] + \cdots + S[f_{n-1}(\alpha)]. \tag{5}$$

The comparison of coefficients of like powers of x in (4) and (5) gives

$$\begin{aligned} S[f_1(\alpha)] &= (n-1)a_1, \\ S[f_2(\alpha)] &= (n-2)a_2, \\ &\vdots \\ S[f_{n-1}(\alpha)] &= a_{n-1}. \end{aligned} \tag{6}$$

But by (1) and (3)

$$\begin{aligned} S[f_1(\alpha)] &= s_1 + na_1, \\ S[f_2(\alpha)] &= s_2 + a_1s_1 + na_2, \\ &\vdots \\ S[f_{n-1}(\alpha)] &= s_{n-1} + a_1s_{n-2} + a_2s_{n-3} + \cdots + na_{n-1}. \end{aligned}$$

Consequently from (6) one obtains the following system of formulae due to Newton:

$$\begin{aligned} 0 &= s_1 + a_1, \\ 0 &= s_2 + a_1s_1 + 2a_2, \\ 0 &= s_3 + a_1s_2 + a_2s_1 + 3a_3, \\ &\vdots \\ 0 &= s_{n-1} + a_1s_{n-2} + a_2s_{n-3} + \cdots + (n-1)a_{n-1}. \end{aligned} \tag{7}$$

This system of formulae can, however, be continued still further; for, since

$$S[f(\alpha)] = 0, \qquad S[\alpha f(\alpha)] = 0, \qquad S[\alpha^2 f(\alpha)] = 0, \ldots,$$

it follows that

$$\begin{aligned} 0 &= s_n + a_1s_{n-1} + a_2s_{n-2} + \cdots + a_n, \\ 0 &= s_{n+1} + a_1s_n + a_2s_{n-1} + \cdots + a_ns_1, \\ 0 &= s_{n+2} + a_1s_{n+1} + a_2s_n + \cdots + a_ns_2, \quad \ldots \end{aligned} \tag{8}$$

By the formulae (7), (8) the problem is now solved of representing, one after another, the functions $s_1, s_2, \ldots$ up to any height as whole rational functions of the symmetric fundamental functions; for example

$$\begin{aligned} s_1 &= -a_1, \\ s_2 &= a_1^2 - 2a_2, \\ s_3 &= -a_1^3 + 3a_1a_2 - 3a_3, \\ s_4 &= a_1^4 - 4a_1^2a_2 + 4a_1a_3 + 2a_2^2 - 4a_4. \end{aligned} \tag{9}$$

The manner of formation of these expressions shows that the coefficients in these representations are whole numbers.

Conversely, by means of the formulae (7) and (8), one can express the symmetric fundamental functions $a_1, a_2, a_3, \ldots$ rationally by the power sums $s_1, s_2, s_3, \ldots$. This representation is, however, far less simple in so far as the coefficients are not whole but fractional numbers, for example

$$a_1 = -s_1, \qquad 2a_2 = s_1^2 - s_2, \qquad 6a_3 = -s_1^3 + 3s_1 s_2 - 2s_3. \tag{10}$$

One can also continue the system of formulae (8) in the opposite direction, if one forms the equations

$$S[\alpha^{-1} f(\alpha)] = 0, \qquad S[\alpha^{-2} f(\alpha)] = 0, \ldots.$$

In this way one obtains a means of expressing the power sums s_ν also for negative exponents ν by the fundamental functions. These sums of negative powers likewise belong to the symmetric functions, although no longer to the whole, but to the fractional, ones. They pass into whole functions only after multiplication by powers of the product $\alpha_1 \alpha_2 \cdots \alpha_n$. For completeness we put down here the first two of these formulae:

$$\begin{aligned} 0 &= a_{n-1} + a_n s_{-1}, \\ 0 &= 2a_{n-2} + a_{n-1} s_{-1} + a_n s_{-2}. \end{aligned} \tag{11}$$

§ 43. Proof of the principal theorem for two variables.

We now pass to the proof of the fundamental theorem of the theory of symmetric functions, that they are all rationally expressible by the symmetric fundamental functions. Since we use complete induction as the means of proof, we first derive the theorem under the assumption that only two independent variables α, β are given, but in a way which at the same time will bring out the guiding thought for the general proof.

We denote the symmetric fundamental functions by

$$a = -(\alpha + \beta), \qquad b = \alpha\beta \tag{1}$$

and accordingly put

$$f(x) = (x - \alpha)(x - \beta) = x^2 + ax + b. \tag{2}$$

Let $S(\alpha, \beta)$ be any whole rational and symmetric function of α and β. We can substitute for β from (1) the value $-(\alpha + a)$, and obtain, if we arrange according to powers of α,

$$S(\alpha, \beta) = S(\alpha, -\alpha - a) = A_0 \alpha^m + A_1 \alpha^{m-1} + \cdots + A_m, \tag{3}$$

where the coefficients $A_0, A_1, \ldots, A_m$ depend only on a and on the coefficients still occurring in S. We note expressly, however, that if in $S(\alpha, \beta)$ no fractional numerical coefficients occur, then no fractions occur in the coefficients $A_0, A_1, \ldots, A_m$ either.

We now put

$$\Phi(x) = A_0 x^m + A_1 x^{m-1} + \cdots + A_{m-1} x + A_m \tag{4}$$

and divide $\Phi(x)$ by $f(x)$, by § 3. We obtain a quotient Q and a remainder which, with respect to x, is at most of the first degree; thus

$$\Phi(x) = Q f(x) + A + Bx. \tag{5}$$

Here A and B are now whole functions of a and b, and they contain no fractional numerical coefficients if none occur in S. If now we put $x = \alpha$, it follows from (5) and (3), since $f(\alpha)$ vanishes, that

$$S(\alpha, \beta) = A + B\alpha. \tag{6}$$

But since $S(\alpha, \beta)$ and likewise A, B are symmetric, it follows by interchanging α and β that

$$S(\alpha, \beta) = A + B\beta. \tag{7}$$

From this one concludes, since α and β are independent variables, that $B = 0$, and consequently

$$S(\alpha, \beta) = A, \tag{8}$$

which proves the fundamental theorem in this case.

§ 44. General proof of the principal theorem.

We now suppose that the fundamental theorem has been proved for symmetric functions of $n-1$ variables, and we derive it, by a procedure entirely analogous to that applied in § 43, for n variables.

Let again

$$S = S(\alpha_1, \alpha_2, \ldots, \alpha_n) \tag{1}$$

be a whole symmetric function of the n variables $\alpha_1, \alpha_2, \ldots, \alpha_n$. If we arrange it according to powers of α_1, and accordingly put

$$S = S_0\alpha_1^\mu + S_1\alpha_1^{\mu-1} + \cdots + S_{\mu-1}\alpha_1 + S_\mu, \tag{2}$$

then the coefficients $S_0, S_1, \ldots, S_\mu$ are whole symmetric functions of the $n-1$ variables $\alpha_2, \ldots, \alpha_n$.

If we denote the symmetric fundamental functions of these latter variables by

$$a_1', a_2', \ldots, a_{n-1}',$$

then, by our supposition, the coefficients $S_0, S_1, \ldots, S_\mu$ can be expressed rationally by these. But by § 42, (2) and (3),

$$\begin{aligned} a_1' &= f_1(\alpha_1) = \alpha_1 + a_1, \\ a_2' &= f_2(\alpha_1) = \alpha_1^2 + a_1\alpha_1 + a_2, \\ a_3' &= f_3(\alpha_1) = \alpha_1^3 + a_1\alpha_1^2 + a_2\alpha_1 + a_3, \end{aligned}$$

and so on. Thus the coefficients $S_0, S_1, \ldots, S_\mu$ can be expressed wholly and rationally by

$$\alpha_1, a_1, a_2, \ldots, a_n.$$

If, therefore, after these expressions have been introduced, we arrange (2) anew according to powers of α_1, we obtain

$$S = A_0\alpha_1^m + A_1\alpha_1^{m-1} + \cdots + A_{m-1}\alpha_1 + A_m, \tag{3}$$

where in general m is an exponent different from μ, and where the coefficients $A_0, A_1, \ldots, A_m$ depend wholly and rationally on $a_1, a_2, \ldots, a_n$. We put again

$$\Phi(x) = A_0x^m + A_1x^{m-1} + \cdots + A_{m-1}x + A_m \tag{4}$$

and divide $\Phi(x)$ by

$$\begin{aligned} f(x) &= x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \\ &= (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n). \end{aligned}$$

There results a quotient and a remainder which, with respect to x, is at most of degree $n-1$. We therefore put

$$\Phi(x) = Qf(x) + \psi(x), \tag{5}$$

$$\psi(x) = C_0x^{n-1} + C_1x^{n-2} + \cdots + C_{n-2}x + C_{n-1}, \tag{6}$$

and here $C_0, C_1, \ldots, C_{n-1}$ are whole rational functions of $a_1, a_2, \ldots, a_n$, in which, if S in its original form contains no fractional coefficients, no fractions occur either.

But now, since $f(\alpha_1)$ vanishes and $S = \Phi(\alpha_1)$, it follows from (5) that

$$S = \psi(\alpha_1). \tag{7}$$

Since S is symmetric, α_1 may here be replaced by $\alpha_2, \alpha_3, \ldots, \alpha_n$. This gives the following n equations:

$$\begin{aligned} C_0\alpha_1^{n-1} + C_1\alpha_1^{n-2} + \cdots + C_{n-2}\alpha_1 + (C_{n-1} - S) &= 0, \\ C_0\alpha_2^{n-1} + C_1\alpha_2^{n-2} + \cdots + C_{n-2}\alpha_2 + (C_{n-1} - S) &= 0, \\ &\vdots \\ C_0\alpha_n^{n-1} + C_1\alpha_n^{n-2} + \cdots + C_{n-2}\alpha_n + (C_{n-1} - S) &= 0. \end{aligned} \tag{8}$$

If we regard this system as a system of homogeneous linear equations with the n unknowns

$$C_0, C_1, \ldots, C_{n-2}, C_{n-1} - S,$$

then its determinant is

$$\begin{vmatrix} \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \alpha_2^{n-1} & \alpha_2^{n-2} & \cdots & \alpha_2 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix},$$

which, by § 22, formula (12), is equal to the product of all differences

$$\alpha_1 - \alpha_2, \quad \alpha_1 - \alpha_3, \ldots$$

and is therefore different from zero. Consequently, by Theorem II in § 24,

$$C_0 = 0, \quad C_1 = 0, \ldots, \quad C_{n-2} = 0,$$

and

$$S = C_{n-1}, \tag{9}$$

which contains the fundamental theorem to be proved.

One reaches the same result also by Theorem II, § 30; for since the function of degree $n-1$ in x,

$$C_0 x^{n-1} + C_1 x^{n-2} + \cdots + C_{n-2} x + C_{n-1} - S,$$

vanishes for $x = \alpha_1, \alpha_2, \ldots, \alpha_n$, that is, for more than $n-1$ values, all its coefficients must be zero.

The proof which we have here given for the fundamental theorem in connection with Cauchy[11] at the same time offers a means, in special cases, actually to compute the expression of a symmetric function by the fundamental functions. The other proofs known for the theorem offer the same possibility. We shall communicate here a second proof, which leads to a method of computation that is often simpler.[12]

§ 45. Second proof of the theorem on symmetric functions.

Let S be a whole symmetric function of the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$. The individual terms of this function are all of the form

$$M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n},$$

where M is a coefficient independent of the α's. These terms shall now be ordered in a definite manner. After the order of the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ has been fixed, of two terms

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n}, \qquad A' = M'\alpha_1^{\nu_1'}\alpha_2^{\nu_2'}\cdots\alpha_n^{\nu_n'},$$

A shall be called the higher if the first of the differences

$$\nu_1 - \nu_1', \quad \nu_2 - \nu_2', \ldots, \nu_n - \nu_n'$$

which is different from zero has a positive value; thus if either $\nu_1 > \nu_1'$, or $\nu_1 = \nu_1', \nu_2 > \nu_2'$, or $\nu_1 = \nu_1', \nu_2 = \nu_2', \nu_3 > \nu_3'$, and so on.

Since we suppose that all terms in which all exponents $\nu_1, \nu_2, \ldots, \nu_n$ agree have been united into one term, it is thereby decided, for any two terms, which is the higher; and if A is higher than A', and A' higher than A'', then A is also higher than A''.

[11] Cauchy, *Exercices de mathématiques*, fourth year.

[12] Waring, *Meditationes algebraicae*. Gauss, *Demonstratio nova altera theorematis omnem functionem algebraicam rationalem integram unius variabilis in factores reales primi vel secundi gradus resolvi posse.* Werke, vol. III, p. 36.

Let now, according to this ordering,

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n}$$

be the highest term of our function S. It follows that

$$\nu_1 \geq \nu_2$$

must hold; for if $\nu_1 < \nu_2$, then the term

$$M\alpha_1^{\nu_2}\alpha_2^{\nu_1}\cdots\alpha_n^{\nu_n},$$

which must also occur in S because of symmetry, would stand higher in the order than A, contrary to the supposition. In the same way it follows that $\nu_2 \geq \nu_3$ must hold; for if $\nu_2 < \nu_3$, then

$$M\alpha_1^{\nu_1}\alpha_2^{\nu_3}\alpha_3^{\nu_2}\cdots\alpha_n^{\nu_n}$$

would stand higher than A, and so forth. We therefore conclude that the exponents in A,

$$\nu_1, \nu_2, \nu_3, \ldots, \nu_n,$$

form a decreasing, or at least never increasing, sequence of numbers; that is, the differences

$$\nu_1 - \nu_2, \quad \nu_2 - \nu_3, \ldots, \nu_{n-1} - \nu_n$$

are all positive, or at least not negative.

We conclude secondly that in no term of the function S can an exponent higher than ν_1 occur; for otherwise there would also occur a term in which α_1 had this higher exponent, and that would be of higher order than A, contrary to the supposition. Thus the terms that can occur at all in a symmetric function are, apart from the coefficients, completely determined by the highest term; and the number of possible terms is finite when the highest term is given.

We can denote the order of the symmetric function by the exponent sequence of its highest term,

$$(\nu_1, \nu_2, \ldots, \nu_n).$$

Apart from the case in which all exponents are zero, so that the function is a constant, the lowest possible order is

$$(1, 0, \ldots, 0),$$

to which corresponds the single symmetric function

$$M(\alpha_1 + \alpha_2 + \cdots + \alpha_n) = -Ma_1.$$

The next higher order is

$$(1, 1, 0, \ldots, 0),$$

to which corresponds the symmetric function

$$Ma_1 + M'a_2,$$

and so in the first n orders we obtain the fundamental functions themselves and their linear combinations, and no others.

The next following order is characterized by

$$(2, 0, 0, \ldots, 0),$$

and contains the linear combinations of the fundamental functions with the sum of the squares.

In all these cases the representability of the symmetric functions has already been proved; and we shall now carry out the general proof by deriving the representation of the function S, of order $(\nu_1, \nu_2, \ldots, \nu_n)$, by the fundamental functions under the supposition that it is already known for functions of lower order, thus by the procedure of complete induction.

We note for this purpose that, by multiplying two symmetric functions S and S', whose highest terms are

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n}, \qquad A' = M'\alpha_1^{\nu_1'}\alpha_2^{\nu_2'}\cdots\alpha_n^{\nu_n'},$$

there arises a symmetric function whose highest term is the product

$$AA' = MM'\alpha_1^{\nu_1+\nu_1'}\alpha_2^{\nu_2+\nu_2'}\cdots\alpha_n^{\nu_n+\nu_n'}.$$

For suppose that in SS' there were a higher term than AA', arising from the multiplication of the two terms

$$B = N\alpha_1^{\mu_1}\alpha_2^{\mu_2}\cdots\alpha_n^{\mu_n}, \qquad B' = N'\alpha_1^{\mu_1'}\alpha_2^{\mu_2'}\cdots\alpha_n^{\mu_n'},$$

so that

$$BB' = NN'\alpha_1^{\mu_1+\mu_1'}\alpha_2^{\mu_2+\mu_2'}\cdots\alpha_n^{\mu_n+\mu_n'},$$

and at the same time not $B = A$ and $B' = A'$. Then the first of the differences

$$\mu_1 - \nu_1 + \mu_1' - \nu_1', \quad \mu_2 - \nu_2 + \mu_2' - \nu_2', \ldots, \quad \mu_n - \nu_n + \mu_n' - \nu_n'$$

which is not zero would be positive; this is impossible, since the first non-vanishing one among the differences

$$\mu_1 - \nu_1, \quad \mu_2 - \nu_2, \ldots, \mu_n - \nu_n$$

and among the differences

$$\mu_1' - \nu_1', \quad \mu_2' - \nu_2', \ldots, \mu_n' - \nu_n'$$

is, by supposition, negative. By repeating this conclusion one obtains the more general theorem that the highest term of a product of several factors is obtained by multiplying the highest terms of the individual factors.

The highest terms of the symmetric fundamental functions

$$a_1, a_2, a_3, \ldots, a_n$$

are now, apart from the sign,

$$\alpha_1, \quad \alpha_1\alpha_2, \quad \alpha_1\alpha_2\alpha_3, \ldots, \alpha_1\alpha_2\alpha_3\cdots\alpha_n;$$

and if therefore we form the product

$$P = \pm M a_1^{\nu_1-\nu_2} a_2^{\nu_2-\nu_3}\cdots a_{n-1}^{\nu_{n-1}-\nu_n} a_n^{\nu_n},$$

we obtain a symmetric function whose highest term is likewise A, as in S.

The difference

$$S - P = S'$$

is therefore again a symmetric function whose highest term is lower than that of S, and which, by the supposition, we can represent rationally by the fundamental functions. Since P is represented in the same way, the goal is reached. Here again it is clear that through the representation by means of the fundamental functions no fractions are introduced if originally none are contained in S.

This method of computing symmetric functions at the same time gives information about the degree of the whole rational function of $a_1, a_2, \ldots, a_n$ by which a definite symmetric function of the $\alpha_1, \alpha_2, \ldots, \alpha_n$ is expressed. Namely, if $(\nu_1, \nu_2, \ldots, \nu_n)$ is the order of the symmetric function, measured by the exponent sequence of the first term, then ν_1 is the degree of P with respect to all $a_1, a_2, \ldots, a_n$, and in the developed expression for S no term of higher degree can occur. Nor can the term P be destroyed by any of the following terms, since P is completely determined by the order of S, and the order of S', and of all following symmetric functions, is lower than the order of S.

If therefore, in place of $a_1, a_2, \dots, a_n$, we put

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \dots, \frac{a_n}{a_0},$$

then $a_0^{\nu_1} S$ is a whole homogeneous function of degree ν_1 of $a_0, a_1, \dots, a_n$, and one cannot transform S into a whole function of the a's by multiplication with a lower power of a_0.

§ 46. Discriminants.

The theorems obtained above teach how to express symmetric functions of the roots of an equation rationally by the coefficients of the equation. The results were first derived under the supposition that the roots, or, what by the third section is equivalent to this, the coefficients were independent variables; but they cannot become false when definite numerical values are substituted for them, since at least for whole symmetric functions no denominators occur which could vanish.

We first make some applications to special symmetric functions which are important for what follows. Already in § 19 it was remarked that the product of differences

$$P = (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3) \cdots (\alpha_1 - \alpha_n)(\alpha_2 - \alpha_3) \cdots (\alpha_2 - \alpha_n) \cdots (\alpha_{n-1} - \alpha_n) \tag{1}$$

changes only its sign when two elements are interchanged. The square of this product will therefore remain unchanged under all interchanges of two of the variables α, and hence also under all permutations; that is, it is a symmetric function of the variables. The order of this symmetric function is, as one sees from (1),

$$(2n-2, 2n-4, \dots, 2, 0).$$

If therefore we take the function $f(x)$, whose roots are $\alpha_1, \alpha_2, \dots, \alpha_n$, in the form

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n,$$

so that the highest power of x has not necessarily the coefficient 1, then P^2 is a whole rational function of

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \dots, \frac{a_n}{a_0}, \tag{2}$$

which, by multiplication by a_0^{2n-2}, passes into a homogeneous function of $a_0, a_1, \dots, a_n$. We call this homogeneous function, which is therefore defined by

$$a_0^{2n-2} P^2 = D, \tag{3}$$

the discriminant of the function $f(x)$. If we put $a_0 = 1$, we obtain the non-homogeneous form of the discriminant.[13]

The vanishing of the discriminant is the necessary and sufficient condition that two of the values $\alpha_1, \alpha_2, \dots, \alpha_n$ become equal.

We can set up a second expression for the discriminant. By § 29, (4),

$$\begin{aligned}
f'(\alpha_1) &= a_0(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3) \cdots (\alpha_1 - \alpha_n), \\
f'(\alpha_2) &= a_0(\alpha_2 - \alpha_1)(\alpha_2 - \alpha_3) \cdots (\alpha_2 - \alpha_n), \\
&\vdots \\
f'(\alpha_n) &= a_0(\alpha_n - \alpha_1)(\alpha_n - \alpha_2) \cdots (\alpha_n - \alpha_{n-1}),
\end{aligned}$$

[13]The discriminant is, apart from the factor $(-1)^{\frac{n(n-1)}{2}} a_0^{2n-2}$, the same as what Gauss calls the determinant of the function (loc. cit., art. 6).

and consequently, since here every factor $(\alpha_h - \alpha_k)$ occurs twice with opposite signs,

$$D = (-1)^{\frac{n(n-1)}{2}} a_0^{n-2} f'(\alpha_1) f'(\alpha_2) \cdots f'(\alpha_n). \tag{5}$$

For actually representing the discriminant we first have a general means. By § 22,

$$(-1)^{\frac{n(n-1)}{2}} P = \begin{vmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \cdots & \alpha_n^{n-1} \end{vmatrix}, \tag{6}$$

and if, by § 27, we represent the square of this determinant by multiplication according to vertical rows as a new determinant, and put

$$s_m = \alpha_1^m + \alpha_2^m + \cdots + \alpha_n^m,$$

then

$$D = a_0^{2n-2} \begin{vmatrix} s_0 & s_1 & s_2 & \cdots & s_{n-1} \\ s_1 & s_2 & s_3 & \cdots & s_n \\ \vdots & \vdots & \vdots & & \vdots \\ s_{n-1} & s_n & s_{n+1} & \cdots & s_{2n-2} \end{vmatrix}. \tag{7}$$

The quantities s_m, however, are the power sums which we have already expressed by the coefficients in § 42; only in place of $a_1, a_2, \ldots, a_n$ the quotients (2) are to be put.

Thus we find, for example, for $n = 2$

$$D = a_0^2(s_0 s_2 - s_1^2) = a_1^2 - 4a_0 a_2, \tag{8}$$

and for $n = 3$

$$D = a_0^4(s_0 s_2 s_4 - s_0 s_3^2 + 2s_1 s_2 s_3 - s_1^2 s_4 - s_2^3). \tag{9}$$

§ 46. Discriminants

Here, according to § 42, one is to set

$$\begin{gathered} s_0 = 3, \qquad a_0 s_1 = -a_1, \\ a_0^2 s_2 = a_1^2 - 2a_0 a_2, \\ a_0^3 s_3 = -a_1^3 + 3a_0 a_1 a_2 - 3a_0^2 a_3, \\ a_0^4 s_4 = a_1^4 - 4a_0 a_1^2 a_2 + 4a_0^2 a_1 a_3 + 2a_0^2 a_2^2, \end{gathered}$$

from which one obtains

$$D = a_1^2 a_2^2 + 18a_0 a_1 a_2 a_3 - 4a_0 a_2^3 - 4a_1^3 a_3 - 27a_0^2 a_3^2. \tag{10}$$

§ 47. Discriminants of forms of the third and fourth orders

For functions of the third and fourth orders, the solutions of the equations of these two degrees (§ 35, 37) give us a simple means for computing the discriminants.

First, for the equation of the third order, the three roots of

$$x^3 + ax + b = 0 \tag{1}$$

were represented in the form

$$\begin{aligned} \alpha &= u + v, \\ \beta &= \varepsilon u + \varepsilon^2 v, \\ \gamma &= \varepsilon^2 u + \varepsilon v, \end{aligned} \tag{2}$$

where

$$\varepsilon = \frac{-1+\sqrt{-3}}{2}, \qquad \varepsilon^2 = \frac{-1-\sqrt{-3}}{2} \tag{3}$$

are the complex third roots of unity, and u, v are determined by § 35, (8). From (3) one obtains

$$\varepsilon - \varepsilon^2 = \sqrt{-3}, \qquad 1 - \varepsilon = \varepsilon^2\sqrt{-3}, \qquad 1 - \varepsilon^2 = -\varepsilon\sqrt{-3}, \tag{4}$$

and from (2)

$$\begin{aligned} \alpha - \beta &= \sqrt{-3}\,(\varepsilon^2 u - \varepsilon v), \\ \alpha - \gamma &= -\sqrt{-3}\,(\varepsilon u - \varepsilon^2 v), \\ \beta - \gamma &= \sqrt{-3}\,(u - v). \end{aligned}$$

Hence, by multiplication,

$$(\alpha-\beta)(\alpha-\gamma)(\beta-\gamma) = 3\sqrt{-3}\,(u^3 - v^3). \tag{5}$$

Squaring, one obtains for the discriminant of the cubic equation (1), according to § 35, (6),

$$D = -(27b^2 + 4a^3). \tag{6}$$

From this one obtains the discriminant of the general function

$$a_0x^3 + a_1x^2 + a_2x + a_3 \tag{7}$$

by putting for a and b the expressions of § 34, (4), replacing a_1, a_2, a_3 there by

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \quad \frac{a_3}{a_0},$$

and multiplying by a_0^4 [§ 46, (3)]. This gives

$$a = \frac{3a_0a_2 - a_1^2}{3a_0^2}, \qquad b = \frac{2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3}{27a_0^3},$$

and consequently

$$D = -\frac{1}{27a_0^2}\left\{(2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3)^2 + 4(3a_0a_2 - a_1^2)^3\right\}. \tag{8}$$

If one carries out the square and the cube, one returns to formula (10) of the preceding paragraph; it is noteworthy, however, that formula (8) apparently contains the denominator $27a_0^2$, which cancels when the expansion is carried out.

In exactly the same way we may proceed for biquadratic equations. If $\alpha, \beta, \gamma, \delta$ are the roots of the biquadratic equation

$$x^4 + ax^2 + bx + c = 0, \tag{9}$$

then, by § 37, (5), we obtain

$$\begin{aligned} \alpha - \beta &= v + w, & \gamma - \delta &= v - w, \\ \alpha - \gamma &= w + u, & \delta - \beta &= w - u, \\ \alpha - \delta &= u + v, & \beta - \gamma &= -u + v. \end{aligned} \tag{10}$$

Accordingly the discriminant of equation (9) is

$$D = (v^2 - w^2)^2(w^2 - u^2)^2(u^2 - v^2)^2,$$

that is, D is equal to the discriminant of the cubic resolvent § 37, (4),

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0, \tag{11}$$

whose roots are u^2, v^2, w^2.

If, however, this discriminant is formed by formula (10), § 46, with

$$a_0 = 1, \quad a_1 = 2a, \quad a_2 = a^2 - 4c, \quad a_3 = -b^2,$$

one obtains

$$D = -4a^3b^2 + 144acb^2 + 16a^4c - 128a^2c^2 + 256c^3 - 27b^4. \tag{12}$$

If, on the other hand, formula (8) is used to form the discriminant of the cubic equation (11), one obtains

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2. \tag{13}$$

From this one derives the discriminant of the general fourth-degree function

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 \tag{14}$$

by expressing a, b, c, according to § 34, (4), by

$$-\frac{3a_1^2}{8} + a_2, \qquad \frac{a_1^3}{8} - \frac{a_1a_2}{2} + a_3, \qquad -\frac{3a_1^4}{256} + \frac{a_1^2a_2}{16} - \frac{a_1a_3}{4} + a_4,$$

then replacing a_1, a_2, a_3, a_4 by

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \quad \frac{a_3}{a_0}, \quad \frac{a_4}{a_0},$$

and multiplying by a_0^6. After carrying out this simple calculation, set, for abbreviation,

$$A = a_2^2 - 3a_1a_3 + 12a_0a_4, \tag{15}$$
$$B = 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3. \tag{16}$$

Then (13) gives

$$27D = 4A^3 - B^2. \tag{17}$$

The quantities A and B, which will occupy us further in the next section, are called the first and second invariants of the biquadratic function f. Thus, by (17), one may express the discriminant rationally through these quantities, though not in such a way that the coefficients become integral. If the calculation prescribed in (17) is carried out in order to represent D itself through the coefficients a_0, a_1, a_2, a_3, a_4, then the factor 27 must still cancel. We shall not write down here the long expression, whose calculation offers no difficulty.

§ 48. Resultants

Another application of the theory of symmetric functions, which also contains the formation of discriminants as a special case, is the formation of resultants.

Let

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n, \tag{1}$$
$$\varphi(x) = b_0x^m + b_1x^{m-1} + b_2x^{m-2} + \cdots + b_m \tag{2}$$

be two integral rational functions of degrees n and m, and let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be the roots of the first, so that one may write

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n). \tag{3}$$

Now the product

$$\varphi(\alpha_1)\varphi(\alpha_2) \cdots \varphi(\alpha_n)$$

is a symmetric function of the roots of $f(x)$ and hence can be expressed integrally and rationally through

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \ldots, \frac{a_n}{a_0}.$$

The order of this symmetric function is $(m, m \ldots m)$ (§ 45), and therefore

$$R = a_0^m \varphi(\alpha_1)\varphi(\alpha_2) \cdots \varphi(\alpha_n) \tag{4}$$

is an integral rational and homogeneous function of order m in $a_0, a_1, \ldots, a_n$. It is also, as its expression shows, an integral homogeneous function of order n in $b_0, b_1, \ldots, b_m$. It is called the resultant of the two functions $f(x)$ and $\varphi(x)$. This designation is justified by a second representation.

Let the roots of $\varphi(x)$ be denoted by $\beta_1, \beta_2, \ldots, \beta_m$, so that

$$\varphi(x) = b_0(x - \beta_1)(x - \beta_2) \cdots (x - \beta_m). \tag{5}$$

Then, by (4),

$$R = a_0^m b_0^n \prod_{i,k} (\alpha_i - \beta_k), \tag{6}$$

where the product sign $\prod$ refers to all indices $i = 1, 2, \ldots, n$, $k = 1, 2, \ldots, m$.

This expression now shows that R, apart from the factor $(-1)^{mn}$, depends on $\varphi(x)$ in the same way as on $f(x)$; hence one can also set

$$(-1)^{mn} R = b_0^n f(\beta_1) f(\beta_2) \cdots f(\beta_m). \tag{7}$$

Accordingly, discriminants are a special case of resultants, obtained by taking $\varphi(x)$ to be the derivative $f'(x)$.

The vanishing of the resultant of $f(x)$ and $\varphi(x)$ is the necessary and sufficient condition that the two equations

$$f(x) = 0 \quad \text{and} \quad \varphi(x) = 0$$

have a common root.

The equation obtained by setting the resultant equal to zero is also called the final equation from $f(x) = 0$ and $\varphi(x) = 0$, and the formation of the final equation is the elimination of x.

For the computation of resultants one may use the algorithm of the greatest common divisor (§ 6). Namely, if $m \leq n$, then one can determine an integral function Q and likewise a function $\psi(x)$, of degree lower than m, such that

$$f(x) = Q\varphi(x) + \psi(x).$$

Thus, putting x equal to one of the roots β_i,

$$f(\beta_i) = \psi(\beta_i),$$

and hence

$$b_0^n \prod_i f(\beta_i) = b_0^n \prod_i \psi(\beta_i). \tag{8}$$

If m' is the degree of $\psi(x)$, then

$$b_0^{m'} \prod_i \psi(\beta_i) = R' \tag{9}$$

is the resultant of $\varphi(x)$ and $\psi(x)$, and therefore

$$R = b_0^{n-m'} R'. \tag{10}$$

The formation of the resultant R is thereby reduced to the formation of R'. One can now divide $\varphi(x)$ again by $\psi(x)$, and so reduce the formation of R' to the formation of a resultant of functions of ever lower degree, until one finally reaches a constant, that is, a function of the coefficients a, b, which must be identical with the resultant sought.

The computation of resultants is usually quite lengthy. We shall record only the simplest example, the resultant of two quadratic functions:

$$f(x) = a_0x^2 + a_1x + a_2, \qquad \varphi(x) = b_0x^2 + b_1x + b_2.$$

Here one may directly compute the product

$$R = a_0^2(b_0\alpha_1^2 + b_1\alpha_1 + b_2)(b_0\alpha_2^2 + b_1\alpha_2 + b_2),$$

and obtains, after setting

$$-a_0(\alpha_1 + \alpha_2) = a_1, \qquad a_0\alpha_1\alpha_2 = a_2,$$

$$R = a_0^2b_2^2 + a_2^2b_0^2 - b_0b_1a_1a_2 - a_0a_1b_1b_2 + b_0b_2a_1^2 + a_0a_2b_1^2 - 2a_0b_0a_2b_2, \tag{11}$$

or, equivalently,

$$R = (a_0b_2 - a_2b_0)^2 - (a_0b_1 - a_1b_0)(a_1b_2 - a_2b_1). \tag{12}$$

The individual terms of the resultant of the two functions $f(x)$ and $\varphi(x)$ of degrees n and m have, apart from a numerical integral factor, the form

$$a_0^{\nu_0}a_1^{\nu_1}\cdots a_n^{\nu_n}b_0^{\mu_0}b_1^{\mu_1}\cdots b_m^{\mu_m}, \tag{13}$$

where, as follows from the degrees determined above,

$$\nu_0 + \nu_1 + \cdots + \nu_n = m, \qquad \mu_0 + \mu_1 + \cdots + \mu_m = n. \tag{14}$$

But we can state still another relation among these exponents. Since, by (6), R is a homogeneous function of degree nm in the $n + m$ variables α, β, and since further a_0 is of degree zero, a_1 of degree one, a_2 of degree two, and so on in the α, and in general a_k and b_k are of degree k in these variables, it follows that

$$\nu_1 + 2\nu_2 + 3\nu_3 + \cdots + n\nu_n + \mu_1 + 2\mu_2 + 3\mu_3 + \cdots + m\mu_m = nm. \tag{15}$$

This is a relation from which an important conclusion follows, one that we shall discuss somewhat more fully in the following paragraph.

§ 49. Elimination. Bezout's theorem

Suppose that in the two functions

$$\begin{aligned} f(x) &= a_0x^n + a_1x^{n-1} + \cdots + a_n, \\ \varphi(x) &= b_0x^m + b_1x^{m-1} + \cdots + b_m \end{aligned} \tag{1}$$

the coefficients a, b are themselves again integral rational functions of a quantity y. Then the resultant will also become an integral rational function of y, which we shall denote by $F(y)$. The formation of this equation is the elimination of x from the two equations (1).

The equation

$$F(y) = 0 \tag{2}$$

has as roots all those values of y for which the two equations

$$f(x) = 0, \qquad \varphi(x) = 0 \tag{3}$$

have a common root. If condition (2) is satisfied, one finds the values of x satisfying the two equations (3) by seeking the greatest common divisor of $f(x)$ and $\varphi(x)$. This common divisor is of the first degree and hence determines x rationally when there is only one such common root; otherwise it is of correspondingly higher degree, and the determination of x still requires the solution of an equation of higher degree.

If only one common root x belongs to each root of the resultant, then the number of value-pairs x, y satisfying equations (3) is equal to the degree of the resultant with respect to y. Thus, if the coefficients a are of degree ν, and the coefficients b of degree μ, with respect to y, then by the degree determination of the preceding paragraph the degree of the resultant is

$$n\nu + m\mu; \tag{5}$$

this therefore is the number of common root-pairs. This number can, however, still be modified by the possibility that the coefficients are so constituted that the highest power of y disappears from the final equation. One would then be able to save the agreement of the number with (5) only by arbitrarily adding infinitely distant roots. Multiple roots also raise difficulties. One partly avoids these inconveniences by taking as the basis the homogeneous form of the equations, a form in which the elimination problem arises especially naturally in geometry.

Namely, if the function $f(x)$ has arisen from a homogeneous function of order n in the three variables x, y, z (a ternary form) by arranging it according to powers of x, then the coefficients $a_0, a_1, a_2, \dots, a_n$ are homogeneous forms in the two variables y, z, of the degree indicated by the index; thus a_0 is a constant, a_1 a linear form, a_2 a quadratic form, and so forth. The same applies to the coefficients $b_0, b_1, \dots, b_m$ of the function $\varphi(x)$.

If x, y, z denote triangular coordinates in the plane, then $f(x) = 0$, $\varphi(x) = 0$ are the equations of curves of orders n and m, and the common roots of these two equations, that is the ratios $x : y : z$, denote the points of intersection of the two curves. The final equation obtained by eliminating x is a homogeneous equation whose degree follows from (15), § 48 as mn.

A diminution of the degree cannot occur here; the final equation is always a homogeneous equation of degree nm in the two unknowns y, z. The only exception to which one must attend is the case in which the resultant vanishes identically (for all y, z), so that infinitely many common roots are present. Geometrically, this case means that the two curves have a common curve-part and not merely individual points in common.

If the equations $f = 0$, $\varphi = 0$ have only a finite number of common roots, then in the expressions

$$y_1 = y + \beta x, \qquad z_1 = z + \gamma x$$

we can choose the constants β, γ so that, if to one value of the ratio $y : z$ there correspond several different values of x satisfying equations (1), then the corresponding values of $y_1 : z_1$ come out different from one another.

We may then regard f and φ also as homogeneous functions of degrees m and n in the variables x, y_1, z_1, and the resultant then becomes a homogeneous function of degree mn, $R(y_1, z_1)$, in y_1 and z_1. If in it we put

$$y_1 = b' + b\lambda, \qquad z_1 = c' + c\lambda,$$

then we obtain an equation of degree mn in λ, and, provided that $R(y_1, z_1)$ does not vanish identically, we may choose the constants b, b', c, c' so that the coefficient of the highest power of λ, namely $R(b, c)$, is non-zero. Then to each root of the resultant there corresponds only one system of values of the ratio $x : y : z$, and the number of common roots of f and φ is at most equal to mn. To be able to state the theorem that the number of common roots is always equal to mn, one must still make a convention according to which certain value-pairs are to be counted more than once.

The theorem which, in geometrical garb, says that two curves of orders m and n meet in mn points, is called Bezout's theorem.

§ 50. Elimination from several equations

The principles of elimination that we have applied in the preceding discussion to two equations may be extended without difficulty to several equations with a corresponding number of unknowns.

First take the equations

$$f(x) = 0, \qquad \varphi(x) = 0, \qquad \psi(x) = 0, \tag{1}$$

of degrees m, n, p, whose coefficients a_i, b_i, c_i are for the present to be regarded as independent variables.

We apply the algorithm of the greatest common divisor to two of them, say to $f(x)$ and $\varphi(x)$, and obtain a last remainder which is a rational function of the a and b and, after clearing denominators, agrees with the resultant R of $f(x)$ and $\varphi(x)$, and a next-to-last remainder which is a linear function of x. The condition for the existence of a common root of $f = 0$ and $\varphi = 0$ is the vanishing of the resultant; under this condition the next-to-last remainder gives a rational expression for the common root in terms of the coefficients.

If one substitutes this expression in $\psi(x)$, then, after all denominators have been removed, one obtains an integral rational function of the coefficients a, b, c, which we shall denote by S; its vanishing, together with $R = 0$, contains the necessary and sufficient condition that the three equations (1) have a common root.

Now replace the coefficients a, b, c of the functions (1) by integral rational functions of a second unknown y, whose coefficients will be denoted by α, β, γ and for the present may be regarded as independent variables. Then R and S become integral rational functions of α, β, γ and y, and if, as equations in y, we consider

$$R = 0, \qquad S = 0, \tag{2}$$

then, according to the rules of the preceding paragraph, we may form their resultant T. If it does not vanish identically, this will be an integral rational function of the coefficients α, β, γ, and the equation

$$T = 0 \tag{3}$$

is the necessary and sufficient condition that the system of equations (1) can be satisfied by one or more value-pairs x, y. T is called the resultant of the system (1). Thus one may form from the three equations (1), by eliminating the unknowns x and y, a final equation for z.

If the α, β, γ are now replaced again by integral rational functions of a third unknown z, then T passes into an equation in z. The functions f, φ, ψ become functions of the three unknowns x, y, z. The degree of equation (3) gives the number of systems of values x, y, z for which the three equations (1) are simultaneously satisfied.

If, however, the coefficients are so constituted that T vanishes identically for all values of z, then there are infinitely many systems of values x, y, z simultaneously satisfying equations (1). If one regards x, y, z as Cartesian coordinates, then equations (1) each represent a surface. The degree of the final equation $T = 0$ with respect to z gives the number of intersection points of these three surfaces. If, however, T is identically zero, the three surfaces have either a common surface part or a common curve.

The degree of the final equation can be determined in the following way, which requires no lengthy considerations about the degrees of the functions R, S, T. To make ourselves independent of exceptional cases, we introduce homogeneous variables by adjoining a fourth variable u, and thus consider three homogeneous quaternary equations

$$F(x, y, z, u) = 0, \qquad \Phi(x, y, z, u) = 0, \qquad \Psi(x, y, z, u) = 0 \tag{4}$$

of orders m, n, p.

Assume the special case in which each of these three functions splits into linear factors. It is then clear that, if the number of common solutions is not infinite, their number must be equal to the product mnp; for we may set one linear factor of F, one of Φ, and one of Ψ simultaneously equal to zero, and thereby obtain mnp systems of three linear equations, which, if independent of one another, yield the same number of systems of values for the ratios $x : y : z : u$.

At first sight, it is questionable to conclude that in the general case the degree of the final equation of (4) is just as large. But this concern can be entirely removed by the following simple consideration, due to a remark of F. Klein.

Imagine the final equation from equations (4) first to have been formed under the assumption that the coefficients in equations (4) are independent variables. Then the resultant obtained after eliminating x, y is certainly not identically zero, since even for special systems of coefficients, for example when F, Φ, Ψ split into linear factors, it does not vanish identically. We therefore obtain a homogeneous final equation in the two unknowns z, u of some degree, which is to be determined.

If the coefficients are specialized in any way, it is possible that the final equation becomes identically satisfied; but if this case does not occur, then its order cannot change, since all its terms have the same order. Since under the specialization assumed above, expressed by the splitting of the functions F, Φ, Ψ, the identical vanishing does not occur but instead a final equation of order mnp appears, this must also be the degree of the final equation in the general case.

Thus Bezout's theorem is proved for the case of three homogeneous quaternary equations.

That we have restricted ourselves here to three equations was done solely in the interest of simplicity. The extension required for the case of n equations in $n + 1$ homogeneous unknowns follows of itself and may be left to the reader.

For our general problem we shall only add that these considerations show that the problem of algebra, insofar as it concerns the solution of algebraic equations of any kind, is thereby reduced to the treatment of a chain of equations each with one unknown.

§ 51. Decomposable and indecomposable functions

Among integral rational functions of several variables one must distinguish decomposable and indecomposable functions.

An integral rational function of any variables is called decomposable if it can be represented as a product of at least two integral rational functions of the same variables; if it cannot be represented as such a product, it is called indecomposable.

If a function is decomposable, then the degree of each factor is lower than the degree of the function itself. A linear function is therefore always indecomposable, and every integral rational function can be decomposed into a finite number of indecomposable functions.

We say that an integral function W is divisible by another w if a third integral function w' exists such that

$$W = ww'.$$

It follows that if U is divisible by w and V is any integral function, then the product UV is also divisible by w; and if $U, U', U'', \dots$ are divisible by w, while $V, V', V'', \dots$ are arbitrary integral rational functions, then $UV + U'V' + U''V'' + \cdots$ is also divisible by w.

If W is divisible by w, one also says that w divides W.

Two integral rational functions U, V that are not divisible by one and the same integral rational function are called relatively prime, or prime to one another.

We prove the following theorem:

I. If U, V, v are integral rational functions of any variables, if U and v are relatively prime, and if UV is divisible by v, then V is divisible by v.

This theorem corresponds exactly to a familiar fundamental theorem from the theory of integers: if a product of two integers is divisible by a third integer that is prime to one factor, then the other factor must be divisible by that integer.

We prove it by complete induction. If U, V, v depend on only one variable x, the theorem is correct; for by § 6 (Theorem I), when U and v are relatively prime, one can determine two other integral functions P and p of x such that

$$PU + pv = 1.$$

Multiplication by V gives

$$PUV + pVv = V,$$

and from this it is seen that if UV is divisible by v, then V must also be divisible by v.

We now assume that the theorem to be proved has been established for functions of n or fewer variables, and derive from this its validity for functions of $n+1$ variables.

For this it is necessary to draw from the theorem I, assumed to be true, some consequences whose final result will give the validity of the theorem for the next higher number of variables.

If v is an indecomposable function, then another function U of the same variables is either relatively prime to v or divisible by v. It follows by I that a product UV of two functions can be divisible by v only when one of the two factors is divisible by v. The same holds for a product of several functions, and thus from I follows the theorem:

II. A product of several integral rational functions is divisible by an indecomposable function v only when at least one of the factors of the product is divisible by v.

If an integral rational function U is decomposed into indecomposable factors in two ways,

$$U = vv'v'' \cdots = ww'w'' \cdots ,$$

then by II at least one of the factors $v, v', v'', \ldots$ must be divisible by w, say v. But since v too is indecomposable, v can differ from w only by a constant factor.

Thus, if c is this constant factor,

$$cv'v'' \cdots = w'w'' \cdots ,$$

from which it follows that one of the functions $v', v'', \ldots$, say v', is divisible by w' and hence differs from w' only by a constant factor.

We therefore obtain as a second consequence of Theorem I:

III. An integral rational function can, apart from constant factors, be decomposed into indecomposable factors in only one way.

From this arises the concept of the greatest common divisor of two or more integral rational functions $U, V, \ldots$. By this one understands the product of all indecomposable factors that occur in the decompositions of each of the functions $U, V, \ldots$, or the function of highest possible degree that divides all the functions $U, V, \ldots$. By III this function is completely determined, up to a constant factor, for every system of functions $U, V, \ldots$.

Several functions $U, V, W, \ldots$ are called relatively prime if no two of them have a common divisor.

All these definitions and theorems are exactly analogous to well-known theorems of elementary number theory, where they arise as consequences of the algorithm of the greatest common divisor; only here the indecomposable functions take the role of prime numbers.

We record one more such theorem:

IV. If u, v are relatively prime and if Uu, Uv are divisible by w, then U is divisible by w.

For if one decomposes the integral rational functions U, u, v, w into their indecomposable factors, then some factor of w, since it cannot occur in both u and v, must occur in U. Removing it from U and w, one argues in the same way for the next factor of w, and so on.

Now consider integral rational functions of one variable t,

$$f(t) = u_0 t^m + u_1 t^{m-1} + \cdots + u_{m-1} t + u_m,$$

whose coefficients are integral rational functions of n variables x, independent of t.

If the coefficients $u_0, u_1, \ldots, u_m$ have no common divisor, then $f(t)$ is called primitive; otherwise the greatest common divisor of the coefficients $u_0, u_1, \ldots, u_m$ is called the divisor of the function $f(t)$ (compare § 2).

Still assuming the validity of I, we conclude:

V. The product of two primitive functions $f(t)$ and $\varphi(t)$ is again a primitive function; and the divisor of a product of two imprimitive functions is equal to the product of the divisors of the two factors.

The proof is exactly as for the corresponding theorem in § 2.

Let

$$\begin{aligned} f(t) &= u_0 t^m + u_1 t^{m-1} + \cdots + u_m, \\ \varphi(t) &= v_0 t^\mu + v_1 t^{\mu-1} + \cdots + v_\mu \end{aligned} \tag{1}$$

be two primitive functions, and let

$$F(t) = U_0 t^{m+\mu} + U_1 t^{m+\mu-1} + \cdots + U_{m+\mu} \tag{2}$$

be their product. Then, if r and s denote any two indices from the rows $0, 1, \ldots, m$ and $0, 1, 2, \ldots, \mu$ (§ 2),

$$U_{r+s} = u_r v_s + u_{r-1} v_{s+1} + \cdots + u_{r+1} v_{s-1} + \cdots . \tag{3}$$

Now let w be any indecomposable function that does not divide all the u_r and also does not divide all the v_s. We choose r and s in (3) so that u_r is the first u not divisible by w, so that $u_{r-1}, u_{r-2}, \ldots$ are divisible by w, and similarly v_s is the first v not divisible by w. Then U_{r+s} cannot be divisible by w, since all terms except the first are divisible by w; that is, $F(t)$ is primitive.

It follows immediately, if p and q are arbitrary integral rational functions of the x, that pq is the divisor of the product of the two imprimitive functions $pf(t)$ and $q\varphi(t)$, which proves the second part of theorem V. It follows further:

VI. If an integral rational function $F(t)$ of the $n+1$ variables x and t is decomposable into two factors that are integral with respect to t and at least rational with respect to the x, then it is also decomposable into two functions that are integral and rational in x and t.

For by hypothesis there is an integral rational function w of the x alone, and two integral rational functions of x and t, $f_1(t), \varphi_1(t)$, such that

$$wF(t) = f_1(t)\varphi_1(t).$$

By (V), w must divide the product of the divisors of $f_1(t)$ and $\varphi_1(t)$, and hence we also obtain

$$F(t) = f(t)\varphi(t),$$

where $f(t), \varphi(t)$ are again integral rational functions of x and t, and with respect to t have the same degrees as $f_1(t)$ and $\varphi_1(t)$, as was to be proved.

From this we conclude further that two functions $F(t), f(t)$ which, considered as integral rational functions of the $n+1$ variables x and t, are relatively prime in the sense defined above, must also prove to be relatively prime when considered as functions of t alone and treated by the algorithm of the greatest common divisor (§ 6). For if they had a common divisor that was integral with respect to t but fractional with respect to the x, then one could determine an integral function T of x and t and an integral function P of the x alone such that

$$PF(t) = TF_1(t), \qquad Pf(t) = Tf_1(t),$$

where F_1, f_1 are integral functions without a common divisor. By IV, P would have to occur in the divisor of T, and $F(t)$ and $f(t)$ could not be relatively prime. Therefore by § 6 one can determine two integral rational functions Q, q of x and t, and an integral rational function X of the x alone, such that the identity

$$QF(t) + qf(t) = X \tag{4}$$

holds.

Now let $\Phi(t)$ be another integral rational function of x and t. From (4), by multiplication with $\Phi(t)$, one obtains

$$Q\Phi(t)F(t) + q\Phi(t)f(t) = X\Phi(t),$$

and if $\Phi(t)F(t)$ is divisible by $f(t)$, then $X\Phi(t)$ is consequently also divisible by $f(t)$.

Thus, if $\varphi(t)$ and $\psi(t)$ again denote two integral functions of x and t, we may put

$$\begin{aligned} X\Phi(t) &= \varphi(t)f(t), \\ F(t)\Phi(t) &= \psi(t)f(t). \end{aligned} \tag{5}$$

Multiplying the second of these equations by X and substituting from the first the expression $\varphi(t)f(t)$ for $X\Phi(t)$, one may cancel $f(t)$ and obtains

$$\varphi(t)F(t) = X\psi(t).$$

By IV, X must therefore occur both in the divisor of $\varphi(t)f(t)$ and in the divisor of $\varphi(t)F(t)$. Since the functions $f(t)$ and $F(t)$, and hence also their divisors, are relatively prime, X must occur in the divisor of $\varphi(t)$. Thus, putting

$$\varphi(t) = X\varphi_1(t),$$

one obtains from (5)

$$\Phi(t) = \varphi_1(t)f(t),$$

that is, $\Phi(t)$ is divisible by $f(t)$. Hence:

VII. If $F(t)$ and $f(t)$ are relatively prime and $\Phi(t)F(t)$ is divisible by $f(t)$, then $\Phi(t)$ is divisible by $f(t)$.

But this is nothing other than Theorem I for functions of $n+1$ variables, and so I is proved in general.

§ 52. Tschirnhausen transformation

We now make another important application of the theory of symmetric functions and the formation of resultants, namely to the transformation of algebraic equations due to Tschirnhausen.[14]

Let

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n \tag{1}$$

be any function of the nth degree, and let

$$y = \frac{\chi(x)}{\psi(x)} \tag{2}$$

[14] Acta eruditorum, Leipzig 1683.

be any rational function, integral or fractional, of x, subject only to the assumption that $f(x)$ and $\psi(x)$ are to have no common divisor. By the concluding theorem of § 6 one can then determine functions $F(x)$ and $\varphi(x)$, the latter of degree at most $n-1$, such that

$$\chi(x) = F(x)f(x) + \varphi(x)\psi(x), \tag{3}$$

and consequently, as soon as x is put equal to a root of the equation

$$f(x) = 0, \tag{4}$$

one has

$$\frac{\chi(x)}{\psi(x)} = \varphi(x).$$

Since we shall consider the values of y only for those values of x that are roots of (4), we lose no generality if from the outset we put

$$y = \varphi(x),$$

therefore equal to an integral function of degree $n-1$. Thus let

$$y = \varphi(x) = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_{n-1}x^{n-1}, \tag{5}$$

where the coefficients α remain for the moment wholly undetermined.

Let the roots of equation (4) be denoted by

$$x_1, x_2, \ldots, x_n. \tag{6}$$

Then from (5) there arise the corresponding values of y:

$$y_1 = \varphi(x_1), \qquad y_2 = \varphi(x_2), \ldots, y_n = \varphi(x_n), \tag{7}$$

and the product

$$\Phi(y) = (y - y_1)(y - y_2)\cdots(y - y_n) \tag{8}$$

is a symmetric function of the quantities (6), and can therefore be expressed rationally through the coefficients of $f(x)$.

Represented in this way, let $\Phi(y)$ have the expression

$$\Phi(y) = y^n + b_1 y^{n-1} + b_2 y^{n-2} + \cdots + b_{n-1}y + b_n. \tag{9}$$

The coefficients b_ν are determined by the y_i, according to § 7, by means of the formulas

$$b_1 = -\sum y_1, \qquad b_2 = \sum y_1 y_2, \qquad b_3 = -\sum y_1 y_2 y_3, \ldots,$$

from which it follows that b_ν is an integral rational and homogeneous function of degree ν in the variables $\alpha_0, \alpha_1, \ldots, \alpha_{n-1}$.

$\Phi(y)$ is nothing other than the resultant of the two equations

$$f(x) = 0, \qquad \varphi(x) - y = 0, \tag{10}$$

and conversely, by § 49, if y becomes equal to one of the values $y_1, y_2, \ldots, y_n$, one can also express the common root of the two equations (10) rationally through the coefficients of (10), that is, rationally through a_i, α_i, y, provided that only one such common root is present.

This will happen when the n values $y_1, y_2, \ldots, y_n$ are distinct. Then one obtains an expression

$$x = \beta_0 + \beta_1 y + \beta_2 y^2 + \cdots + \beta_{n-1}y^{n-1}$$

that is valid when one of the values (6) is put for x and the corresponding value (7) for y. The quantities β_i are rationally composed from the a_i and α_i.

If, however, several of the values $y_1, y_2, \ldots, y_n$ coincide, then determining the corresponding values of x still requires solving an equation of the corresponding degree. But this degree is always less than n, since the function $\varphi(x)$ can take the same value y for at most $n-1$ values of x. Thus, when the equation

$$\Phi(y) = 0 \tag{11}$$

has been completely solved, the equation (1) has thereby also been completely solved if (11) has n distinct roots, or at least reduced to the solution of an equation of lower degree if (11) has equal roots.

Accordingly we regard (11) as a transformation of equation (1), and the formation of (11) from (1) is called the Tschirnhausen transformation of equation (1).

The purpose of such a transformation is to determine the coefficients α_i so that $\Phi(y)$ obtains a simpler form than $f(x)$.

The coefficients b_ν can be formed when the power sums of the y_i are known (§ 42).

Let the sums of the mth powers of the x_i be denoted by s_m, and those of the y_i by σ_m, so that

$$s_m = Sx_i^m, \qquad \sigma_m = Sy_i^m.$$

Then σ_m can be formed from the s_m in the following way. First expand y^m by the polynomial theorem (§ 12), putting

$$y^m = \sum \frac{\Pi(m)}{\Pi(\nu_0)\Pi(\nu_1)\cdots\Pi(\nu_{n-1})} \alpha_0^{\nu_0}\alpha_1^{\nu_1}\cdots\alpha_{n-1}^{\nu_{n-1}} x^{\nu_1+2\nu_2+\cdots+(n-1)\nu_{n-1}}, \tag{12}$$

where the sum extends over all non-negative integral values of $\nu_0, \nu_1, \ldots, \nu_{n-1}$ satisfying the condition

$$\nu_0 + \nu_1 + \cdots + \nu_{n-1} = m. \tag{13}$$

If in (12) one then puts $x_1, x_2, \ldots, x_n$ for x and forms the sum of the values thereby obtained for $y_1^m, y_2^m, \ldots, y_n^m$, it follows that

$$\sigma_m = \sum \frac{\Pi(m)}{\Pi(\nu_0)\Pi(\nu_1)\cdots\Pi(\nu_{n-1})} s_{\nu_1+2\nu_2+\cdots+(n-1)\nu_{n-1}} \alpha_0^{\nu_0}\alpha_1^{\nu_1}\cdots\alpha_{n-1}^{\nu_{n-1}}. \tag{14}$$

Thus σ_m is represented as an integral homogeneous function of order m in $\alpha_0, \alpha_1, \ldots, \alpha_{n-1}$. A somewhat simplified notation is obtained from the second representation of forms (§ 15). Let $\mu_1, \mu_2, \ldots, \mu_m$ run independently through the sequence of numbers $0, 1, \ldots, n-1$. If in one combination the index 0 occurs ν_0 times, the index 1 occurs ν_1 times, the index 2 occurs ν_2 times, and so on, the index $n-1$ occurs ν_{n-1} times, then

$$\mu_1 + \mu_2 + \cdots + \mu_m = \nu_1 + 2\nu_2 + \cdots + (n-1)\nu_{n-1},$$

and accordingly the expression for σ_m can be represented as

$$\sigma_m = \sum_\mu s_{\mu_1+\mu_2+\cdots+\mu_m} \alpha_{\mu_1}\alpha_{\mu_2}\cdots\alpha_{\mu_m}. \tag{15}$$

For example,

$$\begin{aligned} \sigma_1 &= \alpha_0 s_0 + \alpha_1 s_1 + \cdots + \alpha_{n-1}s_{n-1}, \\ \sigma_2 &= \sum_{i,k} s_{i+k}\alpha_i\alpha_k, \\ \sigma_3 &= \sum_{h,i,k} s_{h+i+k}\alpha_h\alpha_i\alpha_k, \end{aligned} \tag{16}$$

where h, i, k run through the sequence of numbers $0, 1, 2, \ldots, n-1$.

§ 53. Application to the cubic and biquadratic equations

The aim that Tschirnhausen already had in view in this transformation was to determine the substitution coefficients $\alpha_0, \alpha_1, \ldots, \alpha_{n-1}$ so that in the transformed equation $\Phi(y) = 0$ so many terms vanish that the equation becomes solvable. This is easily achieved for equations of the third and fourth degrees, although it does not seem simple, and not possible without extensive calculations, to derive by this method the formulas otherwise known.

First take the cubic equation

$$x^3 + ax^2 + bx + c = 0, \tag{1}$$

which we transform by the substitution

$$y = \alpha + \beta x + \gamma x^2. \tag{2}$$

In order to reduce the equation for y,

$$y^3 + Ay^2 + By + C = 0, \tag{3}$$

to a pure cubic equation, we have to determine the ratios $\alpha : \beta : \gamma$ from the two equations

$$A = 0, \qquad B = 0, \tag{4}$$

of which the first is linear and the second quadratic.

Equations (4) are equivalent to

$$\sigma_1 = 0, \qquad \sigma_2 = 0, \tag{5}$$

and hence, by (16) of the preceding paragraph, give

$$\alpha s_0 + \beta s_1 + \gamma s_2 = 0,$$
$$\alpha^2 s_0 + 2\alpha\beta s_1 + 2\alpha\gamma s_2 + \beta^2 s_2 + 2\beta\gamma s_3 + \gamma^2 s_4 = 0,$$

where $s_0 = 3$. If the latter equation is multiplied by s_0 and the square of the former is subtracted, one obtains

$$\beta^2(s_0 s_2 - s_1^2) + 2\beta\gamma(s_0 s_3 - s_1 s_2) + \gamma^2(s_0 s_4 - s_2^2) = 0. \tag{6}$$

The discriminant of this quadratic equation,

$$(s_0 s_3 - s_1 s_2)^2 - (s_0 s_2 - s_1^2)(s_0 s_4 - s_2^2),$$

when expanded gives

$$-s_0(s_0 s_2 s_4 - s_0 s_3^2 - s_1^2 s_4 - s_2^3 + 2 s_1 s_2 s_3),$$

and hence, if D denotes the discriminant of the cubic equation (1) [§ 46, (9)], is equal to $-3D$. The solution of (6) therefore gives

$$\frac{\beta}{\gamma} = \frac{-(s_0 s_3 - s_1 s_2) + \sqrt{-3D}}{s_0 s_2 - s_1^2}. \tag{7}$$

The equation (3) has thereby been reduced to a pure equation, whose solution requires only the extraction of a cube root. The choice of sign for the square root occurring in (7) is left to us.

To use the Tschirnhausen transformation for the biquadratic equation, one may proceed, for instance, by determining $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ in the transformed equation

$$y^4 + b_1 y^3 + b_2 y^2 + b_3 y + b_4 = 0 \tag{8}$$

so that

$$b_1 = 0, \qquad b_3 = 0. \tag{9}$$

Then (8) passes into a quadratic equation for y^2, from whose roots the square roots must still be drawn; thus after the equations (9) have been solved, a square root has still to be drawn twice.

If by means of the first equation (9) one eliminates α_0 from the second, then a homogeneous cubic equation in $\alpha_1, \alpha_2, \alpha_3$ arises. One may therefore choose one of the two ratios $\alpha_1 : \alpha_2 : \alpha_3$ arbitrarily, for example put $\alpha_3 = 0$, and then obtains a cubic equation to determine the other. Thus the solution of the biquadratic equation has been reduced to that of a cubic and to square roots.

§ 54. The Tschirnhausen transformation of the equation of the fifth degree

If one wants to apply the Tschirnhausen transformation to the equation of the fifth degree

$$x^5 + a_1x^4 + a_2x^3 + a_3x^2 + a_4x + a_5 = 0, \tag{1}$$

one must put

$$y = \alpha_0 + \alpha_1x + \alpha_2x^2 + \alpha_3x^3 + \alpha_4x^4, \tag{2}$$

and obtains the transform

$$y^5 + b_1y^4 + b_2y^3 + b_3y^2 + b_4y + b_5 = 0. \tag{3}$$

The most immediate thought would be to determine the ratios of the five unknowns α so that

$$b_1 = 0, \qquad b_2 = 0, \qquad b_3 = 0, \qquad b_4 = 0 \tag{4}$$

would hold, whereby (3) would be brought back to a pure equation; this was probably also the aim Tschirnhausen had in view. But among the four equations (4), the first is linear, the second quadratic, the third of the third degree, and the fourth of the fourth degree. The final equation derivable from them will therefore be, by § 50, of degree $2 \cdot 3 \cdot 4 = 24$, and hence no benefit for the solution of the equation of the fifth degree can be drawn from it. One therefore seeks first to satisfy only the two equations

$$b_1 = 0, \qquad b_2 = 0, \tag{5}$$

whereby equation (3) passes into a form that, following F. Klein's lectures on the icosahedron, is called a principal equation of the fifth degree.

To satisfy the two equations (5), we have at our disposal the four ratios $\alpha_1 : \alpha_2 : \alpha_3 : \alpha_4$, and we can still use these quantities in many ways to simplify the equation of the fifth degree.

We shall return to this subject more fully in a later section, and confine ourselves here to indicating the aims in general. For clarity we use a geometrical formulation, which, however, is not essential for understanding the algebraic theory as it will be given later.

If, with the help of the linear equation $b_1 = 0$, we express one of the five quantities α, say α_0, in terms of the others, then we may regard the four remaining quantities $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ as homogeneous coordinates of a point in space. The equation $b_2 = 0$ then represents a surface of the second order, $b_3 = 0$ a surface of the third order, $b_4 = 0$ a surface of the fourth order, and $b_5 = 0$ a surface of the fifth order.

If one wanted to set b_2, b_3, b_4 simultaneously equal to zero and thereby reduce the given equation to a pure one, then one would have to determine one of the 24 intersection points of three surfaces of the second, third, and fourth orders; this depends on an equation of the 24th degree.

Instead of proceeding in this way, one seeks to determine a straight line on the surface $b_2 = 0$. On every surface of the second order there is one or two systems of such straight lines, real or imaginary, and one can determine one of these lines by square roots, as will be explained later. The determination of an intersection point of one of these straight lines with the surface $b_3 = 0$ then leads to an equation of the third degree. If, moreover, by determining a common factor of the α one makes the coefficient $b_4 = 1$, the equation for y obtains the form

$$y^5 + y + b = 0, \tag{6}$$

so that y is determined as a function of one variable b.

This equation form is usually named after the English mathematician Jerrard, who made it known in 1834. It had, however, already been published much earlier, in 1786, by E. J. Bring in Lund.[15] We shall therefore call it the Bring-Jerrard form.

[15] Compare F. Klein, Vorlesungen über das Ikosaeder, p. 143.

Just as well, one could make $b_4 = 0$ by an equation of the fourth degree and would obtain a second normal form of the equation of the fifth degree:

$$y^5 + y^2 + b = 0, \tag{7}$$

which has the same justification as the Bring-Jerrard form.

For the solution of the equation of the fifth degree, of course, nothing is gained by these considerations. Their use consists in reducing the equation of the fifth degree to a normal form depending on only one variable coefficient. This can be achieved in infinitely many different ways, and is achieved most simply, namely without solving an equation of the third or fourth degree, if one regards the equation of the fifth degree $b_5 = 0$ itself as a transformation of the given equation of the fifth degree. For once the equation $b_5 = 0$ has been solved, one of the five values of y is equal to 0, and the given equation of the fifth degree is at the same time solved.

§ 55. Introduction of the linear transformation

When, in any functions depending on the m variables $x_1, x_2, \dots, x_m$, we substitute for these variables linear expressions depending on m new variables $x'_1, x'_2, \dots, x'_m$,

$$\begin{aligned} x_1 &= \alpha_1^{(1)} x'_1 + \alpha_1^{(2)} x'_2 + \cdots + \alpha_1^{(m)} x'_m, \\ x_2 &= \alpha_2^{(1)} x'_1 + \alpha_2^{(2)} x'_2 + \cdots + \alpha_2^{(m)} x'_m, \\ &\cdots \\ x_m &= \alpha_m^{(1)} x'_1 + \alpha_m^{(2)} x'_2 + \cdots + \alpha_m^{(m)} x'_m, \end{aligned} \tag{1}$$

we call this a linear substitution, and the result a linear transformation.

The quantity

$$r = \begin{vmatrix} \alpha_1^{(1)} & \alpha_1^{(2)} & \cdots & \alpha_1^{(m)} \\ \alpha_2^{(1)} & \alpha_2^{(2)} & \cdots & \alpha_2^{(m)} \\ \vdots & \vdots & & \vdots \\ \alpha_m^{(1)} & \alpha_m^{(2)} & \cdots & \alpha_m^{(m)} \end{vmatrix} \tag{2}$$

is called the substitution determinant. It must always be different from zero, since otherwise (1) would express a dependence among the variables $x_1, x_2, \dots, x_m$.

Under a linear substitution, every homogeneous function of the variables x passes into a homogeneous function of the same degree in the variables x'.

Thus, for example, every linear function

$$y = \sum_i a_i x_i$$

passes into a similar function

$$y = \sum_i a'_i x'_i,$$

where

$$a'_k = \sum_i a_i \alpha_i^{(k)}. \tag{3}$$

If we consider m such linear functions

$$y_\nu = \sum_i a_{\nu,i} x_i \qquad \nu = 1, 2, \dots, m,$$

we obtain m transformed functions

$$y_\nu = \sum_i a'_{\nu,i} x'_i \qquad \nu = 1, 2, \dots, m.$$

The determinants of the systems y_ν, y'_ν,

$$\Delta = \sum \pm a_{1,1} a_{2,2} \cdots a_{m,m}, \qquad \Delta' = \sum \pm a'_{1,1} a'_{2,2} \cdots a'_{m,m},$$

stand, by the multiplication theorem for determinants, in the relation

$$\Delta' = r\Delta, \tag{4}$$

so that one cannot vanish without the other.

§ 56. Quadratic forms

Of particular importance is the transformation of homogeneous functions of the second degree, or quadratic forms. We denote them, as already in the first section, by

$$\varphi(x_1, x_2, \ldots, x_m) = \sum_{i,k} a_{i,k} x_i x_k, \qquad a_{i,k} = a_{k,i}. \tag{1}$$

By the substitution § 55, (1), φ passes into

$$\Phi(x'_1, x'_2, \ldots, x'_m) = \sum_{i,k} a'_{i,k} x'_i x'_k. \tag{2}$$

Put $x + \xi$ for x, and similarly $x' + \xi'$ for x', so that the ξ' stand in the same dependence on the ξ as the x' do on the x. If, for abbreviation, we set

$$\frac{1}{2}\varphi'(x_i) = u_i, \qquad \frac{1}{2}\Phi'(x'_i) = u'_i,$$

then, by comparison of the terms of the first dimension (§ 16), we obtain

$$\sum \xi_i u_i = \sum \xi'_i u'_i, \tag{3}$$

and hence, as in § 55, (3),

$$u'_k = \sum_i \alpha_i^{(k)} u_i. \tag{4}$$

Now

$$u_k = \sum_i a_{k,i} x_i, \qquad u'_k = \sum_i a'_{k,i} x'_i, \tag{5}$$

and consequently (4) gives us the transformation of m linear functions. The coefficients of the given system are $a'_{k,h}$, and those of the transformed system, obtained by substituting the first system (5) into (4), are

$$b_{k,h} = \sum_i \alpha_i^{(k)} a_{i,h}. \tag{6}$$

The determinant of the first system

$$H' = \sum \pm a'_{1,1} a'_{2,2} \cdots a'_{m,m}$$

is, by § 55, (4),

$$= r \sum \pm b_{1,1} b_{2,2} \cdots b_{m,m},$$

and, by (6), the determinant of the $b_{k,h}$ is

$$= r \sum \pm a_{1,1} a_{2,2} \cdots a_{m,m}.$$

If we therefore put

$$H = \sum \pm a_{1,1} a_{2,2} \cdots a_{m,m}, \tag{7}$$

we have the important theorem

$$H' = r^2 H. \tag{8}$$

The determinant H is called the determinant of the form $\varphi(x)$.

It therefore changes, under the linear transformation of the function φ, only by the factor r^2, and it cannot vanish for the transformed form if it does not vanish for the original one.

The vanishing of the determinant H means that the function $\varphi(x)$ may be regarded as a function of fewer than m variables that are linear functions of the x.

For, first, if by a linear transformation $\varphi(x)$ passes into the function $\Phi(x')$, which is independent of one of the variables, say x_1', then H' vanishes, since all the elements in one of the rows and columns vanish; consequently, by (8), $H = 0$ as well.

Second, however, H is the determinant of the system of linear equations

$$\varphi'(x_i) = 0 \qquad i = 1, 2, \ldots, m, \tag{9}$$

and, if this determinant vanishes, then (§ 24, III) there is a system of quantities

$$x_1^{(0)}, x_2^{(0)}, \ldots, x_m^{(0)}, \tag{10}$$

whose elements do not all vanish and which, when substituted for the x, satisfy equations (9).

Now if x_i, x_i' are any two systems of quantities, and λ is likewise an arbitrary quantity, the following identities hold:

$$\begin{aligned} \varphi(x + \lambda x') &= \varphi(x) + \lambda \sum x_i \varphi'(x_i') + \lambda^2 \varphi(x'), \\ \varphi(x') &= \frac{1}{2} \sum x_i' \varphi'(x_i'), \end{aligned} \tag{11}$$

[§ 16, (12) to (15)]. Thus, when for x_i' one substitutes the values $x_i^{(0)}$, one obtains

$$\varphi(x_i + \lambda x_i^{(0)}) = \varphi(x), \tag{12}$$

quite independently of λ. If we now assume that $x_1^{(0)}$ is different from zero and set

$$\lambda = -\frac{x_1}{x_1^{(0)}},$$

and further set

$$\begin{aligned} y_2 &= x_2 x_1^{(0)} - x_1 x_2^{(0)}, \\ y_3 &= x_3 x_1^{(0)} - x_1 x_3^{(0)}, \\ &\cdots \\ y_m &= x_m x_1^{(0)} - x_1 x_m^{(0)}, \end{aligned} \tag{13}$$

then it follows that

$$x_1^{(0)2} \varphi(x) = \varphi(0, y_2, y_3, \ldots, y_m), \tag{14}$$

so that $\varphi(x)$ is expressed through the $m - 1$ variables $y_2, y_3, \ldots, y_m$.

§ 57. Transformation of quadratic forms into a sum of squares

We at once make an important application of the theorem just proved to the discriminant of the function (11), regarded as a function of the second degree in λ, without now assuming that the determinant of the function φ vanishes. This discriminant is

$$\psi(x) = \left[\sum x_i \varphi'(x_i')\right]^2 - 4\varphi(x)\varphi(x'). \tag{1}$$

If, in this expression, we put $x_i + \lambda x'_i$ in place of x_i, then, after substituting for $\varphi(x + \lambda x')$ the expression § 56, (11) and setting $\sum x'_i \varphi'(x'_i) = 2\varphi(x')$, we get

$$\psi(x + \lambda x') = \psi(x),$$

which shows that $\psi(x)$ depends on only $m - 1$ linear combinations of the m variables. If for $x'_1, x'_2, \ldots, x'_m$ we put any special values for which $\varphi(x')$ does not vanish, and assume that x'_1 is not zero, then

$$x_1'^2 \psi(x_1, x_2, \ldots, x_m) = \psi(0, x_2 x'_1 - x_1 x'_2, \ldots, x_m x'_1 - x_1 x'_m),$$

hence it is a function of the $m - 1$ variables

$$y_2 = x_2 x'_1 - x_1 x'_2, \ldots, y_m = x_m x'_1 - x_1 x'_m.$$

This may also be confirmed by the fact that

$$\psi'(x_1), \psi'(x_2), \ldots, \psi'(x_m)$$

all vanish simultaneously for $x_1, x_2, \ldots, x_m = x'_1, x'_2, \ldots, x'_m$, so that the determinant of the function ψ must vanish.

Formula (1) now contains a very important result, most clearly seen when we give this formula the following form. We set

$$\psi(x) = -4\varphi(x')\chi(y_2, \ldots, y_m), \qquad \sum x_i \varphi'(x'_i) = 2\sqrt{\varphi(x')}\, Y_1, \tag{2}$$

and obtain

$$\varphi(x) = Y_1^2 + \chi(y_2, y_3, \ldots, y_m), \tag{3}$$

so that $\varphi(x)$ is represented as the sum of the square of a linear function Y_1 and a function χ of $m - 1$ variables.

From this follows the important theorem: A quadratic form in m variables can always be represented as a sum of squares of m or fewer linear functions.

This theorem is obviously true for a function of one variable, since such a function is itself a square. From (3), however, the truth of the theorem for a function of m variables follows if its truth is assumed for a function of $m - 1$ variables.

As (2) shows, the representation is possible in infinitely many different ways, because the entirely arbitrary quantities x' still occur in this formula.

A representation by fewer than m squares is possible only when the determinant of the function φ vanishes.

Since in the expression (2) for Y_1 a square root occurs, Y_1 may be real or imaginary even when the coefficients of φ and the x and x' are assumed real. But if, in the case where Y_1 is imaginary, we put iY_1 in place of Y_1, then we can also state our theorem as follows:

A real quadratic form in m variables can be represented as a sum of at most m positive or negative squares of real linear functions of the x.[16]

§ 58. Law of inertia of quadratic forms

For the decomposition of a quadratic form into squares, the following theorem now holds. Sylvester gave it the name of the law of inertia of quadratic forms (Philos. Mag. 1852).

[16] For the general theory of the linear transformation of forms of the second degree, besides the investigations of Jacobi and Hesse (Hesse, "Vorlesungen über analytische Geometrie des Raumes", third edition, edited by Gundelfinger, Leipzig 1876), especially the works of Weierstrass and Kronecker (Monthly Reports of the Berlin Academy, 1868, 1874) are important. A detailed historical account of the development of the question is given in the "Bericht über den gegenwärtigen Stand der Invariantentheorie" by Franz Meyer (first annual report of the Deutsche Mathematiker-Vereinigung, Berlin 1892).

In whatever way one decomposes a real quadratic form $\varphi(x)$ into a sum of positive and negative squares of linear functions, the number of positive and negative squares, and therefore also their total number, is always the same, provided that there is no linear dependence among these linear functions.

The proof of this important theorem is very simple.

Let

$$\begin{aligned}\varphi(x) &= Y_1^2 + Y_2^2 + \cdots + Y_\nu^2 - Y_1'^2 - Y_2'^2 - \cdots - Y_{\nu'}'^2 \\ &= Z_1^2 + Z_2^2 + \cdots + Z_\mu^2 - Z_1'^2 - Z_2'^2 - \cdots - Z_{\mu'}'^2\end{aligned} \tag{1}$$

be two decompositions of $\varphi(x)$ into squares. Thus $Y_1, Y_2, \ldots, Y_\nu, Y_1', Y_2', \ldots, Y_{\nu'}'$ are homogeneous linear functions of x among which no linear relation with constant coefficients exists, so that $\nu + \nu' \leq m$. The same assumptions are made about the functions $Z_1, Z_2, \ldots, Z_\mu, Z_1', Z_2', \ldots, Z_{\mu'}'$, so that $\mu + \mu' \leq m$ as well.

Suppose that

$$\nu < \mu.$$

Then we can subject the variables x to the linear equations

$$\begin{aligned} &Y_1 = 0, \quad Y_2 = 0, \ldots, Y_\nu = 0, \\ &Z_1' = 0, \quad Z_2' = 0, \ldots, Z_{\mu'}' = 0,\end{aligned} \tag{2}$$

whose number $\nu + \mu'$ is less than m.

If all the functions $Z_1, Z_2, \ldots, Z_\mu$ were linearly dependent on the functions $Y_1, Y_2, \ldots, Y_\nu, Z_1', Z_2', \ldots, Z_{\mu'}'$, then from these μ equations one could, by eliminating the ν variables $Y_1, Y_2, \ldots, Y_\nu$, derive a linear equation among $Z_1, Z_2, \ldots, Z_\mu, Z_1', \ldots, Z_{\mu'}'$, which by hypothesis is not to exist. Thus the vanishing of all $Z_1, Z_2, \ldots, Z_\mu$ is not a necessary consequence of equations (2), and we may add to equations (2) a non-homogeneous equation, for instance

$$Z_1 = 1.$$

Then, for the m unknowns x, we have a system of m or fewer equations that are independent of one another and hence do not contradict one another.

But then the second representation (1) gives a positive value for $\varphi(x)$, while the first gives a negative or vanishing value, a contradiction. Hence

$$\nu \geq \mu,$$

and, since one similarly concludes that

$$\mu \geq \nu,$$

only $\nu = \mu$ remains. In the same way one may prove that $\nu' = \mu'$, and this proves the theorem completely.

§ 59. Transformation of forms of the nth degree

When a form $F(x)$ of the nth degree in m variables is transformed by a linear substitution [§ 55, (1)] into $\Phi(x')$, the linear function of the variables $\xi_1, \xi_2, \ldots, \xi_m$

$$\sum_i \xi_i F'(x_i) \tag{1}$$

is simultaneously transformed, by the same substitution applied to the ξ_i, into

$$\sum_i \xi_i' \Phi'(x_i'), \tag{2}$$

because both expressions represent the coefficient of the first power of t in the expansion of

$$F(x_i + t\xi_i) = \Phi(x_i' + t\xi_i')$$

in powers of t (§ 16). Likewise, the quadratic function

$$\sum_{i,k} \xi_i \xi_k F''(x_i x_k) \tag{3}$$

is transformed into

$$\sum_{i,k} \xi_i' \xi_k' \Phi''(x_i' x_k'). \tag{4}$$

If we apply the first result to a system of m functions $F_1(x), F_2(x), \ldots, F_m(x)$ of arbitrary degrees and denote by Δ the determinant

$$\Delta = \begin{vmatrix} F_1'(x_1) & F_1'(x_2) & \cdots & F_1'(x_m) \\ F_2'(x_1) & F_2'(x_2) & \cdots & F_2'(x_m) \\ \vdots & \vdots & & \vdots \\ F_m'(x_1) & F_m'(x_2) & \cdots & F_m'(x_m) \end{vmatrix},$$

while Δ' denotes the corresponding determinant formed from the functions $\Phi_1, \Phi_2, \ldots, \Phi_m$, then by § 55, (4), again denoting the substitution determinant by r, we obtain

$$\Delta' = r\Delta. \tag{5}$$

The function Δ is called the functional determinant of the m functions $F_1, F_2, \ldots, F_m$.

Likewise, from the transformation (3), (4), if H denotes the determinant

$$H = \begin{vmatrix} F''(x_1, x_1) & F''(x_1, x_2) & \cdots & F''(x_1, x_m) \\ F''(x_2, x_1) & F''(x_2, x_2) & \cdots & F''(x_2, x_m) \\ \vdots & \vdots & & \vdots \\ F''(x_m, x_1) & F''(x_m, x_2) & \cdots & F''(x_m, x_m) \end{vmatrix}$$

and H' the corresponding determinant for the transformed function Φ, we obtain by § 56, (8)

$$H' = r^2 H. \tag{6}$$

H is called the Hessian determinant[17] of the function F. In both formulas (5) and (6), r denotes the substitution determinant.

§ 60. Invariants and covariants

In the linear transformation of homogeneous functions, certain functions of the coefficients always occur which, when the coefficients of the given form are replaced by the coefficients of the transformed form, change in no other way than by acquiring a factor that is a power of the substitution determinant. Such functions also occur under the simultaneous transformation of a system of several forms; they depend on the coefficients of all forms of the system. Examples are, for a system of n linear forms, the determinant of the system, and, for a quadratic form, the determinant of that form.

These functions are called invariants of the form, or, when they depend on several forms, simultaneous invariants of the system.

[17] Named after Hesse, who frequently used this determinant in geometrical investigations, especially in the theory of curves of the third order and in elimination problems, and recognized its significance.

Thus, if $I(a)$ is a function of the coefficients a of a form $F(x)$, and if a' are the coefficients of the transformed form $\Phi(x')$, then $I(a)$ is called an invariant if the identical relation

$$I(a') = r^{\lambda} I(a) \tag{1}$$

holds. We shall call the exponent λ the weight of the invariant. If this exponent is zero, then I is called an absolute invariant.

One usually considers only integral rational invariants, that is, such functions I as depend rationally on the coefficients a and are moreover integral homogeneous functions of them. Simultaneous invariants are to be homogeneous with respect to the coefficients of each of the forms on which they depend.

Besides invariants, there are also functions that, in addition to the coefficients, contain the variables themselves and otherwise have the same properties as the invariants (1), namely the "invariant property." Thus, if $C(x, a)$ is such a function, then

$$C(x', a') = r^{\lambda} C(x, a). \tag{2}$$

Such functions are called covariants of the form $F(x)$. The functional determinant is a simultaneous covariant of a system of m functions, and the Hessian determinant of a function of degree higher than the second is a covariant of a single form. Among covariants, too, one chiefly considers integral rational and homogeneous functions of x.

If m is the number of variables x, n the degree of the function $F(x)$, and ν and μ the degrees of $C(x, a)$ with respect to x and to a, then the numbers λ, μ, ν, m, n satisfy the relation

$$n\mu = m\lambda + \nu. \tag{3}$$

This is proved by comparing the degree of the right and left sides of (2) with respect to the substitution coefficients. Namely, the a' are of degree n in the substitution coefficients, r is of degree m, and x is of degree one; this gives (3). The corresponding formula for invariants is obtained by putting $\nu = 0$, just as invariants may in general be regarded as a special case of covariants.

We shall briefly discuss a general rule for the formation of invariants and covariants, which we have already used in the special cases of § 59.

If, according to § 16, (4), we arrange the function

$$F(x_1 + t\xi_1, x_2 + t\xi_2, \ldots, x_m + t\xi_m)$$

by powers of t, then for the coefficient of t^{ν} we obtain

$$F_{\nu}(x, \xi) = \frac{1}{\Pi(\nu)} \sum_{\alpha_1+\alpha_2+\cdots+\alpha_m=\nu} \frac{\Pi(\nu)}{\Pi(\alpha_1)\cdots\Pi(\alpha_m)} \xi_1^{\alpha_1}\xi_2^{\alpha_2}\cdots\xi_m^{\alpha_m} \mathrm{D}_{\alpha_1,\alpha_2,\ldots,\alpha_m}(F). \tag{4}$$

If we apply the linear transformation (1), § 55, simultaneously to the variables ξ and x, then $F_{\nu}(x, \xi)$ passes into $\Phi_{\nu}(x', \xi')$, which is obtained from F_{ν} by replacing at the same time all letters x, ξ, a by x', ξ', a' and F by Φ. Hence:

I. If $F_{\nu}(x, \xi)$ is regarded as a form of the νth degree in ξ and one forms an invariant of this form, whose coefficients therefore still depend on the x, one obtains a covariant of the form F.

Since the variables x and ξ undergo the same transformation, it follows similarly that:

II. If one forms a covariant of the function $F_{\nu}(x, \xi)$, regarded as a function of ξ, and then replaces the ξ in it by the x, one obtains a covariant of the original function $F(x)$.

The same theorems also hold for simultaneous invariants and covariants.

For the functions $F_{\nu}(x, \xi)$ one has the theorem

$$F_{\nu}(x, \xi) = F_{n-\nu}(\xi, x), \tag{5}$$

which follows by comparing corresponding powers of t on both sides of the identity

$$F(x_1 + t\xi_1, \dots, x_m + t\xi_m) = t^n F(t^{-1}x_1 + \xi_1, \dots, t^{-1}x_m + \xi_m).$$

The functions

$$P_\nu(x, \xi) = \frac{\Pi(\nu)\Pi(n-\nu)}{\Pi(n)} F_\nu(x, \xi)$$

are called the polars of the function $F(x)$; more precisely, $P_\nu(x, \xi)$ is called the νth polar. They too satisfy the theorem

$$P_\nu(x, \xi) = P_{n-\nu}(\xi, x).$$

The constant factor $\Pi(\nu)\Pi(n-\nu) : \Pi(n)$ is added so that, when the form $F(x)$ is written with polynomial coefficients, the functions P_ν have coefficients as simple as possible.

For the binary form $f(x, y)$, to which in what follows we shall mainly apply these definitions, the following expressions for the polars result:

$$P_1(x, \xi) = \frac{1}{n}\{\xi f'(x) + \eta f'(y)\},$$

$$P_2(x, \xi) = \frac{1}{n(n-1)}\{\xi^2 f''(x, x) + 2\xi\eta f''(x, y) + \eta^2 f''(y, y)\},$$

and in general

$$\begin{aligned} &n(n-1)\cdots(n-\nu+1)P_\nu(x, \xi) \\ &\quad = \xi^\nu \frac{\partial^\nu f}{\partial x^\nu} + \nu\xi^{\nu-1}\eta \frac{\partial^\nu f}{\partial x^{\nu-1}\partial y} + \cdots + \eta^\nu \frac{\partial^\nu f}{\partial y^\nu}. \end{aligned}$$

§ 61. Linear transformation of binary forms

An integral rational function of degree n in x,

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1}x + a_n, \tag{1}$$

is converted into a binary form when x is replaced by $x : y$ and the result is multiplied by y^n:

$$f(x, y) = a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \cdots + a_n y^n. \tag{2}$$

If the roots of $f(x)$ are known, then $f(x)$ can be decomposed into n linear factors:

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n), \tag{3}$$

and from this it follows that

$$f(x, y) = a_0(x - \alpha_1 y)(x - \alpha_2 y)\cdots(x - \alpha_n y). \tag{4}$$

But if $\alpha_1, \alpha_2, \dots, \alpha_n$ are themselves replaced by ratios $\alpha_1 : \beta_1, \alpha_2 : \beta_2, \dots, \alpha_n : \beta_n$, one obtains

$$f(x, y) = A(x\beta_1 - y\alpha_1)(x\beta_2 - y\alpha_2)\cdots(x\beta_n - y\alpha_n), \tag{5}$$

provided that

$$a_0 = A\beta_1\beta_2\cdots\beta_n.$$

For the factors of $f(x, y)$ that will now occur more frequently, we introduce the abbreviated notation

$$x\beta_1 - y\alpha_1 = (x\beta_1). \tag{6}$$

Similarly, we shall put

$$x_1 y_2 - x_2 y_1 = (x_1 y_2), \tag{7}$$

where x_1, y_1 and x_2, y_2 are two arbitrary pairs of variables. If we use the inhomogeneous representation, we shall write still more briefly

$$\begin{gathered} x-\alpha_1=(0,1), \quad x-\alpha_2=(0,2), \ldots, x-\alpha_n=(0,n), \\ \alpha_1-\alpha_2=(1,2), \ldots, \alpha_i-\alpha_k=(i,k). \end{gathered} \tag{8}$$

If, in the form (2), we make the linear substitution

$$x=\alpha x'+\beta y', \qquad y=\gamma x'+\delta y' \tag{9}$$

with non-zero determinant

$$r=\alpha\delta-\beta\gamma, \tag{10}$$

then the form $f(x,y)$ passes into a form $F(x',y')$ of degree n, and

$$F(x',y')=a_0'x'^n+a_1'x'^{n-1}y'+a_2'x'^{n-2}y'^2+\cdots+a_n'y'^n. \tag{11}$$

The coefficients $a_0', a_1', \ldots, a_n'$ of the transformed form are obtained as linear and homogeneous expressions in the coefficients $a_0, a_1, \ldots, a_n$, and as homogeneous expressions of degree n in the substitution coefficients $\alpha, \beta, \gamma, \delta$. Thus, by setting $x'=1, y'=0$ or $x'=0, y'=1$, one obtains

$$a_0'=f(\alpha,\gamma), \qquad a_n'=f(\beta,\delta). \tag{12}$$

The remaining coefficients a' can be expressed through the derivatives of the function $f(x,y)$, for example

$$(n-1)a_1'=\beta f'(\alpha)+\delta f'(\gamma),$$

which we shall not carry out further.

If we apply the transformation (9) simultaneously to the two pairs of variables x, y and α_k, β_k, then we obtain the transformed linear factors of $f(x,y)$, namely

$$(x'\beta_k')=r(x\beta_k), \tag{13}$$

as follows either from the multiplication rule for determinants or by direct calculation. These factors $(x\beta_k)$ are therefore covariants of $f(x,y)$, though irrational ones, since they do not depend rationally on the coefficients of $f(x)$. Similarly one obtains

$$(\alpha_h'\beta_k')=r(\alpha_h\beta_k), \tag{14}$$

according to which the determinants $(\alpha_h\beta_k)$ are to be regarded as irrational invariants. We shall now explain how rational invariants and covariants may be formed from them.

For this purpose it is best to consider the linear transformation in the non-homogeneous form

$$x=\frac{\alpha x'+\beta}{\gamma x'+\delta} \tag{15}$$

and to apply it to the function $f(x)$ in (1). Then, if

$$F(x')=a_0'x'^n+a_1'x'^{n-1}+a_2'x'^{n-2}+\cdots+a_n'$$

is put, one obtains

$$F(x')=(\gamma x'+\delta)^n f(x). \tag{16}$$

We also write this equation as

$$F(x')=(\alpha x'+\beta)^n\left(a_0+a_1\frac{1}{x}+a_2\frac{1}{x^2}+\cdots\right),$$

from which, if $x' = -\delta : \gamma$, so that $x = \infty$, one obtains

$$r^n a_0 = (-\gamma)^n F\left(-\frac{\delta}{\gamma}\right).$$

Or, if $\alpha_1', \alpha_2', \dots, \alpha_n'$ depend on $\alpha_1, \alpha_2, \dots, \alpha_n$ in the same way as x' depends on x, so that

$$F(x') = a_0'(x' - \alpha_1')(x' - \alpha_2') \cdots (x' - \alpha_n')$$

is put, it follows that

$$r^n a_0 = a_0'(\gamma\alpha_1' + \delta)(\gamma\alpha_2' + \delta) \cdots (\gamma\alpha_n' + \delta). \tag{17}$$

Now, by the transformation (15),

$$\begin{aligned} x - \alpha_i &= \frac{r(x' - \alpha_i')}{(\gamma x' + \delta)(\gamma\alpha_i' + \delta)}, \\ \alpha_i - \alpha_k &= \frac{r(\alpha_i' - \alpha_k')}{(\gamma\alpha_i' + \delta)(\gamma\alpha_k' + \delta)}. \end{aligned} \tag{18}$$

Now form a product P from any of the factors

$$(0,1), (0,2), \dots, (0,n),$$

$$(1,2), (1,3), \dots, (n-1,n),$$

but in such a way that each of the indices $1, 2, \dots, n$ occurs altogether equally often, say μ times. The index 0, that is, the variable x, is to occur ν times, and the total number of factors is to be q. Since each factor contains two indices, one has

$$\nu + n\mu = 2q. \tag{19}$$

Then from (17) and (18), if P' is obtained from P by replacing x, α_i by x', α_i', one obtains

$$a_0'^{\mu} P' = a_0^{\mu} r^{n\mu - q} (\gamma x' + \delta)^{\nu} P. \tag{20}$$

We now form

$$C(x, y, a) = a_0^{\mu} y^{\nu} \sum P\left(\frac{x}{y}\right), \tag{21}$$

where the sum is taken over all values obtained from P by permuting the indices $1, 2, 3, \dots, n$ with one another in all possible ways. This sum is then a symmetric function of the roots of (1), and may therefore be expressed as an integral rational function of the ratios $a_1 : a_0, a_2 : a_0, \dots, a_n : a_0$, with no term exceeding the μth order. Thus $C(x, y, a)$ is an integral homogeneous function of $a_0, a_1, \dots, a_n$ of order μ, and an integral homogeneous function of x, y of order ν. Since, by (19), (20), it satisfies the condition

$$C(x', y', a') = r^{\lambda} C(x, y, a), \tag{22}$$

$$\lambda = n\mu - q, \qquad 2\lambda = n\mu - \nu, \tag{23}$$

it is a covariant, and in the special case where $\nu = 0$ an invariant.

The weight λ is, as the second expression shows, always positive or zero, since $n\mu$ is at least equal to ν. The extreme value $n\mu = \nu$, or $\lambda = 0$, occurs only when every one of the indices $1, 2, 3, \dots, n$ occurs in P only in connection with 0, that is, when P is a power of $f(x)$ itself.

§ 62. Binary cubic forms

A binary quadratic form gives no invariant constructions other than the discriminant. We therefore do not dwell on it, and pass at once to the consideration of cubic forms

$$f(x,y) = a_0x^3 + a_1x^2y + a_2xy^2 + a_3y^3. \tag{1}$$

Here we form first, as a quadratic covariant, the Hessian determinant. We shall always observe the rule that forms with integral numerical coefficients are written without a common factor. Hence in the determinant formed from the second derivatives of (1) we must suppress the factor 4, and we obtain as first covariant

$$H = \begin{vmatrix} 3a_0x + a_1y & a_1x + a_2y \\ a_1x + a_2y & a_2x + 3a_3y \end{vmatrix},$$

or

$$H = A_0x^2 + A_1xy + A_2y^2, \tag{2}$$

where, for abbreviation,

$$A_0 = 3a_0a_2 - a_1^2, \qquad A_1 = 9a_0a_3 - a_1a_2, \qquad A_2 = 3a_1a_3 - a_2^2. \tag{3}$$

The discriminant of this quadratic form is an invariant of the given cubic form. It contains the factor 3 in all numerical coefficients, and accordingly we put

$$3D = 4A_0A_2 - A_1^2. \tag{4}$$

Then D agrees with the discriminant of the cubic form [§ 46, (10)]:

$$D = a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2. \tag{5}$$

A further cubic covariant is obtained by forming the functional determinant of $f(x,y)$ and $H(x,y)$:

$$Q(x,y) = f'(x)H'(y) - f'(y)H'(x), \tag{6}$$

or

$$Q = \begin{vmatrix} 3a_0x^2 + 2a_1xy + a_2y^2 & 2A_0x + A_1y \\ a_1x^2 + 2a_2xy + 3a_3y^2 & A_1x + 2A_2y \end{vmatrix}. \tag{7}$$

No numerical factor can be cancelled from it. The expanded coefficients of x^3, x^2y, xy^2, y^3, whose formation is straightforward, are, in order,

$$\begin{aligned} &27a_0^2a_3 - 9a_0a_1a_2 + 2a_1^3, \\ &27a_0a_1a_3 - 18a_0a_2^2 + 3a_1^2a_2, \\ &-27a_0a_2a_3 + 18a_1^2a_3 - 3a_1a_2^2, \\ &-27a_0a_3^2 + 9a_1a_2a_3 - 2a_2^3. \end{aligned}$$

The behaviour of H, D, Q under a linear transformation follows from § 60, (2), (3). If H', D', Q' denote the corresponding expressions for a transformed form, then

$$H' = r^2H, \qquad D' = r^6D, \qquad Q' = r^3Q. \tag{8}$$

The covariants can be used to transform the cubic form into a normal form which also gives the solution of the cubic equation. We choose the normal form

$$f(x,y) = F(\xi,\eta) = \xi^3 + \eta^3, \tag{9}$$

where ξ, η are linear functions of x, y. This form, as will follow at once, can always be obtained when D does not vanish, that is, when the equation $f = 0$ has no two equal roots. The solutions of $f = 0$ are then obtained from the linear equations

$$\xi + \eta = 0, \qquad \xi + \varepsilon\eta = 0, \qquad \xi + \varepsilon^2\eta = 0,$$

where ε is a third root of unity.

To form H', D', Q' for the normal form (9), we put $a_0' = a_3' = 1$ and $a_1' = a_2' = 0$ and obtain

$$H' = 9\xi\eta, \qquad D' = -27, \qquad Q' = 27(\xi^3 - \eta^3).$$

If the normal form (9) is obtained from the general form $f(x, y)$ by a linear transformation, then (8) gives

$$\begin{aligned} 9\xi\eta &= r^2 H, \qquad -27 = r^6 D, \\ 27(\xi^3 - \eta^3) &= r^3 Q, \qquad \xi^3 + \eta^3 = f. \end{aligned} \tag{10}$$

It follows that

$$r^6 = -\frac{27}{D}, \qquad 2\xi^3 = \frac{r^3 Q}{27} + f, \qquad 2\eta^3 = -\frac{r^3 Q}{27} + f.$$

Thus, if in the first equation we put $r^3 = -9/\sqrt{-3D}$, then

$$\begin{aligned} 6\xi^3\sqrt{-3D} &= Q + 3\sqrt{-3D}\, f, \\ 6\eta^3\sqrt{-3D} &= -Q + 3\sqrt{-3D}\, f. \end{aligned} \tag{11}$$

Hence ξ and η are determined by two cube roots, of which one is determined by the other through the first equation (10).

This also proves that, when D is non-zero, the normal form can be obtained by a linear transformation. For under this hypothesis H splits into two distinct linear factors; if these are chosen, up to constant factors, as the new variables ξ, η of a linear transformation, then $A_0' = 0$, $A_2' = 0$. From the evident identities one then obtains $a_1' = 0$, $a_2' = 0$, unless at the same time $A_1' = 0$, which is excluded by the non-vanishing of D. Thus the transformed form of f contains only the cubes of ξ and η.

If the first equation (10) is cubed, and ξ^3, η^3, r^6 are eliminated by means of (11) and the second equation (10), then the covariants satisfy the identity

$$4H^3 + Q^2 + 27Df^2 = 0. \tag{12}$$

To carry out the transformation to the normal form, that is, actually to find ξ, η, one factors the function H into its linear factors:

$$4A_0 H = (2A_0 x + A_1 y + \sqrt{-3D}\, y)(2A_0 x + A_1 y - \sqrt{-3D}\, y).$$

Thus ξ, η differ from these two factors of H only by constant factors. If these two factors are denoted by h, k, we put

$$\begin{aligned} 2\xi &= h(2A_0 x + A_1 y - \sqrt{-3D}\, y), \\ 2\eta &= k(2A_0 x + A_1 y + \sqrt{-3D}\, y). \end{aligned} \tag{13}$$

Multiplication, using the first two equations (10), gives

$$3hkA_0\sqrt[3]{-D} = 1, \tag{14}$$

and (11), by comparison of the coefficients of x^3, gives

$$\begin{aligned} 6h^3 A_0^3\sqrt{-3D} &= q_0 + 3\sqrt{-3D}\, a_0, \\ 6k^3 A_0^3\sqrt{-3D} &= -q_0 + 3\sqrt{-3D}\, a_0, \end{aligned} \tag{15}$$

where q_0 is the coefficient of x^3 in Q, namely

$$q_0 = 27a_0^2 a_3 - 9a_0 a_1 a_2 + 2a_1^3. \tag{16}$$

The signs of $\sqrt{-3D}$ are to be taken in accordance with the signs in (11), as in the formulae (13); this is seen most simply by doing the calculation in the special case $a_0 = 0$, $a_3 = 0$.

§ 63. The full system of forms of the binary cubic form

We now show that the invariants and covariants of the cubic form are exhausted by the functions D, f, H, Q; in other words, that all invariants and covariants of a cubic form can be expressed rationally in terms of these four forms.

Let

$$C = C(x, y, a)$$

be a covariant of degree ν in the variables and of degree μ in the coefficients, and suppose that it has only integral numerical coefficients. Then for any transformed form

$$C' = C(\xi, \eta, a') = r^{\lambda} C(x, y, a), \tag{1}$$

where

$$\lambda = \frac{3\mu - \nu}{2}. \tag{2}$$

We now understand ξ, η to be the variables of the normal form considered above. If ε is a third root of unity, then

$$\xi = \varepsilon \xi', \qquad \eta = \varepsilon^2 \eta'$$

is a linear substitution of determinant 1 which carries the normal form § 62, (9) into itself. Hence C' cannot change when ξ, η are replaced by $\varepsilon\xi, \varepsilon^2\eta$; it can therefore depend rationally only on $\xi\eta, \xi^3, \eta^3$.

Interchanging ξ and η corresponds to a substitution of determinant -1. Consequently C' either remains fixed or changes sign according as λ is even or odd. Thus C' is a sum of terms of the form

$$M(\xi\eta)^{\alpha}(\xi^{3\beta} \pm \eta^{3\beta}),$$

where M is an integer factor, and where the upper or lower sign is to be taken according as λ is even or odd. In the latter case the expression is divisible by $\xi^3 - \eta^3$, and the quotient can be expressed integrally and rationally in terms of $\xi^3 + \eta^3$ and $\xi\eta$. Thus, putting $\rho = 0$ or 1 according as λ is even or odd, we get for C' an expression of the form

$$C'(\xi, \eta) = (\xi^3 - \eta^3)^{\rho} \sum M(\xi\eta)^{\alpha}(\xi^3 + \eta^3)^{\beta},$$

where the M are integral coefficients and α, β run through positive integers subject to

$$3\rho + 3\beta + 2\alpha = \nu. \tag{3}$$

From (1), after multiplication by a sufficiently high power of 3 and substitution of $\xi\eta$, $\xi^3 + \eta^3$, $\xi^3 - \eta^3$ by H, f, Q according to § 62, (10), and of r by D, one obtains

$$3^{\sigma} C = Q^{\rho} \sum M D^{\gamma} H^{\alpha} f^{\beta}, \tag{4}$$

where

$$6\gamma = \lambda - 3\rho - 2\alpha. \tag{5}$$

For each value of γ there is, by (5), a definite value of α, and, by (3), a definite value of β; likewise a value of α or β determines the other two exponents. Hence in (4) each exponent of D, H, or f occurs only once.

That γ is an integer follows easily from (2), (3), (5). Since, by definition, $\lambda - 3\rho$ is even, 3γ is an integer; and (3) gives divisibility by 3, whence γ itself is an integer.

It must also be non-negative. For if negative powers of D occurred on the right side of (4), multiply both sides by a sufficiently high power of D so that no negative powers remain, but the right side is still not divisible by D. In the resulting equation put

$$a_0 = 0, \qquad a_1 = 1, \qquad a_2 = 0, \qquad a_3 = 0. \tag{6}$$

Then

$$D = 0, \qquad H = -x^2, \qquad f = x^2 y, \qquad Q = 2x^3. \tag{7}$$

The left side would vanish while the right side would not, a contradiction.

It remains to observe that no power of 3 can remain on the left side of (4) unless it occurs in all the factors M, and hence can be cancelled. The theorem proved in § 2 says that the product of two primitive integral rational functions of arbitrary variables is again primitive. Applying this, it is enough to prove that an expression

$$\sum M D^\gamma H^\alpha f^\beta,$$

where the integers M have no common divisor, cannot become imprimitive, and in particular cannot acquire the factor 3, after Q, D, H, f have been written out in terms of a, x, y. Since Q is primitive, it suffices to prove this for

$$\sum M D^\gamma H^\alpha f^\beta. \tag{8}$$

We may omit all terms whose M is already divisible by 3, and we may also assume that the expression is not divisible by D, so that it contains a term with $\gamma = 0$. If, under these assumptions, the expanded expression (8) had the divisor 3, then substituting (6), (7), that is, putting $D = 0$, $H = -x^2$, $f = x^2 y$, would reduce it to the single term with $\gamma = 0$ and $2\alpha = \lambda$. Its coefficient M would then have to be divisible by 3, contrary to the assumption.

We have therefore proved: every integral covariant of the cubic form can be represented integrally, rationally, and with integral coefficients by one of the two expressions

$$\sum M D^\gamma H^\alpha f^\beta, \qquad Q \sum M D^\gamma H^\alpha f^\beta.$$

The invariants are included as the special case of covariants and are all powers of D.

§ 64. Biquadratic forms

We now turn to the biquadratic form

$$f(x, y) = a_0 x^4 + a_1 x^3 y + a_2 x^2 y^2 + a_3 x y^3 + a_4 y^4. \tag{1}$$

First we have the Hessian determinant as a covariant of the fourth order; to remove a numerical factor, we divide it by 3:

$$H = \frac{1}{3} \begin{vmatrix} 12a_0x^2 + 6a_1xy + 2a_2y^2 & 3a_1x^2 + 4a_2xy + 3a_3y^2 \\ 3a_1x^2 + 4a_2xy + 3a_3y^2 & 2a_2x^2 + 6a_3xy + 12a_4y^2 \end{vmatrix}. \tag{2}$$

Thus

$$H = A_0 x^4 + A_1 x^3 y + A_2 x^2 y^2 + A_3 x y^3 + A_4 y^4,$$

where

$$\begin{aligned} A_0 &= 8a_0a_2 - 3a_1^2, & A_4 &= 8a_2a_4 - 3a_3^2, \\ A_1 &= 24a_0a_3 - 4a_1a_2, & A_3 &= 24a_1a_4 - 4a_2a_3, \\ A_2 &= 48a_0a_4 + 6a_1a_3 - 4a_2^2. \end{aligned}$$

There is also a covariant of the sixth degree,

$$T = \frac{1}{12} \begin{vmatrix} f'(x) & f'(y) \\ H'(x) & H'(y) \end{vmatrix}. \tag{3}$$

The invariants of the biquadratic form are most simply formed from the differences of the roots, as in § 61. We obtain an invariant of second order and one of third order in the coefficients:

$$A = \frac{1}{2} a_0^2 \big((12)^2(34)^2 + (13)^2(24)^2 + (14)^2(23)^2 \big), \tag{4}$$

$$B = a_0^3 \sum (12)^2(34)^2(13)(42). \tag{5}$$

These expressions take a clearer form if

$$U = (12)(34), \qquad V = (13)(42), \qquad W = (14)(23) \tag{6}$$

is put; then

$$\begin{aligned} A &= \frac{1}{2}a_0^2(U^2 + V^2 + W^2), \\ B &= a_0^3\{U^2(V - W) + V^2(W - U) + W^2(U - V)\} \\ &= a_0^3(W - V)(U - W)(V - U). \end{aligned} \tag{7}$$

The formulae necessary for calculating these quantities were already developed in § 47. From the formulae there, one obtains

$$U = v^2 - w^2, \qquad V = w^2 - u^2, \qquad W = u^2 - v^2,$$

where u^2, v^2, w^2 are the roots of the cubic resolvent

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0.$$

Consequently

$$\begin{aligned} A &= a_0^2\{(u^2 + v^2 + w^2)^2 - 3(v^2w^2 + w^2u^2 + u^2v^2)\} \\ &= a_0^2(a^2 + 12c), \\ B &= a_0^3(2u^2 - v^2 - w^2)(2v^2 - w^2 - u^2)(2w^2 - u^2 - v^2) \\ &= a_0^3(3u^2 + 2a)(3v^2 + 2a)(3w^2 + 2a) \\ &= a_0^3(2a^3 - 72ac + 27b^2). \end{aligned}$$

Thus A, B have the same meaning as in § 47 and are, in terms of the coefficients a_0, a_1, a_2, a_3, a_4,

$$A = a_2^2 - 3a_1a_3 + 12a_0a_4, \tag{8}$$

$$B = 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3. \tag{9}$$

The discriminant of the biquadratic equation, a third invariant but one dependent on A and B, is

$$D = a_0^6U^2V^2W^2, \tag{10}$$

and at the cited place it was found that

$$27D = 4A^3 - B^2. \tag{11}$$

Using the formula of § 67, $2\lambda = n\mu - \nu$, we get for the invariant constructions found so far, denoting those for a transformed form by primes,

$$H' = r^2H, \qquad T' = r^3T, \qquad A' = r^4A, \qquad B' = r^6B, \qquad D' = r^{12}D. \tag{12}$$

§ 65. Solution of the biquadratic equation

We now seek to construct a normal form by a linear substitution of determinant 1. For its derivation we suppose the given biquadratic form decomposed into linear factors:

$$f(x, y) = a_0(x - \alpha y)(x - \beta y)(x - \gamma y)(x - \delta y). \tag{1}$$

Put

$$x\sqrt{\alpha - \beta} = \beta\xi - \alpha\eta, \qquad y\sqrt{\alpha - \beta} = \xi - \eta, \qquad r = 1. \tag{2}$$

Then

$$\begin{aligned} x - \alpha y = -\sqrt{\alpha - \beta}\,\xi, \qquad x - \beta y = -\sqrt{\alpha - \beta}\,\eta, \\ (x - \gamma y)\sqrt{\alpha - \beta} = (\beta - \gamma)\xi - (\alpha - \gamma)\eta, \\ (x - \delta y)\sqrt{\alpha - \beta} = (\beta - \delta)\xi - (\alpha - \delta)\eta. \end{aligned} \tag{3}$$

Thus $f(x, y)$ becomes

$$F(\xi, \eta) = a_1'\xi^3\eta + a_2'\xi^2\eta^2 + a_3'\xi\eta^3. \tag{4}$$

The coefficients a_1', a_2', a_3' are readily expressed by (3) through $\alpha, \beta, \gamma, \delta$:

$$\begin{aligned} a_1' &= a_0(\beta - \gamma)(\beta - \delta), \\ a_3' &= a_0(\alpha - \gamma)(\alpha - \delta), \\ a_2' &= -a_0\{(\alpha - \gamma)(\beta - \delta) + (\alpha - \delta)(\beta - \gamma)\}. \end{aligned} \tag{5}$$

By interchanging the roots one can obtain the normal form (4) in six ways, but only three distinct values of a_2' arise. If, as above,

$$U = (\alpha - \beta)(\gamma - \delta), \quad V = (\alpha - \gamma)(\delta - \beta), \quad W = (\alpha - \delta)(\beta - \gamma), \quad U + V + W = 0, \tag{6}$$

then these three values are

$$a_2' = a_0(V - W), \qquad a_2'' = a_0(W - U), \qquad a_2''' = a_0(U - V). \tag{7}$$

Knowing a_2', a_2'', a_2''', one also knows

$$3a_0U = a_2''' - a_2'', \qquad 3a_0V = a_2' - a_2''', \qquad 3a_0W = a_2'' - a_2'. \tag{8}$$

Hence, if two of a_2', a_2'', a_2''' are equal, one of U, V, W vanishes; that is, two roots of the biquadratic equation are equal and the discriminant vanishes.

Now put

$$\alpha\beta + \gamma\delta = u, \qquad \alpha\gamma + \delta\beta = v, \qquad \alpha\delta + \beta\gamma = w. \tag{9}$$

Then

$$a_0(u + v + w) = a_2,$$

the sum of the products of two roots of $f(x, y)$, and

$$U = v - w, \qquad V = w - u, \qquad W = u - v.$$

Thus

$$3a_0u = a_2 - a_2', \qquad 3a_0v = a_2 - a_2'', \qquad 3a_0w = a_2 - a_2'''. \tag{10}$$

So, along with a_2', a_2'', a_2''', the quantities u, v, w are known. Once u, v, w are known, the solution of the biquadratic equation is reduced to square roots. Indeed,

$$\alpha\beta + \gamma\delta = u, \qquad a_0\alpha\beta\gamma\delta = a_4,$$

and therefore

$$2\alpha\beta = v_1 + \sqrt{\frac{a_0u^2 - 4a_4}{a_0}}, \qquad 2\gamma\delta = v_1 - \sqrt{\frac{a_0u^2 - 4a_4}{a_0}}.$$

The remaining products $\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta$ are obtained similarly, and then, for example,

$$\alpha^2 = \frac{\alpha\beta \cdot \alpha\gamma}{\beta\gamma}$$

can be calculated.

It remains to form the cubic equation whose roots are a_2', a_2'', a_2'''. This comes directly from the invariants. Forming A', B' from the transformed form (4), one finds from § 64, (8), (9), (12), since $r = 1$,

$$A = a_2'^2 - 3a_1'a_3', \qquad B = 2a_2'^3 - 9a_1'a_2'a_3', \qquad D = a_1'^2 a_2'^2 a_3'^2 - 4a_1'a_3'^3. \tag{11}$$

Eliminating $a_1'a_3'$ gives

$$a_2'^3 - 3Aa_2' + B = 0.$$

Thus a_2', a_2'', a_2''' are the roots of the cubic equation in z

$$z^3 - 3Az + B = 0. \tag{12}$$

This is probably the simplest form of the cubic resolvent of the biquadratic equation.

§ 66. The covariants

If the covariant H of § 64, (2) is formed for the transformed form

$$f(x, y) = F(\xi, \eta) = a_1'\xi^3\eta + a_2'\xi^2\eta^2 + a_3'\xi\eta^3, \tag{1}$$

then

$$-H = 3a_1'^2\xi^4 + 4a_1'a_2'\xi^3\eta - (6a_1'a_3' - 4a_2'^2)\xi^2\eta^2 + 4a_2'a_3'\xi\eta^3 + 3a_3'^2\eta^4, \tag{2}$$

and hence

$$H + 4a_2'f = -3(a_1'\xi^2 - a_3'\eta^2)^2. \tag{3}$$

Thus this combination, apart from the factor -3, is the square of a quadratic form; the same holds if a_2' is replaced by a_2'' or a_2'''. We therefore write

$$H + 4a_2'f = -3\psi_1^2, \qquad H + 4a_2''f = -3\psi_2^2, \qquad H + 4a_2'''f = -3\psi_3^2. \tag{4}$$

No two of ψ_1, ψ_2, ψ_3 can have a common divisor when the discriminant D is assumed non-zero. In that case a_2', a_2'', a_2''' are also distinct. If ψ_1 and ψ_2 vanished, then H and f would vanish; any common divisor of ψ_1 and ψ_2 would therefore be a common divisor of H and f. But ξ was an arbitrary linear divisor of f. From (2) this can divide H only when $a_3' = 0$, which again means that the discriminant D vanishes [§ 65, (11)].

The functional determinant T of f and H is the same as the functional determinant of f and $H + \lambda f$, whatever λ may be. Taking $\lambda = 4a_2'$, one obtains

$$T = -\frac{1}{2}\psi_1\{F'(\xi)\psi_1'(\eta) - F'(\eta)\psi_1'(\xi)\}.$$

Thus T is divisible by ψ_1, and for the same reasons by ψ_2 and ψ_3. Hence, up to a factor independent of ξ, η, T is identical with $\psi_1\psi_2\psi_3$.

Taking the product of the three equations (4), and using the cubic equation § 65, (12) satisfied by a_2', a_2'', a_2''', gives

$$H^3 - 48AHf^2 - 64Bf^3 = cT^2.$$

The constant c is still to be determined. By § 64, (12), c is invariant under a linear transformation, and its value is found from the normal form (1), (2) by equating the terms of highest power in ξ. This gives $c = -27$. Hence

$$H^3 - 48AHf^2 - 64Bf^3 = -27T^2. \tag{5}$$

The functions ψ_1, ψ_2, ψ_3 are themselves covariants, though with coefficients not rational in the coefficients of f. They are easily expressed through the roots $\alpha, \beta, \gamma, \delta$ of $f = 0$. Putting $y = 1$ for simplicity, one obtains from § 65, (3), (5)

$$\begin{aligned}\frac{\psi_1}{a_0} &= (\gamma-\beta)(x-\alpha)(x-\delta)-(\alpha-\delta)(x-\beta)(x-\gamma)\\ &= (\delta-\beta)(x-\alpha)(x-\gamma)-(\alpha-\gamma)(x-\beta)(x-\delta),\\ \frac{\psi_2}{a_0} &= (\delta-\gamma)(x-\alpha)(x-\beta)-(\alpha-\beta)(x-\gamma)(x-\delta)\\ &= (\beta-\gamma)(x-\alpha)(x-\delta)-(\alpha-\delta)(x-\gamma)(x-\beta),\\ \frac{\psi_3}{a_0} &= (\beta-\delta)(x-\alpha)(x-\gamma)-(\alpha-\gamma)(x-\delta)(x-\beta)\\ &= (\gamma-\delta)(x-\alpha)(x-\beta)-(\alpha-\beta)(x-\delta)(x-\gamma).\end{aligned} \tag{6}$$

The quantities $\psi_1^2, \psi_2^2, \psi_3^2$ may be regarded as the roots of a cubic equation whose coefficients are covariants. From (4) this equation is

$$\left(z+\frac{H+4a_2'f}{3}\right)\left(z+\frac{H+4a_2''f}{3}\right)\left(z+\frac{H+4a_2'''f}{3}\right)=0.$$

By § 65, (12), and using (5) for T^2, this yields

$$z^3+Hz^2+\frac{1}{3}(H^2-16Af^2)z-T^2=0. \tag{7}$$

It remains of interest to form the discriminant of this equation. It is most simply obtained from

$$\Delta=(\psi_1^2-\psi_2^2)^2(\psi_1^2-\psi_3^2)^2(\psi_2^2-\psi_3^2)^2.$$

Substituting (4), and then replacing $(a_2'-a_2'')^2(a_2'-a_2''')^2(a_2''-a_2''')^2$ by the discriminant of § 65, (12), namely

$$27(4A^3-B^2)=3^6D,$$

where D is the discriminant of f, one obtains

$$\Delta=2^{12}Df^6. \tag{8}$$

§ 67. The full invariant system of the binary biquadratic form

We next prove that the constructions A, B exhaust the system of independent invariants of the binary biquadratic form. Usually one is content to show that every invariant is a rational integral function of A, B. As in § 63 for covariants of the cubic form, however, we shall also touch the question of numerical coefficients; here a new phenomenon appears.

The invariant D, for example, is rationally expressible by A, B through § 64, (11). But the numerical coefficients are not all integers: they have denominator 27, even though in the expanded function D all coefficients are integers.

We therefore now consider as integral invariants the integral rational homogeneous functions of degree μ in the five variables a_0, a_1, a_2, a_3, a_4,

$$I(a_0,a_1,a_2,a_3,a_4)=I(a), \tag{1}$$

whose coefficients are integers and which possess the invariant property

$$I'=r^\lambda I, \tag{2}$$

where I' or $I(a')$ is the same function of the coefficients of a transformed form.

The quantities A, B, D are such invariants, and they satisfy

$$27D = 4A^3 - B^2. \tag{3}$$

Our aim is to prove that all integral invariants are integral rational functions of these three.

If I is formed in the normal form (4) of § 65, then from (2), since $r = 1$,

$$I(a) = I(0, a_1', a_2', a_3', 0).$$

This function can depend only on $a_1'a_3'$ and a_2'; for the substitution

$$\xi' = \lambda\xi, \qquad \eta' = \lambda^{-1}\eta,$$

of determinant 1 leaves the normal form § 65, (4), and in it a_2', unchanged, while a_1', a_3' go into $\lambda^{-2}a_1', \lambda^2 a_3'$, with arbitrary λ.

Putting $a_2' = z$, we therefore have

$$I = \varphi(a_1'a_3', z), \tag{4}$$

where φ is an integral rational function of the two arguments. But by § 65, (11),

$$3a_1'a_3' = z^2 - A.$$

Multiplying (4) by a suitable power of 3, say 3^ν, gives

$$3^\nu I = \chi(A, z), \tag{5}$$

with χ again integral in A, z. Since § 65, (12) gives

$$z^3 = 3Az - B,$$

all higher powers of z can be expressed through the first and second powers. Hence

$$3^\nu I = \chi(A, B, z), \tag{6}$$

where χ is at most quadratic in z. But the left hand side remains unchanged when z is replaced by the three distinct values a_2', a_2'', a_2'''. Therefore χ cannot actually contain z (§ 30, II), and so

$$3^\nu I = \chi(A, B), \tag{7}$$

where χ is an integral rational function.

Substituting, in (7),

$$B^2 = 4A^3 - 27D,$$

we see that (7) has one of the two forms

$$3^\nu I = \Phi(A, D), \qquad 3^\nu I = B\Phi(A, D), \tag{8}$$

according as the degree of I is even or odd.

If I is a primitive function of the a, meaning that its numerical coefficients have no common divisor, then the coefficients of Φ can have no common divisor except possibly a power of 3. We must show that all coefficients are divisible by 3^ν, or equivalently that, if $\Phi(A, D)$ is assumed primitive, then $\nu = 0$.

Thus the question is: can a primitive function $\Phi(A, D)$, or $B\Phi(A, D)$, become divisible by 3 after replacing A, B, D by their expressions in the a? Since B is primitive, the theorem of § 2 implies that $B\Phi(A, D)$ can acquire the factor 3 only if $\Phi(A, D)$ does. It remains only to ask whether the primitive function $\Phi(A, D)$ can acquire the factor 3 when A, D are written out.

This is not the case. First omit from $\Phi(A, D)$ all terms whose coefficients are already divisible by 3; the remaining terms would still have to acquire the factor 3 under substitution. We may also assume that $\Phi(A, D)$ is not divisible by D, since removing such a factor would not remove the property in question. Now put all a_i except a_2 equal to zero and put $a_2 = 1$. Then $D = 0$, $A = 1$. Hence $\Phi(1, 0)$ would have to be divisible by 3. But this is precisely the coefficient of the highest power of A in $\Phi(A, D)$, and by assumption it is not divisible by 3. We have proved:

Every integral invariant of the biquadratic form can be expressed rationally and integrally in terms of A, D, B, and in one of the two forms

$$\Phi(A, D), \qquad B\Phi(A, D).$$

Conversely, it is clear that every such expression, if homogeneous in the coefficients a, represents an integral invariant.

Sixth Section.

Tschirnhausen transformation.

§ 68. Hermite's form of the Tschirnhausen transformation

The basic idea of the Tschirnhausen transformation was already introduced in the fourth section. The problem was to transform an algebraic equation of degree n,

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0, \tag{1}$$

by means of a substitution of degree $n - 1$,

$$y = \alpha_0 + \alpha_1x + \alpha_2x^2 + \cdots + \alpha_{n-1}x^{n-1}, \tag{2}$$

in order to obtain, through the arbitrary coefficients α, means for simplifying the equation.

Hermite greatly simplified this problem, and connected it with invariant theory, by giving the substitution (2) a special form. In § 4 we met a sequence of functions $f_0, f_1, f_2, \ldots, f_{n-1}$ by which the powers of x up to the $(n - 1)$st can be expressed rationally. These are the functions used by Hermite to represent the substitution (2).

Here x denotes a root of (1), while t denotes an indeterminate. By § 4,

$$x^\nu = t^{\nu-1}f_0(x) + t^{\nu-2}f_1(x) + \cdots + f_{\nu-1}(x), \tag{3}$$

and

$$\begin{aligned} f_0(x) &= a_0x + a_1, \\ f_1(x) &= a_0x^2 + a_1x + a_2, \\ &\vdots \\ f_{n-1}(x) &= a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n. \end{aligned} \tag{4}$$

We take (2) in the form

$$y = t_{n-1}f_0(x) + t_{n-2}f_1(x) + \cdots + t_1f_{n-2}(x) + t_0f_{n-1}(x), \tag{5}$$

where $t_{n-1}, t_{n-2}, \ldots, t_1, t_0$ replace the α as indeterminate quantities and are not to be confused with powers of t. The expression for y is obtained from (3) by replacing t^k by t_k.

As in § 42, let S prefixed to a function denote the sum over all roots x of (1). Then

$$S[f_0(x)] = (n - 1)a_1, \qquad S[f_1(x)] = (n - 2)a_2, \qquad \ldots, \qquad S[f_{n-1}(x)] = a_{n-1}, \tag{6}$$

and hence

$$S(y) = na_0 t_{n-1} + (n-1)a_1 t_{n-2} + \cdots + 2a_{n-2}t_1 + a_{n-1}t_0. \tag{7}$$

This is the expression obtained from $f'(t)$ by replacing t^k by t_k.

Using (7) to eliminate t_{n-1} from (5) gives

$$y - \frac{1}{n}S(y) = t_{n-2}F_0(x) + t_{n-3}F_1(x) + \cdots + t_0 F_{n-2}(x), \tag{8}$$

where

$$F_\nu(x) = f_\nu(x) - \frac{n-\nu-1}{na_0}a_{\nu+1}f_0(x). \tag{9}$$

Thus

$$S[F_0(x)] = S[F_1(x)] = \cdots = S[F_{n-2}(x)] = 0. \tag{10}$$

If

$$F(t,x) = \frac{f(t)}{t-x} - \frac{1}{n}f'(t), \tag{11}$$

then the right hand side of (8) is obtained from $F(t,x)$ by replacing t^k by t_k.

If, from the outset, y is taken in the form

$$y = t_{n-2}F_0(x) + t_{n-3}F_1(x) + \cdots + t_0 F_{n-2}(x), \tag{12}$$

then

$$S(y) = 0 \tag{13}$$

is identically satisfied. The usefulness of this form will appear from the following considerations.

§ 69. Invariant property of the Tschirnhausen transformation

We must now examine the influence of a linear transformation applied to $f(x)$ on the Tschirnhausen transformation. In $f(x)$ make the linear substitution

$$x = \frac{\alpha\xi + \beta}{\gamma\xi + \delta}, \qquad \alpha\delta - \beta\gamma = r. \tag{1}$$

Then

$$\varphi(\xi) = (\gamma\xi + \delta)^n f\left(\frac{\alpha\xi + \beta}{\gamma\xi + \delta}\right), \tag{2}$$

so that $\varphi(\xi)$ is an integral rational function of degree n, and $\varphi(\xi) = 0$ is the transform of $f(x) = 0$.

From $\varphi(\xi)$ form $H(\tau, \xi)$ exactly as $F(t,x)$ was formed from $f(x)$:

$$H(\tau, \xi) = \frac{\varphi(\tau)}{\tau - \xi} - \frac{1}{n}\varphi'(\tau). \tag{3}$$

On the other hand, in $F(t,x)$ perform, simultaneously with (1), the substitution

$$t = \frac{\alpha\tau + \beta}{\gamma\tau + \delta}. \tag{4}$$

Then

$$(\gamma\tau + \delta)^n f(t) = \varphi(\tau), \tag{5}$$

$$(\gamma\xi + \delta)(\gamma\tau + \delta)(t - x) = r(\tau - \xi), \tag{6}$$

and differentiating (5) gives

$$r(\gamma\tau + \delta)^{n-1} f'(t) = (\gamma\tau + \delta)\varphi'(\tau) - n\gamma\varphi(\tau). \tag{7}$$

From (5), (6), (7), subtraction yields

$$r(\gamma\tau+\delta)^{n-2}F(t,x) = H(\tau,\xi). \tag{8}$$

More explicitly,

$$H(\tau,\xi) = r\{(\alpha\tau+\beta)^{n-2}F_0 + (\alpha\tau+\beta)^{n-3}(\gamma\tau+\delta)F_1 + \cdots + (\gamma\tau+\delta)^{n-2}F_{n-2}\}. \tag{9}$$

Now replace in $F(t,x)$ the powers t^k by arbitrary variables t_k, and similarly replace in $H(\tau,\xi)$ the powers τ^k by τ_k:

$$Y(t,x) = t_{n-2}F_0 + t_{n-3}F_1 + \cdots + t_0F_{n-2}. \tag{10}$$

The Tschirnhausen substitutions are

$$y = Y(t,x), \qquad \eta = H(\tau,\xi). \tag{11}$$

Equation (9), as an identity in τ, remains valid when the indicated powers are expanded and then τ^k is replaced by τ_k. Hence

$$H(\tau,\xi) = rY(t,x), \tag{12}$$

provided the t are made functions of the τ as follows:

$$t_{n-2} = (\alpha\tau+\beta)^{n-2}, \quad t_{n-3} = (\alpha\tau+\beta)^{n-3}(\gamma\tau+\delta), \quad \ldots, \quad t_0 = (\gamma\tau+\delta)^{n-2}, \tag{13}$$

with τ^k replaced by τ_k after the powers have been expanded.

Let z be another variable. Before carrying out the power expansion in (13), multiply these equations in order by the coefficients of the symbolic form

$$T(z) = t_{n-2}z^{n-2} - (n-2)t_{n-3}z^{n-3} + \frac{(n-2)(n-3)}{2}t_{n-4}z^{n-4} - \cdots + t_0. \tag{14}$$

Writing

$$\Theta(z) = \tau_{n-2}z^{n-2} - (n-2)\tau_{n-3}z^{n-3} + \frac{(n-2)(n-3)}{2}\tau_{n-4}z^{n-4} - \cdots + \tau_0, \tag{15}$$

one obtains the identity

$$(\gamma z+\delta)^{n-2}T\left(\frac{\alpha z+\beta}{\gamma z+\delta}\right) = r^{n-2}\Theta(z). \tag{16}$$

Thus $r^{n-2}\Theta(z)$ is the transform of $(\gamma z+\delta)^{n-2}T(z)$ under the same linear substitution which transforms $(\gamma z+\delta)^n f(z)$ into $\varphi(z)$.

Denote by t' the coefficients of this transformed function, that is, the quantities $r^{n-2}\tau_i$. Since (12) is linear in r, it gives

$$H(t',\xi) = r^{n-1}Y(t,x). \tag{17}$$

Putting successively for x the n roots of $f(x)$ gives n functions from $Y(t,x)$. Every homogeneous rational symmetric function of these n functions is rationally expressible in terms of the coefficients of $f(x)$. If it is denoted by $K(t,a)$, then (17), with a' denoting the coefficients of $\varphi(\xi)$, yields

$$K(t',a') = r^{\nu(n-1)}K(t,a), \tag{18}$$

where ν is its degree. Hermite's theorem follows:

The coefficients of the equation obtained from $f(x) = 0$ by the Tschirnhausen transformation

$$y = Y(t,x)$$

and all symmetric functions of the n values y_i are simultaneous invariants of the two functions f and T.

§ 70. Further details on Hermite's theorem

The previous paragraph showed that, if

$$y = t_{n-2}F_0(x) + t_{n-3}F_1(x) + \cdots + t_0F_{n-2}(x), \tag{1}$$

then the symmetric functions of the n values $y_1, y_2, \ldots, y_n$, obtained by substituting the n roots of $f(x) = 0$ for x, are simultaneous invariants of two forms of degrees n and $n-2$, namely $f(z)$ and $T(z)$.

A rational homogeneous symmetric function of order ν,

$$P(y_1, y_2, \ldots, y_n),$$

is by (1) a homogeneous function of degree ν in the variables t. It is likewise homogeneous of degree ν in the coefficients a, since (1) is explicitly linear in the a and the symmetric functions of the roots x depend only on the ratios $a_1 : a_0, a_2 : a_0, \ldots, a_n : a_0$. A power of a_0 might still remain in the denominator. We now show that this does not occur.

Put

$$P(y_1, y_2, \ldots, y_n) = K(t, a)$$

and choose a power a_0^λ so that $K(t, a)$ is integral in the a and no longer vanishes identically when $a_0 = 0$. It will follow that $\lambda = 0$ if we show that, when $a_0 = 0$, all the y remain finite.

Let a_0 tend to zero, with the other coefficients fixed. One root x then becomes infinite, as seen in § 40. Nevertheless the corresponding value of y remains finite, as follows from the identity in t

$$F(t, x) = \frac{f(t)}{t - x} - \frac{1}{n}f'(t) = t^{n-2}F_0(x) + t^{n-3}F_1(x) + \cdots + F_{n-2}(x). \tag{2}$$

If $a_0 = 0$ and x becomes infinite, then $F_0(x), F_1(x), \ldots, F_{n-2}(x)$ become the coefficients of $f'(t)$, namely

$$\frac{n-1}{n}a_1, \frac{n-2}{n}a_2, \ldots, \frac{1}{n}a_{n-1}.$$

Thus all y remain finite, and $P(y_1, y_2, \ldots, y_n)$ is an integral rational invariant of degree ν both in the t and in the a.

If the equation in y is written

$$y^n + p_2y^{n-2} + p_3y^{n-3} + \cdots + p_n = 0, \tag{3}$$

then p_ν is such an invariant of degree ν. Another invariant, of degree $n(n-1)$, is the discriminant Δ of (3), the square of

$$\Pi = (y_1 - y_2)(y_1 - y_3)\cdots(y_2 - y_3)\cdots(y_{n-1} - y_n). \tag{4}$$

To understand its formation, recall that y is obtained by replacing t^k in $f(t)/(t - x)$ by t_k. Therefore $y_i - y_k$ is obtained from

$$y_i - y_k = (x_i - x_k)\left(\frac{f(t)}{(t - x_i)(t - x_k)}\right)_t, \tag{5}$$

with the same replacement. The quotient is an integral rational function of t of degree $n-2$. Let the result after replacing t^k by t_k be θ_{ik}. Then

$$y_i - y_k = (x_i - x_k)\theta_{ik}. \tag{6}$$

Thus, putting

$$\Theta = \theta_{12}\theta_{13}\cdots\theta_{1n}\theta_{23}\cdots\theta_{2n}\cdots\theta_{n-1,n}, \tag{7}$$

we obtain a decomposable form of degree $\frac{1}{2}n(n-1)$ in the variables t, rational in the coefficients a because it is symmetric in the roots x. If D is the discriminant of $f(x)$, as in § 46, (3), then

$$\Delta = D\Theta^2. \tag{8}$$

Consequently Θ is an invariant which no longer contains a_0 in the denominator. Since Δ has degree $n(n-1)$ in the coefficients a, while D has degree $2n-2$, it follows that Θ has degree $\frac{1}{2}(n-1)(n-2)$ in the coefficients a.

The form Θ is decomposable of degree $\frac{1}{2}n(n-1)$ in the variables t, since by (7) it splits into linear factors, although these factors are not rational in the a.

In § 52 we already noted that in the Tschirnhausen transformation x is rationally expressible in y; the same holds for every rational function of x. Let $\varphi(x)$ be such a function, possibly involving the coefficients a and t, but always assumed integral and homogeneous in both sets of variables. We can write

$$\varphi(x) = C_0 + C_1 y + C_2 y^2 + \cdots + C_{n-1} y^{n-1}, \tag{9}$$

where $C_0, C_1, \ldots, C_{n-1}$ are rational in a and t. Substituting $x = x_1, \ldots, x_n$ and $y = y_1, \ldots, y_n$ gives a system of linear equations for the C with determinant

$$\begin{vmatrix} 1 & y_1 & y_1^2 & \cdots & y_1^{n-1} \\ 1 & y_2 & y_2^2 & \cdots & y_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & y_n & y_n^2 & \cdots & y_n^{n-1} \end{vmatrix}, \tag{10}$$

that is, $\sqrt{\Delta}$. The numerator of C_ν is obtained from this determinant by replacing the entries of the $(\nu+1)$st column with $\varphi(x_1), \ldots, \varphi(x_n)$. When everything is expressed in terms of the x_i, this numerator has the factor $\sqrt{D}$, since it vanishes when two x_i coincide. Thus

$$C_\nu = \frac{Q_\nu}{\Theta},$$

where Q_ν is an integral rational function of the t and a, possibly still containing a power of a_0 in the denominator. Hence

$$\Theta\varphi(x) = Q_0 + Q_1 y + Q_2 y^2 + \cdots + Q_{n-1} y^{n-1}. \tag{11}$$

The coefficients Q_ν are not invariants, and in most cases their calculation is difficult.

§ 71. Transformation of the cubic equation

Hermite's theorem (§ 69) is sufficient for carrying out the transformation of the cubic equation without further calculation. Using homogeneous variables z_1, z_2, we must form simultaneous invariants of the cubic form

$$f(z_1, z_2) = a_0 z_1^3 + a_1 z_1^2 z_2 + a_2 z_1 z_2^2 + a_3 z_2^3 \tag{1}$$

and the linear form

$$T = -t_0 z_1 + t_1 z_2. \tag{2}$$

A linear substitution on T gives another linear form

$$T = -t_0' z_1' + t_1' z_2',$$

where

$$t_0' = \alpha t_0 - \gamma t_1, \qquad t_1' = -\beta t_0 + \delta t_1. \tag{3}$$

This substitution is obtained from that of the variables by replacing z_1, z_2 with t_1, t_0. Therefore every simultaneous invariant of f and T gives a covariant of f, and all such simultaneous invariants are obtained from the covariants of f by replacing z_1, z_2 with t_1, t_0. These covariants were completely determined in § 62, 63.

It is then easy to form the cubic equation obtained from the substitution § 68, (12), namely

$$y = t_1(a_0 x + \tfrac{1}{3}a_1) + t_0\left(\tfrac{2}{3}a_1 x^2 + \tfrac{1}{3}a_2 x + \tfrac{1}{3}a_3\right). \tag{5}$$

Writing the transformed equation as

$$y^3 + P_2 y + P_3 = 0,$$

we see that P_2 and P_3 are covariants of f, respectively of the second and third order in both t and a. Thus, by § 62 and 63, they can differ from $H(t_1, t_0)$ and $Q(t_1, t_0)$ only by numerical factors. A special choice, $t_1 = 1$, $t_0 = 0$, $a_0 = 1$, $a_1 = 0$, gives $y = x$, hence $P_2 = a_2$, $P_3 = a_3$; while § 62 gives $H(1, 0) = 3a_2$, $Q(1, 0) = 27a_3$. Therefore

$$P_2 = \frac{1}{3}H(t_1, t_0), \qquad P_3 = \frac{1}{27}Q(t_1, t_0),$$

and the equation for y is

$$y^3 + \frac{1}{3}H(t_1, t_0)y + \frac{1}{27}Q(t_1, t_0) = 0. \tag{6}$$

Its discriminant is, using § 62, (12),

$$\Delta = Df(t_1, t_0)^2. \tag{8}$$

This agrees with the general results of the preceding paragraph.

To base the solution of the cubic on this, one must express x rationally in y. Put

$$a_0 f(t_1, t_0)x = Q_0 + Q_1 y + Q_2 y^2. \tag{9}$$

The coefficients are found by writing this equation for the three roots and using the symmetric functions of the roots of a cubic. One obtains $Q_0 = 0$ and, after the indicated cyclic summations,

$$Q_1 = \frac{1}{2}a_0 t^2 + \frac{1}{3}a_1 t + \frac{1}{2}a_2, \qquad Q_2 = -t.$$

Putting $t_1 = t$, $t_0 = 1$, hence passing to the inhomogeneous notation, gives

$$(3a_0 x + a_1)f(t) = -tH(t) + yf'(t) - 3y^2, \tag{10}$$

and the equation for y is

$$y^3 + \frac{1}{3}H(t)y + \frac{1}{27}Q(t) = 0. \tag{11}$$

To solve the cubic, determine t from the quadratic equation $H(t) = 0$. For abbreviation put

$$\Omega = \sqrt{-3D}. \tag{12}$$

Then (7) gives

$$Q(t) = 3\Omega f(t), \tag{13}$$

and (11) gives

$$y = -\frac{1}{3}\sqrt[3]{Q(t)}. \tag{14}$$

The expression (10) then yields

$$(3a_0 x + a_1)f(t) = yf'(t) - 3y^2. \tag{15}$$

Substitution into the formulae of § 62 shows the agreement of this result with Cardano's formula. In particular, when $a_0 = 1$ and $a_1 = 0$, one obtains

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}. \tag{17}$$

§ 72. General execution of the transformation

In carrying out the Tschirnhausen transformation in Hermite's form in general, one can proceed another significant step. First consider the general substitution § 68, (5)

$$y = t_{n-1}f_0 + t_{n-2}f_1 + t_{n-3}f_2 + \cdots + t_0 f_{n-1}, \tag{1}$$

from which the more special form § 68, (12) is obtained by imposing on the variables t the single linear condition

$$S(y) = na_0 t_{n-1} + (n-1)a_1 t_{n-2} + \cdots + a_{n-1}t_0 = 0, \tag{2}$$

which, for the moment, is not yet to be imposed.

By the functions $f_0, f_1, f_2, \ldots, f_{n-1}$, all powers of x, and hence all rational functions of x, can be represented linearly and homogeneously. If we can find this representation for the various powers of y, then the equation of degree n in y can be formed by eliminating the f_i.

The same goal is reached more simply by representing the n products yf_s linearly in the f_i. If

$$yf_s = E_{0,s}f_0 + E_{1,s}f_1 + E_{2,s}f_2 + \cdots + E_{n-1,s}f_{n-1} \quad (s = 0, 1, \ldots, n-1), \tag{3}$$

then elimination of $f_0, f_1, \ldots, f_{n-1}$ gives

$$\begin{vmatrix} E_{0,0} - y & E_{1,0} & E_{2,0} & \cdots & E_{n-1,0} \\ E_{0,1} & E_{1,1} - y & E_{2,1} & \cdots & E_{n-1,1} \\ \vdots & \vdots & \vdots & & \vdots \\ E_{0,n-1} & E_{1,n-1} & E_{2,n-1} & \cdots & E_{n-1,n-1} - y \end{vmatrix} = 0. \tag{4}$$

This is an equation of degree n for y, the desired transformed equation. The problem is therefore reduced to determining the coefficients $E_{r,s}$, which are linear functions of the variables t. In particular,

$$E_{0,0} = t_{n-1}, \qquad E_{1,0} = t_{n-2}, \qquad \ldots, \qquad E_{n-1,0} = t_0. \tag{5}$$

To compute the remaining E, use the relations between the functions f following from § 68, (4):

$$\begin{aligned} xf_0 &= f_1 - a_1, \\ xf_1 &= f_2 - a_2, \\ &\vdots \\ xf_{n-2} &= f_{n-1} - a_{n-1}, \\ xf_{n-1} &= -a_n. \end{aligned} \tag{6}$$

Continue the sequence of variables $t_0, t_1, \ldots, t_{n-1}$ by defining $t_n, t_{n+1}, \ldots$ through

$$\begin{aligned} a_0 t_n + a_1 t_{n-1} + a_2 t_{n-2} + \cdots + a_n t_0 &= 0, \\ a_0 t_{n+1} + a_1 t_n + a_2 t_{n-1} + \cdots + a_n t_1 &= 0, \\ a_0 t_{n+2} + a_1 t_{n+1} + a_2 t_n + \cdots + a_n t_2 &= 0. \end{aligned} \tag{7}$$

Multiplying the equations (6) successively by $t_{n-1}, t_{n-2}, \ldots, t_0$ and adding gives

$$xy = t_n f_0 + t_{n-1} f_1 + \cdots + t_1 f_{n-1}.$$

Proceeding similarly with $t_n, t_{n-1}, \ldots, t_1$ gives

$$x^2 y = t_{n+1} f_0 + t_n f_1 + \cdots + t_2 f_{n-1}.$$

Thus

$$
\begin{aligned}
y &= t_{n-1}f_0 + t_{n-2}f_1 + \cdots + t_0 f_{n-1}, \\
xy &= t_n f_0 + t_{n-1}f_1 + \cdots + t_1 f_{n-1}, \\
x^2 y &= t_{n+1}f_0 + t_n f_1 + \cdots + t_2 f_{n-1}, \\
&\;\vdots \\
x^s y &= t_{n+s-1}f_0 + t_{n+s-2}f_1 + \cdots + t_s f_{n-1}.
\end{aligned}
\tag{8}
$$

Multiplying these equations by $a_s, a_{s-1}, \dots, a_0$ and adding yields

$$
yf_s = E_{0,s}f_0 + E_{1,s}f_1 + \cdots + E_{n-1,s}f_{n-1},
$$

where, by (7),

$$
\begin{aligned}
E_{0,s} &= a_0 t_{n+s-1} + a_1 t_{n+s-2} + \cdots + a_s t_{n-1} \\
&= -a_{s+1}t_{n-2} - \cdots - a_n t_{s-1}, \\
E_{1,s} &= a_0 t_{n+s-2} + a_1 t_{n+s-3} + \cdots + a_s t_{n-2} \\
&= -a_{s+1}t_{n-3} - \cdots - a_n t_{s-2}, \\
&\;\vdots \\
E_{s-1,s} &= a_0 t_n + a_1 t_{n-1} + \cdots + a_s t_{n-s} \\
&= -a_{s+1}t_{n-s-1} - \cdots - a_n t_0, \\
E_{s,s} &= a_0 t_{n-1} + a_1 t_{n-2} + \cdots + a_s t_{n-s-1}, \\
E_{s+1,s} &= a_0 t_{n-2} + a_1 t_{n-3} + \cdots + a_s t_{n-s-2}, \\
&\;\vdots \\
E_{n-1,s} &= a_0 t_s + a_1 t_{s-1} + \cdots + a_s t_0.
\end{aligned}
\tag{9}
$$

It should still be noted that the superfluous variables $t_n, t_{n+1}, \dots$ introduced above have already been eliminated again in the second form of $E_{0,s}, E_{1,s}, \dots, E_{s-1,s}$, and that the variable t_{n-1} occurs only in $E_{s,s}$.

§ 73. The Bezoutian

When equation (4) of the preceding paragraph is expanded it has the form

$$
y^n + P_1 y^{n-1} + P_2 y^{n-2} + \cdots + P_n = 0,
$$

or, if t_{n-1} is chosen so that $S(y) = 0$,

$$
y^n + P_2 y^{n-2} + \cdots + P_n = 0. \tag{1}
$$

Then

$$
-2P_2 = S(y^2)
$$

is a function of the second degree in the t's and also in the a's. This function is especially important for many applications and has received from Sylvester the name Bezoutian of the function $f(x)$, in honour of the French mathematician Bezout, who already in the preceding century gave the first correct developments of elimination.

By the formulas of the preceding paragraph this function can be computed rather simply. Beside the variables $t_{n-1}, t_{n-2}, \dots, t_0$ we introduce a second independent system of variables $\tau_{n-1}, \tau_{n-2}, \dots, \tau_0$ and put

$$
\begin{aligned}
y &= t_{n-1}f_0 + t_{n-2}f_1 + \cdots + t_0 f_{n-1}, \\
z &= \tau_{n-1}f_0 + \tau_{n-2}f_1 + \cdots + \tau_0 f_{n-1}.
\end{aligned}
\tag{2}
$$

Instead of forming $S(y^2)$ at once, we first compute $S(yz)$; $S(y^2)$ is then obtained by putting the τ's equal to the corresponding t's.

Multiplying the formula

$$yf_s = E_{0,s}f_0 + E_{1,s}f_1 + \cdots + E_{n-1,s}f_{n-1} \tag{3}$$

of the preceding paragraph by τ_{n-s-1} and summing with respect to s from 0 to $n-1$, one obtains

$$yz = f_0 \sum_{s=0}^{n-1} E_{0,s}\tau_{n-s-1} + f_1 \sum_{s=0}^{n-1} E_{1,s}\tau_{n-s-1} + \cdots + f_{n-1} \sum_{s=0}^{n-1} E_{n-1,s}\tau_{n-s-1}. \tag{4}$$

Now by § 68, (6)

$$Sf_0 = na_0, \qquad Sf_1 = (n-1)a_1, \qquad \ldots, \qquad Sf_{n-1} = a_{n-1},$$

and therefore

$$\begin{aligned} S(yz) = na_0 \sum_{s=0}^{n-1} E_{0,s}\tau_{n-s-1} + (n-1)a_1 \sum_{s=0}^{n-1} E_{1,s}\tau_{n-s-1} \\ + \cdots + a_{n-1} \sum_{s=0}^{n-1} E_{n-1,s}\tau_{n-s-1}. \end{aligned}$$

In this expression the coefficient of τ_{n-1} is

$$na_0E_{0,0} + (n-1)a_1E_{1,0} + \cdots + a_{n-1}E_{n-1,0},$$

and hence, by (3), is

$$a_0S(y).$$

Since $S(z)$ begins with the term $na_0\tau_{n-1}$, the difference

$$S(yz) - \frac{1}{n}S(y)S(z) \tag{5}$$

contains no term multiplied by τ_{n-1}. Since this expression is also symmetric in t and τ, it also contains no t_{n-1}; in forming it we may therefore simply put t_{n-1} and τ_{n-1} equal to zero.

Put

$$S(y) = na_0t_{n-1} + \varphi(t),$$

where

$$\varphi(t) = (n-1)a_1t_{n-2} + (n-2)a_2t_{n-3} + \cdots + a_{n-1}t_0. \tag{6}$$

Then

$$\begin{aligned} S(yz) - \frac{1}{n}S(y)S(z) = na_0 \sum_{s=1}^{n-1} E_{0,s}\tau_{n-s-1} + (n-1)a_1 \sum_{s=1}^{n-1} E_{1,s}\tau_{n-s-1} \\ + \cdots + a_{n-1} \sum_{s=1}^{n-1} E_{n-1,s}\tau_{n-s-1} - \frac{1}{n}\varphi(t)\varphi(\tau). \end{aligned} \tag{7}$$

Here it is to be observed that $t_{n-1} = 0$ is to be assumed; thus

$$E_{s,s} = a_1t_{n-2} + \cdots + a_st_{n-s-1}$$

is to be put, while the remaining E's are determined by the formulas (9) of the preceding paragraph.

The expression on the right of (7) is a bilinear function of t and τ. Let its expanded form be

$$\sum_{h=0}^{n-2}\sum_{k=0}^{n-2} B_{h,k}t_h\tau_k,$$

with $B_{i,k} = B_{k,i}$. The Bezoutian is then obtained by putting $\tau = t$, namely

$$B = \sum_{h=0}^{n-2} \sum_{k=0}^{n-2} B_{h,k} t_h t_k, \tag{8}$$

where the coefficients $B_{i,k}$ are quadratic functions of the a's. From (7) one obtains

$$S(yz) - \frac{1}{n} S(y)S(z) = \sum_{h=0}^{n-2} \sum_{k=0}^{n-2} B_{h,k} t_h \tau_k, \tag{9}$$

$$S(y^2) = \frac{1}{n}[S(y)]^2 + B. \tag{10}$$

By formula (8) of § 68,

$$\begin{aligned} y - \frac{1}{n} S(y) &= t_{n-2} F_0 + t_{n-3} F_1 + \cdots + t_0 F_{n-2}, \\ z - \frac{1}{n} S(z) &= \tau_{n-2} F_0 + \tau_{n-3} F_1 + \cdots + \tau_0 F_{n-2}. \end{aligned}$$

Multiplying these two expressions gives

$$yz - \frac{1}{n}[yS(z) + zS(y)] + \frac{1}{n^2} S(y)S(z) = \sum_{h,k=0}^{n-2} \tau_h t_k F_{n-h-2} F_{n-k-2}.$$

Summing this formula over all roots x, that is, taking the sum denoted by S, gives

$$S(yz) - \frac{1}{n} S(y)S(z) = \sum_{h,k=0}^{n-2} \tau_h t_k S(F_{n-h-2} F_{n-k-2}).$$

Comparison with (9) yields

$$B_{h,k} = S(F_{n-h-2} F_{n-k-2}). \tag{11}$$

Let Δ denote the determinant of the function B:

$$\Delta = \sum \pm B_{0,0} B_{1,1} \cdots B_{n-2,n-2}.$$

By the multiplication theorem for determinants, and using

$$SF_0 = 0, \qquad SF_1 = 0, \qquad \ldots, \qquad SF_{n-2} = 0,$$

if

$$\Phi = \begin{vmatrix} 1 & F_0(x_1) & F_1(x_1) & \cdots & F_{n-2}(x_1) \\ 1 & F_0(x_2) & F_1(x_2) & \cdots & F_{n-2}(x_2) \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & F_0(x_n) & F_1(x_n) & \cdots & F_{n-2}(x_n) \end{vmatrix},$$

then (11) gives

$$\Phi^2 = n\Delta. \tag{12}$$

Now using the expressions (9) of § 68 for $F_0, F_1, \ldots, F_{n-2}$, and the theorem that in a determinant one may add to one column another column multiplied by an arbitrary factor (§ 22, VII), Φ is transformed into

$$a_0^{n-1} \begin{vmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{vmatrix},$$

the square of which is, by § 46, the discriminant of $f(x)$. Thus (12) gives the important theorem:

The determinant of the Bezoutian of $f(x)$ is the nth part of the discriminant of $f(x)$.

By these formulas the computation of the Bezoutian is no longer difficult and is quite simple. For $n = 3$ one obtains the result already to be inferred from the formulas of § 71,

$$B = -\frac{2}{3} H(t_1, t_0). \tag{13}$$

To illustrate the preceding discussion, we write out the complete formula system for the computation of the Bezoutian of the biquadratic form, leaving to the reader the execution of the only slightly longer calculation for the form of the fifth degree. For the form of the fourth degree we have, from (7) and § 72, (9),

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4,$$

$$\begin{array}{ll|l}
E_{0,1} = & -a_2t_2 - a_3t_1 - a_4t_0 & 4a_0t_2 \\
E_{1,1} = & a_1t_2 & 3a_1t_2 \\
E_{2,1} = & a_0t_2 + a_1t_1 & 2a_2t_2 \\
E_{3,1} = & a_0t_1 + a_1t_0 & a_3t_2 \\
\hline
E_{0,2} = & -a_3t_2 - a_4t_1 & 4a_0t_1 \\
E_{1,2} = & -a_3t_1 - a_4t_0 & 3a_1t_1 \\
E_{2,2} = & a_1t_2 + a_2t_1 & 2a_2t_1 \\
E_{3,2} = & a_0t_2 + a_1t_1 + a_2t_0 & a_3t_1 \\
\hline
E_{0,3} = & -a_4t_2 & 4a_0t_0 \\
E_{1,3} = & -a_4t_1 & 3a_1t_0 \\
E_{2,3} = & -a_4t_0 & 2a_2t_0 \\
E_{3,3} = & a_1t_2 + a_2t_1 + a_3t_0 & a_3t_0.
\end{array}$$

These expressions are to be multiplied by the factors written to their right and added; then one must add

$$-\frac{1}{4}\varphi(t)\varphi(\tau) = -\frac{1}{4}(3a_1t_2 + 2a_2t_1 + a_3t_0)(3a_1\tau_2 + 2a_2\tau_1 + a_3\tau_0),$$

and collect the coefficients of $\tau_h t_k$. This gives

$$\begin{aligned}
B_{2,2} &= -\frac{1}{4}(8a_0a_2 - 3a_1^2), & B_{0,1} &= -\frac{1}{2}(6a_1a_4 - a_2a_3), \\
B_{1,1} &= -4a_0a_4 - 2a_1a_3 + a_2^2, & B_{1,2} &= -\frac{1}{2}(6a_0a_3 - a_1a_2), \\
B_{0,0} &= -\frac{1}{4}(8a_2a_4 - 3a_3^2), & B_{0,2} &= -\frac{1}{4}(16a_0a_4 - a_1a_3).
\end{aligned} \tag{14}$$

In the same way one may proceed with a form of the fifth degree, and obtains for the coefficients of the Bezoutian

$$\begin{aligned}
& & 5B_{0,1} &= 3a_3a_4 - 15a_2a_5, \\
5B_{0,0} &= 4a_4^2 - 10a_3a_5, & 5B_{0,2} &= 2a_2a_4 - 20a_1a_5, \\
5B_{1,1} &= 6a_3^2 - 10a_2a_4 - 20a_1a_5, & 5B_{0,3} &= a_1a_4 - 25a_0a_5, \\
5B_{2,2} &= 6a_2^2 - 10a_1a_3 - 20a_0a_4, & 5B_{1,2} &= 4a_2a_3 - 15a_1a_4 - 25a_0a_5, \\
5B_{3,3} &= 4a_1^2 - 10a_0a_2, & 5B_{1,3} &= 2a_1a_3 - 20a_0a_4, \\
& & 5B_{2,3} &= 3a_1a_2 - 15a_0a_3.
\end{aligned}$$

§ 74. Transformation of the equation of the fifth degree

These developments will now be applied in order to carry out the transformation of the equation of the fifth degree as it was sketched in § 54.

In infinitely many ways a system of values t_0, t_1, t_2, t_3 can be so determined that the Bezoutian B vanishes. In general this requires the solution of a quadratic equation. For example, we may put

$$t_2 = 0, \qquad t_3 = 0$$

and determine the ratio $t_0 : t_1$ from the quadratic equation

$$B_{0,0}t_0^2 + 2B_{0,1}t_0t_1 + B_{1,1}t_1^2 = 0.$$

These values of t then contain the square root

$$\sqrt{B_{0,1}^2 - B_{0,0}B_{1,1}}.$$

Instead one can make other determinations and obtains other square roots. In the choice of this square root there is something arbitrary and indeterminate.

Among these various possible determinations we choose one and thereby transform the given equation of the fifth degree into a principal equation, that is, into one in which the third and fourth powers of the unknown do not occur; or we suppose, following Klein's procedure, that the equation to be transformed is already from the outset a principal equation

$$f(x) = a_0x^5 + a_3x^2 + a_4x + a_5 = 0. \tag{1}$$

We must, however, expressly emphasize that by § 70, (8), under this preliminary Tschirnhausen transformation the discriminant of the given equation is changed only by a factor Θ^2, where Θ is rational in the applied t's and hence also in the square root used in determining them.

Under the supposition (1), the coefficients of the Bezoutian are, by the formulas of the preceding paragraph,

$$\begin{array}{ll} 5B_{0,0} = 4a_4^2 - 10a_3a_5, & 5B_{0,1} = 3a_3a_4, \\ 5B_{1,1} = 6a_3^2, & 5B_{0,2} = 0, \\ 5B_{2,2} = -20a_0a_4, & 5B_{0,3} = -25a_0a_5, \\ 5B_{3,3} = 0, & 5B_{1,2} = -25a_0a_5, \\ 5B_{1,3} = -20a_0a_4, & 5B_{2,3} = -15a_0a_3. \end{array} \tag{2}$$

Computing the determinant from these entries, one obtains without difficulty, by § 73, (12), the discriminant of the principal equation of the fifth degree:

$$\begin{aligned} D = a_0^5(5^5a_0^3a_5^4 + 2^8a_0a_4^5 + 2250a_0a_3^2a_4a_5^2 \\ - 1600a_0a_3a_4^3a_5 + 108a_3^5a_5 - 27a_3^4a_4^2). \end{aligned} \tag{3}$$

Since here $B_{3,3} = 0$, we already have a system of values of the t for which the Bezoutian vanishes, namely

$$t_0 = 0, \qquad t_1 = 0, \qquad t_2 = 0.$$

We now take our Tschirnhausen substitution in the form [§ 68, (12)]

$$y = t_3F_0 + t_2(\alpha_2F_1 + \alpha_1F_2 + \alpha_0F_3), \tag{4}$$

and choose $\alpha_0, \alpha_1, \alpha_2$ so that the equation of the fifth degree for y becomes, independently of t_3, t_2, a principal equation, that is, so that $S(y^2)$ vanishes identically for all t_2, t_3.

Since $S(F_0^2) = B_{3,3}$ vanishes here, the requirement is satisfied, by § 73, (11), if $\alpha_0, \alpha_1, \alpha_2$ are chosen so that

$$\alpha_0^2B_{0,0} + \alpha_1^2B_{1,1} + \alpha_2^2B_{2,2} + 2\alpha_0\alpha_1B_{0,1} + 2\alpha_0\alpha_2B_{0,2} + 2\alpha_1\alpha_2B_{1,2} = 0, \tag{5}$$

$$\alpha_0B_{0,3} + \alpha_1B_{1,3} + \alpha_2B_{2,3} = 0. \tag{6}$$

By eliminating one of the three quantities α, these equations lead to a quadratic equation for the ratio of the other two. In itself there is no need to exclude any special case, not even the case in which $B_{0,3}, B_{1,3}, B_{2,3}$ all vanish; one would then merely have to choose any solution of equation (5) for the α's.

In order not to be prolix, however, we assume that $B_{0,3}$ is non-zero. The cases thereby excluded, in which a root of the given equation becomes zero or infinite, are of no real interest here, since they reduce to an equation of lower degree. The treatment remains the same if one assumes instead that another of $B_{0,3}, B_{1,3}, B_{2,3}$ is non-zero.

To treat equations (5), (6) symmetrically, and especially to recognize which square root is required for their solution, we proceed as follows. Put, for abbreviation,

$$\begin{aligned} B_0 &= \alpha_0 B_{0,0} + \alpha_1 B_{0,1} + \alpha_2 B_{0,2}, \\ B_1 &= \alpha_0 B_{1,0} + \alpha_1 B_{1,1} + \alpha_2 B_{1,2}, \\ B_2 &= \alpha_0 B_{2,0} + \alpha_1 B_{2,1} + \alpha_2 B_{2,2}. \end{aligned} \tag{7}$$

Then equations (5) and (6) can be written as

$$\begin{aligned} \alpha_0 B_0 + \alpha_1 B_1 + \alpha_2 B_2 &= 0, \\ \alpha_0 B_{0,3} + \alpha_1 B_{1,3} + \alpha_2 B_{2,3} &= 0. \end{aligned} \tag{8}$$

We multiply the second equation by an undetermined factor α_3 and add it to the first, obtaining

$$\alpha_0(B_0 + \alpha_3 B_{0,3}) + \alpha_1(B_1 + \alpha_3 B_{1,3}) + \alpha_2(B_2 + \alpha_3 B_{2,3}) = 0. \tag{9}$$

This equation is satisfied if we put

$$B_0 + \alpha_3 B_{0,3} = 0, \qquad B_1 + \alpha_3 B_{1,3} = \sigma\alpha_2, \qquad B_2 + \alpha_3 B_{2,3} = -\sigma\alpha_1. \tag{10}$$

From these three equations, together with (6), the unknowns $\alpha_0, \alpha_1, \alpha_2, \alpha_3, \sigma$ are to be determined. Written out, the equations are

$$\begin{aligned} \alpha_0 B_{0,0} + \alpha_1 B_{0,1} + \alpha_2 B_{0,2} + \alpha_3 B_{0,3} &= 0, \\ \alpha_0 B_{1,0} + \alpha_1 B_{1,1} + \alpha_2 (B_{1,2} - \sigma) + \alpha_3 B_{1,3} &= 0, \\ \alpha_0 B_{2,0} + \alpha_1 (B_{2,1} + \sigma) + \alpha_2 B_{2,2} + \alpha_3 B_{2,3} &= 0, \\ \alpha_0 B_{3,0} + \alpha_1 B_{3,1} + \alpha_2 B_{3,2} &= 0. \end{aligned} \tag{11}$$

Eliminating $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ gives a quadratic equation for σ; after solving it the ratios $\alpha_0 : \alpha_1 : \alpha_2 : \alpha_3$ can be determined from linear equations.

The quadratic equation for σ is, in determinant form,

$$\begin{vmatrix} B_{0,0} & B_{0,1} & B_{0,2} & B_{0,3} \\ B_{1,0} & B_{1,1} & B_{1,2} - \sigma & B_{1,3} \\ B_{2,0} & B_{2,1} + \sigma & B_{2,2} & B_{2,3} \\ B_{3,0} & B_{3,1} & B_{3,2} & 0 \end{vmatrix} = 0.$$

Since the left hand side is unchanged when σ is replaced by $-\sigma$, it is a pure quadratic equation, and by § 73, (12) it gives

$$B_{0,3}^2 \sigma^2 - 5D = 0,$$

where D is the discriminant of the given equation. Hence by (2) we obtain

$$5 a_0 a_5 \sigma = \sqrt{5D}, \tag{12}$$

with either choice of sign.

§ 75. Normal form of the equation of the fifth degree

As was remarked earlier, the principal aim of these considerations is to produce a normal form of the equation of the fifth degree which depends on only one undetermined coefficient, one parameter. One such normal form is the Bring-Jerrard form. To obtain it, according to (4) of the preceding paragraph one must put

$$y = t_3 F_0 + t_2(\alpha_2 F_1 + \alpha_1 F_2 + \alpha_0 F_3), \tag{1}$$

so that identically

$$S(y) = 0, \qquad S(y^2) = 0,$$

and then determine the ratio $t_3 : t_2$ from the cubic equation

$$S(y^3) = 0.$$

This cubic equation can indeed be formed, although its expression is long. The formulas required for it were computed by Cayley.

It is the result of a remarkable investigation of Gordan that another normal form, the Brioschi normal form, can be obtained without any new irrationality. We shall derive this result at the close of the discussion of the Tschirnhausen transformation.

We retain the assumptions of the preceding paragraph and put, for the moment,

$$u = F_0, \qquad v = \alpha_2 F_1 + \alpha_1 F_2 + \alpha_0 F_3, \tag{2}$$

where $\alpha_0, \alpha_1, \alpha_2$ are to have the values determined in the preceding paragraph.

By formula § 73, (4), the three functions

$$u^2, \qquad uv, \qquad v^2$$

can be represented linearly and homogeneously in terms of f_0, f_1, f_2, f_3, f_4; and since

$$S(u^2) = 0, \qquad S(uv) = 0, \qquad S(v^2) = 0, \tag{3}$$

these expressions are also linear and homogeneous in F_0, F_1, F_2, F_3.

The calculation is easily carried out from the formulas of §§ 72, 73, but will not be pursued here, since only the underlying idea is needed. We record only that the coefficients in the expressions for u^2, uv, v^2 are linear in the coefficients of the original equation of the fifth degree and quadratic in $\alpha_0, \alpha_1, \alpha_2$. If we now eliminate F_0, F_1, F_2, F_3 from these expressions using (2), we obtain a relation of the form

$$pu^2 + 2quv + rv^2 = au + bv, \tag{4}$$

where p, q, r, a, b depend rationally on the coefficients a_i and α_i, and in any case do not all vanish simultaneously.

For the moment put

$$pu^2 + 2quv + rv^2 = \varphi(u, v).$$

Then by § 57,

$$4\varphi(u', v')\varphi(u, v) - [u'\varphi'(u) + v'\varphi'(v)]^2$$

is the square of a linear function of u, v, so that $\varphi(u, v)$ is thereby decomposed into the sum of two squares. Choosing $u' = b$, $v' = -a$, one obtains

$$\varphi(b, -a)\varphi(u, v) = [b(pu + qv) - a(qu + rv)]^2 + (pr - q^2)(au + bv)^2.$$

Thus, with the abbreviations

$$\begin{aligned} pb^2 - 2qab + ra^2 &= m, \\ bp - aq &= a', \\ bq - ar &= b', \\ q^2 - pr &= c, \end{aligned} \tag{5}$$

we get

$$m(pu^2 + 2quv + rv^2) = (a'u + b'v)^2 - c(au + bv)^2. \tag{6}$$

By (4) this equation may also be written as

$$m(au + bv) = (a'u + b'v)^2 - c(au + bv)^2. \tag{7}$$

Now put

$$y = \frac{a'u + b'v}{au + bv}, \tag{8}$$

and form the equation of the fifth degree whose roots are the five values of y:

$$y^5 + c_1 y^4 + c_2 y^3 + c_3 y^2 + c_4 y + c_5 = 0. \tag{9}$$

The coefficients c can then be further determined as follows.

From (8) one obtains

$$y + \sqrt{c} = \frac{a'u + b'v + \sqrt{c}(au + bv)}{au + bv},$$

and hence, by (7),

$$y + \sqrt{c} = \frac{m}{a'u + b'v - \sqrt{c}(au + bv)}, \qquad \frac{m}{y + \sqrt{c}} = a'u + b'v - \sqrt{c}(au + bv).$$

It follows from (3) that

$$S\left(\frac{1}{y + \sqrt{c}}\right) = 0, \qquad S\left(\frac{1}{(y + \sqrt{c})^2}\right) = 0.$$

If from (9) we derive the equation whose roots are the values

$$\xi = -\frac{1}{y + \sqrt{c}},$$

that is, if we put

$$y = \frac{1 - \xi\sqrt{c}}{\xi},$$

then the resulting equation in ξ must have vanishing coefficients of ξ^4 and ξ^3, whichever sign is assigned to the square root $\sqrt{c}$. This yields four equations among the coefficients c_ν of (9).

The equation in ξ is

$$\begin{aligned}(1 - \xi\sqrt{c})^5 + c_1\xi(1 - \xi\sqrt{c})^4 + c_2\xi^2(1 - \xi\sqrt{c})^3 \\ + c_3\xi^3(1 - \xi\sqrt{c})^2 + c_4\xi^4(1 - \xi\sqrt{c}) + c_5\xi^5 = 0.\end{aligned}$$

Setting the coefficients of ξ^4 and ξ^3 equal to zero gives

$$\begin{aligned}5\sqrt{c}^4 - 4c_1\sqrt{c}^3 + 3c_2\sqrt{c}^2 - 2c_3\sqrt{c} + c_4 = 0, \\ -10\sqrt{c}^3 + 6c_1\sqrt{c}^2 - 3c_2\sqrt{c} + c_3 = 0.\end{aligned}$$

Because of the two possible signs of $\sqrt{c}$, these equations split into the four equations

$$\begin{aligned}5c^2 + 3c_2 c + c_4 = 0, \\ 4c_1 c + 2c_3 = 0, \\ 10c + 3c_2 = 0, \\ 6c_1 c + c_3 = 0.\end{aligned}$$

The second and fourth give

$$c_1 = 0, \qquad c_3 = 0,$$

and then the third and first give

$$c_2 = -\frac{10}{3}c, \qquad c_4 = -5c^2.$$

Thus the equation for y assumes the form

$$y^5 - \frac{10}{3}cy^3 + 5c^2y + c_5 = 0. \tag{10}$$

This equation still depends on the two parameters c_5 and c. It can be reduced to an equation with one parameter by the substitution

$$y = \sqrt{\frac{c}{3}}\, z, \qquad \gamma = c_5\sqrt{\frac{3^5}{c^5}}, \tag{11}$$

which gives the simple and elegant form

$$z^5 - 10z^3 + 45z + \gamma = 0. \tag{12}$$

This is the Brioschi normal form. The substitution (11), however, has the disadvantage that it still contains a square root, whereas in formulas (8) and (10) only the two square roots discussed in the preceding paragraph occur. But, by a remark of Klein, one can also pass rationally from (10) to a normal form containing only one parameter.

For example, putting

$$y = \frac{3c_5}{c^2}\frac{1}{z}, \qquad \gamma = \frac{9c_5^2}{c^5}, \tag{13}$$

one obtains from (10) the equation

$$z^5 + 15z^4 - 10\gamma z^2 + 3\gamma^2 = 0. \tag{14}$$

In this transformation it has been tacitly assumed that the quantities denoted in (5) by m and c are not zero.

If $m = 0$, then in formula (4) the expression $pu^2 + 2quv + rv^2$ is divisible by $au + bv$; hence at least for some of the roots the equation

$$au + bv = 0$$

must hold, that is, an equation of degree not higher than the fourth.

If $c = 0$, so that $pu^2 + 2quv + rv^2$ is the square of a linear function, then (7) shows that, on putting

$$y = a'u + b'v,$$

one has

$$S(y) = 0, \qquad S(y^2) = 0, \qquad S(y^3) = 0,$$

and therefore the Bring-Jerrard form can be produced rationally.

We may finally summarize the results of these considerations as follows:

The principal equation of the fifth degree can be transformed into a normal form with one parameter by a transformation whose only irrationality is the square root of the discriminant.

BOOK TWO.

THE ROOTS.

Seventh Section.

Reality of the Roots.

§ 76. Generalities on the reality of roots of equations and on the discriminant

In this section we shall occupy ourselves with the question how many roots of an algebraic equation are real. The coefficients of the equation are assumed here to be real numbers. We shall call such equations, for short, real equations, and begin with some general considerations.

The real equation of degree n

$$f(x) = 0$$

has, as we saw in the third section, n roots, which are either real or imaginary, that is, of the form $\xi + \mathrm{i}\eta$, with real ξ, η. The number n of roots is obtained in general only when, in certain circumstances, one counts a root repeatedly, namely $(m+1)$ times when, together with $f(x)$, the first m derivatives of $f(x)$ vanish. Since a complex expression of the form $X + \mathrm{i}Y$ is equal to zero only when the two real components X and Y vanish separately, so that with $X + \mathrm{i}Y$ also $X - \mathrm{i}Y$ vanishes, and since with real coefficients, if $f(\xi + \mathrm{i}\eta) = X + \mathrm{i}Y$, one obtains $f(\xi - \mathrm{i}\eta) = X - \mathrm{i}Y$, it follows from $f(\xi + \mathrm{i}\eta) = 0$ that also $f(\xi - \mathrm{i}\eta) = 0$. Thus to every imaginary root $\xi + \mathrm{i}\eta$ there belongs a second imaginary root, distinct from it, $\xi - \mathrm{i}\eta$, that is, the conjugate imaginary root. Since with the derivatives $f'(\xi + \mathrm{i}\eta), f''(\xi + \mathrm{i}\eta), \dots$ the conjugate quantities $f'(\xi - \mathrm{i}\eta), f''(\xi - \mathrm{i}\eta), \dots$ also vanish, conjugate imaginary roots have the same multiplicity.

It follows that the imaginary roots always occur in an even number, and that an equation of odd degree must always have at least one real root.

Our next aim will be to determine the number of real roots, without solving the equation, directly from the values of certain rational functions of the coefficients of the equation.

A very important role is played here by the discriminant, and we must first discuss its significance for our question.

In § 46 we defined the discriminant as the product of the squares of all differences of roots,

$$(x_1 - x_2)^2 (x_1 - x_3)^2 \cdots (x_{n-1} - x_n)^2,$$

multiplied further by a_0^{2n-2}, when a_0 is the coefficient of the highest power of the unknown in the given equation. We showed there how the discriminant can be computed as a rational function of the coefficients. The factor a_0^{2n-2} is always positive and could here, without essential restriction of generality, be assumed equal to 1. As before, we shall use the letter D for the discriminant.

The discriminant vanishes if and only if two of the roots are equal.

Suppose that $f(x)$ has been freed of repeated factors by separating off the greatest common divisor of $f(x)$ and $f'(x)$, which is done by rational calculation. Then D will not vanish.

If x_1 and x_2 are real, then $(x_1 - x_2)^2$ is positive. If x_1 is real and x_2 imaginary, then there is a root x_2' conjugate to x_2, and the product

$$(x_1 - x_2)(x_1 - x_2')$$

is positive as the product of two conjugate imaginary quantities, and therefore its square is positive as well.

If x_1 and x_2 are both imaginary but not conjugate, then the product

$$(x_1 - x_2)(x_1' - x_2')$$

and its square are positive.

Finally, if x_1 and x_2 are conjugate imaginary roots, then their difference is purely imaginary and its square is negative.

Thus in the product by which we have defined the discriminant there occur exactly as many negative factors as there are pairs of conjugate imaginary roots. We conclude from this the important fundamental theorem:

If $f(x) = 0$ is a real equation without repeated roots, then the discriminant is positive or negative according as the number of pairs of conjugate imaginary roots is even or odd.

The case in which only real roots are present belongs to those in which the discriminant is positive.

For quadratic and cubic equations, in which only one pair of imaginary roots can occur, this main theorem already completes the distinction of the possible cases with respect to reality of the roots, that is, the determination.

§ 77. Discussion of the quadratic and cubic equations

For the quadratic equation

$$a_0x^2 + a_1x + a_2 = 0$$

we have (§ 46)

$$D = a_1^2 - 4a_0a_2 \begin{cases} > 0, & \text{real roots,} \\ < 0, & \text{imaginary roots.} \end{cases}$$

For the cubic equation

$$a_0x^3 + a_1x^2 + a_2x + a_3 = 0$$

one has

$$D = a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2,$$

and

$$\begin{array}{ll} D > 0, & \text{three real roots,} \\ D < 0, & \text{one real and two imaginary roots.} \end{array}$$

The case $D = 0$ also gives no further distinctions here; for then the two roots of the quadratic equation are equal and real, and of the three roots of the cubic equation two are equal and all three are real. Indeed, an imaginary root cannot occur here twice, since then the conjugate root would also occur twice; nor can the single root be imaginary, since then a second would have to be present.

For the cubic equation it remains only to find the condition that all three roots be equal.

In that case the left hand side of the cubic equation must be a complete cube, hence

$$a_0x^3 + a_1x^2 + a_2x + a_3 = a_0(x - \alpha)^3.$$

Comparison of the two sides of this identity gives

$$a_1 = -3a_0\alpha, \qquad a_2 = 3a_0\alpha^2, \qquad a_3 = -a_0\alpha^3,$$

from which, by eliminating α, one obtains

$$a_1^2 - 3a_0a_2 = 0, \qquad a_1a_2 - 9a_0a_3 = 0. \tag{1}$$

As a consequence, multiplying the first of these equations by a_2 and the second by a_1 and subtracting, one obtains

$$a_2^2 - 3a_1a_3 = 0;$$

and if these conditions are fulfilled, then conversely

$$f(x) = a_0\left(x + \frac{a_1}{3a_0}\right)^3,$$

which expresses the equality of all three roots.

For the cubic equation

$$x^3 + a_1x^2 + a_2x + a_3 = 0$$

we can decide completely not only on the reality but also on the signs of the roots.

Let α, β, γ be the three roots. Then

$$a_1 = -(\alpha + \beta + \gamma), \qquad a_2 = \alpha\beta + \alpha\gamma + \beta\gamma, \qquad a_3 = -\alpha\beta\gamma. \tag{2}$$

If now $D < 0$, so that only one root, say α, is real, then $\beta\gamma$ is positive and α is negative or positive according as a_3 is positive or negative. If, however, D is positive, so that all three roots are real, then if α, β, γ are positive, (2) gives

$$a_1 < 0, \qquad a_2 > 0, \qquad a_3 < 0. \tag{3}$$

These conditions are necessary for all three roots to be positive. They are also sufficient. For if one or three roots are negative, then $a_3 > 0$. But if two roots, say β, γ, are negative, then either

$$a_1 \geq 0 \qquad \text{or} \qquad \alpha > -(\beta + \gamma).$$

In the latter case one has

$$a_2 < \beta\gamma - (\beta + \gamma)^2 = -\beta^2 - \beta\gamma - \gamma^2,$$

so that a_2 is negative.

If finally one root is equal to 0, then necessarily a_3 vanishes.

Replacing x by $-x$, one concludes that for three negative roots the necessary and sufficient condition is

$$a_1 > 0, \qquad a_2 > 0, \qquad a_3 > 0. \tag{4}$$

Under the assumption of a positive discriminant we therefore obtain the following table:

a_1	a_2	a_3	number of positive roots
$-$	$+$	$-$	3
$-$	$-$	$-$	1
$+$	$+$	$-$	1
$+$	$-$	$-$	1
$+$	$+$	$+$	0
$-$	$-$	$+$	2
$-$	$+$	$+$	2
$+$	$-$	$+$	2

where the last column gives the number of positive roots. We can state the result of this consideration as follows:

For positive discriminant, the number of positive roots of the cubic equation is equal to the number of variations of sign in the sequence

$$1, a_1, a_2, a_3,$$

where by a variation of sign we mean the succession of a positive and a negative quantity, or of a negative and a positive quantity.

If one of the two quantities a_1, a_2 vanishes, then, since not all roots can have the same sign, we have one or two positive roots.

If $a_3 = 0$, then the discriminant reduces to

$$a_2^2(a_1^2 - 4a_2),$$

and if this is positive, the cubic equation has one vanishing root and two further real roots. These are positive when

$$a_1 < 0, \qquad a_2 > 0,$$

negative when

$$a_1 > 0, \qquad a_2 > 0,$$

and one of them is positive and the other negative when

$$a_1 \gtreqless 0, \qquad a_2 < 0.$$

§ 78. Discussion of the biquadratic equation

For equations of the fourth and fifth degrees there are either no, two, or four imaginary roots. If the discriminant is negative, there are two imaginary roots. With positive discriminant, either four or no imaginary roots may be present. To distinguish these two cases will be our next task. But first we turn to an elementary consideration of the biquadratic equation, which for simplicity we take in the shortened form

$$x^4 + ax^2 + bx + c = 0, \tag{1}$$

from which we can easily return to the general form.

Let the roots be denoted by $\alpha, \beta, \gamma, \delta$, and put, as in § 47,

$$\begin{aligned} \alpha - \beta &= v + w, & \gamma - \delta &= v - w, \\ \alpha - \gamma &= w + u, & \delta - \beta &= w - u, \\ \alpha - \delta &= u + v, & \beta - \gamma &= u - v. \end{aligned} \tag{2}$$

Then u^2, v^2, w^2 are the roots of the cubic resolvent

$$y^3 + 2ay^2 + (a^2 - 4c)y - b^2 = 0, \tag{3}$$

and the discriminant D of this cubic equation, determined by

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2, \tag{4}$$

is at the same time the discriminant of the biquadratic equation (1), and the signs of u, v, w are constrained by

$$uvw = -b \tag{5}$$

[§ 37, (3)].

If the discriminant is negative, then equation (3), like (1), has two conjugate imaginary roots.

If D is positive and all four roots $\alpha, \beta, \gamma, \delta$ are real, then u, v, w are also real, and therefore their squares, the roots of (3), are positive.

If all four roots are imaginary, for instance α conjugate to β and γ conjugate to δ, it follows from (2) that v and w are purely imaginary and u is real. In this case equation (3) therefore has one positive and two negative roots.

In both cases, however, if b vanishes, one of u, v, w may also be equal to zero.

Taking account of the results of the preceding paragraph on the cubic equation, we arrive at the following result:

The necessary and sufficient condition for the existence of four distinct real roots is

$$D > 0, \qquad a < 0, \qquad a^2 - 4c > 0. \tag{6}$$

In all other cases in which D is positive, the equation has four imaginary roots.

We can also carry out the discussion completely for $D = 0$, that is, in the case of equality of two roots.

Suppose $\gamma = \delta$. Since we have assumed $\alpha + \beta + \gamma + \delta = 0$, (2) gives

$$2v = 2w = \alpha - \beta, \qquad 2u = \alpha + \beta - 2\gamma = 2(\alpha + \beta),$$

and hence, from (3),

$$\begin{aligned} -4a &= 2(\alpha + \beta)^2 + (\alpha - \beta)^2, \\ 16(a^2 - 4c) &= (\alpha - \beta)^2[8(\alpha + \beta)^2 + (\alpha - \beta)^2]. \end{aligned} \tag{7}$$

First consider the case where also $\alpha = \beta$. From (7) we obtain the condition $a^2 - 4c = 0$, and the first equation (7) shows that a is negative when α and γ are real, and positive when α and γ are conjugate (and then, because $\alpha + \gamma = 0$, purely imaginary). Therefore:

The biquadratic equation has two pairs of equal real roots when

$$D = 0, \qquad a < 0, \qquad a^2 - 4c = 0, \tag{8}$$

and two pairs of equal imaginary roots when

$$D = 0, \qquad a > 0, \qquad a^2 - 4c = 0. \tag{9}$$

If, however, α is different from β, then γ must be real, and (7) shows that if α and β are real, a is negative and $a^2 - 4c$ positive, whereas if α and β are conjugate imaginary, either a is positive or $a^2 - 4c$ is negative. For in this case $(\alpha - \beta)^2$ is negative, and if $2(\alpha + \beta)^2 + (\alpha - \beta)^2$ is positive, then certainly $8(\alpha + \beta)^2 + (\alpha - \beta)^2$ must also be positive.

Thus we can supplement (6) by saying:

The necessary and sufficient condition for the existence of four real roots, of which two, but not more, may be equal, is

$$D \geq 0, \qquad a < 0, \qquad a^2 - 4c > 0. \tag{10}$$

The necessary and sufficient condition for three equal roots, which are necessarily all real, is by (2)

$$u^2 = v^2 = w^2.$$

Thus the cubic resolvent (3) must have three equal roots, and for this the necessary and sufficient conditions, by § 77 (1), are

$$a^2 + 12c = 0, \qquad 2a^3 - 8ac + 9b^2 = 0. \tag{11}$$

By § 47, (15), (16), the two invariants A and B of the biquadratic equation are

$$A = a^2 + 12c, \qquad B = 2a^3 - 72ac + 27b^2.$$

Thus equations (11) can also be written

$$A = 0, \qquad B + 4aA = 0,$$

so that (11) is equivalent to

$$A = 0, \qquad B = 0.$$

By eliminating c from (11), one may also derive

$$8a^3 + 27b^2 = 0. \tag{12}$$

Finally, the condition for equality of all four roots is

$$a = 0, \qquad b = 0, \qquad c = 0. \tag{13}$$

These relations can be made quite vivid by a geometric interpretation. Although geometric considerations do not properly belong to our plan, we shall not omit occasional indications of them for the reader.

If a, b, c are interpreted as rectangular coordinates in space, then every point of space may be regarded as the carrier of a certain biquadratic equation of the form (1), $f(x) = 0$. All equations for which a fixed number x is a root are represented by points of a plane $[f(x) = 0]$. The equation $D = 0$ is the equation of a curved surface of the fifth degree, the discriminant surface, generated by the lines of intersection of the planes $f(x) = 0$, $f'(x) = 0$; it is therefore a developable surface. It is the envelope of all planes $f(x) = 0$.

The surface has a cuspidal edge consisting of two branches, represented by equations (11), (12), and a double line determined by $b = 0$, $a^2 - 4c = 0$, hence of parabolic shape. On the side of negative a the surface intersects itself in this parabola. On the side of positive a the parabola continues as an isolated line.

The discriminant surface divides all space into three compartments, of which two are connected only along the double parabola, and these compartments contain the points corresponding to no, two, and four real roots. Let us temporarily call these three compartments [0], [2], [4]. Then [0] borders [4] only along the double parabola, while [0] and [4] each border [2] along parts of the surface. We shall denote by $[0, 2]$ and $[4, 2]$ the boundary surfaces of [0], [2], and of [0], [4]. The isolated part of the parabola continues into the interior of [0]. The cuspidal edges lie on the part of the surface that separates [4] from [2].

At the origin of coordinates the three regions of space and their boundary surfaces and boundary lines meet. The points of the surface parts $[0, 2]$ represent equations with two equal and two imaginary roots; the points of $[2, 4]$ represent equations with two equal and two distinct real roots.

On the double parabola there are twice two equal roots, namely real roots on the part where [0] borders [4], and imaginary roots on the isolated part.

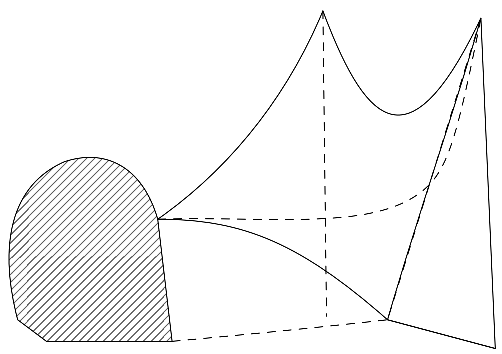


Fig. 5.

The points of the cuspidal edges represent equations with three equal roots, and the origin of coordinates represents the equation with four equal roots.

Kronecker was the first to point out the discriminant surface and its significance for the discussion of the biquadratic equation. A model of the surface was constructed by Kerschensteiner.

The analytic expression for the conditions for reality of the roots can be put into a form in which only the covariants of the biquadratic form occur.

The starting point is given by the expressions for the functions ψ_1, ψ_2, ψ_3 given in § 66, (6). If in those expressions we interchange α with β, then ψ_1, ψ_2, ψ_3 pass into $\psi_1, -\psi_3, -\psi_2$; and if at the same time we interchange α with β and γ with δ, then ψ_1, ψ_2, ψ_3 pass into $\psi_1, -\psi_2, -\psi_3$. From this the following results.

Put any real value for the variable x. If the roots $\alpha, \beta, \gamma, \delta$ of the biquadratic form are all real, then ψ_1, ψ_2, ψ_3 are also real.

If α, β are conjugate imaginary and γ, δ real, then interchanging α and β is equivalent to replacing i by $-$i. Thus in this case ψ_1 is real, while ψ_2, ψ_3 are conjugate imaginary.

Finally, if α, β and γ, δ are two pairs of conjugate imaginary roots, then α is interchanged with β and γ with δ when i passes into $-$i; therefore in this case ψ_1 is real and ψ_2, ψ_3 are purely imaginary.

If the four roots are real, then $\psi_1^2, \psi_2^2, \psi_3^2$ are real and positive. If two roots are real, then ψ_1^2 is real and positive, while ψ_2^2, ψ_3^2 are conjugate imaginary. If four roots are imaginary, then ψ_1^2 is positive, while ψ_2^2, ψ_3^2 are real and negative.

Now in § 66, (7) we set up the cubic equation whose roots are $\psi_1^2, \psi_2^2, \psi_3^2$, and also formed its discriminant. In the preceding paragraph we learned a criterion for the number of positive roots of a cubic equation, and from it the following result is obtained:

If $D < 0$, then the biquadratic equation has two real and two imaginary roots.

If $D > 0$, then in order that four real roots should be present there must be three variations of sign in the sequence

$$1, \qquad H, \qquad H^2 - 16Af^2, \qquad -I^2,$$

that is, one must have

$$H < 0, \qquad H^2 - 16Af^2 > 0.$$

In the three other possible sign combinations all four roots are imaginary. Here an arbitrary real value may be put for x.

§ 79. The Bezoutiant and its significance for reality of roots

For the investigation of the reality of the roots of an arbitrary real equation in more general cases, one may use with advantage the function which in § 73 we called the Bezoutiant of the equation.

Let

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n = 0 \tag{1}$$

be any real equation of degree n. The Bezoutiant was defined as follows. Put

$$y = t_{n-1} f_0(x) + t_{n-2} f_1(x) + \cdots + t_0 f_{n-1}(x), \tag{2}$$

where $t_0, t_1, \ldots, t_{n-1}$ denote indeterminate variables and

$$f_0(x) = 1, \qquad f_1(x) = x + a_1, \qquad f_2(x) = x^2 + a_1 x + a_2, \quad \ldots$$

Then [§ 73, (10)]

$$S(y^2) = \frac{1}{n}[S(y)]^2 + B, \tag{3}$$

where B is a quadratic form in the $n-1$ variables $t_0, t_1, \ldots, t_{n-2}$, with coefficients composed rationally from the coefficients a_i. This function B we have called the Bezoutiant, and have actually formed it in the cases $n = 4$ and $n = 5$.

In the sum on the left hand side of (3),

$$y_1^2 + y_2^2 + \cdots + y_n^2, \tag{4}$$

the $y_1, y_2, \ldots, y_n$ are linear functions of the variables $t_0, t_1, \ldots, t_{n-1}$; hence in (3) we have a transformation of the quadratic form

$$\frac{1}{n}[S(y)]^2 + B = \Phi(t_0, t_1, \ldots, t_{n-1}) \tag{5}$$

into a sum of n squares.

For the moment we make no restrictive assumptions about equality or distinctness of the roots of $f(x)$. We must first examine to what extent the linear functions $y_1, y_2, \ldots, y_n$ are independent. In this respect the following theorem holds.

If $x_1, x_2, \ldots, x_\nu$ are distinct roots of equation (1), then $y_1, y_2, \ldots, y_\nu$ are linearly independent.

For suppose that there exists among $y_1, y_2, \ldots, y_\nu$ a linear relation, identical with respect to t,

$$\alpha_1 y_1 + \alpha_2 y_2 + \cdots + \alpha_\nu y_\nu = 0,$$

whose coefficients α are independent of the t and are not all zero. This would split into the following equations:

$$\begin{aligned}
\alpha_1 f_0(x_1) + \alpha_2 f_0(x_2) + \cdots + \alpha_\nu f_0(x_\nu) &= 0,\\
\alpha_1 f_1(x_1) + \alpha_2 f_1(x_2) + \cdots + \alpha_\nu f_1(x_\nu) &= 0,\\
&\cdots\\
\alpha_1 f_{n-1}(x_1) + \alpha_2 f_{n-1}(x_2) + \cdots + \alpha_\nu f_{n-1}(x_\nu) &= 0.
\end{aligned}$$

Since $\nu \leq n$, the determinant of the first ν of these equations would have to vanish, so that

$$\begin{vmatrix} f_0(x_1) & f_1(x_1) & \cdots & f_{\nu-1}(x_1)\\ f_0(x_2) & f_1(x_2) & \cdots & f_{\nu-1}(x_2)\\ \vdots & \vdots & & \vdots\\ f_0(x_\nu) & f_1(x_\nu) & \cdots & f_{\nu-1}(x_\nu)\end{vmatrix} = 0,$$

or

$$\begin{vmatrix} 1 & x_1 + a_1 & x_1^2 + a_1 x_1 + a_2 & \cdots\\ 1 & x_2 + a_1 & x_2^2 + a_1 x_2 + a_2 & \cdots\\ \vdots & \vdots & \vdots & \\ 1 & x_\nu + a_1 & x_\nu^2 + a_1 x_\nu + a_2 & \cdots\end{vmatrix} = 0.$$

By repeated use of the theorem on addition of columns, § 22, (VII), this determinant can be brought to the form

$$\begin{vmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{\nu-1}\\ 1 & x_2 & x_2^2 & \cdots & x_2^{\nu-1}\\ \vdots & \vdots & \vdots & & \vdots\\ 1 & x_\nu & x_\nu^2 & \cdots & x_\nu^{\nu-1}\end{vmatrix} = \prod_{1\leq r<s\leq\nu}(x_r - x_s),$$

and hence it cannot vanish when, as assumed, $x_1, x_2, \ldots, x_\nu$ are distinct.

When, therefore, in the sum (4) all mutually equal terms are collected, there remain as many squares of linearly independent functions as equation (1) has distinct roots.

If x_1 is a real root of (1), then y_1^2 is a positive square in the sum (4). But if x_1 and x_2 are conjugate imaginary, then also y_1 and y_2 split into two conjugate imaginary components

$$y_1 = u + v\mathrm{i}, \qquad y_2 = u - v\mathrm{i},$$

where u and v are real linear functions of the t. Therefore

$$y_1^2 + y_2^2 = 2u^2 - 2v^2.$$

Thus by (4) the function $\Phi(t)$ defined by (5) is decomposed into a sum of squares whose number equals the number of distinct roots of $f(x) = 0$, and among which there are as many negative squares as there are pairs of imaginary roots among these distinct roots.

By the law of inertia for quadratic forms (§ 58), these numbers remain the same under any other real linear transformation of the quadratic form into a sum of squares. Now $S(y)$ is a real linear function of the t, and B no longer contains the variable t_{n-1}. Therefore formula (5) gives the following theorem:

I. If the Bezoutiant B is decomposed by a real linear transformation into a sum of π positive and ν negative squares, not reducible to a smaller number, then $\pi + \nu + 1$ is the number of distinct roots of $f(x) = 0$; and ν is the number of pairs of conjugate imaginary roots among them, while $\pi - \nu + 1$ is the number of real roots.

If the determinant of B, and hence also the discriminant of $f(x)$ (§ 73), is non-zero, then $\pi + \nu + 1 = n$, and ν is less than or at most equal to $\frac{1}{2}n$.

Thus the Bezoutiant is a quadratic form of a special nature, expressed precisely by the fact that among the squares into which it can be decomposed at most $\frac{1}{2}n$ can be negative.

§ 80. The inertia of forms of the second degree

By the theorem of the preceding paragraph we have been led to the investigation of homogeneous functions of the second degree with real coefficients. For these functions we encountered in § 58, under the name of the law of inertia, a theorem which forms the basis of what follows. The theorem asserted that however one transforms a quadratic form in m variables into a sum of positive and negative squares of linearly independent linear functions, and this can be done in infinitely many different ways, the number of positive and likewise the number of negative squares is always the same.

Let π denote the number of positive and ν the number of negative squares. Then $\pi + \nu$ is at most equal to m.

If we denote the difference $m - \pi - \nu$ by ρ, then, when the general quadratic form in m variables is represented as a sum of m squares, ρ may be regarded as the number of squares with vanishing coefficients.

The three numbers π, ν, ρ, none of which can be negative and whose sum equals the number of variables,

$$\pi + \nu + \rho = m, \tag{1}$$

are therefore invariant for a given quadratic form.

We must seek means of determining the numbers π, ν, ρ from the coefficients of the quadratic form. Applying these means to the Bezoutiant, we obtain criteria for inferring the number of real roots of an equation from its coefficients.

Thus let, as in § 56,

$$\varphi(x_1, x_2, \ldots, x_m) = \sum_1^m a_{i,k} x_i x_k, \qquad a_{i,k} = a_{k,i}, \tag{2}$$

be a quadratic form in m variables with real coefficients $a_{i,k}$.

The determinant

$$R = \sum \pm a_{1,1} a_{2,2} \cdots a_{m,m} \tag{3}$$

indicates, by its vanishing, that φ can be transformed by a linear transformation into a function of fewer than m variables, so that ρ is greater than zero.

The determinant R is symmetric. Among its subdeterminants of various orders there occur certain ones which are again symmetric determinants, namely those obtained from R by deleting rows and columns that meet in diagonal terms. Thus, if $\alpha, \beta, \gamma, \ldots$ denote any of the indices $1, 2, \ldots, m$, they are denoted by

$$\sum \pm a_{\alpha,\alpha} a_{\beta,\beta} a_{\gamma,\gamma} \cdots . \tag{4}$$

We shall call these the principal subdeterminants of R.

We shall determine the number of principal subdeterminants.

The number of μ-rowed principal subdeterminants is equal to the number of ways of selecting from the sequence of indices $1, 2, 3, \ldots, m$ groups of μ distinct indices, hence equal to the number of combinations of m elements taken μ at a time; this number is the binomial coefficient B_μ^m. Thus the number of all principal subdeterminants, if we count the determinant R itself and also the unit as a zero-row determinant, is

$$1 + m + \frac{m(m-1)}{1 \cdot 2} + \cdots + m + 1 = 2^m.$$

But usually only a small part of these is actually to be considered.

The indices $1, 2, 3, \ldots, m$ can be arranged in $\Pi(m) = 1 \cdot 2 \cdot 3 \cdots m$ different ways. If from any one of these arrangements we form the determinant

$$\begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,m} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,m} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,m} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m,1} & a_{m,2} & a_{m,3} & \cdots & a_{m,m} \end{vmatrix},$$

then a sequence of principal subdeterminants can be derived from it, as indicated by the strokes: each preceding determinant is obtained from the following one by removing the last row and column. Thus

$$R_0 = 1, \qquad R_1 = a_{1,1}, \qquad R_2 = a_{1,1}a_{2,2} - a_{1,2}^2,$$

$$R_3 = \begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix}, \qquad \ldots, \qquad R_m = R.$$

Such a system, when arranged in descending order

$$R_m, R_{m-1}, \ldots, R_1, R_0,$$

will be called a chain of principal subdeterminants. There are $\Pi(m)$ such chains, in all of which the first term is R and the last term is 1.

§ 81. Quadratic forms with vanishing determinant

If the determinant R of the quadratic form $\varphi(x_1, x_2, \ldots, x_m)$ defined in (3), § 80, vanishes, then, as we already saw in § 57, φ can be expressed by fewer than m independent linear functions of x. If we denote by $m - \rho$ the least number of variables through which φ can be expressed, then ρ has the same meaning as in the preceding paragraph.

If among the principal subdeterminants of R there is a k-rowed one which is not zero, say

$$R_k = \sum \pm a_{1,1}a_{2,2} \cdots a_{k,k},$$

then, putting

$$x_{k+1} = 0, \qquad x_{k+2} = 0, \qquad \ldots \qquad x_m = 0,$$

one obtains

$$\varphi(x_1, x_2, \ldots, x_k, 0, 0, 0),$$

a quadratic form in the k variables $x_1, x_2, \ldots, x_k$ whose determinant R_k is non-zero, and which therefore cannot be expressed by fewer than k variables. Still less can the function φ in the unrestricted variables $x_1, x_2, \ldots, x_m$ depend on fewer than k variables. Hence

$$\rho \leq m - k. \tag{1}$$

Suppose now that the form $\varphi(x_1, x_2, \ldots, x_m)$ can be expressed by k mutually independent linear forms in x, which we denote by

$$\begin{aligned} \xi_1 &= \beta_1^{(1)} x_1 + \beta_2^{(1)} x_2 + \cdots + \beta_k^{(1)} x_k + \beta_{k+1}^{(1)} x_{k+1} + \cdots + \beta_m^{(1)} x_m, \\ &\cdots \\ \xi_k &= \beta_1^{(k)} x_1 + \beta_2^{(k)} x_2 + \cdots + \beta_k^{(k)} x_k + \beta_{k+1}^{(k)} x_{k+1} + \cdots + \beta_m^{(k)} x_m. \end{aligned} \tag{2}$$

Since $\xi_1, \xi_2, \ldots, \xi_k$ are to be linearly independent, at least one of the k-rowed determinants formed from the matrix

$$\begin{matrix} \beta_1^{(1)} & \beta_2^{(1)} & \cdots & \beta_m^{(1)} \\ \cdots & \cdots & & \cdots \\ \beta_1^{(k)} & \beta_2^{(k)} & \cdots & \beta_m^{(k)} \end{matrix}$$

must be non-zero. For if all of them were zero, the unknowns $h_1, h_2, \ldots, h_k$, not all of which are to vanish, could be determined from the equations

$$h_1\beta_s^{(1)} + h_2\beta_s^{(2)} + \cdots + h_k\beta_s^{(k)} = 0, \qquad s = 1, 2, 3, \ldots, m$$

(§ 24), and the $\xi_1, \xi_2, \ldots, \xi_k$ would satisfy an equation

$$h_1\xi_1 + h_2\xi_2 + \cdots + h_k\xi_k = 0,$$

and hence would not be independent.

We may therefore, without restricting generality, assume that the determinant

$$\sum \pm \beta_1^{(1)}\beta_2^{(2)} \cdots \beta_k^{(k)}$$

is non-zero; then the equations (2) can be solved for the variables $x_1, x_2, \ldots, x_k$.

We give these solutions the form

$$\begin{aligned} x_1 &= y_1 + \alpha_{k+1}^{(1)}x_{k+1} + \cdots + \alpha_m^{(1)}x_m, \\ x_2 &= y_2 + \alpha_{k+1}^{(2)}x_{k+1} + \cdots + \alpha_m^{(2)}x_m, \\ &\cdots \\ x_k &= y_k + \alpha_{k+1}^{(k)}x_{k+1} + \cdots + \alpha_m^{(k)}x_m, \end{aligned} \tag{3}$$

where the y_s are homogeneous linear combinations of the ξ_s, determined by the equations

$$\begin{aligned} \beta_1^{(1)}y_1 + \beta_2^{(1)}y_2 + \cdots + \beta_k^{(1)}y_k &= \xi_1, \\ \beta_1^{(2)}y_1 + \beta_2^{(2)}y_2 + \cdots + \beta_k^{(2)}y_k &= \xi_2, \\ &\cdots \\ \beta_1^{(k)}y_1 + \beta_2^{(k)}y_2 + \cdots + \beta_k^{(k)}y_k &= \xi_k. \end{aligned}$$

The α are new coefficients, derived in an easily surveyable way from the β, but their formation is of no further importance here.

By hypothesis, the function $\varphi(x_1, x_2, \ldots, x_m)$ can be expressed by $\xi_1, \xi_2, \ldots, \xi_k$, and therefore also by $y_1, y_2, \ldots, y_k$ alone. Denote this expression by $\Phi(y_1, y_2, \ldots, y_k)$. Then, by means of (3), we have the identity

$$\varphi(x_1, x_2, \ldots, x_k, x_{k+1}, \ldots, x_m) = \Phi(y_1, y_2, \ldots, y_k), \tag{4}$$

and if in it we put $x_{k+1}, x_{k+2}, \ldots, x_m = 0$, then

$$\Phi(x_1, x_2, \ldots, x_k) = \varphi(x_1, x_2, \ldots, x_k, 0, 0, \ldots, 0), \tag{5}$$

which, since the names of the variables are immaterial, determines Φ completely. We may therefore write

$$\begin{aligned} \Phi(y_1, y_2, \ldots, y_k) &= \varphi(y_1, y_2, \ldots, y_k, 0, 0, \ldots, 0) \\ &= \varphi(x_1, x_2, \ldots, x_m)'. \end{aligned} \tag{6}$$

Thus Φ is obtained from φ by setting the last $m - k$ variables (for the ordering chosen here) equal to zero:

$$\Phi(x_1, \ldots, x_k) = \sum_1^k a_{r,s}x_rx_s. \tag{7}$$

The determinant of Φ is one of the principal subdeterminants of R; it is obtained by deleting from R the last $m-k$ rows and columns.

Suppose that φ cannot be expressed by fewer than k variables, so that $k = m - \rho$. Then the determinant of this function Φ cannot vanish.

From this and from what was proved above, we obtain the theorem:

II. If all principal subdeterminants of R of more than k rows vanish, while among the principal subdeterminants of k rows at least one is non-zero, then

$$\rho = m - k.$$

For (7) shows that if $\rho = m - k$, then at least one k-rowed principal subdeterminant must be non-zero; and (1) shows that if a principal subdeterminant of more than k rows is still non-zero, then

$$\rho < m - k.$$

If π, ν, ρ are the numbers of positive, negative, and vanishing squares into which φ can be decomposed, then π, ν are the numbers of positive and negative squares into which the form Φ determined by (7) can be decomposed. Thus the determination of π, ν still needs to be carried out only for this latter function, whose determinant is non-zero.

§ 82. Quadratic forms with non-vanishing determinant

In the investigation of the forms $\varphi(x_1, x_2, \ldots, x_m)$ with non-vanishing determinant, we make use of a theorem on determinants that we proved at the end of § 28, and which we recall here.

If A is a determinant of m rows, and if $A_i^{(k)}$ are its first and $A_{i,k}^{h,l}$ its second minors, then

$$A_1^{(1)} A_i^{(i)} - A_1^{(i)} A_i^{(1)} = A A_{1,i}^{1,i}. \tag{1}$$

If A is a non-zero symmetric determinant, then $A_1^{(i)} = A_i^{(1)}$, and we may write (1) in the form

$$A_1^{(1)} A_i^{(i)} - \left(A_1^{(i)}\right)^2 = A A_{1,i}^{1,i}. \tag{2}$$

Here i may denote any one of the indices $2, 3, \ldots, m$. If $A_1^{(1)} = 0$ and $A_{1,i}^{1,i} = 0$, it follows that also $A_1^{(i)} = 0$; from this we conclude that the quantities

$$A_1^{(1)}, \quad A_{1,2}^{1,2}, \quad A_{1,3}^{1,3}, \ldots, A_{1,m}^{1,m}$$

cannot all vanish at the same time, for then

$$A_1^{(2)}, \quad A_1^{(3)}, \ldots, A_1^{(m)},$$

and hence, contrary to the assumption, A itself would vanish (§ 22).

We now apply this to the determinant $R = R_m$ of our function φ, which is assumed non-zero. It follows that although the first principal minor R_{m-1} may vanish, not all second principal minors R_{m-2} can vanish. The same argument can be applied when we replace R by a non-zero $R_{m-1}, R_{m-2}, \ldots$. We therefore obtain the following important theorem:

III. If R is non-zero, the indices $1, 2, \ldots, m$ can be ordered so that in the chain of principal minors

$$R_m, R_{m-1}, \ldots, R_1, R_0 \tag{3}$$

no two consecutive terms vanish.

The determinant relation (2), when one puts

$$A = R_{k+1}, \qquad (k = 1, 2, \ldots, m-1),$$

gives an equation between three successive terms of the chain (3), which we may write as

$$R_k S_k - T_k^2 = R_{k-1} R_{k+1}, \tag{4}$$

where S_k and T_k are certain minors of R, and hence integral rational functions of the coefficients $a_{i,k}$.

More precisely, R_k, S_k, T_k are first minors of R_{k+1}, namely

$$R_k = \frac{\partial R_{k+1}}{\partial a_{k+1,k+1}}, \qquad S_k = \frac{\partial R_{k+1}}{\partial a_{k,k}}, \qquad T_k = \frac{\partial R_{k+1}}{\partial a_{k,k+1}}.$$

We shall suppose in what follows that an ordering of the indices corresponding to theorem (3) has been chosen. Then, if R_k vanishes, R_{k-1} and R_{k+1} are non-zero, and (4) shows that they have opposite signs. Thus:

IV. If an interior term of a chain of principal minors vanishes, then the two adjacent terms have opposite signs.

§ 83. Number of positive and negative squares

To determine the number of positive and negative squares of a form with non-vanishing determinant, we first suppose that no term vanishes in the chain of principal minors

$$R_m, R_{m-1}, R_{m-2}, \ldots, R_1, R_0 = 1.$$

We can then choose the coefficient λ so that the determinant of the form

$$\psi = \varphi(x_1, x_2, x_3, \ldots, x_m) - \lambda x_m^2 \tag{1}$$

vanishes. Its determinant is in fact

$$\begin{vmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,m} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,m} \\ \cdots & \cdots & & \cdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,m} - \lambda \end{vmatrix} = R_m - \lambda R_{m-1},$$

and it vanishes when

$$\lambda = \frac{R_m}{R_{m-1}}$$

is put.

The function ψ can then be expressed by means of $m-1$ variables y, and by formula (6), § 81, one obtains

$$\psi(x_1, x_2, \ldots, x_m) = \psi(y_1, y_2, \ldots, y_{m-1}, 0),$$

or, by (1),

$$\varphi(x_1, x_2, \ldots, x_m) = \frac{R_m}{R_{m-1}} x_m^2 + \varphi(y_1, y_2, \ldots, y_{m-1}, 0). \tag{2}$$

Thus the investigation of the function φ in m variables is reduced to that of

$$\varphi_1(y) = \varphi(y_1, y_2, \ldots, y_{m-1}, 0)$$

in $m-1$ variables, whose determinant is equal to R_{m-1} and is therefore non-zero. According as $R_m : R_{m-1}$ is positive or negative, the function $\varphi(x)$ has one positive or one negative square more than $\varphi_1(y)$.

Applying the same procedure to $\varphi_1(y)$ and to the following functions gives the theorem:

The number of positive and negative terms of the sequence

$$\frac{R_m}{R_{m-1}}, \quad \frac{R_{m-1}}{R_{m-2}}, \quad \dots, \quad \frac{R_1}{R_0} \tag{3}$$

agrees with the number of positive and negative squares into which the function $\varphi(x)$ can be decomposed.

We shall now give this theorem a somewhat different form, but first introduce the following terminology. If a sequence of non-zero real numbers is given in a definite order, their signs may change in various ways. If two quantities of the same sign follow one another, there is a permanence of sign; but if a quantity is followed by another with the opposite sign, there is a variation of sign.

Now consider from this point of view the sequence of quantities

$$R_m, R_{m-1}, R_{m-2}, \dots, R_1, R_0.$$

In passing from R_k to R_{k-1} one obtains a permanence or a variation of sign according as the quotient $R_k : R_{k-1}$ is positive or negative.

We may therefore also state the following theorem:

V. If π is the number of positive squares and ν the number of negative squares of φ, then π is equal to the number of permanences, and ν to the number of variations, in the chain

$$R_m, R_{m-1}, R_{m-2}, \dots, R_1, R_0. \tag{4}$$

This form of the theorem has the advantage that it carries over to the case where individual interior terms in (4) vanish, provided that no two consecutive terms are zero, which by § 82, III can always be assumed. If, for instance, $R_k = 0$, while R_{k+1} and R_{k-1} are non-zero, then by § 82, theorem IV, R_{k-1} and R_{k+1} have opposite signs. In the sequence

$$R_{k+1}, \quad R_k, \quad R_{k-1}$$

there is therefore one variation and one permanence of sign, no matter whether we replace the vanishing R_k by a positive or by a negative quantity.

If we replace the chain (4)

$$R_m, R_{m-1}, \dots, R_1, R_0,$$

in which individual terms vanish, by another chain

$$R'_m, R'_{m-1}, \dots, R'_1, R'_0, \tag{5}$$

in which no term vanishes, and in which the non-vanishing terms of (4) are matched by terms of the same sign, then (4) and (5) have the same number of variations, whatever signs the R' corresponding to the vanishing R may have.

Let $\psi(x)$ now be a second arbitrary quadratic form in the variables x, and form

$$\varphi' = \varphi + \varepsilon\psi, \tag{6}$$

where ε is a coefficient still to be determined. We shall show at once that ε can be so chosen that φ' and φ have the same number of positive and negative squares, that the chain of principal minors R'_k of φ' contains no vanishing term, and finally that a non-vanishing R_k corresponds to an R'_k of the same sign. Then the numbers π, ν for φ and φ' can be determined from (4) as well as from (5), and the number of variations, which is the same in both, gives the number ν of negative squares.

To prove that ε can be chosen in this way, suppose that φ has somehow been transformed into a sum of squares,

$$\varphi = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_m y_m^2,$$

and that ψ, expressed in the variables $y_1, y_2, \ldots, y_m$, has the form

$$\psi = \sum \beta_{i,k} y_i y_k.$$

The numbers π, ν for the function φ' are then determined, by theorem V, from the chain of principal minors of

$$\begin{vmatrix} \lambda_1 + \varepsilon\beta_{1,1} & \varepsilon\beta_{1,2} & \cdots & \varepsilon\beta_{1,m} \\ \varepsilon\beta_{2,1} & \lambda_2 + \varepsilon\beta_{2,2} & \cdots & \varepsilon\beta_{2,m} \\ \cdots & \cdots & & \cdots \\ \varepsilon\beta_{m,1} & \varepsilon\beta_{m,2} & \cdots & \lambda_m + \varepsilon\beta_{m,m} \end{vmatrix}.$$

But ε can be taken so small that these principal minors agree in sign with

$$\lambda_1\lambda_2\lambda_3\cdots\lambda_m, \quad \lambda_1\lambda_2\lambda_3\cdots\lambda_{m-1}, \quad \ldots, \quad \lambda_1\lambda_2, \quad \lambda_1, \quad 1.$$

Thus the number of positive and negative squares of φ' is equal to the number of positive and negative coefficients λ; none of these vanishes. Hence the numbers π and ν are the same for φ and φ'.

Now let the coefficients of ψ, expressed in the original variables, be $b_{i,k}$, so that

$$\psi = \sum b_{i,k} x_i x_k.$$

Then one of the principal minors of φ' is

$$R'_k = \begin{vmatrix} a_{1,1} + \varepsilon b_{1,1} & \cdots & a_{1,k} + \varepsilon b_{1,k} \\ \cdots & & \cdots \\ a_{k,1} + \varepsilon b_{k,1} & \cdots & a_{k,k} + \varepsilon b_{k,k} \end{vmatrix},$$

and is therefore an integral rational function of degree k in ε,

$$R'_k = R_k + \varepsilon M_1 + \varepsilon^2 M_2 + \cdots + \varepsilon^k M_k,$$

where $M_1, M_2, \ldots, M_k$ depend rationally on the $a_{i,k}, b_{i,k}$. In particular,

$$M_k = \sum \pm b_{1,1} b_{2,2} \cdots b_{k,k},$$

and the $b_{i,k}$ can always be chosen so that M_k is non-zero. By § 31 one can therefore choose ε so that, if R_k is non-zero, R'_k has the same sign as R_k, and if R_k vanishes, R'_k does not vanish.

Combining what has just been proved with the results of § 81, we obtain the following general rule for determining the number of negative squares, also in the case of a vanishing determinant.

VI. To determine the numbers π, ν, ρ of the function $\varphi(x)$, arrange the variables $x_1, x_2, \ldots, x_m$ so that in the chain of principal minors

$$R_m, R_{m-1}, \ldots, R_1, R_0 \tag{7}$$

as small as possible a number of initial terms vanish, and so that among the following terms no two adjacent terms vanish. Then ρ is the number of vanishing initial terms, ν the number of variations of sign, and

$$\pi = m - \nu - \rho.$$

Finally, it should be remarked that the number of variations in the sequence (7) can also be counted from right to left; that is, one finds the same number of variations when the sequence is written in the reverse order.

§ 84. Application to the Bezoutian

The detailed discussion of the inertia of quadratic forms has had only one purpose for us: to determine the number of real and imaginary roots of an algebraic equation by counting the positive and negative squares of the Bezoutian. If n is the degree of the equation, its Bezoutian is a quadratic form in $n-1$ variables, and in § 73 we constructed it completely for $n = 3, 4, 5$ and learned the methods by which one may also compute it in other cases.

We have already noted that the Bezoutian is not a general quadratic form. It has the special property that the number ν of its negative squares can never be larger than $\frac{1}{2}n$. This property must find its expression in certain inequalities between the coefficients, but these are not yet known. Only in the case $n = 3$ can we give the algebraic character of this restriction completely.

For the cubic equation

$$f(x) = a_0x^3 + a_1x^2 + a_2x + a_3 = 0$$

the Bezoutian, by § 73, (13), is nothing other than the Hessian covariant

$$-\frac{2}{3}H(t_1, t_0) = -\frac{2}{3}\left(A_0t_1^2 + A_1t_1t_0 + A_2t_0^2\right).$$

But if D denotes the discriminant, so that

$$3D = 4A_0A_2 - A_1^2,$$

then the relation [§ 62, (4), (12)]

$$4H^3 + Q^2 + 27Df^2 = 0$$

holds. Therefore, if we put $t_0 = 0$, we obtain

$$-4A_0^3 + q_0^2 + 27Da_0^2 = 0, \tag{1}$$

where

$$q_0 = 27a_0^2a_3 - 9a_0a_1a_2 + 2a_1^3.$$

This relation shows that A_0 and D cannot both be positive, and therefore that the form $-H$ cannot be decomposed into two negative squares.

We shall now apply the general theorem of the preceding paragraph to the biquadratic equation, which for simplicity we take in the form

$$x^4 + ax^2 + bx + c = 0. \tag{2}$$

To form the coefficients of the Bezoutian, we must replace, in the formulas § 73, (14),

$$a_0, a_1, a_2, a_3, a_4$$

by

$$1, 0, a, b, c.$$

Thus we obtain

$$\begin{array}{lll} B_{2,2} = -2a, & B_{1,1} = a^2 - 4c, & B_{0,0} = -\frac{1}{4}(8ac - 3b^2), \\ B_{0,1} = \frac{1}{2}ab, & B_{2,0} = -4c, & B_{1,2} = -3b. \end{array}$$

We order the determinant as follows:

$$\begin{vmatrix} -2a & -3b & -4c \\ -3b & a^2 - 4c & \frac{1}{2}ab \\ -4c & \frac{1}{2}ab & -\frac{1}{4}(8ac - 3b^2) \end{vmatrix},$$

and obtain the chain of principal minors

$$D, \quad -2a^3 + 8ac - 9b^2, \quad -2a, \quad 1, \tag{3}$$

where D denotes the discriminant, which by § 64, (11) is determined by

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2. \tag{4}$$

The criterion that all four roots be real is therefore

$$D > 0, \qquad -2a^3 + 8ac - 9b^2 > 0, \qquad -2a > 0. \tag{5}$$

This criterion seems different, and in any case less simple, than the one established in § 78,

$$D > 0, \qquad a < 0, \qquad a^2 - 4c > 0. \tag{6}$$

We shall now show, however, that the two say exactly the same thing.

It is first immediately clear that (6) must hold when (5) holds, for

$$-2a(a^2 - 4c) - 9b^2$$

cannot be positive if a and $a^2 - 4c$ are negative. Conversely, to see that (6) implies (5), we must take the value of D into consideration.

For brevity put

$$\alpha = -2a, \qquad \beta = -6a^3 + 24ac - 27b^2, \qquad \gamma = 3a^2 - 12c. \tag{7}$$

Then the conditions (5) are

$$\alpha > 0, \qquad \beta > 0, \qquad D > 0,$$

while the conditions (6) are

$$\alpha > 0, \qquad \gamma > 0, \qquad D > 0.$$

It remains to prove that if α, γ, and D are positive, then β must also be positive. To this end we express D in (4), by means of (7), in terms of α, β, γ and obtain

$$27D = 4(\alpha^2 - \gamma)^3 - \left[2\alpha(\alpha^2 - \gamma) - \beta\right]^2.$$

If D is positive, then necessarily $\alpha^2 - \gamma$ is positive. Put

$$\alpha^2 - \gamma = \delta^2.$$

Then $\alpha^2 > \delta^2$, hence, if δ is taken positive,

$$\alpha > \delta,$$

and

$$27D = 4\delta^6 - (2\alpha\delta^2 - \beta)^2.$$

Thus D cannot be positive when β is negative, for if β is negative, then

$$2\alpha\delta^2 - \beta > 2\alpha\delta^2 > 2\delta^3,$$

and therefore $4\delta^6 - (2\alpha\delta^2 - \beta)^2$ is negative.

This directly proves that the criteria (5) and (6) have exactly the same meaning.

According to the geometric interpretation given in § 78, $\beta = 0$ denotes a surface of third order that passes through the parabola $b = 0, \gamma = 0$, and, insofar as negative values of a are concerned, lies entirely in the part of space in which D is negative.

To have before us an example of a quintic equation, consider

$$x^5 + x^2 + a = 0.$$

Thus in our general expressions § 73, or also § 74, (2), (3), we have to put

$$a_0, a_1, a_2, a_3, a_4, a_5 = 1, 0, 0, 1, 0, a,$$

and for the determinant of the Bezoutian we obtain

$$\begin{vmatrix} -2a & 0 & 0 & -5a \\ 0 & \frac{6}{5} & -5a & 0 \\ 0 & -5a & 0 & -3 \\ -5a & 0 & -3 & 0 \end{vmatrix},$$

and for the discriminant

$$D = 108a + 3125a^4.$$

A chain of principal minors is

$$D, \quad 50a^3, \quad \frac{-12a}{5}, \quad -2a, \quad 1. \tag{8}$$

The discriminant is negative when

$$0 < -a < \sqrt[3]{\frac{108}{3125}}, \tag{9}$$

and positive otherwise, in particular for all positive a. One sees that when the discriminant is positive, (8) always has two variations of sign, while for a negative discriminant, where a is also negative, it has one variation.

Our equation therefore has two imaginary roots as long as a lies in the interval (9), and four imaginary roots when a lies outside this interval; it never has four real roots.

A purely numerical example is provided by the equation, arising from the complex multiplication of elliptic functions,

$$x^5 - x^3 - 2x^2 - 2x - 1 = 0.$$

For the determinant of the Bezoutian we obtain from § 73

$$\begin{vmatrix} -\frac{4}{5} & -\frac{3}{5} & \frac{4}{5} & 5 \\ -\frac{3}{5} & \frac{4}{5} & \frac{33}{5} & 8 \\ \frac{4}{5} & \frac{33}{5} & \frac{46}{5} & 6 \\ 5 & 8 & 6 & 2 \end{vmatrix},$$

and the chain of principal minors

$$\frac{47^2}{5}, \quad \frac{94}{5}, \quad -1, \quad -\frac{4}{5}, \quad 1.$$

Thus the discriminant of our equation is 47^2. The equation therefore has two pairs of imaginary roots and one real root. If the discriminant is known, the second term of this chain, $94 : 5$, need not be computed; indeed, it is already enough to know the negative sign of the penultimate term in order to decide the reality of the roots as above.

Eighth Section.

Sturm's Theorem.

§ 85. Sturm's problem

The question treated in the preceding section, namely the number of real roots of an equation, is a special case of a more general problem. This problem will form the subject of the present section, and it is the foundation of all methods for the approximate numerical calculation of roots of equations.

The question is this: how many real roots of a real numerical equation lie between two given real numerical values a and b?

Once this has been decided, the next task is to narrow the bounds a, b until only one root of the given equation lies between them, and finally to bring them so close together that either of them may be regarded as an approximate value of that root.

If we take $a = -\infty, b = +\infty$, or at least a negative and b positive so large that no roots can lie beyond these bounds, then this task coincides with the one treated in the preceding section: the determination of the number of all real roots.

We first show how the general problem can be reduced to the special one, so that the question just posed is answered, at least in principle, by the considerations of the preceding section. Suppose first that we wish to determine the number of positive roots of an equation $f(x) = 0$. Put $x = y^2$. Then to every real value of y there corresponds a positive value of x, and conversely to every positive value of x there correspond two real values of y with opposite signs. The number of positive roots of $f(x) = 0$ is therefore half the number of real roots of $f(y^2) = 0$.

Further, put

$$y = \frac{x - a}{b - x}.$$

Then, as x goes from a to b, y runs through positive values from 0 to infinity; and if under this substitution $f(x)$ is transformed into $F(y)$, the equation $f(x) = 0$ has as many roots between a and b as $F(y) = 0$ has positive roots, hence half as many as $F(z^2) = 0$ has real roots. Thus our problem is indeed reduced to determining the number of real roots of a certain other equation, whose degree is twice as large as the degree of the given equation. But this reduction does not give the solution in the simplest form, and we must seek a simpler answer to the question.

§ 86. Sturm chains

The following consideration leads to a solution of the problem set in the preceding paragraph.

Let α and β be any two real numbers with $\alpha < \beta$. Let $f(x)$ be a given integral rational function of x, and suppose that we are to determine how many of its roots lie in the interval Δ

$$\alpha < x < \beta.$$

We assume that α, β themselves are not roots of $f(x) = 0$, so that $f(\alpha)$ and $f(\beta)$ are non-zero. For the moment we also assume that $f(x)$ has no multiple root, at least in the interval Δ, so that for no value of this interval can $f(x)$ and $f'(x)$ vanish simultaneously.

Suppose that in some way we can construct a sequence of $m + 1$ continuous functions

$$f(x), f_1(x), f_2(x), \ldots, f_m(x), \tag{1}$$

with the following properties:

1. In the interval Δ, no two consecutive functions f_ν vanish simultaneously.

2. The last of them, $f_m(x)$, does not vanish anywhere in Δ, and hence keeps an unchanging sign.

3. If an intermediate term, say $f_\nu(x)$, vanishes for some x in Δ, then for this x the two adjacent functions $f_{\nu-1}(x)$ and $f_{\nu+1}(x)$ have opposite signs.

4. If $f(x)$ vanishes in the interval Δ, then $f_1(x)$ has, for this value of x, the same sign as $f'(x)$.

We shall call such a sequence (1) a Sturm chain. We shall see later how it is constructed; first we draw some consequences from the definition.

For every value of x for which none of the functions in (1) vanishes, each of these functions has a definite sign. We count one variation of sign whenever, in passing along the chain from left to right, a positive term is followed by a negative one or a negative term by a positive one. Let $V(x)$ denote the number of such variations for a specified value of x. If an intermediate term $f_\nu(x)$ vanishes, then by 3. in passing from $f_{\nu-1}(x)$ to $f_{\nu+1}(x)$ we must count one variation.

Let α, β be two values for which none of the functions f_ν vanishes, and let x increase continuously from α to β. A change in the number of variations can occur only when one of the functions in (1) changes its sign, that is, passes through the value zero, by the assumed continuity.

If this is an intermediate function, then the number of variations is not changed, by 3. Indeed, in

$$f_{\nu-1}, \quad f_\nu, \quad f_{\nu+1}$$

there is always one variation before and after f_ν passes through zero, since $f_{\nu+1}$ and $f_{\nu-1}$ have opposite signs.

But if $f(x)$ passes through zero, then, by 4., one variation is lost. If $f(x)$ passes from negative to positive values, then (compare § 32) $f'(x)$ and hence $f_1(x)$ are positive, so the signs change as follows:

$$\begin{array}{cc} f(x) & f_1(x) \\ - & + \\ + & + \end{array}$$

If $f(x)$ passes from positive to negative values, then $f'(x)$ and $f_1(x)$ are negative, so the signs are

$$\begin{array}{cc} f(x) & f_1(x) \\ + & - \\ - & - \end{array}$$

and in both cases a variation has been lost.

Since $f_m(x)$, by 2., does not change sign, it follows that $V(\alpha) - V(\beta)$ is equal to the number of roots of $f(x) = 0$ lying between α and β, or:

The number of roots of $f(x) = 0$ between α and β is equal to the excess of the number of variations of the chain for $x = \alpha$ over the number of variations for $x = \beta$.

If for $x = \alpha$ or $x = \beta$ one or more of the intermediate functions in (1) vanish, then, by 3., it is immaterial whether we replace the vanishing value by a positive or by a negative one.

§ 87. First example: spherical functions

Before passing to the general methods by which Sturm chains are formed, we shall consider a few examples in which such chains arise from special circumstances.

We first consider the so-called spherical functions, which are frequently used in mathematical physics and mechanics.

By spherical functions one understands a system of integral rational functions of increasing degree defined as follows:

$$
\begin{aligned}
P_0(x) &= 1,\\
P_1(x) &= x,\\
P_2(x) &= \frac{3}{2}\left(x^2 - \frac{1}{3}\right),\\
P_3(x) &= \frac{5}{2}\left(x^3 - \frac{3}{5}x\right),\\
&\dots\\
P_n(x) &= \frac{1\cdot 3\cdots(2n-1)}{1\cdot 2\cdots n}\left(x^n - \frac{n(n-1)}{2(2n-1)}x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2\cdot 4(2n-1)(2n-3)}x^{n-4} - \cdots\right).
\end{aligned}
$$

Thus $P_n(x)$ is an integral rational function of degree n, containing only even or only odd powers according as n is even or odd. By use of the symbol $\Pi(n)$, the general expression for the spherical functions may also be written

$$P_n(x) = \sum_\mu \frac{(-1)^\mu}{2^n}\frac{\Pi(2n-2\mu)x^{n-2\mu}}{\Pi(\mu)\Pi(n-2\mu)\Pi(n-\mu)}, \tag{1}$$

where the summation over μ is to be extended from $\mu = 0$ as far as $n - 2\mu$ remains non-negative.

Between these functions there hold the following relations, which, after substituting expression (1), are verified by simply comparing the coefficients of equal powers of x:

$$
\begin{aligned}
nP_n(x) - (2n-1)xP_{n-1}(x) + (n-1)P_{n-2}(x) &= 0,\\
P_1(x) - xP_0(x) &= 0,
\end{aligned} \tag{2}
$$

$$(1-x^2)P_n'(x) + nxP_n(x) - nP_{n-1}(x) = 0. \tag{3}$$

From the last equation it follows, for $x = \pm 1$, that

$$\pm P_n(\pm 1) = P_{n-1}(\pm 1),$$

and hence

$$P_n(1) = 1, \qquad P_n(-1) = (-1)^n. \tag{4}$$

From (2) and (3) it follows that the functions P_n, taken downward from any chosen m,

$$P_m, \quad P_{m-1}, \quad P_{m-2}, \dots, P_0 \tag{5}$$

have the properties of a Sturm chain in the interval from -1 to $+1$. For if P_n and P_{n-1} vanished simultaneously for some x, then by (2) also P_{n-2}, hence P_{n-3}, and so on down to P_0 would vanish, which is impossible since $P_0 = 1$. Thus condition § 86, 2. is also satisfied for any interval.

If $P_{n-1} = 0$, then by (2) P_n and P_{n-2} have opposite signs, as condition § 86, 3. requires. Equation (3) shows that $P_n(x)$ and $P_n'(x)$ can never vanish simultaneously, since otherwise $P_{n-1}(x)$ would also vanish; and finally (3) also shows that if $P_n(x)$ vanishes and $-1 < x < 1$, then $P_n'(x)$ and $P_{n-1}(x)$ have the same sign.

Now put $x = -1, +1$ in (5). By (4), at $x = -1$ all signs in (5) are alternating, while at $x = +1$ there is no variation. Hence:

The equation $P_m(x) = 0$ has m real roots between $x = -1$ and $x = +1$, and since P_m is of degree m, these are all its roots.

We can draw a somewhat stronger conclusion. Let $\Delta = (\alpha, \beta)$ be an interval containing two, and not more than two, roots ξ, η of $P_m = 0$. Then, in passing from α to β, two variations are lost in the chain (5). We choose α so close to ξ and β so close to η that $P_{m-1}(\alpha)$ has the same sign as

$P_{m-1}(\xi)$, and $P_{m-1}(\beta)$ the same sign as $P_{m-1}(\eta)$. If P_m passes through ξ from negative to positive, then it passes through η from positive to negative; therefore $P'_m(\xi)$, and consequently $P_{m-1}(\xi)$ and $P_{m-1}(\alpha)$, are positive, while $P'_m(\eta)$, and consequently $P_{m-1}(\eta)$ and $P_{m-1}(\beta)$, are negative. The opposite happens if P_m passes through ξ from positive to negative. Thus in passing from α to β in

$$P_m, \quad P_{m-1}$$

one variation is lost. Hence, in the same passage, and consequently also in the passage from ξ to η, in the chain

$$P_{m-1}, \quad P_{m-2}, \dots, P_0$$

one variation must likewise be lost. We conclude:

Between two successive roots of $P_m = 0$ there lies one and only one root of $P_{m-1} = 0$.

§ 88. Second example

As a second example we consider an equation that, for brevity, we shall call the secular equation, because the investigation of the secular perturbations of the planets first led to it.[18] It is also known elsewhere under this name, but it occurs in many other investigations as well, for example in the determination of the principal axes of a surface of second degree and in the theory of small oscillations. Its general analytic significance lies in the fact that it furnishes a special, the so-called orthogonal, transformation of a quadratic form into a sum of squares. We shall take the equation here as it stands, without relation to any particular application, and discuss it by means of our general theorems.

Let $a_{i,k}$, as i and k run through the indices $1, 2, 3, \dots, n$, be any system of real quantities, and suppose that

$$a_{i,k} = a_{k,i}. \tag{1}$$

We consider the symmetric determinant

$$L_n(x) = \begin{vmatrix} a_{1,1} - x & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} - x & \cdots & a_{2,n} \\ \cdots & \cdots & & \cdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} - x \end{vmatrix}, \tag{2}$$

which is an integral rational function of degree n in x, with coefficient $(-1)^n$ at x^n. The equation $L_n(x) = 0$ is the object of our consideration.

We form the chain of principal minors taken with alternating signs,

$$L_n(x), \quad -L_{n-1}(x), \quad L_{n-2}(x), \dots, \quad (-1)^{n-1} L_1(x), \quad (-1)^n, \tag{3}$$

and shall show that, provided no two consecutive quantities in (3) vanish simultaneously, this is a Sturm chain.

This is a simple consequence of the formula that we already used in § 82 for a similar purpose, and which we may write as

$$L_k S_k - T_k^2 = L_{k-1} L_{k+1}, \tag{4}$$

where S_k and T_k are integral rational functions of x.

We have only to show that the requirements § 86, 1. to 4. are satisfied. Requirement 1. has been included in the assumption, and 2. is satisfied because the last element is equal to ± 1. From (4) it follows that if $L_k = 0$, then L_{k-1} and L_{k+1} have opposite signs, so condition 3. is fulfilled.

[18]Laplace, Histoire de l'Académie des Sciences 1772. Among more recent works one may compare Dziobek, Theorie der Planetenbewegung, Leipzig 1888.

Only condition 4. remains. Applying formula (4) with $k = n-1$, we obtain

$$\frac{\partial L_n}{\partial a_{n,n}} \frac{\partial L_n}{\partial a_{n-1,n-1}} - \left(\frac{\partial L_n}{\partial a_{n,n-1}}\right)^2 = L_n L_{n-2}.$$

It follows that, when $L_n = 0$,

$$\frac{\partial L_n}{\partial a_{n,n}}, \qquad \frac{\partial L_n}{\partial a_{n-1,n-1}}$$

have the same sign; similarly one concludes that all principal minors of L_n,

$$\frac{\partial L_n}{\partial a_{i,i}},$$

have the same sign.

But the derivative of L_n (compare § 22) is

$$L_n'(x) = -\sum_{i=1}^{n} \frac{\partial L_n}{\partial a_{i,i}},$$

and therefore, for a value x at which $L_n(x)$ vanishes, it has the sign opposite to that of L_{n-1}. This is precisely what requirement 4. demands. Hence:

I. If α, β are two real values with $\alpha < \beta$, then the number of roots of $L_n(x)$ lying between α and β is equal to the excess of the number of variations of the chain (3) for $x = \alpha$ over the number of variations for $x = \beta$.

The highest power of x that occurs in $L_k(x)$ is, as already remarked,

$$(-1)^k x^k.$$

If the absolute value of x is sufficiently large, the sign of this term determines the sign of $L_k(x)$. Thus, if for α we take a sufficiently large negative value and for β a sufficiently large positive value, then in (3), for $x = \alpha$, all sign transitions are variations, while for $x = \beta$ all are permanences. Thus there are n roots between α and β, and we obtain:

II. The equation $L_n(x) = 0$ has only real roots.

These two theorems, however, are of only limited applicability as long as we have not freed ourselves from the assumption that in the chain of the L_k no two consecutive terms may vanish simultaneously. This restriction can be removed by the following simple consideration.

Suppose that for some value of x, L_k vanishes but L_{k-1} does not. We can choose the quantities $a_{k+1,i}$, which do not occur in L_k, so that L_{k+1} does not vanish for this value of x; for L_{k+1} cannot vanish identically for all $a_{k+1,i}$, since it contains the term

$$-a_{k+1,k}^2 L_{k-1}$$

[§ 23, (12)].

Therefore, if in the chain

$$L_n, \quad -L_{n-1}, \quad L_{n-2}, \ldots, \quad \pm L_1, \quad \mp 1 \tag{5}$$

several consecutive terms vanish for some value of x, then by changing the $a_{i,k}$ we may derive another sequence

$$L_n', \quad -L_{n-1}', \quad L_{n-2}', \ldots, \quad \pm L_1', \quad \mp 1 \tag{6}$$

in which no two consecutive terms vanish for any value x of the interval $\alpha \ldots \beta$.

At the same time the $a_{i,k}'$, on which the L' depend, can be taken so that they differ from the $a_{i,k}$ by less than any prescribed quantity ω.

If now the numbers α, β are chosen so that no term of the sequence (5) vanishes for $x = \alpha$ or $x = \beta$, then ω can be taken so small that corresponding terms of (5) and (6) have the same signs. By our theorem, the number of roots of $L'_n = 0$ between α and β is determined by the signs of the sequence (6).

On the other hand, ω can also be taken so small that the roots of $L'_n = 0$ differ arbitrarily little from those of $L_n = 0$ (§ 40).

A double root of $L_n = 0$ may, it is true, pass into two simple roots of $L'_n = 0$; but since $L'_n = 0$ has no imaginary roots, these are real. The same applies if $L_n = 0$ has multiple roots.

Thus the theorems I, II remain valid when the assumption that no consecutive terms in the chain of the L_k vanish is dropped; only the multiple roots must then be counted according to their multiplicity.

§ 89. The Sturm functions

After these special examples we turn to the procedure by which a Sturm chain is obtained in all cases.[19] The construction of these functions is in principle extraordinarily simple, although in practical execution it is usually not feasible.

We restrict ourselves here to equations without repeated roots; or we assume that before the application, $f(x)$ has been freed from every common factor with its derivative $f'(x)$.

If we then put

$$f_1(x) = f'(x), \tag{1}$$

then condition § 86, 4. is certainly satisfied. Now we proceed as if we were finding the greatest common divisor of $f(x)$ and $f_1(x)$, but each time reversing the sign of the remainder. Thus by division we form the equations

$$\begin{aligned} f &= q_1 f_1 - f_2, \\ f_1 &= q_2 f_2 - f_3, \\ &\cdots \\ f_{m-2} &= q_{m-1} f_{m-1} - f_m, \end{aligned} \tag{2}$$

where the $q_1, q_2, \ldots, q_{m-1}$, and likewise $f_1, f_2, \ldots, f_m$, are integral rational functions of x. The degrees of the functions $f, f_1, f_2, \ldots, f_m$ decrease, and therefore one can continue the operation until f_m is constant, or at least no longer vanishes in the interval under consideration. That f_m cannot be zero is a consequence of the assumption that f and f_1 have no common divisor.

That in the sequence

$$f, \quad f_1, \quad f_2, \ldots, \quad f_m \tag{3}$$

one actually has a Sturm chain follows immediately by going through the criteria § 86, 1. to 4. For if two consecutive functions of (3) vanished simultaneously, then by (2) all following ones would also vanish; this is impossible because f_m is non-zero.

If $f_\nu(x) = 0$, then from (2) one obtains

$$f_{\nu-1}(x) = -f_{\nu+1}(x),$$

which satisfies all the requirements of § 86.

It is, of course, permissible to multiply the functions in the sequence (3) by positive, for instance constant, factors without ceasing to have a Sturm chain.

[19] Sturm, Mém. sur la résolution des èquations numériques. Mém. de l'Académie de Paris. Sav. ètrang. VI, 1835. Extract in Bull. de Ferussac XI, 1829.

For $n = 2$ we may therefore take the following as the Sturm functions:

$$\begin{aligned} f(x) &= x^2 + ax + b, \\ f_1(x) &= 2x + a, \\ f_2(x) &= a^2 - 4b. \end{aligned}$$

§ 90. Hermite's solution of Sturm's problem

Another path to the solution of Sturm's problem, leading to simpler results, at least as regards carrying out the calculation in detail, was taken by Hermite; he attaches the question to the law of inertia for quadratic forms and to the Tschirnhausen transformation.[20]

Let

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0 \tag{1}$$

be the given equation, and let

$$x_1, x_2, \ldots, x_n \tag{2}$$

be its roots, which we assume to be distinct from one another. We use the functions defined in § 68,

$$f_0(x), f_1(x), f_2(x), \ldots, f_{n-1}(x), \tag{3}$$

and, as there, put, with $t_0, t_1, \ldots, t_{n-1}$ independent variables,

$$y = t_{n-1}f_0(x) + t_{n-2}f_1(x) + \cdots + t_1f_{n-2}(x) + t_0f_{n-1}(x). \tag{4}$$

Let $y_1, y_2, \ldots, y_n$ be the values that y assumes for $x = x_1, x_2, \ldots, x_n$.

If α is any real value, set

$$H_\alpha = (x_1 - \alpha)y_1^2 + (x_2 - \alpha)y_2^2 + \cdots + (x_n - \alpha)y_n^2. \tag{5}$$

This is a quadratic form in the n variables t, and we first determine the number of its negative and positive terms. If x_1 is a real root, and hence y_1 is real as well, the square $(x_1 - \alpha)y_1^2$ is positive or negative according as x_1 is greater or smaller than α. But if x_1, x_2 form an imaginary pair, then y_1, y_2 are also conjugate imaginary, and

$$(x_1 - \alpha)y_1^2 + (x_2 - \alpha)y_2^2$$

splits into one positive and one negative square. This follows as in § 79 by putting

$$y_1\sqrt{x_1 - \alpha} = u + \mathrm{i}v, \qquad y_2\sqrt{x_2 - \alpha} = u - \mathrm{i}v.$$

It follows that the number N_α of negative squares in H_α is equal to the number of imaginary pairs, increased by the number of real roots that are less than α. If we therefore take a second real number $\beta > \alpha$, form H_β, and denote by N_β the number of its negative squares, then the difference

$$N_\beta - N_\alpha$$

is equal to the number of real roots between α and β.

Thus one obtains a means of determining this number, that is, of solving Sturm's problem, once the coefficients of the function H_α have been represented as functions of the coefficients of $f(x)$ and of α, and the number N_α has then been examined.

§ 91. Determination of Hermite's form H

To determine Hermite's form H we can use the means learned in the sixth section. In § 72 we derived the formula

$$yf_s = E_{0,s}f_0 + E_{1,s}f_1 + \cdots + E_{n-1,s}f_{n-1}. \tag{1}$$

Using the relations

$$xf_0 = f_1 - a_1, \quad xf_1 = f_2 - a_2, \quad \ldots, \quad xf_{n-1} = -a_n,$$

[20] Hermite, Remarques sur le théorème de M. Sturm. Comptes rendus of the Paris Academy, vol. 36 (1853).

we get

$$xyf_s = E_{0,s}f_1 + E_{1,s}f_2 + \cdots + E_{n-2,s}f_{n-1} - a_1E_{0,s} - a_2E_{1,s} - \cdots - a_nE_{n-1,s}. \qquad (2)$$

We therefore determine one more row of functions $E_{-1,s}$ by the equation

$$a_0E_{-1,s} + a_1E_{0,s} + a_2E_{1,s} + \cdots + a_nE_{n-1,s} = 0 \qquad (3)$$

and obtain from (1) and (2)

$$\begin{aligned}(x-\alpha)yf_s = {} & (E_{-1,s} - \alpha E_{0,s})f_0 + (E_{0,s} - \alpha E_{1,s})f_1 + \cdots \\ & + (E_{n-2,s} - \alpha E_{n-1,s})f_{n-1}.\end{aligned} \qquad (4)$$

The function $E_{-1,s}$ can, by the relation [§ 72, (7)]

$$a_0t_{n+s} + a_1t_{n+s-1} + \cdots + a_nt_s = 0,$$

be expressed in exactly the same way as the remaining E's. We therefore obtain the following system of formulae [§ 72, (9)]:

$$\begin{aligned}
E_{-1,s} &= a_0t_{n+s} + a_1t_{n+s-1} + \cdots + a_st_n = -a_{s+1}t_{n-1} - \cdots - a_nt_s,\\
E_{0,s} &= a_0t_{n+s-1} + a_1t_{n+s-2} + \cdots + a_st_{n-1} = -a_{s+1}t_{n-2} - \cdots - a_nt_{s-1},\\
E_{1,s} &= a_0t_{n+s-2} + a_1t_{n+s-3} + \cdots + a_st_{n-2} = -a_{s+1}t_{n-3} - \cdots - a_nt_{s-2},\\
&\cdots\\
E_{s-1,s} &= a_0t_n + a_1t_{n-1} + \cdots + a_st_{n-s} = -a_{s+1}t_{n-s-1} - \cdots - a_nt_0,\\
E_{s,s} &= a_0t_{n-1} + a_1t_{n-2} + \cdots + a_st_{n-s-1},\\
E_{s+1,s} &= a_0t_{n-2} + a_1t_{n-3} + \cdots + a_st_{n-s-2},\\
&\cdots\\
E_{n-1,s} &= a_0t_s + a_1t_{s-1} + \cdots + a_st_0.
\end{aligned} \qquad (5)$$

Introduce a second system of variables τ and put

$$z = \tau_{n-1}f_0 + \tau_{n-2}f_1 + \cdots + \tau_0f_{n-1}.$$

Then (4), using the formulae [§ 68, (6)], gives

$$\begin{aligned}
S(f_s) &= (n-s)a_s,\\
S[(x-\alpha)yz] &= na_0\sum_{s=0}^{n-1}(E_{-1,s} - \alpha E_{0,s})\tau_{n-s-1} otag \qquad (1)\\
&\quad + (n-1)a_1\sum_{s=0}^{n-2}(E_{0,s} - \alpha E_{1,s})\tau_{n-s-1} otag \qquad (2)\\
&\quad + \cdots + a_{n-1}\sum_{s=0}^{0}(E_{n-2,s} - \alpha E_{n-1,s})\tau_{n-s-1}. \qquad (6)
\end{aligned}$$

This passes directly into the function H when $z = y$ is put.

For example, in the case $n = 3$ one obtains

$$\begin{array}{ll}
E_{-1,0} = -a_1t_2 - a_2t_1 - a_3t_0, & E_{0,0} = a_0t_2,\\
E_{-1,1} = -a_2t_2 - a_3t_1, & E_{0,1} = -a_2t_1 - a_3t_0,\\
E_{-1,2} = -a_3t_2, & E_{0,2} = -a_3t_1,\\
E_{1,0} = a_0t_1, & E_{2,0} = a_0t_0,\\
E_{1,1} = a_0t_2 + a_1t_1, & E_{2,1} = a_0t_1 + a_1t_0,\\
E_{1,2} = -a_3t_0, & E_{2,2} = a_0t_2 + a_1t_1 + a_2t_0,
\end{array}$$

and from these the coefficients of the form H are readily computed from (6):

$$\begin{aligned}
H_{2,2} &= \quad -a_0(a_1 + 3a_0\alpha),\\
H_{1,1} &= \quad -3a_0a_3 - a_1a_2 - 2(a_1^2 - a_0a_2)\alpha,\\
H_{0,0} &= \quad -a_2a_3 + (2a_1a_3 - a_2^2)\alpha,\\
H_{1,0} &= \quad -2a_1a_3 + (3a_0a_3 - a_1a_2)\alpha,\\
H_{2,1} &= \quad -2a_0(a_2 + a_1\alpha),\\
H_{0,2} &= \quad -a_0(3a_3 + a_2\alpha).
\end{aligned}$$

§ 92. The determinant of Hermite's form

For the question of the number of negative squares of the form H, knowledge of its determinant is important. This determinant can be computed in general as follows.

Put

$$H = \sum_{i,k=0}^{n-1} h_{i,k} t_i t_k.$$

The coefficients $h_{i,k}$ whose determinant is to be formed are given by formula § 91, (6). Thus $h_{i,k}$ is the coefficient of $t_i t_k$ in that formula, namely

$$h_{i,k} = S\big((x-\alpha) f_{n-i-1} f_{n-k-1}\big).$$

By the multiplication theorem the determinant formed from these n^2 quantities can be written as

$$\Delta = \begin{vmatrix} (x_1-\alpha)f_0(x_1) & (x_1-\alpha)f_1(x_1) & \cdots & (x_1-\alpha)f_{n-1}(x_1) \\ \vdots & \vdots & & \vdots \\ (x_n-\alpha)f_0(x_n) & (x_n-\alpha)f_1(x_n) & \cdots & (x_n-\alpha)f_{n-1}(x_n) \end{vmatrix} \begin{vmatrix} f_0(x_1) & \cdots & f_0(x_n) \\ f_1(x_1) & \cdots & f_1(x_n) \\ \vdots & & \vdots \\ f_{n-1}(x_1) & \cdots & f_{n-1}(x_n) \end{vmatrix}.$$

Since

$$a_0(x_1-\alpha)(x_2-\alpha)\cdots(x_n-\alpha) = (-1)^n f(\alpha)$$

and since the remaining determinant of the $f_i(x_k)$ is a product of two determinants whose square is $a_0^{2n-2}D$, where D denotes the discriminant of $f(x)$ (§ 46), we get

$$\Delta = (-1)^n a_0 f(\alpha) D. \tag{1}$$

Because D is not to vanish, Δ vanishes only when α is put equal to one of the roots of $f(x) = 0$; this case is also to be excluded.

Let a chain of principal minors of Δ, beginning with the lowest, be denoted by

$$\Delta_1(\alpha), \Delta_2(\alpha), \ldots, \Delta_n(\alpha).$$

Then by § 83, V, the number of sign changes in

$$1, \Delta_1(\alpha), \Delta_2(\alpha), \ldots, \Delta_n(\alpha) \tag{2}$$

is equal to the number N_α of negative squares in H; and if we count the corresponding number N_β in

$$1, \Delta_1(\beta), \Delta_2(\beta), \ldots, \Delta_n(\beta),$$

then $N_\beta - N_\alpha$ is the number of roots of $f(x)$ lying between α and β.

For the cubic equation the expressions are obtained from the end of the preceding paragraph. We form the chain

$$a_0, \quad -H_{2,2}, \quad -\frac{H_{2,2}H_{1,1} - H_{1,2}^2}{a_0}, \quad -\Delta,$$

which plainly has the same number of sign changes as (2), and thereby obtain the four functions

$$\begin{aligned}
&a_0,\\
&-3a_0\alpha - a_1,\\
&2a_2a_0\alpha^2 + \alpha(2a_1a_2 - 9a_0a_3 - a_1^2) - 3a_0a_3 - a_1a_2,\\
&f(\alpha)D.
\end{aligned}$$

As a check, take $\alpha = -\infty$, $\beta = +\infty$, and also put $a_0 = 1$. The signs are then determined by

$$\begin{array}{llll}
1, & +1, & a_1^2 - 3a_2, & D,\\
1, & -1, & a_1^2 - 3a_2, & -D.
\end{array}$$

In counting one must note that, when D is positive, $a_1^2 - 3a_2$ cannot be negative [§ 47, (8)].

§ 93. Elements of the theory of characteristics

Sturm's theorems are generalized in an extraordinary way by Kronecker's theory of characteristics, which pursues for systems of equations with arbitrarily many variables the same aim as Sturm's theorem. We restrict ourselves here to the simplest case, which we shall use for enclosing the complex roots of an equation; and we freely use geometrical intuition and the notation of differential calculus.[21]

We first consider two real functions $\varphi(x, y)$ and $\psi(x, y)$, and interpret the real variables x, y as rectangular coordinates in a plane. The equations

$$\varphi(x, y) = 0, \qquad \psi(x, y) = 0 \tag{1}$$

then represent two curves, which we call simply the curve φ and the curve ψ. We assume first that these curves are closed and do not extend to infinity. We further assume that φ is negative inside the curve φ and positive outside it; similarly, ψ is negative inside ψ and positive outside it.

Let, as Fig. 6 shows, ϑ be the angle made with the positive x-axis by the normal n drawn at any point of φ in the direction in which φ increases, that is, outwards. If $\varphi'(x)$ and $\varphi'(y)$ denote the partial derivatives of φ, and the square root is taken positive, then

$$\cos\vartheta = \frac{\varphi'(x)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}, \qquad \sin\vartheta = \frac{\varphi'(y)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}.$$

Draw the tangent t so that it has to this normal n the same position as the positive y-axis has to the positive x-axis, and denote by ξ the angle it makes with the positive x-axis. Then

$$\cos\xi = \frac{-\varphi'(y)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}, \qquad \sin\xi = \frac{\varphi'(x)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}.$$

If the advance on φ in the direction t is called positive, and if dx, dy are the projections of this advance, then dx and dy are proportional to, and have the same signs as, $-\varphi'(y)$ and $\varphi'(x)$. If θ is any other function, then

$$d\theta = \theta'(x)dx + \theta'(y)dy$$

has the same sign as the functional determinant

$$[\varphi, \theta] = \varphi'(x)\theta'(y) - \varphi'(y)\theta'(x). \tag{2}$$

With the usual convention for the direction of the coordinates, the positive direction of advance is the one for which the interior of the region lies on the left.

[21] Monatsberichte der Berliner Akademie, March and August 1869, February 1873, February 1878.

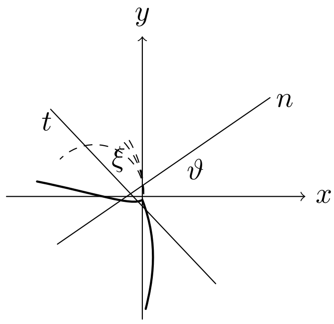


Fig. 6.

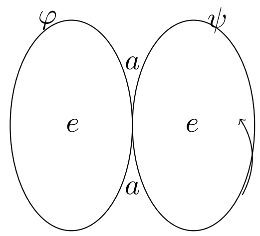


Fig. 7.

If now the positive direction of advance on φ, at an intersection point of the curves φ, ψ, leads into the interior of ψ, then $d\psi$, and hence also $[\varphi, \psi]$, is negative; in the opposite case it is positive (Fig. 7). We call an intersection point of φ and ψ an exit point $A(\varphi, \psi)$ or an entry point $E(\varphi, \psi)$ according as the positive direction on φ leads from the interior of ψ to the exterior, or from the exterior to the interior. Thus

$$\begin{array}{lll} \text{at an } A(\varphi, \psi) & \text{one has} & [\varphi, \psi] > 0, \\ \text{at an } E(\varphi, \psi) & \text{one has} & [\varphi, \psi] < 0. \end{array} \tag{4}$$

The number of $A(\varphi, \psi)$ is equal to the number of $E(\varphi, \psi)$; and if we interchange φ and ψ, the $A(\varphi, \psi)$ and $E(\varphi, \psi)$ pass into $E(\psi, \varphi)$ and $A(\psi, \varphi)$.

The two functions φ, ψ determine in a unique way a plane region characterized by the fact that in it the product $\varphi\psi$ is negative. We shall call it the inner region (φ, ψ) or (ψ, φ). In Fig. 7 it is the hatched region.

§ 94. Characteristic of a system of three functions

We now add to the two functions φ, ψ a third function $f(x, y)$. The curve represented by the equation $f = 0$, or the curve f, is also to be closed in the finite part of the plane; and we shall further suppose that it passes through none of the intersection points of φ and ψ.

We give the three functions a definite cyclic order

$$f, \quad \varphi, \quad \psi,$$

so that f follows again after ψ, and with this order we associate a definite, say positive, direction of advance on each of the three curves. We stipulate that with the reversed order

$$f, \quad \psi, \quad \varphi$$

the opposite, i.e. negative, direction of advance is associated on each curve.

In the first order we traverse the curve f in the positive sense and consider the intersection points of f and φ. Such a point is called an exit point $A(f; \varphi, \psi)$ when the positive direction of f leads from the interior of the inner region (φ, ψ) to the exterior, and an entry point $E(f; \varphi, \psi)$ when it leads from the exterior to the interior.

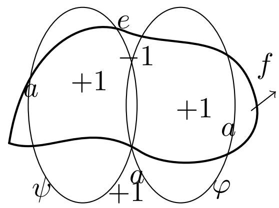


Fig. 8.

In Fig. 8 the points a are exit points and the point e is an entry point. The total number of entry and exit points agrees with the number of intersections of f and φ, and is therefore, since both curves are closed, an even number. If a is the number of points $A(f;\varphi,\psi)$ and e the number of points $E(f;\varphi,\psi)$, then $e-a$ is also even, and

$$\frac{1}{2}(e-a)=k$$

is an integer; following Kronecker it is called the characteristic of the system of functions f,φ,ψ. In the case of Fig. 8 it is equal to -1.

If we cyclically permute f,φ,ψ, we get three determinations of the characteristic; if we reverse the order and, according to the convention just made, replace the positive directions by the negative ones, we get three more. If the symbols $A(f;\varphi,\psi)$ and $E(f;\varphi,\psi)$ are now also used for the numbers of the corresponding points, these determinations are

$$\begin{array}{ll}
1. & \frac{1}{2}\,[E(f;\varphi,\psi)-A(f;\varphi,\psi)],\\
2. & \frac{1}{2}\,[E(\varphi;\psi,f)-A(\varphi;\psi,f)],\\
3. & \frac{1}{2}\,[E(\psi;f,\varphi)-A(\psi;f,\varphi)],\\
4. & \frac{1}{2}\,[E(f;\psi,\varphi)-A(f;\psi,\varphi)],\\
5. & \frac{1}{2}\,[E(\psi;\varphi,f)-A(\psi;\varphi,f)],\\
6. & \frac{1}{2}\,[E(\varphi;f,\psi)-A(\varphi;f,\psi)].
\end{array}$$

The fundamental theorem is that these six determinations give the same number.

To prove it, traverse the curve f in the positive sense and consider all its intersections with φ and ψ. The path along f must enter the inner region (φ,ψ) as often as it leaves it, because it returns to its starting point. But the curve f enters at the points $E(f;\varphi,\psi)$ and $A(f;\psi,\varphi)$ and leaves at the points $A(f;\varphi,\psi)$ and $E(f;\psi,\varphi)$. Thus

$$E(f;\varphi,\psi)+A(f;\psi,\varphi)=A(f;\varphi,\psi)+E(f;\psi,\varphi),$$

which proves the agreement of 1. and 4.

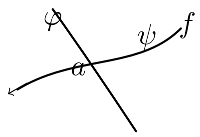


Fig. 9.

Second, if we traverse the curve φ in the negative sense and look at its intersections with f, every point $E(f;\varphi,\psi)$ is also a point $E(\varphi;f,\psi)$, and every point $A(f;\varphi,\psi)$ is also a point $A(\varphi;f,\psi)$. In Fig. 9, where the point a denotes any intersection of f and φ, this point is an $E(f;\varphi,\psi)$ and an $E(\varphi;f,\psi)$ if ψ is positive at a, and an $A(f;\varphi,\psi)$ and an $A(\varphi;f,\psi)$ if ψ is negative at a.

This gives the agreement of 1. with 6. Reapplying the first argument gives the agreement of 6. with 2.; then the second gives that of 2. with 5.; and finally the first gives that of 5. with 3. Hence all six expressions agree. We are therefore entitled to call the number so determined simply the characteristic of the system of functions (f,φ,ψ).

§ 95. Relation of the characteristic to the intersection points

It must now be observed that the designation of the intersection points of two curves as entry or exit points in § 93 is essentially different from that in § 94. There only the relation of the points to the two intersecting curves was involved, while in § 94 the relation to a third curve also enters. This is expressed completely in the notation $E(\varphi,\psi)$, $A(\varphi,\psi)$ and $E(f;\varphi,\psi)$, $A(f;\varphi,\psi)$. We must now examine more closely how the two modes of designation are related.

For clarity we choose a somewhat simpler figure than before, which at the same time illustrates all possible relations (Fig. 10).

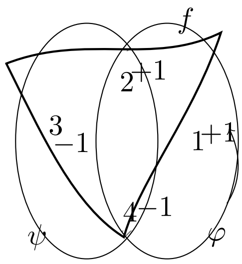


Fig. 10.

At a point $E(\varphi,\psi)$ the curve φ enters the region of the curve ψ; that is, ψ passes from positive to negative values. If at the same time f is positive, then this point is an $E(\varphi;\psi,f)$, as is point 1 in the figure; if f is negative, it is an $A(\varphi;\psi,f)$, as is point 3. Thus:

A point $E(\varphi,\psi)$ is an	$E(\varphi;\psi,f)$,	if $f>0$,	(point 1 of Fig. 10),
	$A(\varphi;\psi,f)$,	if $f<0$,	(point 3 of Fig. 10),
A point $A(\varphi,\psi)$ is an	$E(\varphi;\psi,f)$,	if $f<0$,	(point 2 of Fig. 10),
	$A(\varphi;\psi,f)$,	if $f>0$,	(point 4 of Fig. 10).

Since at the points $E(\varphi,\psi)$ the functional determinant $[\varphi,\psi]$ is negative, and at $A(\varphi,\psi)$ positive (§ 93), we can summarize the result as follows: $E(\varphi;\psi,f)$ is the number of intersections of φ,ψ at which $[\varphi,\psi]f<0$, and $A(\varphi;\psi,f)$ is the number of intersections at which $[\varphi,\psi]f>0$. Hence the characteristic is half the excess of the first number over the second.

This theorem can be expressed in the following way, where it appears wholly independent of magnitudes and dependent only on the relative positions of the curves and their intersections. Distinguish the intersections of φ,ψ according as they are exit points $A(\varphi,\psi)$ or entry points $E(\varphi,\psi)$ by the symbols α and ε; likewise distinguish them according as they are exterior or interior points with respect to the curve f by a and ϵ. Give the points $\alpha\epsilon$ and εa the character $+1$, and the points αa and $\varepsilon\epsilon$ the character -1. Then the characteristic of the system (f,φ,ψ) is equal to one half of the sum of the characters of all intersection points of φ,ψ.[22]

The theorems and definitions just given remain unchanged when the curves under consideration consist of several closed components; double points of the curves may also occur. It is only necessary that for every direction of advance on one of the curves it be completely determined whether that direction is to be considered positive or negative; that is, every piece of a curve f, φ, or ψ must separate regions in which the corresponding function has opposite signs.

Only the cases must be excluded in which one curve passes through a double point of another, or in which the curves touch one another, or in which the three curves f,φ,ψ pass through one point; for in these cases the designation of a point as an entry or exit point would become doubtful. If the curves extend to infinity, they must, in order to make our theorems applicable, be closed by arbitrarily added curve pieces, as we shall see in the next paragraph.

§ 96. Application of characteristics to enclosing the complex roots of an equation

Let now

$$z=x+y\mathrm{i}$$

be a complex variable and

$$F(z)=\varphi(x,y)+\mathrm{i}\psi(x,y) \tag{1}$$

an integral rational function of z with real or complex coefficients, where $\varphi(x,y)$ and $\psi(x,y)$ are real functions of the real variables x,y. We assume that $F(z)$ and $F'(z)$ do not vanish simultaneously.

[22] Kronecker understands something else by the character of a point.

The function $F(z)$ vanishes only at the intersections of the two curves φ and ψ. Differentiating (1) gives

$$F'(z) = \varphi'(x) + \mathrm{i}\psi'(x) = -\mathrm{i}\varphi'(y) + \psi'(y),$$

hence

$$\psi'(y) = \varphi'(x), \qquad \varphi'(y) = -\psi'(x),$$

and therefore

$$[\varphi, \psi] = \varphi'(x)\psi'(y) - \varphi'(y)\psi'(x) = \varphi'(x)^2 + \varphi'(y)^2.$$

Thus the functional determinant $[\varphi, \psi]$ is never negative and vanishes only where $\varphi'(x)$ and $\varphi'(y)$, and hence also $\psi'(x)$ and $\psi'(y)$, vanish simultaneously.

But this never occurs at an intersection of φ and ψ, for otherwise $F(z)$ and $F'(z)$ would both vanish there. The positive direction of advance is completely determined on every part of the curve φ or ψ by the sign of the function in the adjacent regions; possible double points at which φ, $\varphi'(x)$ and $\varphi'(y)$ vanish simultaneously are no exception.

Since $[\varphi, \psi]$ is positive at all intersections of the curves φ, ψ, all these points are exit points $A(\varphi, \psi)$. It follows that the curves φ and ψ cannot be closed, but must extend to infinity; otherwise an exit point would necessarily be followed by an entry point. Apart from this, the curves φ, ψ here too determine an inner region in which the product $\varphi\psi$ is negative.

We now add to φ, ψ a third function f, whose equation $f = 0$ represents a closed curve, and determine the characteristic of the system exactly as before, by advancing along the curve f:

$$k = \frac{1}{2}[E(f; \varphi, \psi) - A(f; \varphi, \psi)] = \frac{1}{2}[E(f; \psi, \varphi) - A(f; \psi, \varphi)]. \tag{1}$$

Thus, for example, in Fig. 11 the characteristic is $k = 2$.

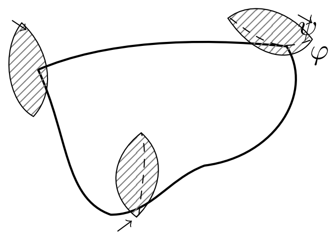


Fig. 11.

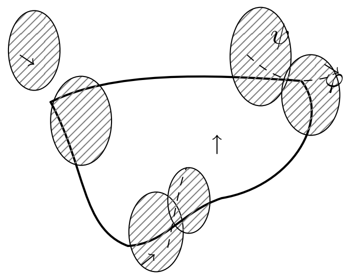


Fig. 12.

To apply our theorems, we must close the curves φ, ψ in some way by arbitrarily drawn connections, all lying outside f. The characteristic of this system is then still determined by (1). As an example, Fig. 12 completes Fig. 11 by drawing such closing connections. The intersections thereby added may be partly exit points $A(\varphi, \psi)$ and partly entry points $E(\varphi, \psi)$. Let their numbers be denoted by

$$A'(\varphi, \psi), \qquad E'(\varphi, \psi),$$

while $A(\varphi, \psi)$ denotes the number of originally present intersections. One must always have

$$A(\varphi, \psi) + A'(\varphi, \psi) = E'(\varphi, \psi). \tag{2}$$

We divide the points $A(\varphi, \psi)$ into two groups $A_a(\varphi, \psi)$ and $A_\epsilon(\varphi, \psi)$, the first lying outside and the second inside f. Then the characters of the intersections are

$$\begin{aligned} A_a(\varphi, \psi) &: -1, \\ A_\epsilon(\varphi, \psi) &: +1, \\ A'(\varphi, \psi) &: -1, \\ E'(\varphi, \psi) &: +1. \end{aligned}$$

The double value of the characteristic is therefore, if we omit the notation (φ, ψ) for simplicity (§ 95),

$$-A_a + A_\epsilon - A' + E' = 2k; \tag{3}$$

adding equation (2) gives

$$k = A_\epsilon. \tag{4}$$

That is, the characteristic is equal to the number of intersections of φ, ψ lying inside f.

We have thus found a means of determining, from the characteristic, the number of roots of the equation

$$F(z) = 0$$

whose geometrical images lie inside an arbitrarily given curve. The arbitrarily added exterior connections play no role in this and may be suppressed entirely.

§ 97. Determination of the characteristic

To make the theorem just proved applicable, we still have to show how to determine characteristics in practice. For this purpose imagine the curve f divided into partial arcs in such a way that in any one of them there lies only one intersection with the curve φ or one intersection with the curve ψ. Let a (Fig. 13) be the beginning, b the end of such an arc, and let ξ be the intersection of f with φ lying in it. Then ξ is a point $E(f; \varphi, \psi)$ if the product $\varphi\psi$ is positive at a, and a point $A(f; \varphi, \psi)$ if $\varphi\psi$ is negative at a; this gives the possibility of determining the characteristic completely from the signs of φ, ψ at the points dividing the arcs.

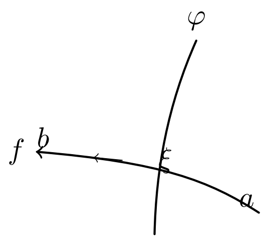


Fig. 13.

How to divide the curve f into such arcs is taught by Sturm's theorem. To this end, think of x and y as functions of a variable t which, along the curve f, runs through all real values from $-\infty$ to $+\infty$ (or also only through the values of a finite interval). Then φ and ψ also become functions of t, and the location of their roots is to be investigated only by Sturm's theorem.

For example, if as in Fig. 14 the curve f is a circle with equation

$$(x - \alpha)^2 + (y - \beta)^2 = \rho^2,$$

then we may put

$$x - \alpha = \rho \frac{1 - t^2}{1 + t^2}, \qquad y - \beta = \rho \frac{2t}{1 + t^2},$$

and t simply runs through all values from $-\infty$ to $+\infty$, while the point x, y moves over the whole circle in the positive sense. After multiplication by a suitable power of $1 + t^2$, φ and ψ become integral rational functions of t, to which Sturm's theorem may be applied.

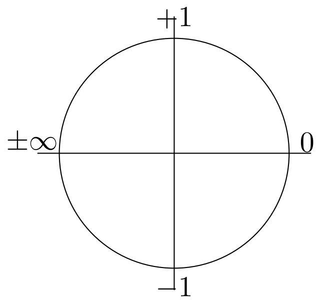


Fig. 14.

§ 98. Gauss's first proof of the fundamental theorem of algebra

Gauss gave three different proofs of the principal theorem of algebra, that an equation of degree n has n roots. The first of them, found in his doctoral dissertation and later further developed and presented differently, rests, although in another guise, on the thought underlying the concept of the characteristic.[23] We therefore close this section with an account of that proof.

Let $z = x + y\mathrm{i}$ and let

$$F(z) = \varphi(x, y) + \mathrm{i}\psi(x, y) \tag{1}$$

be an integral rational function of z of degree n, with real or complex coefficients, in which z^n has coefficient 1; thus

$$F(z) = z^n + a_1 z^{n-1} + a_2 z^{n-2} + \cdots + a_n. \tag{2}$$

We again assume that $F(z)$ and $F'(z)$ have no common divisor. The proof that $F(z)$ vanishes for n values of z rests on the fact that, if for f we choose a circle of sufficiently large radius, the characteristic of the system of functions f, φ, ψ can be determined directly and is found to be n. Then § 96 implies that in the chosen circle there are n points z at which $F(z)$ vanishes.

We use polar coordinates and put, since $a_1, a_2, a_3, \ldots, a_n$ may be complex,

$$a_1 = p_1 e^{\mathrm{i}q_1}, \quad a_2 = p_2 e^{\mathrm{i}q_2}, \quad \ldots, \quad a_n = p_n e^{\mathrm{i}q_n}, \qquad z = Re^{\mathrm{i}\vartheta},$$

where $p_1, p_2, \ldots, p_n$ are positive and the angles $q_1, q_2, \ldots, q_n, \vartheta$ may be chosen in any interval of length 2π. Hence

$$\begin{aligned} \varphi(x, y) &= R^n \cos n\vartheta + p_1 R^{n-1} \cos[(n-1)\vartheta + q_1] + \cdots + p_n \cos q_n, \\ \psi(x, y) &= R^n \sin n\vartheta + p_1 R^{n-1} \sin[(n-1)\vartheta + q_1] + \cdots + p_n \sin q_n, \end{aligned} \tag{3}$$

and we also form the derivatives with respect to ϑ:

$$\begin{aligned} \frac{d\varphi}{d\vartheta} &= -nR^n \sin n\vartheta - (n-1)p_1 R^{n-1} \sin[(n-1)\vartheta + q_1] - \cdots - p_{n-1} \sin(\vartheta + q_{n-1}), \\ \frac{d\psi}{d\vartheta} &= nR^n \cos n\vartheta + (n-1)p_1 R^{n-1} \cos[(n-1)\vartheta + q_1] + \cdots + p_{n-1} \cos(\vartheta + q_{n-1}). \end{aligned} \tag{4}$$

We now divide the circumference of the circle of radius R into $4n$ parts, each of angular size

$$2\omega = 2\frac{\pi}{4n},$$

beginning at $\vartheta = -\omega$. The division points are therefore

$$-\omega, \quad \omega, \quad 3\omega, \quad 5\omega, \ldots, (8n-3)\omega,$$

and the intervals are numbered in order as $1, 2, 3, \ldots, 4n$. Fig. 15 shows this division for $n = 3$.

[23] Gauss, *Demonstratio nova theorematis omnem functionem algebraicam rationalem integram unius variabilis in factores reales primi vel secundi gradus resolvi posse* (1799); Beiträge zur Theorie der algebraischen Gleichungen (1849). Kronecker, Monatsberichte der Berliner Akademie, 21 February 1878.

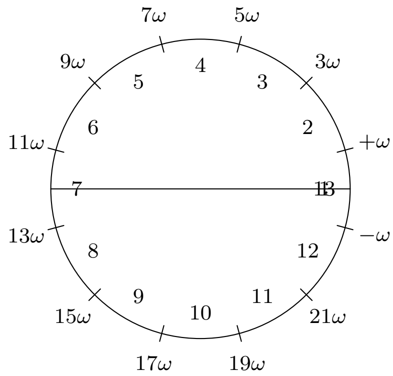


Fig. 15.

In the intervals

$$1, 3, 5, \ldots, (4n-1)$$

the absolute value of $\cos n\vartheta$ is greater than $1 : \sqrt{2}$, and its sign is alternately positive and negative. In the intervals

$$2, 4, 6, \ldots, 4n$$

the absolute value of $\sin n\vartheta$ is greater than $1 : \sqrt{2}$ and likewise alternates in sign.

But R can be taken so large that in the intervals $1, 3, 5, \ldots, (4n-1)$ the signs of φ and $\frac{d\psi}{d\vartheta}$ agree with that of $\cos n\vartheta$, and in the intervals $2, 4, 6, \ldots, 4n$ the signs of ψ and $\frac{d\varphi}{d\vartheta}$ agree with that of $\sin n\vartheta$. Since therefore φ changes sign in the intervals $2, 4, 6, \ldots, 4n$, it must vanish in each of them; and since the derivative $\frac{d\varphi}{d\vartheta}$ does not change sign there, it can vanish only once in each.

Similarly, ψ must vanish once and only once in each of the intervals $1, 3, 5, \ldots, 4n-1$. We have therefore a division of the circumference of the circle of the kind required in the preceding paragraph for determining the characteristic; and since at the division points $\omega, 5\omega, 9\omega, \ldots, (4n-3)\omega$ the product $\varphi\psi$ is positive, only points $E(f; \varphi, \psi)$ occur on the circumference, and their number is $2n$. Thus the number of roots of $F(z)$ is equal to n, as was to be proved.

Ninth Section.

Estimation of the Roots.

§ 99. The Budan–Fourier Theorem

The functions of Sturm chains do solve, completely and without exception, the problem of determining the number of roots of an equation between prescribed limits; but their actual computation is usually difficult and often impracticable. There is a series of theorems for estimating the number of roots between given limits. They supply much simpler rules, but of course they do not answer the question completely: they give only a maximum beyond which the number of roots cannot go. These rules are often very useful and sufficient in applications, and therefore they must not be omitted here.

We first consider a procedure due to Budan and afterwards used and extended by Fourier.[24] In place of a Sturm chain let us consider the sequence of derivatives of a real function $f(x)$ of degree n,

$$f(x), \quad f'(x), \quad f''(x), \ldots, \quad f^{(n)}(x), \tag{1}$$

where $f^{(n)}(x)$ is a constant, which we suppose different from zero and positive. Apart from a positive numerical factor it is the coefficient of x^n in $f(x)$.

Let α and $\beta > \alpha$ again be two real numbers determining the interval (α, β) in which the real roots of $f(x)$ are to be counted. We shall first suppose that, in the interval (α, β), no two members of the sequence (1) vanish at the same time, and that for $x = \alpha$ and $x = \beta$ no member of the sequence vanishes.

Let x increase continuously from α to β. A change in the succession of signs of (1) can occur only when one of the functions passes through zero. If $f^{(\nu)}(x)$ passes through zero for $x = \xi$, then, according as $f^{(\nu+1)}(\xi)$ is positive or negative, $f^{(\nu)}(x)$ passes from negative to positive values or from positive to negative values; hence between $f^{(\nu)}$ and $f^{(\nu+1)}$ one variation of sign is lost in passing through ξ. This also applies when $f^{(\nu)}$ is the original function $f(x)$ itself. If, however, $f^{(\nu)}$ is one of the derivatives, then a function $f^{(\nu-1)}$ precedes it. Since $f^{(\nu-1)}$ is not zero at $x = \xi$ by hypothesis, while $f^{(\nu)}$ changes its sign in passing through ξ, between $f^{(\nu-1)}$ and $f^{(\nu)}$ either one variation is lost or one variation is gained. Thus, if an interior member of the sequence (1) passes through zero, the number of variations remains unchanged or two variations are lost. If $f(x)$ itself passes through zero, one variation is lost.

This gives the theorem:

> **I.** The number of roots of $f(x)$ lying between α and β is at most equal to the number of variations lost between α and β; and, if it is smaller, the difference is an even number.

In formula form this may be stated as follows. If $V(x)$ denotes the number of sign variations displayed by the sequence (1) for a value x, then the number of roots lying between α and β is

$$V(\alpha) - V(\beta) - 2h, \tag{2}$$

where h is a non-negative integer.

The proof of I still requires a supplement for the case in which several consecutive members of (1) vanish simultaneously. Suppose that, for a value ξ of x between α and β, all the members in the sequence

$$f^{(\nu)}(x), \quad f^{(\nu+1)}(x), \ldots, f^{(\nu+\mu-1)}(x), \quad f^{(\nu+\mu)}(x) \tag{3}$$

[24] Budan, *Nouvelle méthode pour la résolution des équations numériques*, submitted to the Paris Academy in 1803. Fourier, *Analyse des équations déterminées.* Paris 1831. For the history of this question compare Lagrange, *Traité de la résolution des équations*, Note VIII (Works, vol. 8).

except the last vanish, and let the last, $f^{(\nu+\mu)}(\xi)$, be positive. We mark off around ξ two intervals δ_1, δ_2, with all values in δ_1 smaller than ξ and all values in δ_2 larger than ξ, and choose them so small that the functions (3) vanish in them only at ξ; thus $f^{(\nu+\mu)}(x)$ remains positive. Since, when $f^{(\nu+\mu)}(x)$ is positive, $f^{(\nu+\mu-1)}(x)$ increases with x, the latter is negative in δ_1 and positive in δ_2. It follows that $f^{(\nu+\mu-2)}(x)$ decreases in δ_1 and increases in δ_2, and is therefore positive in both intervals. Continuing in this way, one obtains the sign sequence

$$\begin{array}{c|ccccc} & f^{(\nu)} & f^{(\nu+1)} & f^{(\nu+2)} & \cdots & f^{(\nu+\mu)} \\ \hline \delta_1 & (-1)^{\mu} & (-1)^{\mu-1} & (-1)^{\mu-2} & \cdots & + \\ \delta_2 & + & + & + & \cdots & + \end{array} \tag{4}$$

for the sequence (3). Thus in passing through ξ a row consisting entirely of variations is transformed into a row of permanences; μ variations are lost in the sequence (3). If $f^{(\nu+\mu)}(\xi)$ is negative, all the signs in (4) are reversed, and the conclusion is otherwise the same.

If now $\nu = 0$, so that $f^{(\nu)}(x)$ is the original function $f(x)$, then f has at ξ a root of multiplicity $\mu+1$, and in the sequence (1) a loss of μ variations also occurs. If, however, $\nu > 0$ and the function preceding $f^{(\nu)}(x)$, namely $f^{(\nu-1)}(x)$, is not zero, then the following is obtained: for even μ no variation is lost between $f^{(\nu-1)}$ and $f^{(\nu)}$; for odd μ one variation is lost or one variation is gained. Hence the loss of variations in passing through ξ in the sequence

$$f^{(\nu-1)}, \quad f^{(\nu)}, \quad f^{(\nu+1)}, \ldots, f^{(\nu+\mu)} \tag{5}$$

is equal to μ for even μ, and to $\mu+1$ or $\mu-1$ for odd μ; it is therefore always an even number and never negative.

This result can be expressed analytically in a form that makes its meaning clearer. Let $V(x)$ denote, as before, the number of variations in (1) for a specified value of x, with the additional convention that, if some members vanish, they are simply omitted in counting variations. Then $V(x)$ may be treated as a function of x, though it changes only by integral jumps and is discontinuous at the values of x for which some members of (1) vanish. Let $V(x-0)$ and $V(x+0)$ denote the values immediately before and immediately after such a point. The preceding discussion gives, in all cases,

$$V(\xi+0) = V(\xi), \tag{6}$$

and, if ξ is a root of $f(x)$ of multiplicity μ,

$$V(\xi-0) = V(\xi) + \mu + 2h, \tag{7}$$

where h is a non-negative integer.

It follows, as long as none of the functions (1) vanishes at $x = \alpha$ or $x = \beta$, that $V(\alpha) - V(\beta)$ differs from the sum of the multiplicities μ, that is from the number of roots in (α, β) counted with multiplicity, by an even, non-negative number. Thus Theorem I is freed from its earlier restriction.

We may also drop the assumption that none of the functions (1) vanishes at α and β, provided that α and β are not roots of $f(x)$. The theorem may then be applied to two values, the first slightly greater than α, the second slightly less than β, enclosing the same roots. If λ is the number of these roots, then

$$V(\alpha+0) - V(\beta-0) = \lambda + 2h.$$

Since

$$V(\alpha+0) = V(\alpha), \qquad V(\beta-0) = V(\beta) + 2h',$$

it follows that

$$V(\alpha) - V(\beta) = \lambda + 2(h+h'),$$

again the expression of our theorem. If, however, α or β itself is a root of $f(x)$, then the theorem is correct only when these limiting values are not counted as belonging to the interval.

The Budan–Fourier theorem does not, like Sturm's theorem, give a completely certain decision about the number of roots in an interval. Yet in many cases it can replace Sturm's theorem. If the interval (α, β) has been narrowed so far that no variation, or only one variation, is lost from α to β, then it follows with certainty that in the first case no root, and in the second case exactly one root, lies in the interval. The separation of the roots can therefore be given completely by Budan's sequence only when several variations are not lost simultaneously in passing through a value ξ. This behaviour cannot be recognized with certainty, however small the interval (α, β) is chosen; consequently, if after a suitable narrowing of the interval the separation has not succeeded, one must pass to another procedure, in the last resort to Sturm's theorem.

§ 100. Newton's Rule

A supplement to the preceding rule, one that gives a closer approximation to the exact fixing of root bounds, is the rule now to be discussed, due to Newton but first proved by Sylvester.[25]

Let again

$$f(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n = 0 \tag{1}$$

be the equation to be examined. With $n(m)$ denoting, as usual, the product $1 \cdot 2 \cdot 3 \cdots m$, and with $f^{(\nu)}(x)$ denoting the νth derivative, put

$$f_\nu(x) = \frac{(n-\nu)!}{n!} f^{(\nu)}(x), \tag{2}$$

$$F_\nu(x) = f_\nu(x)^2 - f_{\nu-1}(x) f_{\nu+1}(x), \tag{3}$$

with the addition that

$$F_0 = F_n = 1, \qquad F_n = f_n^2, \tag{4}$$

so that these end terms, which is essential, are positive. From (2) and (3) one obtains, for the derivatives,

$$f'_\nu(x) = (n-\nu) f_{\nu+1}(x), \tag{5}$$

$$F'_\nu(x) = (n-\nu+1)(f_{\nu-1} F_{\nu+1} + F_{\nu-1} f_{\nu+1}), \tag{6}$$

up to positive numerical factors in the only sense relevant for signs. Since positive factors do not matter, Budan's sequence may be replaced by

$$f_0, \quad f_1, \quad f_2, \ldots, \quad f_n.$$

Newton's rule now uses the double row

$$\begin{array}{cccccc} f_0, & f_1, & f_2, & \ldots, & f_{n-1}, & f_n, \\ F_0, & F_1, & F_2, & \ldots, & F_{n-1}, & F_n. \end{array} \tag{7}$$

For two consecutive columns, say

$$\begin{array}{cc} f_\nu, & f_{\nu+1} \\ F_\nu, & F_{\nu+1}, \end{array}$$

and for a value of x at which none of the four functions vanishes, the following possibilities for the

[25] Newton, *Arithmetica universalis.* Sylvester, *Transactions of the Royal Irish Academy*, vol. 24 (1871); *Philosophical Magazine*, fourth series, vol. 31. Compare also Petersen, *Theorie der algebraischen Gleichungen.* Copenhagen 1878.

succession of signs may occur:

$$\begin{array}{r|cccc|l}
a) & \begin{smallmatrix}++\\++\end{smallmatrix} & \begin{smallmatrix}++\\--\end{smallmatrix} & \begin{smallmatrix}--\\++\end{smallmatrix} & \begin{smallmatrix}--\\--\end{smallmatrix} & PP \\
b) & \begin{smallmatrix}+-\\+-\end{smallmatrix} & \begin{smallmatrix}+-\\-+\end{smallmatrix} & \begin{smallmatrix}-+\\+-\end{smallmatrix} & \begin{smallmatrix}-+\\-+\end{smallmatrix} & VV \\
c) & \begin{smallmatrix}++\\+-\end{smallmatrix} & \begin{smallmatrix}++\\-+\end{smallmatrix} & \begin{smallmatrix}--\\+-\end{smallmatrix} & \begin{smallmatrix}--\\-+\end{smallmatrix} & PV \\
d) & \begin{smallmatrix}+-\\++\end{smallmatrix} & \begin{smallmatrix}+-\\--\end{smallmatrix} & \begin{smallmatrix}-+\\++\end{smallmatrix} & \begin{smallmatrix}-+\\--\end{smallmatrix} & VP.
\end{array}$$

We designate these behaviours as follows:

a) permanence–permanence PP,
b) variation–variation VV,
c) permanence–variation PV,
d) variation–permanence VP.

The two theorems to be proved are the following.

II. The number of roots of $f(x)$ lying between α and β is either exactly equal to, or smaller by an even number than, the number of variation–permanences lost in the double row (7) in passing from α to β.

III. The number of roots of $f(x)$ lying between α and β is either exactly equal to, or smaller by an even number than, the number of permanence–permanences gained in the double row (7) in passing from α to β.

Since these two theorems do not always give the same upper bound for the number of roots, one will use whichever gives the lower bound.

In the proof we make the following hypotheses, from some of which we shall later free ourselves.

1. $f(x)$ has no double or multiple roots.
2. Of the functions $f_\nu(x)$ no two adjacent ones vanish for the same value of x.
3. Of the functions $F_\nu(x)$ no two adjacent ones vanish for the same value of x.
4. A consequence of 2, by formula (3), is that $f_\nu(x)$ and $F_\nu(x)$ do not vanish for the same value of x.
5. At $x = \alpha$ and $x = \beta$ none of the functions f_ν, F_ν vanishes.

The proof of II and III is then as follows. Let x increase continuously from α to β. A change in the sign sequence can occur only when one of the functions f_ν or F_ν passes through zero; it is therefore enough to examine the effect of such a passage.

Suppose first that one of the middle functions f_ν passes through zero, so that $0 < \nu < n$. According as $f_{\nu+1}$ is positive or negative, f_ν increases or decreases. Thus in the first case $f_\nu(x-0)$ is negative and $f_\nu(x+0)$ positive, and in the second case the reverse occurs. Further, (3) shows that, if $f_\nu = 0$, then $F_{\nu-1}$ and $F_{\nu+1}$ are positive, while F_ν has the sign opposite to that of $f_{\nu-1}f_{\nu+1}$. The remaining possibilities are therefore

	$\nu-1$	ν	$\nu+1$		$\nu-1$	ν	$\nu+1$
$f(x-0)$	$+$	$-$	$+$	$f(x-0)$	$-$	$-$	$+$
$f(x+0)$	$+$	$+$	$+$	$f(x+0)$	$-$	$+$	$+$
F	$+$	$-$	$+$	F	$+$	$+$	$+$

and hence

$$\begin{array}{r|l|l} x-0 & VV,\ VV & PP,\ VP \\ x+0 & PV,\ PV & VP,\ PP. \end{array}$$

Thus neither a VP nor a PP is lost or gained.

If $f(x)$ itself passes through zero, then, taking $f_1(x)$ positive, which is enough in view of the preceding remark, we have

$$\begin{array}{c|cc} & 0 & 1 \\ \hline f(x-0) & - & + \\ f(x+0) & + & + \\ \hline F & + & + \end{array}$$

so that a VP changes into a PP: one VP is lost and one PP is gained.

Finally let F_ν pass through zero, necessarily with $0 < \nu < n$. If $F_\nu = 0$, then $f_{\nu-1}$ and $f_{\nu+1}$ must have the same sign; otherwise F_ν would consist of two positive terms and could not vanish. Moreover, by (6) the derivative $F'_\nu(x)$ has the same sign as $f_{\nu-1}f_\nu F_{\nu+1}$. Hence $F_\nu(x-0)$ has the opposite sign and $F_\nu(x+0)$ the same sign as that product. Omitting cases arising merely by simultaneous reversal of signs, one obtains

$$\begin{array}{c|ccc||c|ccc} & \nu-1 & \nu & \nu+1 & & \nu-1 & \nu & \nu+1 \\ \hline f & + & + & + & f & + & + & + \\ F(x-0) & + & - & + & F(x-0) & - & - & + \\ F(x+0) & + & + & + & F(x+0) & - & + & + \end{array}$$

$$\begin{array}{c|ccc||c|ccc} & \nu-1 & \nu & \nu+1 & & \nu-1 & \nu & \nu+1 \\ \hline f & + & - & + & f & + & - & + \\ F(x-0) & + & + & + & F(x-0) & - & + & + \\ F(x+0) & + & - & + & F(x+0) & - & - & + \end{array}$$

and the variations in the four cases are

$$\begin{array}{r|llll} x-0 & PV, PV & PP, PV & VP, VP & VV, VP \\ x+0 & PP, PP & PV, PP & VV, VV & VP, VV. \end{array}$$

In the first case two PP's are gained; in the second and fourth no change occurs; in the third two VP's are lost. Theorems II and III are thereby proved.

The assumptions 2 and 3, that no two adjacent functions f_ν and no two adjacent functions F_ν vanish for the same x, may be removed. If one retains the first assumption, that there are no multiple roots, then one can change the coefficients a_ν by arbitrarily small amounts so that 2 and 3 are satisfied and, at the same time, the roots between α and β move so little that none leaves the interval. Since the roots occur singly, real roots cannot thereby pass into imaginary ones. Whether the theorem, with proper counting, remains valid for multiple roots as well may be left open here.

§ 101. Descartes' Rule

Descartes' rule, also named after Harriot,[26] gives a rule for estimating the number of positive roots. We can regard it as a special case of Budan's theorem by taking $\alpha = 0$ and $\beta = \infty$, or at least so large that

$$f(x), \quad f'(x), \quad f''(x), \ldots, \quad f^{(n)}(x) \tag{1}$$

all have the same sign for $x > \beta$. This sign is the sign of the coefficient of the highest power of x in $f(x)$, and may be assumed positive.

[26] Harriot, *Artis analyticae praxis ad aequationes algebraicas resolvendas.* 1631. Gauss' Werke, vol. III, p. 67. Lagrange, l. c.

If
$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n,$$
then, apart from positive numerical factors, the functions (1) take at $x = 0$ the values
$$a_n, \quad a_{n-1}, \quad a_{n-2}, \ldots, \quad a_0. \tag{2}$$
This gives the theorem:

> **IV.** The number of positive roots of $f(x) = 0$ is equal to, or smaller by an even number than, the number of sign variations in the sequence of coefficients of $f(x)$, zero coefficients being simply omitted and the root $x = 0$, if present, not being counted.

For negative roots one may either apply Budan–Fourier to the interval $(-\infty, 0)$, or replace x by $-x$, that is, change the signs of $a_1, a_3, a_5, \ldots$, and then apply IV.

In the same manner we specialize Newton's criterion of § 100. We observe that
$$f_\nu(0) = \frac{(n-\nu)!\,\nu!}{n!}\,a_{n-\nu},$$
and consequently
$$F_\nu(0) = \frac{\nu!(\nu-1)!(n-\nu)!(n-\nu-1)!}{n!^2}\left\{\nu(n-\nu)a_{n-\nu}^2 - (\nu+1)(n-\nu+1)a_{n-\nu-1}a_{n-\nu+1}\right\}.$$
Furthermore the row of the $f_\nu(x)$ has only permanences for $x = +\infty$ and only variations for $x = -\infty$. Put
$$A_\nu = \nu(n-\nu)a_\nu^2 - (\nu+1)(n-\nu+1)a_{\nu-1}a_{\nu+1}, \qquad A_0 = 1, \quad A_n = 1. \tag{3}$$
Then A_ν agrees, up to a positive factor, with $F_{n-\nu}(0)$, and the double row (7) of § 100, at $x = 0$ and written in reverse order, has the same signs as
$$A_0, \quad A_1, \quad A_2, \ldots, \quad A_n. \tag{4}$$
In passing to $x = +\infty$ all variations in the row $f(x), f_1(x), \ldots, f_n(x)$ are lost, and in passing to $x = -\infty$ all permanences are lost; hence in the double row of the f_ν, F_ν all VP's are lost for $+\infty$ and all PP's for $-\infty$. By § 100, II and III, we obtain:

> **V.** The number of positive roots of $f(x)$ is equal to, or smaller by an even number than, the number of variation–permanences in the double row (4).
>
> **VI.** The number of negative roots of $f(x)$ is equal to, or smaller by an even number than, the number of permanence–permanences in the double row (4).

The sum of the numbers of VP and PP is the number of permanences in the simple row
$$A_0, \quad A_1, \quad A_2, \ldots, \quad A_n. \tag{5}$$
Thus V and VI combine into:

> **VII.** The number of all real roots of $f(x)$ is equal to, or smaller by an even number than, the number of permanences in the row (5).

The total number of variations and permanences in (5) is n, the total number of real and imaginary roots together, provided no member of (5) vanishes. Since the first and last members have the same sign, the number of variations in (5) is even. Thus one may also add:

> **VIII.** The number of imaginary roots of $f(x)$ is at least equal to the number of sign variations in the row (5).

In this form of the last two theorems it is immaterial whether some coefficients $a_1, a_2, \ldots, a_n$ vanish, provided none of the A_ν is zero.

For the cubic

$$a_0x^3 + a_1x^2 + a_2x + a_3 = 0$$

one has

$$A_1 = 2(a_1^2 - 3a_0a_2), \qquad A_2 = 2(a_2^2 - 3a_1a_3).$$

In the row $1, A_1, A_2, 1$ there are two variations when one or both of A_1, A_2 is negative; in these cases the equation has two imaginary roots. If A_1 and A_2 are positive, the decision is not obtained from this theorem alone; one must then also consider

$$B = 9a_0a_3 - a_1a_2$$

and the sign of

$$D = A_1A_2 - B^2$$

(§ 62).

For a biquadratic equation one has the three expressions

$$A_1 = 3a_1^2 - 8a_0a_2, \qquad A_2 = 4a_2^2 - 9a_1a_3, \qquad A_3 = 3a_3^2 - 8a_2a_4.$$

If their signs are, in order, $+, -, +$, then there are four imaginary roots. If all three are positive, four, two, or no imaginary roots may occur; in all other cases there are two or four imaginary roots.

§ 102. Jacobi's Criterion

Jacobi derived from Descartes' rule a criterion for estimating the number of roots that lie between two limits α, β.[27] If

$$f(x) = 0 \tag{1}$$

is the equation of degree n whose roots between α and β are to be counted, Jacobi puts

$$x = \frac{\alpha + \beta y}{1 + y} \tag{2}$$

and forms

$$(1+y)^n f\left(\frac{\alpha + \beta y}{1 + y}\right) = \varphi(y) = b_0y^n + b_1y^{n-1} + \cdots + b_n. \tag{3}$$

Here $b_0, b_1, \ldots, b_n$ are integral homogeneous functions of degree n in α and β, and linear functions of the coefficients of $f(x)$. As x increases from α to β, the variable y also increases, from 0 to ∞. Therefore $f(x)$ has precisely as many roots between α and β as $\varphi(y)$ has positive roots. Descartes' theorem gives:

> **IX.** The number of roots of $f(x)$ between α and β is equal to, or smaller by an even number than, the number of sign variations in the row
>
> $$b_0, \quad b_1, \quad b_2, \ldots, \quad b_n.$$

[27] Jacobi, *Observatiunculae ad theoriam aequationum algebraicarum pertinentes.* Crelle's Journal, vol. 13. Collected Works, vol. III.

Thus, whereas Budan's theorem requires the variations in two rows of values to be counted and compared, Jacobi's criterion requires only the counting of variations in one row. Newton's rule can be transformed in the same way. If

$$B_\nu = \nu(n-\nu)b_\nu^2 - (\nu+1)(n-\nu+1)b_{\nu-1}b_{\nu+1},$$

then one obtains:

> **X.** The number of roots of $f(x)$ between α and β is equal to, or smaller by an even number than, the number of variation–permanences in the double row
>
> $$b_0,\ b_1,\ b_2, \ldots,\ b_n; \qquad B_0,\ B_1,\ B_2, \ldots,\ B_n.$$

§ 103. Geometrical Comparison of the Different Criteria

In order to compare the reach of the different criteria with one another, Klein used a geometrical interpretation, which we now discuss. If one does not wish to operate in spaces of more dimensions and thereby lose geometrical intuition, one must restrict oneself to the first cases. We begin with quadratic equations

$$x^2 + ax + b = 0. \tag{1}$$

Regard a and b as rectangular coordinates in a plane. Then every point of the plane is the image of one and only one quadratic equation, and conversely every quadratic equation with real coefficients has a point of the plane as its image.

The equation

$$a^2 - 4b = 0 \tag{2}$$

represents a parabola. On it lie the images of equations with equal roots. Points inside the parabola, that is, in its hollow, represent equations with imaginary roots; the outside points represent equations with real roots.

Following § 89, form the Sturm chain

$$x^2 + ax + b, \qquad 2x + a, \qquad a^2 - 4b. \tag{3}$$

For a fixed x, the equation

$$x^2 + ax + b = 0$$

is the tangent to the parabola at the point

$$a = -2x, \qquad b = x^2.$$

Moreover $x^2 + ax + b$ is positive in the part of the plane that contains the parabola. This tangent and the parabola divide the plane into four regions, in which the row (3) has $0, 1, 2$ variations as indicated in Fig. 16.

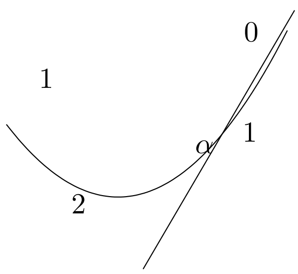


Fig. 16.

If we put $x = \alpha$ and $x = \beta$, we get two such tangents and a division of the plane into regions representing equations that have no, one, or two roots between α and β. In Fig. 17 these four regions are marked by the corresponding numerals. It is the distinction between points from which no tangent, one tangent, or two tangents can be drawn to the parabola with point of contact between α and β. Thus Fig. 17 gives the exact and complete distinction of the cases.

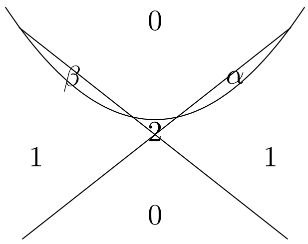


Fig. 17.

Compare this with Budan's theorem. We have to examine the three functions $f(x), f'(x), f''(x)$, hence

$$x^2 + ax + b, \qquad 2x + a, \qquad 1, \tag{4}$$

with respect to sign variations. The two straight lines

$$x^2 + ax + b = 0, \qquad 2x + a = 0$$

divide the plane into three regions in which the row (4) has the variations indicated in Fig. 18.

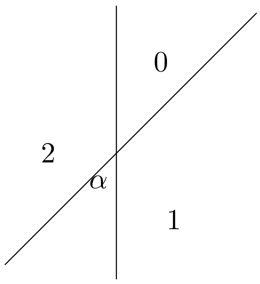


Fig. 18.

Taking again two values $x = \alpha$ and $x = \beta$, we obtain six regions, in which the loss of variations is shown in Fig. 19. In the regions marked 0 and 1, this loss gives the correct number of roots. In the region marked 2, however, the decision between two roots and no root is not made.

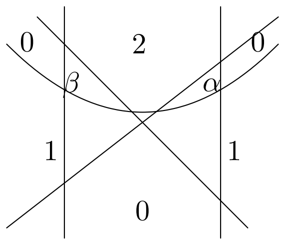


Fig. 19.

Now consider Jacobi's criterion (§ 102). To estimate the number of roots between α and β, the equation (1) is first transformed into

$$(\alpha + \beta y)^2 + a(\alpha + \beta y)(1 + y) + b(1 + y)^2 = 0$$

or

$$(\beta^2 + a\beta + b)y^2 + \{2\alpha\beta + a(\alpha + \beta) + 2b\}y + (\alpha^2 + a\alpha + b) = 0.$$

Thus one must count the variations in

$$\beta^2 + a\beta + b, \qquad 2\alpha\beta + a(\alpha+\beta) + 2b, \qquad \alpha^2 + a\alpha + b.$$

The first and third of these expressions, when set equal to zero, give the two tangents already considered. The middle expression

$$2\alpha\beta + a(\alpha+\beta) + 2b$$

vanishes at $a = -2\alpha$, $b = \alpha^2$, and at $a = -2\beta$, $b = \beta^2$; hence it represents the line joining the two points α, β, and it is negative on the side containing the point of intersection of the two tangents. At that intersection point it is itself equal to $-(\beta-\alpha)^2$.

We therefore obtain five regions with zero, one, or two variations, as shown in Fig. 20.

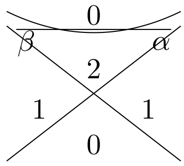


Fig. 20.

One sees that this figure confines the indeterminacy to a much smaller region than the previous one. From this standpoint it does not appear justified that Jacobi gives such decided preference to the Budan–Fourier criterion over his own.[28]

§ 104. Determination of an Upper Bound for the Roots

In the third section we already used the fact that one can always find a positive number greater than the absolute values of all roots of a given equation. What is now required is a determination of such an upper bound that is as simple and at the same time as sharp as possible.

We first consider only real roots and seek a positive number greater than the largest positive root of a given equation $f(x) = 0$. Budan's theorem gives such a determination. Assume the coefficient of the highest power of x in $f(x)$ equal to 1 (or at least positive). The positive number α can always be taken so large that

$$f(\alpha), \quad f'(\alpha), \quad f''(\alpha), \ldots, \quad f^{(n)}(\alpha) \tag{1}$$

are all positive. Then in the row

$$f(x), \quad f'(x), \quad f''(x), \ldots, \quad f^{(n)}(x)$$

from $x = \alpha$ to $x = \infty$ no variation can be lost, and therefore by Theorem I no root of $f(x)$ can lie between α and ∞.

For a lower bound for the negative roots take a positive number β such that

$$(-1)^n f(-\beta), \quad (-1)^{n-1} f'(-\beta), \ldots, \quad f^{(n)}(-\beta) \tag{2}$$

are all positive. Then $f(x) = 0$ has certainly no negative root below $-\beta$.

This determination of bounds already goes back to Newton. Another upper bound, simpler in calculation, was given by Laguerre.[29] Instead of the derivatives he uses the functions

[28] Compare F. Klein, *Geometrisches zur Abzählung der reellen Wurzeln einer algebraischen Gleichung*, in Dyck's catalogue of mathematical models (Munich 1892).

[29] Laguerre, Nouvelles Annales de math., second series, vol. XIX, 1880; Journal de math., third series, vol. IX, 1883.

$f_0(x), f_1(x), f_2(x), \ldots, f_n(x)$, which have already served us several times, especially in the Tschirnhausen transformation.

These functions were defined in § 4 by

$$\begin{aligned}
f_0(x) &= 1,\\
f_1(x) &= x + a_1,\\
f_2(x) &= x^2 + a_1 x + a_2,\\
&\vdots\\
f_{n-1}(x) &= x^{n-1} + a_1 x^{n-2} + a_2 x^{n-3} + \cdots + a_{n-1},\\
f(x) = f_n(x) &= x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1}x + a_n,
\end{aligned}$$

with the recursion

$$f_\nu(x) - x f_{\nu-1}(x) = a_\nu.$$

As shown there,

$$f(x) = (x-\alpha)\{x^{n-1} f_0(\alpha) + x^{n-2} f_1(\alpha) + \cdots + f_{n-1}(\alpha)\} + f(\alpha). \tag{3}$$

This formula immediately shows that if α makes

$$f_0(\alpha), \quad f_1(\alpha), \quad f_2(\alpha), \ldots, \quad f_{n-1}(\alpha), \quad f(\alpha) \tag{4}$$

positive, then no positive x greater than α can make $f(x)$ vanish, since the right side of (3) then consists only of positive terms. Thus such an α is an upper bound for the positive roots of $f(x)$.

Likewise one obtains a lower bound $-\beta$ for the negative roots if $\beta > 0$ is so chosen that

$$f_0(-\beta), \quad -f_1(-\beta), \quad f_2(-\beta), \ldots, \quad (-1)^n f(-\beta) \tag{5}$$

are all positive. This determination is easier in calculation, but it can give less sharp bounds than Newton's. Indeed, differentiating (3) repeatedly with respect to x and then putting $x = \alpha$, under the assumption that all quantities in (4) are positive, gives expressions for $f'(\alpha), f''(\alpha), \ldots$ consisting of positive terms. Hence if (4) is positive, then the derivatives (1) are necessarily all positive; the converse is not necessary.

To find an upper bound for the absolute value of the imaginary roots, valid also for complex coefficients, consider

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n,$$

where $a_1, a_2, \ldots, a_n$ may be real or imaginary. Denote absolute value by $|a|$. If $|x| > 1$, then

$$\left| a_1 + \frac{a_2}{x} + \cdots + \frac{a_n}{x^{n-1}} \right| < |a_1| + |a_2| + \cdots + |a_n| = \alpha.$$

Thus, if $|x|$ is greater than both 1 and α, the expression

$$x^{1-n} f(x) = x + a_1 + \frac{a_2}{x} + \cdots + \frac{a_n}{x^{n-1}}$$

cannot vanish, because the absolute value of x is greater than the absolute value of the sum of all other terms on the right. Hence, according as α is greater or less than 1, if one draws about the origin a circle of radius α or of radius 1, no root of $f(x)$ lies outside this circle.

§ 105. Estimation of the Imaginary Roots

In § 96 we saw how the characteristic of a certain system of curves determines the exact number of complex roots of an equation that lie inside a closed curve. This test too can be simplified when one seeks not the exact number of roots but only an upper bound for it, which in many cases is already enough for exact determination.

Let, as in § 96,

$$F(z) = \varphi(x, y) + \mathrm{i}\psi(x, y) \tag{1}$$

be a real or complex function of the complex argument $z = x + \mathrm{i}y$, with φ and ψ real functions of x and y, and let a closed curve f be drawn. We have:

XI. If the curve f is divided by the curve φ into $2m$ arcs in which φ is alternately positive and negative, then the number of roots of $F(z)$ inside f is at most m. The same holds if f is divided by ψ into $2m$ such arcs.

This is an immediate consequence of the theorem on characteristics II, § 96. It gives

$$k = \frac{1}{2}\{E(f; \varphi, \psi) - A(f; \varphi, \psi)\}.$$

If f is cut by the φ-curve in $2m$ points, then

$$m = \frac{1}{2}\{E(f; \varphi, \psi) + A(f; \varphi, \psi)\},$$

and hence

$$k = m - A(f; \varphi, \psi).$$

Here k is, by the cited theorem, the number of roots of F inside f, and $A(f; \varphi, \psi)$ is certainly non-negative. The same argument applies to

$$k = \frac{1}{2}\{E(f; \psi, \varphi) - A(f; \psi, \varphi)\}.$$

§ 106. Rolle's Theorem

Suppose that $f(x) = 0$ has two real roots α and β, the first of multiplicity a and the second of multiplicity b. Then

$$f(x) = (x - \alpha)^a (x - \beta)^b f_1(x). \tag{1}$$

If $\alpha < \beta$ and no root of $f(x) = 0$ lies between α and β, then $f(x)$ and $f_1(x)$ have constant sign throughout this interval. Differentiating (1), one obtains

$$\frac{(x - \alpha)(x - \beta)f'(x)}{f(x)} = a(x - \beta) + b(x - \alpha) + (x - \alpha)(x - \beta)\frac{f_1'(x)}{f_1(x)}. \tag{2}$$

The right side is negative for $x = \alpha$, namely $a(\alpha - \beta)$, and positive for $x = \beta$, namely $b(\beta - \alpha)$. As x passes from α to β it must therefore pass through zero once, or more generally an odd number of times. On the left side of (2), $(x - \alpha)(x - \beta)$ and $f(x)$ have constant sign, and consequently $f'(x)$ must vanish an odd number of times. This gives Rolle's theorem:

XII. If α and β are two consecutive roots of $f(x) = 0$, then between α and β, endpoints excluded, there lie roots of $f'(x) = 0$ in odd number, hence at least one.

Important consequences follow for the case in which $f(x) = 0$ has only real roots. Let these be ordered by size as

$$\alpha, \beta, \gamma, \ldots,$$

and suppose that α is of multiplicity a, β of multiplicity b, γ of multiplicity c, and so on. Then $f'(x) = 0$ has at α a root of multiplicity $a - 1$, at β a root of multiplicity $b - 1$, at γ a root of multiplicity $c - 1$, and between α and β, between β and γ, and so forth, at least one real root. The total number so obtained is

$$a + b + c + \cdots - 1 = n - 1,$$

and because $f'(x)$ has degree $n - 1$, this minimum number is not exceeded. Thus:

> **XIII.** If $f(x) = 0$ has only real roots, then $f'(x) = 0$ also has only real roots; those which do not coincide with multiple roots of $f(x)$ are simple and separate the roots of $f(x)$.

Repeated application gives the general form:

> **XIV.** If $f(x) = 0$ has only real roots, then $f'(x), f''(x), f'''(x), \ldots$ also have only real roots. If one of these functions $f^{(\nu)}(x)$ has a root α of multiplicity a, and $a > 1$, then α is a root of $f(x)$ of multiplicity $a + \nu$.

These theorems are essentially generalized by applying a linear transformation.[30] The matter is most easily handled in homogeneous notation: let $f(x, y)$ be a homogeneous function of degree n. For an arbitrary value $\xi : \eta$ form the polars (§ 60)

$$\begin{aligned}
P_1(x, \xi) &= \frac{1}{n}\{\xi f'(x) + \eta f'(y)\}, \\
P_2(x, \xi) &= \frac{1}{n(n-1)}\{\xi^2 f''(x, x) + 2\xi\eta f''(x, y) + \eta^2 f''(y, y)\}, \\
P_3(x, \xi) &= \frac{1}{n(n-1)(n-2)}\{\xi^3 f'''(x, x, x) + 3\xi^2\eta f'''(x, x, y) \\
&\qquad + 3\xi\eta^2 f'''(x, y, y) + \eta^3 f'''(y, y, y)\}.
\end{aligned} \tag{3}$$

These functions are unchanged if ξ, η and x, y are transformed simultaneously by a linear transformation, and if f and its derivatives are replaced by the corresponding transformed functions. The reality of the roots is not changed by a real linear transformation. If ξ, η are real, the transformation may be chosen so that $\xi' = 1, \eta' = 0$; after also putting $y = 1$, the polars become the derivatives of $f(x)$ with respect to x, apart from numerical factors.

Thus in XIII and XIV the derivatives $f'(x), f''(x), \ldots$ may be replaced by the polars $P_1(x, \xi), P_2(x, \xi), \ldots$ for arbitrary real ξ, η.

Apply this to the equation obtained by setting the $(n - 2)$nd polar equal to zero:

$$x^2 f''(\xi, \xi) + 2xy f''(\xi, \eta) + y^2 f''(\eta, \eta) = 0. \tag{4}$$

If, as assumed, $f(x, y) = 0$ has only real roots, then this quadratic equation must have real roots. Hence the Hessian determinant

$$f''(\xi, \xi) f''(\eta, \eta) - f''(\xi, \eta)^2 \tag{5}$$

must be negative for all real ξ, η. It can vanish only if (4) has equal roots, and by XIV this is possible only when $f(x, y)$ is the nth power of a linear function. In that exceptional case the determinant (5) vanishes identically. Apart from this case, therefore, for an equation with only real

[30] Laguerre, Nouvelles Annales de Mathématiques, second series, vol. 19, 1880.

roots, the equation obtained by setting the Hessian determinant equal to zero has only imaginary roots.

Returning to inhomogeneous notation, Euler's theorem gives

$$\begin{aligned} xf'(x) + yf'(y) &= nf(x), \\ xf''(x,x) + yf''(x,y) &= (n-1)f'(x), \\ xf''(y,x) + yf''(y,y) &= (n-1)f'(y). \end{aligned} \tag{6}$$

Eliminating $f'(y)$, $f''(x,y)$, $f''(y,y)$, and then putting $y = 1$, one obtains

$$\begin{aligned} f'(y) &= nf(x) - xf'(x), \\ f''(x,y) &= (n-1)f'(x) - xf''(x), \\ f''(y,y) &= n(n-1)f(x) - 2(n-1)xf'(x) + x^2 f''(x). \end{aligned} \tag{7}$$

Thus, if

$$(n-1)H(x,y) = f''(x,x)f''(y,y) - f''(x,y)^2, \tag{8}$$

then

$$H(x,y) = H(x) = nf(x)f''(x) - (n-1)f'(x)^2. \tag{9}$$

The degree of $H(x)$ in x is at most $2n - 4$.

§ 107. Laguerre's Theorems for Equations with Only Real Roots

Laguerre made an application of the preceding theorems to equations with only real roots. Let $f(x) = 0$ be an equation of degree n with n distinct real roots

$$\alpha, \alpha_1, \alpha_2, \ldots.$$

For a variable x we have the familiar formula (§ 14, (11))

$$\frac{f'(x)}{f(x)} = \sum \frac{1}{x - \alpha}, \tag{1}$$

where the sum runs over the different roots. Differentiating once more gives

$$\sum \frac{1}{(x-\alpha)^2} = \frac{f'(x)^2 - f(x)f''(x)}{f(x)^2}. \tag{2}$$

Introducing H from § 106, (9), this becomes

$$\sum \frac{1}{(x-\alpha)^2} = \frac{nf'(x)^2 - H(x)}{nf(x)^2}. \tag{3}$$

A linear transformation in which the value $x = \infty$ corresponds to an arbitrary other real value of the new variable gives a more general formula. Introduce homogeneous variables by replacing x by $x : y$ and α by $\alpha : \beta$. Applying the linear transformation

$$\begin{aligned} x &= ax' + by', \quad & \alpha &= a\alpha' + b\beta', \\ y &= cx' + dy', \quad & \beta &= c\alpha' + d\beta', \end{aligned} \tag{5}$$

or, inversely,

$$rx' = dx - by, \qquad ry' = -cx + ay, \qquad r = ad - bc,$$

and using the covariance of H, one obtains, after dropping primes and writing again f,

$$\sum \left(\frac{\xi\beta - \eta\alpha}{x\beta - y\alpha} \right)^2 = \frac{[\xi f'(x) + \eta f'(y)]^2 - H(x,y)}{nf(x,y)^2}. \tag{8}$$

In this form both sides are unchanged by any simultaneous linear transformation of the three pairs x, y; ξ, η; α, β.

Formula (8) is an identity for every form $f(x, y)$ and for all values of the variables. Now assume that $f(x, y)$ has only real roots, so that the α, β are real, and let also ξ, η and x, y be real. Both sides are then essentially positive. Denote their common value by P and determine $X : Y$ from

$$\frac{(X\eta - Y\xi)^2}{(Xy - Yx)^2} = P,$$

or

$$\Omega = (Xy - Yx)^2 P - (X\eta - Y\xi)^2 = 0. \tag{9}$$

This quadratic equation has two real roots.

If $X = x$, $Y = y$, then Ω is negative. If $X = \alpha$, $Y = \beta$ is any root of $f = 0$, then the corresponding quotient is one term of the sum P and is therefore smaller than P; hence Ω is positive. Ordering the points $\alpha : \beta$ and $x : y$ cyclically on the circle and writing the variable value $X : Y$ as a moving point, Ω changes sign between x and the nearest roots on either side. If X_1, X_2 are the roots of (9), their order is

$$\alpha_1, \quad X_1, \quad x, \quad X_2, \quad \alpha_2. \tag{10}$$

Thus (X_1, X_2) is an interval containing x but no root of $f = 0$; X_1 and X_2 lie closer to the nearest roots α_1, α_2 than the original value x.

The point ξ may have any position. The points X_1, X_2 depend on it, but they always lie in (α_1, x) and (x, α_2) respectively.

The question is how to choose ξ so that X_1 is as close as possible to α_1, or X_2 as close as possible to α_2. Equivalently, with X fixed, (9) is a quadratic equation for ξ; admissible boundary positions occur when its discriminant vanishes. The coefficients of $\xi^2, 2\xi\eta, \eta^2$ in (9) are

$$\frac{f'(x)^2 - y^2 H}{nf^2}(Xy - Yx)^2 - Y^2,$$

$$\frac{f'(x)f'(y) + xyH}{nf^2}(Xy - Yx)^2 + XY,$$

$$\frac{f'(y)^2 - x^2 H}{nf^2}(Xy - Yx)^2 - X^2.$$

Using $xf'(x) + yf'(y) = nf$, the discriminant condition gives

$$\frac{(Xy - Yx)^2}{nf^2}\left\{(n-1)H(Xy - Yx)^2 + [Xf'(x) + Yf'(y)]^2\right\} = 0.$$

Therefore the limiting values of $X : Y$ satisfy

$$[Xf'(x) + Yf'(y)]^2 + (n-1)(Xy - Yx)^2 H = 0. \tag{11}$$

Since H is negative, this quadratic equation has two real roots, determined by

$$Xf'(x) + Yf'(y) = \pm(Xy - Yx)\sqrt{(1-n)H}. \tag{12}$$

Several special results follow on leaving the homogeneous notation. First take $y = 0$, so that $x : y$ is infinite, and put $Y = 1$. Then two values X_1, X_2 are obtained between which all roots of $f(x)$ are contained. If

$$f(x, y) = a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \cdots,$$

then

$$H = \{2na_0a_2 - (n-1)a_1^2\}x^{2n-4}, \tag{13}$$

and these two values are obtained from

$$na_0 X = -a_1 \pm \sqrt{(n-1)^2 a_1^2 - 2n(n-1)a_0 a_2}. \tag{14}$$

If, however, $x : y$ is left arbitrary and we return to the inhomogeneous form by putting $y = 1$, $Y = 1$, and using (7) and (9) of § 106, then (12) gives

$$x - X = \frac{nf(x)}{f'(x) \pm \sqrt{(n-1)^2 f'(x)^2 - n(n-1)f(x)f''(x)}}. \tag{15}$$

These values have the following meaning. If x lies between two roots α_1, α_2 of $f(x)$, then of the two values of X one lies nearer α_1 and the other nearer α_2, both still inside (α_1, α_2). If x is greater than the greatest root or smaller than the least root, then all roots lie between the two values of X, while x lies outside. If x is close to a root, one of the two values of X gives a still more accurate value of that root.

Just as X_1, X_2 determine an interval containing no root, one can also determine an interval in which roots certainly lie. Let $f(x, y)$ again be a form of degree n with only real roots, and let $x : y$, $\xi : \eta$, $\xi' : \eta'$ be three real values, two arbitrary, the third related by

$$[\xi f'(x) + \eta f'(y)]\,[\xi' f'(x) + \eta' f'(y)] - H(\xi y - \eta x)(\xi' y - \eta' x) = 0. \tag{16}$$

This relation is invariant under simultaneous linear transformations. Reducing it to a special position gives

$$\xi' = \frac{\sum(\alpha^2)}{\sum(\alpha)}$$

and equivalently

$$\sum \alpha(\alpha - \xi') = 0.$$

Since all α are real, the products $\alpha(\alpha - \xi')$ cannot all be positive nor all negative. Thus some roots lie between 0 and ξ', and some outside.

By transforming back, one obtains the general conclusion: on the circle carrying the points α, x, ξ, ξ', with ξ, ξ' connected by (16), each of the two arcs into which the circle is divided by ξ and ξ' contains at least one root α. The point x and one of ξ, ξ' are arbitrary; the third is determined by (16).

In particular, let ξ' coincide with one of the two points X determined by (12), say

$$Xf'(x) + Yf'(y) + (Xy - Yx)\sqrt{(1-n)H} = 0. \tag{17}$$

Putting $\xi', \eta' = X, Y$ in (16) and cancelling $Xy - Yx$ gives

$$(n-1)[\xi f'(x) + \eta f'(y)] + (\xi y - \eta x)\sqrt{(1-n)H} = 0, \tag{18}$$

which determines ξ uniquely.

In inhomogeneous form, (17) and (18) give

$$x - X = \frac{nf(x)}{f'(x) + (n-1)\sqrt{f'(x)^2 - \dfrac{n}{n-1}f(x)f''(x)}}, \tag{19}$$

$$x - \xi = \frac{nf(x)}{f'(x) + \sqrt{f'(x)^2 - \dfrac{n}{n-1}f(x)f''(x)}}, \tag{20}$$

where the square roots may be taken with either sign.

The position of the points is then described as follows. There is at least one root α such that

$$\xi, \quad \alpha, \quad X, \quad x \tag{21}$$

occur in this order, and no root lies between x and X. One may choose the root so that no further root lies between α and X. Between α and ξ, however, other roots may occur.

Let now x approach the root α. Then X approaches the same root, and (20) shows that ξ approaches it as well. Hence, when x is sufficiently close to a root, this nearest root lies in the interval (ξ, X), which contains no second root, and the interval closes more and more around α.

As an example take the spherical functions, which we earlier proved (§ 87) to have only real roots, all between -1 and $+1$. The function

$$P_n(x) = \frac{1 \cdot 3 \cdot \ldots \cdot (2n-1)}{1 \cdot 2 \cdot \ldots \cdot n} \left(x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \cdots \right)$$

has

$$P_n(1) = 1, \qquad P_n'(1) = \frac{n(n+1)}{2}, \qquad P_n''(1) = \frac{n(n-1)(n+1)(n+2)}{8}.$$

This is easily proved by induction, using the recursion

$$(1-x^2)P_n'(x) + nxP_n(x) - nP_{n-1}(x) = 0.$$

Putting $x = 1$ in (19) and (20) gives two bounds enclosing the root nearest to 1:

$$X = 1 - \frac{2}{n+1+(n-1)\sqrt{\frac{n(n+1)}{2}}},$$

$$\xi = 1 - \frac{2}{n+1+\sqrt{\frac{n(n+1)}{2}}}.$$

For $n = 7$, for example,

$$X = 0.94967\ldots, \qquad \xi = 0.84952\ldots.$$

The more exact values of the two roots nearest to 1, listed for comparison from Gauss' computation (Gauss, Works, vol. III, p. 195), are

$$0.94911\ldots; \qquad 0.74153\ldots.$$

Thus here only one root lies between X and ξ, and X exceeds the value of that root only in the fourth decimal.

Tenth Section.

Approximate computation of roots.

§ 108. Interpolation. Regula falsi

Once intervals have been separated in which only one root of an algebraic equation lies, it becomes possible to compute the real roots numerically to arbitrary accuracy by narrowing the interval more and more, for example by repeated bisection. If an interval of length 1 is repeatedly divided into ten parts, each new step supplies one further decimal place.

When a root has once been separated from all the others, then in the further narrowing of the interval one has only to take account of the sign of the function $f(x)$ itself. The computations needed for this may be greatly eased by applying the interpolation formula that we learned in the first section.

In § 10, (7), we derived a formula by which a function of degree n, which we shall now denote by $\varphi(\xi)$, can be determined when $n+1$ consecutive values

$$\varphi(0), \quad \varphi(1), \ldots, \varphi(n)$$

are given. This formula is, if

$$B_\nu^{(\xi)} = \frac{\xi(\xi-1)\cdots(\xi-\nu+1)}{1\cdot 2\cdot 3\cdots \nu}, \qquad \Delta_\xi = \varphi(\xi+1) - \varphi(\xi), \tag{1}$$

$$\Delta'_\xi = \Delta_{\xi+1} - \Delta_\xi, \qquad \Delta_\xi^{(\mu-1)} = \Delta_{\xi+1}^{(\mu-2)} - \Delta_\xi^{(\mu-2)} \tag{2}$$

are put,

$$\varphi(\xi) = \varphi(0) + \Delta_0 B_1^{(\xi)} + \Delta'_0 B_2^{(\xi)} + \cdots + \Delta_0^{(n-1)} B_n^{(\xi)}. \tag{3}$$

The value of the formula lies in the fact that the sequence of differences

$$\Delta_0, \quad \Delta'_0, \quad \Delta''_0, \ldots$$

in many cases gives numerical values that decrease so rapidly that one may be content with a few of the first terms of the series (3).

Let $f(x)$ be the function to be investigated, and let (α, β) be the interval in which one of the required roots lies. If m denotes any suitably chosen integer, put

$$\delta = \frac{\beta-\alpha}{m}, \qquad x = \alpha + \delta\xi,$$

and hence

$$\Delta_x = f(x+\delta) - f(x), \qquad \Delta'_x = \Delta_{x+\delta} - \Delta_x, \quad \ldots.$$

Then

$$f(x) = f(\alpha) + \frac{\Delta_\alpha}{\delta}(x-\alpha) + \frac{\Delta'_\alpha}{\delta^2}\frac{(x-\alpha)(x-\alpha-\delta)}{2} + \cdots. \tag{4}$$

One could just as well start from the other end point β of the interval, and would then have to put in (3)

$$x = \beta - \delta\xi.$$

Then one would get

$$\Delta_x = f(x-\delta) - f(x), \qquad \Delta'_x = \Delta_{x-\delta} - \Delta_x, \quad \ldots$$

and

$$f(x) = f(\beta) + \frac{\Delta_\beta}{\delta}(x-\beta) + \frac{\Delta'_\beta}{\delta^2}\frac{(x-\beta)(x-\beta+\delta)}{2} + \cdots. \tag{5}$$

If the equation

$$y = f(x)$$

is interpreted as the equation of a curve in rectangular coordinates x, y, then the relations may be illustrated geometrically. If, for example, equation (4) is applied to the interval from α to $\alpha+\delta$, and only the first difference is used, then in geometrical language the arc of the curve between the two points $\alpha, f(\alpha)$ and $\alpha+\delta, f(\alpha+\delta)$ is replaced by the chord. If the second difference is also taken into account, then the arc through the three consecutive points with abscissas $\alpha, \alpha+\delta, \alpha+2\delta$ is replaced by the arc of a parabola passing through the same points and having its axis parallel to the ordinate axis; this parabola will adhere to the curve more closely than the chord.

The smaller the interval δ is, the less influence the higher differences have. How far one must go in the approximation depends not only on the accuracy with which one wishes to know $f(x)$, but essentially also on the density of the values

$$\alpha, \quad \alpha+\delta, \quad \alpha+2\delta, \ldots$$

for which the function is known. The arrangement of numerical tables, especially logarithm tables, rests on these propositions. Strictly speaking, the functions involved there are not integral rational functions; but for continuous functions generally the same laws hold here. Thus in the tables, beside the values

$$f(\alpha), \quad f(\alpha+\delta), \quad f(\alpha+2\delta), \dots$$

one also finds the first or the first two differences. In the customary seven-place tables the first difference suffices. In Vega's ten-place table, the *Thesaurus logarithmorum*, the second differences are also printed and must be used in quite precise computations.

Our interpolation formulae may be used with advantage to compute the roots of equations, or, more exactly, to improve approximate values found by other means. We may formulate the problem as follows: for a given value of $f(x)$ between $f(\alpha)$ and $f(\alpha+\delta)$, find the corresponding value of x.

We regard $x=\alpha$ as a first approximation. Put

$$f(x) - f(\alpha) = \Delta.$$

Then Δ is a given value of the same sign as

$$\Delta_\alpha = f(\alpha+\delta) - f(\alpha)$$

and absolutely smaller than Δ_α. If further $x-\alpha=u$, formula (4) gives

$$\Delta = \frac{\Delta_\alpha}{\delta} u + \frac{\Delta'_\alpha}{\delta^2}\frac{u(u-\delta)}{2} + \cdots . \tag{6}$$

If one first stops with the first difference, the first correction is

$$u = \delta\,\frac{\Delta}{\Delta_\alpha}. \tag{7}$$

If now

$$u = \delta\,\frac{\Delta}{\Delta_\alpha} + u'$$

is put, then from (6), omitting u' in the second term, follows

$$0 = \frac{\Delta_\alpha}{\delta} u' + \frac{\Delta'_\alpha \Delta(\Delta - \Delta_\alpha)}{2\Delta_\alpha^2},$$

and hence as the second correction

$$u' = \delta\,\frac{\Delta'_\alpha \Delta(\Delta_\alpha - \Delta)}{2\Delta_\alpha^3}. \tag{8}$$

The first correction is obtained by replacing the arc between α and $\alpha+\delta$ by the chord; the second by replacing the arc through $\alpha, \alpha+\delta, \alpha+2\delta$ by a parabola through the same points, but now with its axis parallel to the x-axis.

As an example take the equation

$$f(x) = x^3 - 2x - 2 = 0,$$

which has a real root between 1.7 and 1.8. One computes

x	$f(x)$	Δ_x	Δ'_x
1.7	−0.487	0.719	0.108
1.8	+0.232	0.827	
1.9	1.059		

Since $f(x) = 0$, one must put

$$\Delta = 0.487.$$

The first correction to $\alpha = 1.7$ is, by (7),

$$u = 0.06773,$$

and the second correction by (8) is

$$u' = 0.00164,$$

so that

$$\alpha + u + u' = 1.76937.$$

The more exact value computed in another way is 1.76929... Thus we have a result correct in the first three decimals. Here, however, we have no other means of estimating the accuracy beforehand than the decrease of the differences $\Delta, \Delta', \Delta'', \dots$ If Δ' is so small as to fall outside the limits of the intended accuracy, the use of the first difference gives an exact result for that purpose.

The rule for computing roots just explained is called the *regula falsi*. Its simplest form, suited for the first approximation, is this. If a root x of the equation $f(x) = 0$ lies between α and β, and if $f(\alpha) = -a$, $f(\beta) = b$, then

$$x = \frac{b\alpha + a\beta}{a + b} \tag{9}$$

is an approximate value of x. This is only another way of writing formula (7).

§ 109. Newton's method of approximation

A method that is usually better than interpolation for the approximate computation of roots of an equation goes back to Newton, and was developed and investigated more precisely by Fourier. The method is simply as follows. Let

$$f(x) = 0 \tag{1}$$

be the equation to be solved, and suppose that a value $x = \alpha$ has been found which may be regarded as a certain approximation to a root. Put in (1)

$$x = \alpha + h.$$

Then

$$f(\alpha + h) = f(\alpha) + hf'(\alpha) + \frac{h^2}{1 \cdot 2} f''(\alpha) + \cdots . \tag{2}$$

If h is now determined from

$$f(\alpha) + hf'(\alpha) = 0, \tag{3}$$

which presupposes that $f'(\alpha)$ is not zero, then

$$f(\alpha + h) = \frac{h^2}{1 \cdot 2} f''(\alpha) + \cdots ,$$

and if h is small this value will be small, since only the square and higher powers of h occur. Thus, under suitable hypotheses,

$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\alpha)} \tag{4}$$

may be regarded as a better approximation to the true value of the root. Replacing α by α', one obtains

$$\alpha'' = \alpha' - \frac{f(\alpha')}{f'(\alpha')}$$

as a still better approximation, and so on.

Two questions remain to be answered:

1. Under what hypotheses is α' really a better value than α?
2. How can one estimate the degree of accuracy that has thereby been reached?

Fourier answered these questions, but he makes the following assumption.

A root of $f(x)$ is enclosed in an interval (α, β) which contains no second root. In the interval (α, β) neither $f'(x)$ nor $f''(x)$ vanishes.

The latter assumption, that $f''(x)$ does not vanish in the interval, is made only in order to obtain conditions for applying the method that are easier to express. The usefulness of the method itself, once the interval is sufficiently narrowed, does not depend on it. If $f(x)$ and $f''(x)$ have no common divisor, so that they do not vanish simultaneously, then Fourier's assumption can always be satisfied by narrowing the interval; and if $f(x)$ and $f''(x)$ do have a common divisor, one may first separate it and then apply the approximation method to the individual factors of $f(x)$.

The relations are most easily surveyed geometrically, by interpreting $y = f(x)$ as the equation of a plane curve in rectangular coordinates. The equation

$$y = f(\alpha) + (x - \alpha)f'(\alpha) \tag{3}$$

is the equation of the tangent to the curve at the point $\alpha, f(\alpha)$; thus Newton's method amounts to replacing the curve in the interval (α, β) in first approximation by the tangent at one end point, rather than by its chord as in the interpolation method.

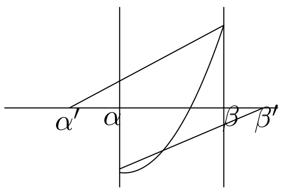


Fig. 21.

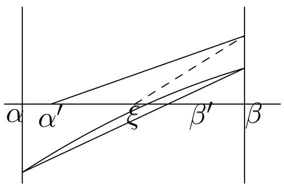


Fig. 22.

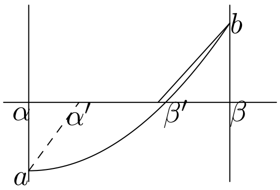


Fig. 23.

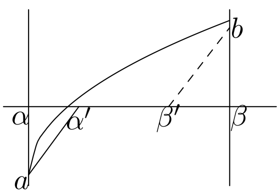


Fig. 24.

If the second differential quotient vanishes in the interval (α, β), so that the curve has a point of inflection, it may happen that both end tangents lead out of the interval; then Newton's method is not applicable (Fig. 21). But it may also happen, when the interval (α, β) has already been sufficiently narrowed, that both tangents lead to interior points of the interval (Fig. 22). In these cases, however, the regula falsi will always give a better approximation.

We now assume that $f'(x)$ and $f''(x)$ do not vanish in (α, β). Then the curve does not change the direction of its curvature, and certainly one of the two end tangents has its intersection with the x-axis inside the interval. This is certainly the case if the tangent is taken at the end point of the interval at which $f(x)$ and $f''(x)$ have the same sign. Figures 23 and 24 illustrate the relation for positive $f'(x)$ and positive, respectively negative, $f''(x)$. Thus in the first case β', and in the second α', is a better approximate value than β or α.

If one wishes at the same time to move the other boundary in order to get a new narrower interval, then, following Fourier, one can draw from the other end point the line parallel to $b\beta'$; in the case of Fig. 23 this is done at α, and one obtains the point α' as the lower boundary of the new interval. In the first case one has

$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\beta)}, \qquad \beta' = \beta - \frac{f(\beta)}{f'(\beta)},$$

and in the second case

$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\alpha)}, \qquad \beta' = \beta - \frac{f(\beta)}{f'(\alpha)}.$$

The interval (α', β') is narrower and contains the required root. It is hardly necessary to consider separately the two other cases in which $f'(x)$ is negative in the interval.

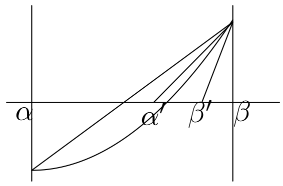


Fig. 25.

To obtain two new limits one may, with still better success, combine Newton's method with interpolation, as Fig. 25 shows. One then obtains the two new limits

$$\alpha' = \alpha - \frac{f(\alpha)(\beta - \alpha)}{f(\beta) - f(\alpha)} = \frac{\alpha f(\beta) - \beta f(\alpha)}{f(\beta) - f(\alpha)}, \qquad \beta' = \beta - \frac{f(\beta)}{f'(\beta)}.$$

We summarize the result in the following proposition.

If an interval (α, β) is bounded in which $f(x)$ changes sign once, while $f'(x)$ and $f''(x)$ do not change sign, and if β is the end point at which $f(\beta)$ and $f''(\beta)$ have the same sign, whether β is smaller or greater than α, then a narrower interval (α', β') with the same properties is obtained by putting

$$\alpha' = \alpha - \frac{f(\alpha)(\beta - \alpha)}{f(\beta) - f(\alpha)}, \qquad \beta' = \beta - \frac{f(\beta)}{f'(\beta)}. \tag{4}$$

To prove these geometrically evident results analytically, suppose that in the interval $f'(x)$ and $f''(x)$ are positive, so that $\alpha < \beta$, and

$$f(\alpha) < 0, \qquad f(\beta) > 0.$$

It is enough to treat this one case, since the three others are handled in exactly the same way.

Form the function

$$\varphi(x) = \frac{f(\beta) - f(x)}{\beta - x}$$

and its first derivative

$$\varphi'(x) = \frac{-(\beta - x)f'(x) + f(\beta) - f(x)}{(\beta - x)^2}.$$

The numerator

$$\psi(x) = -(\beta - x)f'(x) + f(\beta) - f(x)$$

vanishes for $x = \beta$, and its derivative is

$$\psi'(x) = -(\beta - x)f''(x),$$

hence is negative in the interval. Therefore $\psi(x)$ decreases as x increases and, since it vanishes for the greatest value $x = \beta$, remains positive. Consequently $\varphi'(x)$ is positive, and $\varphi(x)$ increases throughout the interval. Hence

$$\frac{f(\beta) - f(\alpha)}{\beta - \alpha} < \frac{f(\beta) - f(x)}{\beta - x} < f'(\beta), \qquad \alpha < x < \beta. \tag{5}$$

With

$$\alpha' = \alpha - \frac{f(\alpha)(\beta - \alpha)}{f(\beta) - f(\alpha)}, \qquad \beta' = \beta - \frac{f(\beta)}{f'(\beta)},$$

one has

$$\beta' - \alpha' = f(\beta)\left\{\frac{\beta - \alpha}{f(\beta) - f(\alpha)} - \frac{1}{f'(\beta)}\right\}, \tag{6}$$

and by (5) this difference is positive. Thus

$$\alpha < \alpha' < \beta' < \beta.$$

Putting $x = \alpha'$ and $x = \beta'$ in (5), and observing that

$$\beta - \alpha' = \frac{(\beta - \alpha)f(\beta)}{f(\beta) - f(\alpha)}, \qquad \beta - \beta' = \frac{f(\beta)}{f'(\beta)},$$

one gets

$$f(\alpha') < 0, \qquad f(\beta') > 0.$$

Thus the required root is contained between α' and β', and the interval (α', β') is smaller than (α, β). Repeating the construction with α', β' in place of α, β, one obtains a new still narrower interval, and so on.

For convergence put, by (6),

$$\frac{\beta' - \alpha'}{\beta - \alpha} = \Phi(\alpha, \beta) = \frac{f(\beta)\{f(\alpha) - f(\beta) + (\beta - \alpha)f'(\beta)\}}{(\beta - \alpha)f'(\beta)\{f(\beta) - f(\alpha)\}}. \tag{7}$$

The numerator and denominator are both divisible by $(\beta - \alpha)^2$; after cancelling this common factor and writing ξ, η for α, β, one obtains an expression

$$\Phi(\xi, \eta) = \frac{\psi_1(\xi, \eta)}{\psi_2(\xi, \eta)},$$

where ψ_1 and ψ_2 are integral rational functions of the two variables ξ, η.

Let x be the root lying in (α, β). Then, by our hypotheses on $f(x)$, the functions ψ_1 and ψ_2 do not vanish when

$$\alpha \le \xi < x, \qquad x < \eta \le \beta. \tag{8}$$

The function $\Phi(\xi, \eta)$ is smaller than 1 for $\xi = \alpha, \eta = \beta$, and likewise for every other interval contained in (α, β) in which $f(\xi)$ is negative and $f(\eta)$ positive, since every such interval could have been used in place of (α, β). Since $\Phi(\xi, \eta)$ is continuous in the two variables as long as they remain in the region (8), it must have there a maximum value, and this must be less than 1, because $\Phi(\xi, \eta)$ remains less than 1 also in the limiting cases $\xi = x, \eta = x$. Hence there is a positive proper fraction Θ such that

$$\Phi(\xi, \eta) < \Theta$$

throughout the region (8). It follows from (7) that

$$\beta' - \alpha' < (\beta - \alpha)\Theta.$$

The same reasoning applied to the step from (α', β') to (α'', β'') gives

$$\beta'' - \alpha'' < (\beta' - \alpha')\Theta',$$

where Θ' has for α', β' the same meaning as Θ for α, β. Since α', β' are in the region (8), Θ' is not greater than Θ, and Θ may be used in its place. Thus

$$\beta'' - \alpha'' < (\beta - \alpha)\Theta^2,$$

and in general

$$\beta^{(\nu)} - \alpha^{(\nu)} < (\beta - \alpha)\Theta^\nu.$$

The intervals therefore decrease at least as fast as the terms of a decreasing geometric progression.

As an example take

$$x^3 - 2x^2 - 2 = 0,$$

which has a root between

$$\alpha = 2.35, \qquad \beta = 2.36.$$

From the formulae (4) one obtains

$$\beta' = 2.35931\ldots, \qquad \alpha' = 2.359298\ldots,$$

so that a value correct in the fourth decimal is

$$2.3593.$$

For this value itself, a more exact computation shows that $f(x)$ is still negative; it can therefore be taken as α'. The next approximation gives

$$2.359304.$$

§ 110. The approximation method of Daniel Bernoulli and related methods

The method for approximately solving an equation that goes back to Daniel Bernoulli rests on the fact that, if one has a sequence of real quantities, the powers of the greatest among them dominate the equal powers of the others more and more as the exponents grow.

Let $\alpha, \beta, \gamma, \ldots$ be arbitrary real or complex quantities, but such that the absolute value of α is greater than the absolute value of all the others; that is, the absolute values of $\beta : \alpha, \gamma : \alpha, \ldots$ are proper fractions. Then

$$\frac{\alpha^m + \beta^m + \gamma^m + \cdots}{\alpha^{m-1} + \beta^{m-1} + \gamma^{m-1} + \cdots} = \alpha \, \frac{1 + \left(\dfrac{\beta}{\alpha}\right)^m + \left(\dfrac{\gamma}{\alpha}\right)^m + \cdots}{1 + \left(\dfrac{\beta}{\alpha}\right)^{m-1} + \left(\dfrac{\gamma}{\alpha}\right)^{m-1} + \cdots}. \tag{1}$$

As m increases, this expression approaches the limit α. If $\alpha, \beta, \gamma, \ldots$ are the roots of an algebraic equation, the left hand side of (1) is the quotient of the mth and $(m-1)$st power sums, and therefore:

The quotient of the mth and $(m-1)$st power sums approaches, as m increases, the root that is absolutely greatest.

Taking m negative and supposing α absolutely smaller than $\beta, \gamma, \ldots$, one similarly obtains:

The quotient of the $-m$th and $-(m+1)$st power sums approaches, as m increases, the root that is absolutely smallest.

Since power sums can be computed from the coefficients as symmetric functions, one need only take m sufficiently large to obtain an approximate value of the absolutely greatest or absolutely smallest root. If among the roots of the equation there are roots with the same absolute value, and this value is the greatest or the smallest, then the method is not applicable; thus, for instance, it is not applicable to real equations in which a pair of imaginary roots has greater absolute value than the real roots. If the greatest root occurs singly, but exceeds the next only slightly, the process gives accuracy only slowly.

Jacobi supplemented Bernoulli's method in one direction.[31] Suppose that among the roots $x_1, x_2, \ldots, x_n$ the roots

$$x_1, x_2, \ldots, x_k$$

have absolute values greater than those of $x_{k+1}, x_{k+2}, \ldots, x_n$, and put

$$\begin{aligned} p_m &= x_1^m + x_2^m + \cdots + x_k^m, \\ p_{m+1} &= x_1^{m+1} + x_2^{m+1} + \cdots + x_k^{m+1}, \\ &\vdots \\ p_{m+k} &= x_1^{m+k} + x_2^{m+k} + \cdots + x_k^{m+k}. \end{aligned} \tag{2}$$

If

$$(x - x_1)(x - x_2) \cdots (x - x_k) = x^k + A_1 x^{k-1} + \cdots + A_k, \tag{3}$$

then the coefficients $A_1, A_2, \ldots, A_k$ satisfy

$$\begin{aligned} p_{m+k} + p_{m+k-1}A_1 + p_{m+k-2}A_2 + \cdots + p_m A_k &= 0, \\ p_{m+k+1} + p_{m+k}A_1 + p_{m+k-1}A_2 + \cdots + p_{m+1}A_k &= 0, \\ &\vdots \\ p_{m+2k-1} + p_{m+2k-2}A_1 + p_{m+2k-3}A_2 + \cdots + p_{m+k-1}A_k &= 0. \end{aligned} \tag{4}$$

If the equation

$$x^k + A_1 x^{k-1} + A_2 x^{k-2} + \cdots + A_k = 0$$

is adjoined to (4) and the A_i are eliminated, one obtains the equation of degree k

$$\begin{vmatrix} x^k & x^{k-1} & \cdots & 1 \\ p_{m+k} & p_{m+k-1} & \cdots & p_m \\ p_{m+k+1} & p_{m+k} & \cdots & p_{m+1} \\ \vdots & \vdots & & \vdots \\ p_{m+2k-1} & p_{m+2k-2} & \cdots & p_{m+k-1} \end{vmatrix} = 0,$$

whose roots are $x_1, x_2, \ldots, x_k$. The quantities $p_m, p_{m+1}, \ldots$ may, with increasing accuracy, be replaced by the corresponding power sums of all the roots as m increases.

For example, when $k = 2$ one gets

$$x^2 + \frac{p_m p_{m+3} - p_{m+1} p_{m+2}}{p_{m+1}^2 - p_m p_{m+2}} x + \frac{p_{m+2}^2 - p_{m+1} p_{m+3}}{p_{m+1}^2 - p_m p_{m+2}} = 0,$$

an equation applicable when, in a real equation, a pair of conjugate roots has the greatest absolute value.

[31] Observatiunculae etc., Works, vol. 3, p. 280.

Applying Bernoulli's method to the sum of negative powers gives, as observed, an approximation to the absolutely smallest root. But by shifting the origin one can make any root the absolutely smallest one, although this requires knowledge of a value already approximated to a certain degree.

A formula sufficient for all cases was given by Fr. Meyer.[32] Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be the roots of $f(x) = 0$, real or imaginary. Seek a point p in the x-plane such that a circle with centre p contains only one of the points $\alpha_1, \alpha_2, \ldots, \alpha_n$, say α_1. Then the absolute values of

$$\frac{p-\alpha_1}{p-\alpha_2}, \quad \frac{p-\alpha_1}{p-\alpha_3}, \ldots, \frac{p-\alpha_1}{p-\alpha_n} \tag{5}$$

are proper fractions. Such points always exist; if necessary they must be found by Sturm's method. Choose also any function $\varphi(x)$ that has no common divisor with $f(x)$, for instance $\varphi(x) = 1$, and form the symmetric functions

$$y_m = \frac{\dfrac{\alpha_1\varphi(\alpha_1)}{(p-\alpha_1)^m} + \dfrac{\alpha_2\varphi(\alpha_2)}{(p-\alpha_2)^m} + \cdots + \dfrac{\alpha_n\varphi(\alpha_n)}{(p-\alpha_n)^m}}{\dfrac{\varphi(\alpha_1)}{(p-\alpha_1)^m} + \dfrac{\varphi(\alpha_2)}{(p-\alpha_2)^m} + \cdots + \dfrac{\varphi(\alpha_n)}{(p-\alpha_n)^m}}. \tag{6}$$

Then y_m tends to α_1 as m increases, and the convergence is faster the smaller the absolute values of the fractions (5) already are. This is immediate if (6) is written as

$$y_m = \frac{\alpha_1\varphi(\alpha_1) + \alpha_2\varphi(\alpha_2)\left(\frac{p-\alpha_1}{p-\alpha_2}\right)^m + \cdots + \alpha_n\varphi(\alpha_n)\left(\frac{p-\alpha_1}{p-\alpha_n}\right)^m}{\varphi(\alpha_1) + \varphi(\alpha_2)\left(\frac{p-\alpha_1}{p-\alpha_2}\right)^m + \cdots + \varphi(\alpha_n)\left(\frac{p-\alpha_1}{p-\alpha_n}\right)^m}.$$

§ 111. Gräffe's method of approximation

A procedure resting on a similar idea to Bernoulli's approximation was given by Gräffe and further developed by Encke.[33] It is especially suited to a practical execution of numerical computations.

Let $\alpha, \beta, \gamma, \ldots$ be the roots of an equation, and let α be absolutely the greatest of them, no other root being equal to α in absolute value. Then

$$\sqrt[m]{\alpha^m + \beta^m + \gamma^m + \cdots} \tag{1}$$

has the limit α as m grows without bound; indeed its convergence to α is even better than for the quotient considered in the previous paragraph,

$$\frac{\alpha^m + \beta^m + \gamma^m + \cdots}{\alpha^{m-1} + \beta^{m-1} + \gamma^{m-1} + \cdots}.$$

This is seen by putting, by the generalized binomial theorem,

$$\sqrt[m]{\alpha^m + \beta^m + \gamma^m + \cdots} = \alpha\sqrt[m]{1 + \left(\frac{\beta}{\alpha}\right)^m + \left(\frac{\gamma}{\alpha}\right)^m + \cdots}$$

$$= \alpha\left[1 + \frac{1}{m}\left(\frac{\beta}{\alpha}\right)^m + \frac{1}{m}\left(\frac{\gamma}{\alpha}\right)^m + \cdots\right].$$

The absolutely smallest root is obtained in the same way as the limit of

$$\frac{1}{\sqrt[m]{\dfrac{1}{\alpha^m} + \dfrac{1}{\beta^m} + \dfrac{1}{\gamma^m} + \cdots}}, \tag{2}$$

[32] Mathematische Annalen, vol. 33.
[33] Crelle's Journal, vol. 22 (1841).

and, if p is arbitrary, the root nearest to p is the limit of

$$p + \frac{1}{\sqrt[m]{\dfrac{1}{(\alpha - p)^m} + \dfrac{1}{(\beta - p)^m} + \cdots}}. \tag{3}$$

By transforming the equation $f(x)$ by the substitutions

$$y = \frac{1}{x}, \qquad y = \frac{1}{x - p},$$

the expressions (2) and (3) are obtained from (1), to which we therefore restrict ourselves.

Applying (1) to a real equation, $\alpha^m + \beta^m + \gamma^m + \cdots$ is a real number; and if the absolutely greatest root α is real, the mth root is to be taken real. If m is even, one must also know, to decide the sign, whether α is positive or negative. If necessary this is decided by substituting the approximated value in the given equation. Likewise, if (1) is applied to an imaginary equation, one must decide which of the several mth roots gives the correct approximation.

The quantities

$$s_m = \alpha^m + \beta^m + \gamma^m + \cdots$$

are particularly easy to compute when m is taken to be successive powers of 2, that is $m = 2, 4, 8, 16, \ldots$, and then give very good results. The number s_2 is the negative coefficient of the $(n-1)$st power of the unknown in the equation whose roots are the squares of the roots of $f(x)$. This equation is easily formed as follows. Collect the terms of $f(x)$ with even and odd powers of x and put

$$f(x) = \varphi(x^2) + x\psi(x^2).$$

Then

$$f_1(x) = \varphi(x)^2 - x\psi(x)^2 = 0$$

is the equation whose roots are the squares of the roots of $f(x)$. Treating $f_1(x)$ in the same manner gives an equation whose roots are the fourth powers of the roots of $f(x)$, and so on. The computation is very simple and after few steps usually leads to a good approximation.

First consider

$$x^3 - 2x - 2 = 0, \tag{4}$$

which has one positive real and two imaginary roots. The real root is greater than 1.7, and since the product of all three roots is 2 and $(1.7)^3 > 2$, the absolute value of each of the two imaginary roots is smaller than the real root. Thus the real root is the absolutely greatest one and Gräffe's method must lead to it. The equations whose roots are the second, fourth, eighth, … powers of the roots of (4) are

$$x^3 - 4x^2 + 4x - 4 = 0,$$

$$x^3 - 8x^2 - 16x - 16 = 0,$$

$$x^3 - 96x^2 - 16^2 = 0,$$

and

$$x = \sqrt[8]{96} = 1.7692\ldots$$

is already correct in the first four decimals. The next step gives no new result, because the first power of the unknown is missing in the last equation; but the following step gives the value

$$x = \sqrt[32]{85032960} = 1.769293\ldots,$$

correct in the sixth decimal.

As a second example take a fifth-degree equation, which also gives occasion to apply several propositions of the preceding paragraph. As before, the example is taken from the theory of complex multiplication of elliptic functions. The equation is

$$x^5 - x^3 - 2x^2 - 2x - 1 = 0. \tag{5}$$

We derive the equation whose roots are the squares of the roots of (5) by replacing x^2 by x in

$$(x^5 - x^3 - 2x)^2 - (2x^2 + 1)^2 = 0.$$

This gives

$$x^5 - 2x^4 - 3x^3 - 1 = 0, \tag{6}$$

and applying the same procedure twice more yields the equations whose roots are the fourth and eighth powers of the roots of (5):

$$x^5 - 10x^4 + 9x^3 - 4x^2 - 1 = 0, \tag{7}$$

$$x^5 - 82x^4 + x^3 - 36x^2 - 8x - 1 = 0. \tag{8}$$

The last of these equations is suited to discussion. By Descartes' theorem (§ 101) it has at least one positive root. If, for equation (8), one forms the functions that occur in Newton's criterion (§ 101),

$$\begin{array}{ccc} 1, & 4a_0^2 - 10a_0a_2, & 6a_2^2 - 12a_1a_3, \\ 6a_2^2 - 12a_3a_1, & 4a_4^2 - 10a_3a_5, & 1, \end{array}$$

one obtains the signs

$$+ \quad + \quad - \quad + \quad - \quad +$$

and (8) therefore has, by § 101, theorem VIII, four imaginary roots. The positive root is certainly greater than 1, since the left side of (8) is negative for $x = 0$ and $x = 1$.

To decide which root has the greatest absolute value, put $x = 1 : z$ in (8) and then, following § 96, 97, investigate how many roots lie inside the unit circle in the z-plane. Thus in

$$z^5 + 8z^4 + 36z^3 - z^2 + 82z - 1$$

put

$$z = \cos\vartheta + \mathrm{i}\sin\vartheta.$$

The real and imaginary parts are

$$\begin{aligned} \varphi &= \cos 5\vartheta + 8\cos 4\vartheta + 36\cos 3\vartheta - \cos 2\vartheta + 82\cos\vartheta - 1, \\ \psi &= \sin 5\vartheta + 8\sin 4\vartheta + 36\sin 3\vartheta - \sin 2\vartheta + 82\sin\vartheta. \end{aligned}$$

It can be shown that ψ is positive when $0 < \vartheta < \pi$ and negative when $0 > \vartheta > -\pi$, so that the unit circle is divided into two segments in which ψ has opposite signs. It follows by § 96 that only one root can lie inside the circle, namely the real root, while the imaginary roots lie outside. Hence, for equation (8), the real root has the greatest absolute value and is therefore found by Gräffe's method.

To prove this property of ψ, put $2\cos\vartheta = \xi$ and use the trigonometric formulae

$$\begin{aligned} \sin 5\vartheta &= \sin\vartheta(\xi^4 - 3\xi^2 + 1), \\ \sin 4\vartheta &= \sin\vartheta(\xi^3 - 2\xi), \\ \sin 3\vartheta &= \sin\vartheta(\xi^2 - 1), \\ \sin 2\vartheta &= \sin\vartheta\,\xi. \end{aligned}$$

This gives

$$\psi = \sin\vartheta\{\xi^4 + \xi^2(8\xi + 33) + (47 - 17\xi)\},$$

from which it is immediately seen that the factor of $\sin\vartheta$ remains positive as long as ξ lies between -2 and $+2$.

Equation (8) itself already gives a fairly good approximation to the real root,

$$x = \sqrt[8]{82} = 1.73471.$$

A more exact value, obtained by going two more steps, is 1.73469, still correct in the fifth decimal.

This method does not enclose the required root between two bounds. The sign that a sufficient degree of accuracy has been reached is that, if s_μ and $s_{\mu'}$ are two successive power sums of the roots used for the computation, then they behave with sufficient accuracy as the μth and μ'th powers of a quantity; equivalently, approximately

$$\mu \log s_{\mu'} - \mu' \log s_\mu = 0.$$

One must carry out this check for several successive terms, because in special cases the relation can appear to be satisfied and then fail again later in the computation.

§ 112. Trigonometric solution of cubic equations

We next discuss a few methods of numerical solution that apply to equations of special forms. Their aim is to make the widely available tables of trigonometric functions and logarithms useful for solving equations. We first turn to cubic equations, which we always assume in the reduced form

$$x^3 + ax + b = 0, \tag{1}$$

where a and b are real numbers. Replacing x by $-x$ amounts to replacing b by $-b$, so we may assume that b is negative. There remain three cases. Put $b = -g$, and according as a is positive or negative put $a = \pm e$. Then

$$\begin{array}{lll} \text{a)} & x^3 + ex - g = 0, & \\ \text{b)} & x^3 - ex - g = 0, & \dfrac{e^3}{27} < \dfrac{g^2}{4}, \\ \text{c)} & x^3 - ex - g = 0, & \dfrac{e^3}{27} > \dfrac{g^2}{4}. \end{array}$$

The limiting cases in which one of $e, g, 4e^3 - 27g^2$ vanishes are excluded.

To apply Cardano's formula (§ 35), set

$$R = \frac{g^2}{4} \pm \frac{e^3}{27}, \tag{2}$$

where the upper sign applies in case a), and the lower in cases b) and c). In cases a) and b) only one root is real; in case c), the irreducible case, all three roots are real. Cardano's formula gives

$$x = \sqrt[3]{\frac{g}{2} + \sqrt{R}} + \sqrt[3]{\frac{g}{2} - \sqrt{R}}. \tag{3}$$

Proceed as follows in the three cases.

a) Introduce an angle ϑ, chosen between 0° and 90°, by

$$\frac{g}{2} = \sqrt{\frac{e^3}{27}}\,\operatorname{cotg}\vartheta, \tag{4}$$

so that

$$\sqrt{R} = \sqrt{\frac{e^3}{27}}\frac{1}{\sin\vartheta},$$

and
$$x = \sqrt{\frac{e}{3}}\left(\sqrt[3]{\operatorname{cotg}\vartheta + \frac{1}{\sin\vartheta}} + \sqrt[3]{\operatorname{cotg}\vartheta - \frac{1}{\sin\vartheta}}\right) = \sqrt{\frac{e}{3}}\left(\sqrt[3]{\operatorname{cotg}\frac{\vartheta}{2}} - \sqrt[3]{\operatorname{tg}\frac{\vartheta}{2}}\right).$$
If an angle φ is now determined from
$$\operatorname{tg}\frac{\vartheta}{2} = \operatorname{tg}^3\varphi, \tag{5}$$
then
$$x = 2\sqrt{\frac{e}{3}}\operatorname{cotg}2\varphi. \tag{6}$$
The angles ϑ, φ and finally x are found from the logarithmic trigonometric tables by formulae (4), (5), (6). To obtain the imaginary roots, replace $\operatorname{tg}\varphi$ by $\varrho\operatorname{tg}\varphi$, where ϱ denotes an imaginary cube root of unity.

b) In the second case determine ϑ, again in the first quadrant, from
$$\frac{g}{2} = \sqrt{\frac{e^3}{27}}\frac{1}{\sin\vartheta}, \tag{7}$$
$$\sqrt{R} = \sqrt{\frac{e^3}{27}}\operatorname{cotg}\vartheta, \qquad x = \sqrt{\frac{e}{3}}\left(\sqrt[3]{\operatorname{cotg}\frac{\vartheta}{2}} + \sqrt[3]{\operatorname{tg}\frac{\vartheta}{2}}\right),$$
$$\operatorname{tg}\frac{\vartheta}{2} = \operatorname{tg}^3\varphi, \tag{8}$$
$$x = 2\sqrt{\frac{e}{3}}\frac{1}{\sin 2\varphi}. \tag{9}$$

c) Finally, in the last case, where three real roots occur, put
$$\frac{g}{2} = \sqrt{\frac{e^3}{27}}\cos\vartheta, \tag{10}$$
$$\sqrt{R} = \mathrm{i}\sqrt{\frac{e^3}{27}}\sin\vartheta,$$
$$x = \sqrt{\frac{e}{3}}\left(\sqrt[3]{\cos\vartheta + \mathrm{i}\sin\vartheta} + \sqrt[3]{\cos\vartheta - \mathrm{i}\sin\vartheta}\right).$$
By Moivre's theorem,
$$x = 2\sqrt{\frac{e}{3}}\cos\frac{\vartheta}{3}. \tag{11}$$
Taking ϑ again in the first quadrant, the other two real roots are
$$-2\sqrt{\frac{e}{3}}\cos\frac{\pi+\vartheta}{3}, \qquad -2\sqrt{\frac{e}{3}}\cos\frac{\pi-\vartheta}{3}.$$
All these formulae are arranged for logarithmic computation.

§ 113. Gauss's method for solving trinomial equations

Gauss gave a method for numerically solving, in a simple way, the roots of an equation containing only three terms. Such equations occur frequently and include, as special cases, all quadratic and all reduced cubic equations.

For such an equation, in the general form
$$x^{m+n} + ax^m + b = 0, \tag{1}$$

only the positive root, if it exists, is first sought. The possible negative root is obtained by replacing x by $-x$. According to the signs of a and b there are three cases, since if both signs are positive no positive root exists. Thus, with e and g positive,

$$\begin{aligned} &\text{a)} \quad x^{m+n} + ex^m - g = 0, \\ &\text{b)} \quad x^{m+n} - ex^m - g = 0, \\ &\text{c)} \quad x^{m+n} - ex^m + g = 0. \end{aligned}$$

In the first two cases Descartes' theorem gives one positive root; in the third there may be two positive roots or none.

Put, with Gauss,

$$\lambda = \frac{g^m}{e^{m+n}},$$

and reduce the three equations a), b), c) by suitable substitutions to

$$\sin^2\Theta + \cos^2\Theta = 1.$$

Use

$$\begin{aligned} &\text{a)} \quad \frac{x^{m+n}}{g} = \sin^2\Theta, \quad \frac{ex^m}{g} = \cos^2\Theta, \quad \lambda = \frac{\sin^{2m}\Theta}{\cos^{2m+2n}\Theta}, \\ &\text{b)} \quad gx^{-m-n} = \sin^2\Theta, \quad ex^{-n} = \cos^2\Theta, \quad \lambda = \frac{\sin^{2m}\Theta}{\cos^{2m+2n}\Theta}, \\ &\text{c)} \quad \frac{x^n}{e} = \sin^2\Theta, \quad \frac{gx^{-m}}{e} = \cos^2\Theta, \quad \lambda = \sin^{2m}\Theta\cos^{2n}\Theta. \end{aligned} \tag{2}$$

The last equation distinguishes the two cases in which equation c) has two positive roots or none. The maximum of $\sin^{2m}\Theta\cos^{2n}\Theta$ is

$$\frac{m^m n^n}{(m+n)^{m+n}},$$

and is reached when $\operatorname{tg}\Theta^2 = m : n$. If λ is below this bound, case c) gives two real roots; otherwise none.

In (2), λ is known, and one has to find from the tables the angle Θ satisfying the equation. If one has as yet no knowledge of the approximate position of this angle, it is expedient for the first approximation to use a table proceeding degree by degree and rounded to two decimals, and then to improve the value thus found with more accurate tables.

Gauss does not use trigonometric tables, but the addition and subtraction logarithms first introduced by him. We briefly show this in one case; the details for practical application are to be sought in Gauss's paper.[34]

The tables of addition and subtraction logarithms, as first introduced and computed by Gauss and as now found in common table works, give, for three numbers greater than 1,

$$a, \qquad b = 1 + \frac{1}{a}, \qquad c = 1 + a,$$

the Briggsian logarithms

$$A, \qquad B, \qquad C.$$

If Θ lies in the first octant, one can put

$$a = \operatorname{cotg}\Theta^2, \qquad b = \frac{1}{\cos\Theta^2}, \qquad c = \frac{1}{\sin\Theta^2}, \quad 0 < \Theta < 45^\circ, \tag{3}$$

and if Θ lies in the second octant,

$$a = \operatorname{tg}\Theta^2, \qquad b = \frac{1}{\sin\Theta^2}, \qquad c = \frac{1}{\cos\Theta^2}, \quad 45^\circ < \Theta < 90^\circ. \tag{4}$$

[34] Beiträge zur Theorie der algebraischen Gleichungen, second part (1849). Gauss, Works, vol. III, p. 85.

Apply this to case b). One must distinguish whether λ is less or greater than 2^m, since that determines whether Θ lies in the first or second octant. Thus:

$$\begin{aligned}
&\alpha)\quad \lambda < 2^m, \quad \lambda = \frac{b^m}{a^n} = \frac{c^m}{a^{m+n}} = \frac{b^{m+n}}{c^n},\\
&\qquad x^{m+n} = gc, \qquad x^n = eb, \qquad x^m = \frac{ga}{e};\\
&\beta)\quad \lambda > 2^m, \quad \lambda = a^{m+n}b^m = a^m c^m = \frac{c^{m+n}}{b^n},\\
&\qquad x^{m+n} = gb, \qquad x^n = ec, \qquad x^m = \frac{g}{ea}.
\end{aligned} \tag{5}$$

In case α) one would put

$$\log \lambda = mB - nA \tag{6}$$

and then seek the corresponding values of A and B from the table. Once these are found,

$$m \log x = A + \log g - \log e. \tag{7}$$

As an example, take

$$x^3 - 2x - 2 = 0,$$

so that $e = g = 2$, $m = 1$, $n = 2$, $\lambda = \frac{1}{2}$. Formula (6) gives

$$2A - B = 0.3010300, \tag{8}$$

and (7)

$$\log x = A. \tag{9}$$

For a first approximate value of A one uses a small excerpt from the table:

A	B
0	0.301
0.1	0.254
0.2	0.212
0.3	0.176

This gives:

A	$2A - B$	error
0.2	0.188	−0.113
0.3	0.424	+0.123

The interpolation method may then be applied by dividing the total error 0.236 between the two values of A in the ratio of the partial errors:

$$A = 0.2 + \frac{0.1}{0.236}\,0.113 = 0.247\ldots.$$

With the help of a seven-place table of Zach one obtains:

A	B	$2A - B$	error
0.247	0.1948581	0.2991419	−0.0018881
0.248	0.1944969	0.3015031	+0.0004731

and by another interpolation

$$A = 0.2477996,$$

or

$$x = 1.769292.$$

§ 114. Computation of the imaginary roots of a trinomial equation

Gauss also gave a method for computing the imaginary roots of a trinomial equation. We briefly discuss it here. We assume real coefficients, although the method also applies in the general case. Put the equation in the form

$$x^{m+n} - ex^m - g = 0, \tag{1}$$

without here restricting e and g to positive values. Let

$$x = re^{\mathrm{i}\vartheta}.$$

Equating the imaginary part in (1) to zero gives

$$r^n = \frac{e \sin m\vartheta}{\sin(n+m)\vartheta}. \tag{2}$$

Two more equations of this kind are obtained by multiplying (1) by x^{-m} and by x^{-m-n}:

$$r^{n+m} = -\frac{g \sin m\vartheta}{\sin n\vartheta}, \tag{3}$$

$$r^m = -\frac{g \sin(n+m)\vartheta}{e \sin n\vartheta}. \tag{4}$$

Any one of these three equations follows from the other two. Eliminating r from two of them gives, with

$$\lambda = \frac{g^m}{e^{n+m}},$$

$$\lambda = (-1)^n \frac{\sin^n n\vartheta \, \sin^m m\vartheta}{\sin^{m+n}(n+m)\vartheta}. \tag{5}$$

Since only one of the two conjugate roots has to be computed, ϑ may be restricted to the first quadrant; under some circumstances r must then be taken negative.

We illustrate how the angle ϑ is obtained from (5) using trigonometric tables, and then the corresponding value of r from one of (2), (3), (4), by the example already used:

$$x^3 - 2x - 2 = 0.$$

Here equation (5) becomes

$$\frac{1}{2} = \frac{\sin^2 2\vartheta \, \sin\vartheta}{\sin^3 3\vartheta}, \tag{6}$$

or

$$3 \log \sin 3\vartheta - 2 \log \sin 2\vartheta - \log \sin\vartheta = 0.3010300. \tag{7}$$

This first shows that ϑ lies between 0 and 60°. After a few trials one finds that ϑ lies between 33° and 34°. If for these two values one computes

$$3 \log \sin 3\vartheta - 2 \log \sin 2\vartheta - \log \sin\vartheta - 0.3010300, \tag{8}$$

then to three decimals one first obtains

$$0.026, \qquad -0.013,$$

and by interpolation

$$\vartheta = 33°40'.$$

One then computes (8) somewhat more accurately, say to five places, for angles near 33°40′, progressing minute by minute. From the change of sign one finds that ϑ lies between

$$33°41' \quad \text{and} \quad 33°42'.$$

Carrying out the calculation to seven places for these two values gives, again by interpolation,

$$\vartheta = 33°41'20.6''.$$

From (3) or (4) one sees that here r is negative, and r is obtained very simply from one of these equations. The Briggsian logarithms are

$$\begin{aligned} \log(-r) &= 0.0266148, \\ \log\cos\vartheta &= 9.9201547 - 10, \\ \log\sin\vartheta &= 9.7440458 - 10. \end{aligned}$$

Thus the two imaginary roots are

$$-0.884646 \pm 0.589740\,\mathrm{i}.$$

Eleventh Section.

Continued fractions.

§ 115. Transformation of rational fractions into continued fractions

If one natural number m is divided by another n according to the ordinary rules, one obtains a quotient and a remainder, positive and smaller than the divisor n. If a denotes the quotient and r the remainder, then

$$m = an + r. \tag{1}$$

All numbers $m, m', m'', \ldots$ that give the same remainder are called congruent modulo n, and Gauss denotes this by

$$m \equiv m' \pmod{n}. \tag{2}$$

Such a formula is called a congruence.

Thus, if we consider the system

$$r = 0, 1, 2, \ldots, n-1,$$

every arbitrary positive or negative integer m is congruent modulo n to one and only one of these numbers. The same property is possessed by every system

$$s = m_0, m_1, m_2, \ldots, m_{n-1}$$

chosen so that all residues r occur among their residues, and each occurs only once. Then every integer m is congruent modulo n to one and only one of the numbers s. We call such a system of numbers a complete residue system modulo n. For odd n one such complete residue system is formed by the numbers

$$\varrho = -\frac{n-1}{2},\ -\frac{n-3}{2}, \ldots, -1, 0, 1, 2, \ldots, \frac{n-1}{2}.$$

The numbers r are called the least, or least positive, residues; the numbers ϱ the absolutely least residues.

To continue the division indicated in (1), denote m, n, r by m, m_1, m_2, and denote the quotient and remainder of the division of m_1 by m_2 by a_1, m_3, and so on. Also put a_0 for a. This gives a sequence of equations

$$\begin{aligned} m &= a_0 m_1 + m_2, \\ m_1 &= a_1 m_2 + m_3, \\ m_2 &= a_2 m_3 + m_4, \\ &\vdots \\ m_{\nu-1} &= a_{\nu-1} m_\nu + m_{\nu+1}, \end{aligned} \tag{3}$$

which may be continued as long as no remainder vanishes, that is, as long as no division is exact. Since

$$m_1 > m_2 > m_3 > \cdots,$$

and since there is only a finite number of positive integers smaller than m_1, the division must terminate after finitely many steps. If m_ν is divisible by $m_{\nu+1}$, then $m_{\nu+1}$ is the greatest common divisor of m and m_1. For, by (3), $m_{\nu+1}$ divides all preceding m_μ, and conversely every common divisor of two adjacent m_μ also divides all following terms, hence also $m_{\nu+1}$.

If therefore m and m_1 have no common divisor other than 1 (that is, if they are relatively prime), then $m_{\nu+1} = 1$, and (3) may be written

$$\frac{m}{m_1} = a_0 + \frac{m_2}{m_1}, \qquad \frac{m_1}{m_2} = a_1 + \frac{m_3}{m_2}, \ldots, \qquad \frac{m_{\nu-1}}{m_\nu} = a_{\nu-1} + \frac{1}{m_\nu}. \tag{4}$$

This gives the expansion of the fraction $m : m_1$, or $m : n$, into a continued fraction. Successively,

$$\frac{m}{n} = a_0 + \cfrac{1}{\cfrac{m_1}{m_2}}, \qquad \frac{m}{n} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\cfrac{m_2}{m_3}}}, \ldots \tag{5}$$

and finally

$$\frac{m}{n} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_{\nu-1} + \cfrac{1}{m_\nu}}}}}. \tag{6}$$

For more convenient printing we also write

$$\frac{m}{n} = \left(a_0, \frac{m_1}{m_2}\right), \qquad \frac{m}{n} = \left(a_0, a_1, \frac{m_2}{m_3}\right), \qquad \frac{m}{n} = (a_0, a_1, a_2, \ldots, a_{\nu-1}, m_\nu). \tag{7}$$

The following is still to be noted. Every positive rational fraction can be put in the form $m : n$ with m and n relatively prime; hence every such fraction can be expanded into a continued fraction of the form (6) or (7). The same holds for negative rational fractions, if negative values are allowed for a_0. The numbers $a_1, a_2, \ldots, a_{\nu-1}, m_\nu$ are then still positive integers. These numbers are completely and unambiguously determined. Only one arbitrariness remains.

By the determination just made, the last partial denominator m_ν is greater than 1, since $m_{\nu+1}$ was assumed to be the first remainder equal to 1. Hence, if a_ν is a positive integer, we may put either

$$m_\nu = a_\nu$$

or

$$m_\nu = a_\nu + \frac{1}{1}$$

and thus obtain the two continued-fraction representations

$$\frac{m}{n} = (a_0, a_1, a_2, \ldots, a_\nu) = (a_0, a_1, a_2, \ldots, a_\nu, 1). \tag{8}$$

I. Every rational fraction can therefore be expanded into a continued fraction having, at will, either an even or an odd number of partial denominators.

The numbers

$$\frac{m_1}{m_2}, \quad \frac{m_2}{m_3}, \quad \frac{m_3}{m_4}, \ldots$$

that occur in (5) are called the final quantities of the continued fraction; the numbers $a_0, a_1, a_2, \dots$ will be called the partial denominators, although a_0 is not actually a denominator.

If m is replaced by a number m' congruent to it modulo n, then the two fractions $m : n$ and $m' : n$ differ only by an integer; their continued-fraction expansions therefore differ only in their first terms a_0.

§ 116. Continued-fraction expansion of irrational numbers

If x is a real irrational number, then there is always one and only one integer a_0 such that x lies between a_0 and $a_0 + 1$. The integer a_0 is positive if $x > 1$, zero if x is a positive proper fraction, and negative if x is negative. Thus put

$$x = a_0 + \frac{1}{x_1}.$$

Then x_1 is a positive improper fraction; so if a_1 is the integer immediately below x_1, then a_1 is positive. Put

$$x_1 = a_1 + \frac{1}{x_2}, \qquad x_2 = a_2 + \frac{1}{x_3}, \quad \dots, \quad x_{n-1} = a_{n-1} + \frac{1}{x_n}.$$

Thus $x_1, x_2, \dots, x_n$ are positive improper fractions, and $a_1, a_2, \dots, a_{n-1}$ are positive integers. In the notation of the preceding paragraph, x is therefore the continued fraction

$$(a_0, a_1, a_2, \dots, a_{n-1}, x_n).$$

If x is irrational, this continued-fraction expansion can be continued without bound; that is, n may be taken arbitrarily large. The following expansion arises from the preceding one by putting

$$x_n = a_n + \frac{1}{x_{n+1}}.$$

Here also the integers $a_0, a_1, a_2, \dots, a_{n-1}$ are called the partial denominators, and x_n the final quantity.

The continued-fraction expansion of rational numbers is contained in this as a special case. By proposition I, a continued fraction can be continued at most by one term when the final quantity is an integer.